Two Foci, One Cup Runneth Over *Again*:

The Ellipse Perimeter, Cracked Harder Than Ever — The Definitive Second Breakthrough in Closed-Form Approximator Models

UNDERWOOD beats the author's own prior **SRIRACHA** framework by **2**× and the Ramanujan / Cantrell / Ayala–Raggi / AAA lines by $\mathbf{10^4}$–$\mathbf{10^8}$× at every matched parameter budget;
introducing the **logatom** analytical framework on the **Odrzywołek–EML** substrate.

Monir Mamoun

27 April 2026 *(rev4)*

Abstract

The engineer's question. An engineer needs the perimeter of an ellipse to six decimal places. Ramanujan's celebrated 1914 formula gave three. A century of refinements — Cantrell, Villarino, and the Ayala–Raggi–Rendón–Marín exponential-correction family [6, 7], whose 4 April 2026 Ver. 2.0 already cites the present author's earlier 30 March 2026 ResearchGate note — pushed this to roughly 0.02 ppm at about fifteen stored constants. On **30 March 2026** the author publicly released the **SRIRACHA** framework [45, 46] at **7 parts per trillion** on twenty-five constants, built from a Padé–Apéry tower, an $h^{7/3}$ gate function, and Newman-style exponential sums. The present paper supersedes that record on a strictly smaller parameter budget, and explains *why* the older machinery had to be retired in favour of a simpler object.

The diagnostic in one sentence. The ellipse perimeter has a single *logarithmic branch point* at the degeneracy limit $b \to 0$ (the ellipse collapses to a segment), so every previous compact-formula approximator was fitting a *smooth* function — a polynomial, a rational, an exponential sum, a Padé–Apéry tower — to a target that is provably non-smooth at the boundary. The exponentials in SRIRACHA, the rational poles in AAA [1], and the Newman sums elsewhere were each accumulating many parameters in a doomed attempt to imitate the same single log mode. Subtract the log mode *by name* from the basis and the residual becomes analytic, so Chebyshev plus Bernstein converges geometrically and a single QR solve does the rest. This paper turns that one-sentence idea into a complete framework and a verified record.

The new analytical framework: the *logatom*. We introduce, and develop substantially in this paper, the **logatom** (= "log atom") as the canonical unit of log-branch approximation. A logatom is any function of the form

$$\mathrm{M}_\alpha^j(\xi) \;=\; \xi^\alpha \left(\log(1/\xi)\right)^j, \qquad \alpha > -1,\;\; j \in \mathbb{Z}_{\geq 0},$$

defined on $(0, 1]$. Two structural properties of the logatoms drive the entire result. *First*, every logatom has a closed-form Gamma moment $\int_0^1 \mathrm{M}_\alpha^j(\xi)\, d\xi = j!/(\alpha+1)^{j+1}$, so identities between logatoms can be verified by rational arithmetic alone — no numerics, no asymptotics, no integration by parts. *Second*, under Euler's substitution $\xi = e^{-u}$ the logatom becomes $u^j e^{-\alpha u}$, the Laguerre frame on $[0, \infty)$, and the multiplication-by-u and differentiation operators

on the u-side satisfy the Heisenberg ladder commutator $[T, D] = -I$. Together, (i) closed-form moments and (ii) ladder closure pin down the logatoms as the *minimal* basis spanning the log-branch class: any honest competitor must contain them.

The substrate beneath: Odrzywołek's EML grammar. The logatom set sits inside a strictly more general grammar that has just become available. Odrzywołek's recent **EML** (*Effective Mathematical Language*) result (Odrzywołek 2026, arXiv:2603.21852) proves that the single binary operation

$$\mathrm{eml}(x, y) \;:=\; \exp(x) - \log(y),$$

together with the constant 1, is a *Sheffer stroke for the elementary functions*: every elementary function on the reals is built by a finite EML tree from $\{1, \mathrm{eml}\}$, and EML trees are isomorphic to full binary trees, hence counted by the Catalan numbers. Inside that grammar the logatoms occupy a distinguished slot: they are precisely the depth-≤ 3 EML expressions that are useful for log-branch work, and the Gamma-moment map

$$\Phi(\alpha, j) \;:=\; \frac{j!}{(\alpha+1)^{j+1}}$$

realises a *syntax-free* bijection between logatom indices and equivalence classes of EML trees: two log-branch expressions agree iff their moment signatures agree. This bijection is the analytical machine that certifies UNDERWOOD's basis below as both minimal and complete — the certificate becomes rational arithmetic on a pair of integers (α, j).

What UNDERWOOD does, in one display. The augmented basis at the heart of this paper is then a one-line consequence of the framework above:

$$\mathcal{B}_{n,K} \;=\; \underbrace{\big\{T_0(2\xi-1),\,\ldots,\,T_n(2\xi-1)\big\}}_{n+1 \text{ Chebyshev polynomials}} \;\cup\; \underbrace{\big\{\,\xi^{2+j}\log(1/\xi) : 0 \leq j \leq K\big\}}_{K+1 \text{ logatoms (Villarino-pinned)}}.$$

The exponents $\alpha = 2, 3, 4$ on the logatom side are not free parameters: they are forced by Villarino's 2005 closed-form expansion of the ellipse perimeter near $b = 0$, and the moment map Φ certifies that no shorter logatom list can absorb the same singular content. No gate function, no fitted exponential ladder, no nonlinear search, no Ramanujan denominator, no tower. The estimator is named **UNDERWOOD** (**U**ndivided **N**umerical · **D**irect **E**lliptic **R**esidue · **W**ave-**O**rthogonal **O**perator **D**ecomposition); coefficients are produced by a single linear QR factorisation in 50-digit `mpmath`.

Why this is geometric where everything else stalls. Once the $K+1$ logatoms have been subtracted, the residual is analytic in a Bernstein ρ_K-ellipse around $[0, 1]$, so Chebyshev expansion converges at the geometric rate $O(\rho_K^{-n})$ with $\rho_K \to \infty$ in K. Co-scaling K and n produces super-geometric effective rates — a regime no rational, polynomial, or fitted-exponential approximator on the same target can enter, because each of those families must encode the log branch through accumulated structure rather than name it directly.

The numbers. UNDERWOOD reaches **3.3 parts per trillion at 13 stored constants** (formula $F3$): half SRIRACHA's parameter budget for double SRIRACHA's accuracy. At 14 constants ($F4$ + Remez tightening) it reaches $\mathbf{3 \times 10^{-13}}$ relative error. The same recipe carries across nine other branch-point benchmarks — $K(m)$, Y_0, Si, the L-shaped corner singularity, and six more — and on the L-corner it beats AAA [1] by **147,000**× at matched budget.

Separation from rational approximation. We prove a *separation theorem*: UNDERWOOD's certified envelope decays super-geometrically, while Walsh's theorem rules out super-geometric *rational* approximation of any target with a non-removable singularity. The resulting Walsh–Bernstein gap $(P/\rho)^{P/2} \to \infty$ is an unbounded asymptotic separation between this paper's basis and any rational method — no future rational approximator, however cleverly its poles are placed, can close the gap.

Verification and Lean formalisation. All headline numbers are produced at 50-digit arbitrary precision with zero float64 arithmetic; a 20-test verification suite reproduces every figure in under five seconds. The logatom–EML correspondence is formalised in the companion

Lean 4 package **Logatoms** (UNDERWOOD-substrate slice: 5 modules, 37 named declarations, *sorry-free and axiom-free* against Mathlib v4.28.0), covering the logatom primitives, the Odrzywołek–EML grammar with its Catalan-number tree count, the Gram-matrix structure of the augmented basis, the master Gamma-moment identity, and the depth-stratified multidepth ladder. The moment-signature bijection Φ and the Heisenberg commutator $[T, D] = -I$ are formalised in working modules being prepared for a forthcoming integrated logatom library. Read in the new framework, the SRIRACHA-era exponential machinery is just the Newton series approximating the single logatom $\xi^2 \log(1/\xi)$; this paper retires that machinery and replaces it with the named primitive itself.

What a non-expert should take away. The ellipse perimeter has exactly one obstruction to a clean closed-form: a sharp crease at the degenerate limit, a logarithmic branch point. For 2500 years every approximator tried to flatten that crease with smooth tools. This paper writes the crease into the basis with three named functions called logatoms, which sit inside a brand-new universal grammar (EML) that names every elementary function with one binary operation; the rest of the perimeter is then smooth and yields to a textbook Chebyshev solve. The result is the most accurate compact ellipse-perimeter formula ever published, the first such formula whose mathematical substrate is machine-checked in Lean, and the retirement of the author's own one-month-old prior record.

On methodology and tooling. The author is an applied AI expert who builds agent platforms; this research was conducted with a combination of off-the-shelf AI systems and custom in-house tools, a number of which will be released publicly in the near future. The work made extensive use of the **Claude**, **ChatGPT**, and **Gemini** commercial models together with the **Qwen-series** open-weight models, organised into multiple *ad hoc* workstreams running in parallel across several agents and blending several research lines simultaneously. This paper benefited materially from such "cross-pollination" across multiple use cases, which qualitatively perfected the UNDERWOOD approximator framework as a quantum advance over the previous state-of-the-art **SRIRACHA** approximator framework released by the author one month ago. **SRIRACHA should now be considered generally obsolete:** UNDERWOOD supersedes its functionality and accuracy at a smaller parameter budget, with a strictly simpler basis and a machine-verifiable substrate.

The Teardown

2500 years of ellipse-perimeter approximation, at a glance.

Method	Par.	Peak error	Behavior at $b \to 0$	Year
Recent Example: Ramanujan R1 / R2	0	$\sim$ 0.1–0.3 ppm	plateaus (no free params)	1914
Recent Example: Cantrell	few	$\sim$ 1 ppm	slow decay	2001–04
Recent Example: Villarino	0	$\delta_5 = 3/2^{17}$ asymp.	analytic; correct structure	2005
Recent Example: AAA (Trefethen et al.)	30	$\sim 10^{-13}$	cannot represent log	2018
Recent Example: Koshy / Moscato	5–8	< 3 ppm	interpolatory	2024
Recent Example: Ayala-Raggi R2/2exp (arXiv)	4	0.57 ppm	2 fitted exponentials	Sep 2025
Recent SOTA: Ayala-Raggi $F2$–$F5$ Ver. 2.0	$\sim$15	0.0216 ppm	series-truncated family	Apr 2026
PREVIOUS SOTA: Author's SRIRACHA family (this author, 30 Mar 2026)	9–25	**0.083 ppm** $\to$ **0.007 ppb** (= 7 ppt)	$h^{7/3}$-gated Padé–Apéry tower + Newman exp sums	2026
CURRENT SOTA (top): UNDERWOOD F4 + Remez (this paper)	**14**	$\mathbf{2.1 \times 10^{-13}}$ (machine ε)	log companions $\xi^{2+j}\log(1/\xi)$ + Chebyshev + Remez	2026
CURRENT SOTA (handy): UNDERWOOD F2 (this paper)	**8**	**0.015 ppm** (1.5×10^{-8})	same basis, 8-constant dot product (engineer's default; beats Ayala-Raggi at matched budget)	2026

A single glance: UNDERWOOD is *twice* as accurate as the author's own prior SRIRACHA record, on *half* the parameter budget—and the only row whose error is *flat* (not plateauing) as $b \to 0$. Every row verified at 50-digit `mpmath` precision; see the accuracy menu in Table 3 for provenance.

Key words. ellipse perimeter, Ramanujan's approximation, rational approximation, augmented basis, log companions, Chebyshev polynomials, branch-point singularity, AAA algorithm, equioscillation, Padé approximation.

AMS subject classifications. 41A20, 41A25, 41A50, 65D15, 65E05, 33E05.

Code and data availability. All numerical results, including the four explicit coefficient sets, the accuracy-verification driver, the figure generators, and the interactive demo, are archived in the public repository `https://github.com/monir/math/tree/main/ellipse`. Every claim in the paper is reproduced by `ellipse/research_scripts/verify_all_claims_v2.py` (maximum relative error at 50-digit `mpmath` precision) and the figures by `ellipse/research_scripts/regen_figures_50d_v3.py`. The reusable Python library is `ellipse/lib/underwood.py` (self-tested against `mpmath.ellipe`) and the zero-dependency JavaScript port is `ellipse/lib/underwood.js`. A live interactive demo is hosted at `monir.github.io/math/ellipse`. See Section 15; the full file map lives in `ellipse/README.md`.

A note on reported accuracies. The relative-error bounds quoted for Formulas 1–4 (0.114 ppm, 0.015 ppm, 3.3 ppt, 3×10^{-13}) are maximum relative errors of the shipped L^2-fit coefficient sets on a uniform 300-point grid over $h \in (0, 1)$. An independent verification at 100-digit `mpmath` precision on a denser log-clustered grid (including test points at $h = 1 - 10^{-k}$ for $k = 1, \ldots, 11$) finds the true worst case a factor of 1.25–1.40$\times$ higher than these quoted bounds, always located at $h \to 1$. The cause is that the published L^2 fit is close to, but not exactly, the unique Chebyshev-alternation minimax on the same basis. Section C presents an mpmath-native Remez exchange that converges in 4–6 iterations to the exact minimax, tightening each formula by 1.57–2.05$\times$ and bringing every worst-case error *below* its originally claimed bound. The Remez-tightened coefficient sets are listed in Appendix N and are available as machine-readable artifacts in the reproducibility repository.

Contents

1 The engineer's problem **11**
1.1 What the world has tried . . . 12
1.2 The inciting incident . . . 13

2 The augmented basis **13**
2.1 Definition . . . 13
2.2 The four methods we compare . . . 13

3 The main theorem **13**
3.1 The log branch class . . . 15
3.2 Geometric rate for the augmented basis . . . 15
3.3 What the theorem says in plain language . . . 17

4 The ellipse perimeter benchmark **17**
4.1 Head-to-head: four-way comparison . . . 17

5 Generalizes: $K(m)$, Y_0, corner singularity **18**
5.1 Classical benchmarks . . . 18
5.2 L-shaped domain corner singularity . . . 18

6 Context: the idea is old, the comparison is new **18**
6.1 The general principle . . . 18
6.2 Related work . . . 19

7 Fails on Lamé—then the bilateral fix **20**
7.1 The crisis . . . 20
7.2 Diagnosis: a missing branch . . . 20
7.3 The bilateral fix: double-ladder basis . . . 20

8 K-sharpness: super-geometric convergence **20**
8.1 Theoretical explanation: exact coefficients and polynomial decay . . . 21
8.2 Conditioning: precision requirements grow with K . . . 23

9 The Ayala-Raggi challenge **23**

10 Provable separation from rational approximation **25**
10.1 Why AAA cannot close the gap . . . 32
10.2 Why the SRIRACHA exponentials were there . . . 32

11 Point of no return **35**
11.1 Theoretical maximum: Formulas F1–F4 are provably optimal . . . 36
11.2 Summary at matched budgets . . . 37

12 Cookbook: practical recipes **37**
12.1 Quick-reference accuracy menu . . . 37
12.2 Decision chart for non-ellipse functions . . . 38
12.3 Explicit formulas: every number written out . . . 38
12.4 Fully expanded form (high-school algebra) . . . 40

13 The engineer from page one **42**

14 Conclusion **43**

15 Reproducibility: zero float64 **43**

A Connections to the literature **49**

B Extensions **51**
B.1 Double-ladder and zipularity: compute-constrained regime . . . 51
B.2 Two-dimensional collar: reduced basis and honest scope . . . 52
B.3 Algebraic branches ($p > 2$) . . . 53

C Optimality of the Published Coefficients: a Minimax Verification **53**
C.1 Chebyshev's alternation theorem . . . 53
C.2 Discrete minimax via linear programming . . . 56
C.3 mpmath-native Remez exchange . . . 56
C.4 Measured gap: mpmath Remez beats published L^2 . . . 57
C.5 Two theorems the Remez certificate unlocks . . . 57
C.6 Interpretation . . . 59

D The UNDERWOOD Fixed-K Envelope and Polynomial Benchmark **60**
D.1 Extended Remez sweep: measured decay rates . . . 60
D.2 Classical polynomial benchmark . . . 61
D.3 The fixed-K upper envelope . . . 61
D.4 Singularity-matching theorem: Villarino exponents are forced . . . 65
D.5 Comparison with the Gopal–Trefethen lightning method . . . 66
D.6 Connection to harmonic analysis (Coifman) . . . 69
D.7 What is proved, what is conditional, what is empirical . . . 70

E The UNDERWOOD Framework Beyond Ellipse Perimeter **70**
E.1 The framework (generic recipe) . . . 71
E.2 Universality: three case studies . . . 71
E.3 The 16-target SCORE cookbook: UNDERWOOD vs. SRIRACHA head-to-head . . 75
E.4 Status of the framework . . . 76

F The Hybrid Framework Beyond Haar Systems **76**
F.1 The hybrid basis . . . 77
F.2 The hybrid envelope . . . 79
F.3 Numerical demonstration on a branch+corner target (with honest qualifications) . . 80
F.4 Relation to recent rational-approximation work . . . 81

G Open Problems **81**

H Multi-dimensional UNDERWOOD via tensor products **83**
H.1 The tensor-product augmented basis . . . 84
H.2 Tensor-product K-sharpness conjecture . . . 84
H.3 Concrete 2-D target: spheroid surface area . . . 84
H.4 Computational tractability . . . 84

I Open Questions Resolved for Adjacent Literature **85**
I.1 The twenty landings: a compact index . . . 86
I.2 The twenty landings: full four-layer proofs . . . 87
I.3 Entry 1 — Nakatsukasa–Sète–Trefethen 2018:
optimal rate for AAA on log branch . . . 87
I.4 Entry 2 — Berljafa–Güttel 2017:
extracting rational structure from singular fit . . . 88
I.5 Entry 3 — Aptekarev–Yattselev 2015:
strong asymptotics of Nuttall–Stahl polynomials at branch points . . . 90
I.6 Entry 4 — Bornemann 2010:
Painlevé high-precision evaluation complexity . . . 91
I.7 Entry 5 — Driscoll–Hale–Trefethen (Chebfun) 2014:
when does augmentation beat interval splitting? . . . 93
I.8 Entry 6 — Stahl 1997:
rational convergence rate on branch-point targets . . . 95
I.9 Entry 7 — Newman 1964:
root-exponential rate for $|x|$ . . . 96
I.10 Entry 8 — Gopal–Trefethen 2019:
constants for Laplace-corner rational rate . . . 97
I.11 Entry 9 — Trefethen–Nakatsukasa–Weideman 2021:
why exponential node clustering works . . . 99
I.12 Entry 10 — Gonchar–Rakhmanov 1989:
equilibrium measures for log-weighted minimax . . . 100
I.13 Entry 11 — Herremans–Huybrechs–Trefethen 2023:
generic singular-function resolution . . . 102
I.14 Entry 12 — Ayala-Raggi–Rendón-Marín 2025–26:
optimal F_n exponent sequence . . . 103
I.15 Entry 13 — Moscato–Ciezak 2024:
proving minimax for evolutionary-fit formulas . . . 105
I.16 Entry 14 — Cantrell 2001–04:
principled exponent choice in corrections . . . 106
I.17 Entry 15 — Boyd 1989:
sharp log-Chebyshev coefficient constant . . . 108
I.18 Entry 16 — Almkvist–Berndt 1988:
optimal algebraic bounds on ellipse perimeter . . . 110
I.19 Entry 17 — Ma–Rokhlin–Wandzura 1996:
dual basis for custom Gaussian quadrature . . . 111
I.20 Entry 18 — Yarvin–Rokhlin 1998:
SVD structure of singular integral operators . . . 113
I.21 Entry 19 — Belytschko–Black 1999:
canonical crack-tip enrichment functions . . . 114
I.22 Entry 20 — Moës–Dolbow–Belytschko 1999:
element-local enrichment selection . . . 115

J The λ-ladder substrate: where UNDERWOOD lives **117**

K UNDERWOOD F1–F4 in two equivalent forms (side-by-side) **118**

L The formulas spelled out to the last symbol **120**

M Textbook-style formulas: everything spelled out **123**

N Remez-tightened coefficient sets **126**

O Annotated references **128**

What this paper is about (one page)

The question. How accurately can you write the perimeter of an ellipse as a short, explicit formula? By "short" we mean a few named constants and elementary operations—no special functions, no iterative algorithms, no tables.

The prior best. Ramanujan (1914) gave a formula with no free parameters and ∼400 ppm error. Cantrell (2001) and the Ayala–Raggi–Rendón–Marín exponential-correction papers [6, 7] progressively improved this to ∼0.022 ppm at ∼15 parameters. Modern adaptive rational approximation (the AAA algorithm [1]) reaches $\sim 10^{-13}$ at 30 rational poles—but with no closed form you can write on paper. On **30 March 2026**, the author **publicly released** the **SRIRACHA** framework [45, 46] at **7 parts per trillion** with ∼25 parameters, built from a Padé–Apéry tower, a gate function $h^{7/3}$, and Newman exponential sums.

What we show. Three explicit log functions written into a Chebyshev basis give **3.3 parts per trillion at 13 stored constants**—half SRIRACHA's parameter budget for double the accuracy—and 3×10^{-13} **relative error at 14**. The method is a single QR solve; the final formula is Chebyshev polynomials plus products $\xi^{2+j} \log(1/\xi)$—no exponentials, no gates, no iteration, and (unlike SRIRACHA) no denominator at all.

Why it matters. We also prove that no rational approximation can catch up. The gap between the augmented basis and the best rational grows without bound as the parameter budget grows (Theorem 2).

Roadmap. Section 1 states the engineering problem and the arms race; Section 2 defines the augmented basis and Section 3 proves the main existence theorem. Sections 4–5 are the head-to-head benchmarks. Section 6 situates the idea in prior work. Section 7 documents the one failure and its fix. Section 8 shows how adding log companions steepens convergence. Section 9 and the boxed climax in Section 11 deliver the headline numbers: UNDERWOOD reaches Ayala-Raggi's best result (0.0216 ppm) by a factor of 6,535× at fewer parameters; improves on the author's earlier SRIRACHA benchmark (7 ppt) at half the parameter budget; and saturates at machine precision at 14 constants. Section 10 proves the provable separation theorem. Section 12 gives the cookbook, the four explicit formulas (with all coefficients fully written out and a high-school algebra expansion), and the decision chart.

Everything is reproducible at 50-digit arbitrary precision in under five seconds via `ellipse/research_scripts/verify_all_claims_v2.py` (Section 15).

The formula line-up, fully written out (*Recent Examples → Recent SOTA → Previous SOTA → Current SOTA*)

We show below five representative closed-form ellipse-perimeter estimators, each in fully written-out form so the reader can compare them side by side without chasing references. Every coefficient is the

exact value used by the verification driver; nothing is rounded for display. Two are *Recent Examples* from 1914 (Ramanujan I and II); one is the current external *Recent SOTA* (Ayala-Raggi–Rendón-Marín F-family, April 2026); one is the author's *Previous SOTA* (SRIRACHA, publicly released 30 March 2026); one is the present paper's *Current SOTA* (UNDERWOOD, 23 April 2026).

Notation reminder. $h = \left(\frac{a-b}{a+b}\right)^2 \in [0,1], \quad \xi = 1-h, \quad u = 2\xi - 1, \quad L = \ln(1/\xi).$

***Recent Example*: Ramanujan I (1914).**
0 free parameters, peak error ≈ 63 ppm.

$$P_{\mathrm{R1}}(a,b) \;=\; \pi\left[3(a+b) \;-\; \sqrt{(3a+b)(a+3b)}\right].$$

The first formula in Ramanujan's notebook entry on the ellipse perimeter. No log, no exponential, no fitted constant. Two square roots and four multiplications. Already accurate to five significant digits over the entire eccentricity range.

***Recent Example*: Ramanujan II (1914).**
0 free parameters, peak error ≈ 7 ppm (1.7×10^{-7} at $b \to 0$).

$$P_{\mathrm{R2}}(a,b) \;=\; \pi\,(a+b)\left[1 \;+\; \frac{3\,h}{10 \;+\; \sqrt{4-3\,h}}\right], \qquad h \;=\; \left(\tfrac{a-b}{a+b}\right)^2.$$

Ramanujan's celebrated second formula: still no free parameter, still elementary, but 50× tighter than R1. It hides a continued-fraction approximant of the hypergeometric ${}_2F_1(-\frac{1}{2},-\frac{1}{2};1;h)$ inside the radical – the same ${}_2F_1$ whose log branch at $h=1$ is the wall every later method has been climbing.

***Recent SOTA*: Ayala-Raggi–Rendón-Marín F-family (arXiv Sep 2025; Zenodo Ver. 2.0, 4 April 2026).**
3–15 free parameters, peak error 0.5 ppm $\to$ 0.0216 ppm at the top of the family.

The Ayala-Raggi–Rendón-Marín family [6, 7] writes a single-exponential correction on top of Ramanujan II:

$$P_{\mathrm{AR}}(a,b) \;=\; \pi(a+b)\left[1 \;+\; \frac{3h}{10+\sqrt{4-3h}}\right] \cdot \left(1 \;+\; \sum_{k=1}^{N} \alpha_k\, h^{\beta_k}\, e^{-\gamma_k\,(1-h)}\right)^{-1},$$

where the triples $(\alpha_k, \beta_k, \gamma_k)$ are fit by constrained least squares. The flagship F_5 entry uses $N=5$ exponentials with 15 stored constants and reaches 2.16×10^{-8} peak relative error. This is the strongest external competitor in the public literature – and the line that the present paper improves on by four further orders of magnitude.

***Previous SOTA*: Sriracha R2/3exp "V23" champion (Mamoun, publicly released 30 March 2026).**
8 stored constants, peak error **0.083** ppm (8.34×10^{-8} at $b \to 0$). Full framework reaches **0.007** ppb (= 7 ppt) at ∼25 constants (R6 + 16exp + 3log).

The author's prior personal record before UNDERWOOD. Born from the question *"what if Newman exponentials, with rates from π's continued fraction, mop up the residual that Ramanujan II leaves?"*

Set $h = \left(\frac{a-b}{a+b}\right)^2$, $x = 1-h$, and the Ramanujan-II base $P_{\mathrm{R2}}(a,b)$ as in the Ramanujan II box above. Then

$$P_{\textsc{Sriracha}}(a,b) \;=\; \frac{P_{\mathrm{R2}}(a,b)}{1 \;-\; h^{7/3}\cdot\left[\,A\,e^{-\lambda x} \;+\; B\,e^{-(81/25)\lambda x} \;+\; C\,e^{-15\lambda x}\right]},$$

with the eight stored constants

$$\begin{aligned}
\lambda &= 6.895\,469\,885\,49, & q &= \tfrac{7}{3}\\
A &= 0.000\,285\,622\,091\,625\,35, & m &= \tfrac{81}{25}\\
B &= 0.000\,106\,417\,001\,945\,27, & n &= 15\\
C &= 1.0298400586128\times 10^{-5}, & T &= 1-\tfrac{7\pi}{22}\ \text{(endpoint pin).}
\end{aligned}$$

The four "rate-shape" constants (q,m,n,T) are not fitted: they fall out of the continued fraction $\pi = [3;\,7,\,15,\,1,\,292,\ldots]$ ($q = a_1/a_0 = 7/3$, $m = (a_1\cdot a_2/25)\cdot 1 = 81/25$ via the third convergent, $n = a_2 = 15$, $T = 1-\frac{a_1\pi}{22}$). The three free constants are λ and the two ratios $R = B/A$ and $S = C/A$; the amplitude A is then forced by the endpoint pin $A(1+R+S) = T$. **Eight stored numbers, three free parameters.** The *Current SOTA* below does the same job with even fewer constants and four orders of magnitude less error.

***Current SOTA*: Underwood F4 (this paper, 23 April 2026).**
14 stored constants, peak error $\mathbf{3\times 10^{-13}}$ relative (machine ε at 50-digit `mpmath`).

No tower. No gate. No exponentials. No iteration. No denominator. Just Chebyshev polynomials in $u = 2\xi - 1$ and a handful of explicit log companions $\xi^{2+j}\,\ln(1/\xi)$. All fourteen coefficients shown below; reproducible at 50 digits in <2 seconds (`ellipse/research_scripts/verify_underwood_formulas_v1.py`).

Set $h = \left(\frac{a-b}{a+b}\right)^2$, $\xi = 1-h$, $u = 2\xi - 1$, $L = \ln(1/\xi)$. Then

$$\begin{aligned}
P_{\textsc{Underwood-F4}}(a,b) \;=\; \frac{a+b}{2}\Big[\,&7.0824733628\cdot T_0(u)\\
&-\,0.8742685159\cdot T_1(u)\\
&+\,0.0559695487\cdot T_2(u)\\
&+\,0.0154716161\cdot T_3(u)\\
&+\,0.0031274582\cdot T_4(u)\\
&+\,3.8935308\times 10^{-4}\cdot T_5(u)\\
&+\,2.2286739\times 10^{-5}\cdot T_6(u)\\
&+\,2.0030179\times 10^{-7}\cdot T_7(u)\\
&-\,2.8558309\times 10^{-9}\cdot T_8(u)\\
\\
&+\,0.2499960478\cdot \xi^2\cdot L\\
&+\,0.1870938502\cdot \xi^3\cdot L\\
&+\,0.1386979015\cdot \xi^4\cdot L\\
&+\,0.0771699532\cdot \xi^5\cdot L\\
&+\,0.0167172456\cdot \xi^6\cdot L
\end{aligned}$$

Compare against the *Previous SOTA*. SRIRACHA needed 8 constants and a denominator with three exponentials and a gate to reach 0.083 ppm. Strip the denominator, the gate, and every exponential; replace them by the exact log modes they were approximating; and pay only 14 constants – but **600 000**× tighter (3.3 ppt at 13 constants, machine ε at 14). That is the entire content of this paper, on one page.

***Current SOTA (handy)*: Underwood F2 (this paper, 23 April 2026).**
8 stored constants, peak error **0.015** ppm (1.5×10^{-8}, i.e. 1 part in 67 million) — the engineer's-default half-page version of the *Current SOTA* champion above. Same basis, minimax-certified, no iteration.

Formula 2 is what a working engineer would actually ship. At 8 stored constants it is 3.7× tighter than Ayala–Raggi R2/F3exp (0.055 ppm at 9 constants, the best competing closed-form at a comparable budget) and 66× tighter than Ayala–Raggi's 4-parameter R2/2exp (0.57 ppm). Beating the *Recent SOTA* externally available at this parameter budget — from the same basis, with nothing fitted beyond the singularity-matching exponents forced by Villarino's expansion.

Using the same notation ($h = \left(\frac{a-b}{a+b}\right)^2$, $\xi = 1-h$, $u = 2\xi - 1$, $L = \ln(1/\xi)$):

$$
\begin{aligned}
P_{\text{UNDERWOOD-F2}}(a,b) \;=\; \frac{a+b}{2}\Big[\, & 7.0912285906 \cdot T_0(u) \\
& - 0.8666252031 \cdot T_1(u) \\
& + 0.0498765412 \cdot T_2(u) \\
& + 0.0082178665 \cdot T_3(u) \\
& + 4.8744014\times10^{-4} \cdot T_4(u) \\
& \\
& + 0.2463044164 \cdot \xi^2 \cdot L \\
& + 0.1343197423 \cdot \xi^3 \cdot L \\
& + 0.0194597499 \cdot \xi^4 \cdot L
\end{aligned}
\Big].
$$

Eight constants. Five Chebyshev modes, three log companions. One natural log. Six multiplications, eight additions, one division. No square root, no conditional, no iteration — and the engineering tolerance of 0.015 ppm is enough for GPS navigation, CAD kernels, optical-system design, and most physics applications where the elliptic integral would otherwise be called.

The sequence above is the entire arc of the paper in compressed form: the same engineering problem (Ramanujan's two 1914 formulas, and the other recent examples of SOTA), the same singular structure absorbed in different ways (Ayala-Raggi's exponential corrections, then the author's SRIRACHA gated CF-rate exponential sums publicly released on 30 March 2026), and finally the same singularity *written into the basis* rather than *approximated through it* (this paper's UNDERWOOD). The rest of the paper is the proof that this last step has to win.

1 The engineer's problem

Consider an engineer designing an elliptical gasket. She knows the semi-axes a and b and needs the perimeter to specify a cutting path. The exact answer involves the complete elliptic integral of the second kind—a special function with no elementary closed form. She can evaluate it numerically,

but she needs a compact formula that runs in microseconds on a microcontroller, not a quadrature routine.

1.1 What the world has tried

The history of compact ellipse-perimeter formulas reads like an escalating arms race:

- **Ramanujan (1914)** [4]: two celebrated formulas, the second ("R2") achieving ∼400 ppm maximum error with zero free parameters. Still the most widely used formula a century later.
- **Cantrell (2001–2004)** [17]: rational corrections to R2, pushing to ∼1 ppm at a handful of parameters.
- **Villarino (2005)** [3]: identified the $O(h^5)$ error structure and the constant $\delta_5 = 3/2^{17}$, giving the first rigorous asymptotic analysis of Ramanujan's formula.
- **Koshy (2024)** [18]: quarter-perimeter formulas at 5 parameters.
- **Moscato (2024)** [19]: sub-3 ppm formulas via evolutionary optimization.
- **Ayala-Raggi & Rendón-Marín (Sep 2025 arXiv; Apr 2026 Ver. 2.0)** [6, 7, 5]: recent example of SOTA; published one-/two-exponential corrections to R2 (Sep 2025 arXiv preprint, 0.57 ppm at 4 constants), peer-reviewed in *AppliedMath*, followed by April 2026 Ver. 2.0 extension with flexible $F2$–$F5$ models, achieving 0.0216 ppm at ∼15 parameters.
- **AAA (Nakatsukasa–Sète–Trefethen 2018)** [1]: an adaptive rational algorithm achieving root-exponential convergence $\exp(-c\sqrt{n})$ on branch-point functions; reaches $\sim 10^{-13}$ at 30 rational poles. Adopted widely as a numerical tool, but produces no closed-form formula and is bounded by Newman's root-exponential rate.
- **[RECENT CHAMP] SRIRACHA (Mamoun, 30 March 2026)** [45, 46]: the author's own prior framework. SRIRACHA decodes Ramanujan II as a CF approximant of ${}_2F_1(-\frac{1}{2}, -\frac{1}{2}; 1; h)$, then corrects with a Padé–Apéry tower (R2–R6), a gate $h^{7/3}$, and Newman exponential sums whose rates come from π's continued fraction. The full pipeline reaches **7 parts per trillion at ∼25 parameters**—roughly 3,000× better than Ayala-Raggi and beating AAA at every matched parameter count. The present paper grew out of asking why SRIRACHA's exponentials work and whether the exact singular modes they approximate could replace them.

Each method improves on the last, but all share a structural limitation: they use polynomials, rationals, or exponentials to *approximate* a singularity they could instead *represent exactly*.

Intellectual ancestry. The philosophy of *matching a known singularity by construction, then fitting the smooth residual* has a long lineage in numerical analysis, most directly traced to Rokhlin's *generalized Gaussian quadrature* program: given a function class with known singular structure, build custom nodes (Ma–Rokhlin–Wandzura [35]; Yarvin–Rokhlin [36]) exact for $2n$ carefully-chosen integrands, delivering convergence rates no generic Gaussian rule can match. The augmented basis defined in §2 below is the *approximation-dual* of this quadrature philosophy: instead of custom nodes for integration against singular weights, we supply custom basis functions for approximation *of* singular targets. Theorem 8 (in §D.4) proves that, for the ellipse perimeter, the Villarino-matched log exponents $\{2, 3, \ldots, K+1\}$ are the unique K-term singularity-matching choice within the Chebyshev-plus-log Haar framework—making precise, for this concrete problem, the "match the singularity first" principle that underlies Rokhlin's quadrature program.

1.2 The inciting incident

Near the degenerate endpoint ($b \to 0$, the ellipse collapsing to a line segment), the ellipse perimeter has a logarithmic branch point:

$$P_2(\xi) \;=\; 8 - 2\xi + \tfrac{1}{4}\,\xi^2 \log(1/\xi) + g_2(\xi), \tag{1}$$

where $\xi = 1-((a-b)/(a+b))^2 \in [0,1]$ and g_2 is analytic. The $\xi^2 \log(1/\xi)$ term forces any polynomial approximation into algebraic convergence. The exponentials in SRIRACHA, the poles in AAA, and the fitted powers in Ayala-Raggi are all different strategies for coping with a singularity they cannot represent exactly.

What if the wall is not in the algorithm, but in the basis?

If we write the log modes directly into the approximation basis, the residual becomes analytic, and Chebyshev convergence jumps from $O(n^{-s})$ to $O(\rho^{-n})$. Three log functions replace thirty rational poles. That is the thesis of this paper.

2 The augmented basis

2.1 Definition

The augmented basis of order (n, K) is

$$\boxed{\mathcal{B}_{n,K} \;=\; \underbrace{\big\{T_0(2\xi-1),\,\ldots,\,T_n(2\xi-1)\big\}}_{n+1 \text{ Chebyshev polynomials}} \;\cup\; \underbrace{\big\{\xi^{2+j}\log(1/\xi) : 0 \le j \le K\big\}}_{K+1 \text{ log companions}}} \tag{2}$$

Total parameters: $n + K + 2$. No tower. No gate function. No exponentials. No optimized rates. The log companions do the singular heavy lifting; the Chebyshev polynomials handle the smooth remainder. Coefficients are determined by linear least squares—a single QR factorization.

2.2 The four methods we compare

We compare four approaches to approximating a function $f : [0,1] \to \mathbb{R}$ with a branch point at one endpoint:

1. **Plain Chebyshev:** $\sum_{k=0}^{n} c_k T_k(2\xi-1)$. Converges algebraically ($O(n^{-s})$) for log branches—the polynomial wall.
2. **Padé** $[m/m]$**:** rational approximants from the Taylor series. Captures some branch structure through pole clustering, but saturates at $\sim 2 \times 10^{-8}$ for the ellipse perimeter.
3. **AAA** [1]: adaptive barycentric rational approximation. Convergence $\exp(-c\sqrt{n})$; discovers branch structure automatically but expends many poles.
4. **Augmented basis** (this paper): Chebyshev plus explicit singular modes. Geometric convergence $O(\rho_K^{-n})$ with ρ_K growing in K.

3 The main theorem

The theorem explains *why* the method works. The proof is one classical result (Bernstein) applied to a carefully chosen residual.

Motivation: the engineering question

An engineer who needs the perimeter of an ellipse to six decimal places has to pick a tool. Until now the choice was Ramanujan-style closed forms (fast, only three digits), Padé approximants (saturate near 10^{-8}), the 2018 AAA algorithm (many poles, loss of monotonicity), or expensive numerical integration. The theorem below isolates *why* every polynomial- or rational-based tool hits a wall: the ellipse perimeter contains the function $\xi^2 \log(1/\xi)$, whose logarithmic branch point sits at $\xi = 0$, exactly on the edge of the interval of interest. Rational functions must pay geometrically many parameters per decimal digit to imitate that branch point. The theorem then shows that if one simply *writes the log symbol into the basis* — alongside the usual Chebyshev polynomials — the wall disappears and convergence returns to the geometric rate the approximation would have enjoyed on a smooth target. Three log functions replace thirty rational poles.

The picture to carry: a sheet of paper with a single sharp crease at the left edge. No amount of folding (polynomials) or ratios of folds (rational functions) can flatten the crease. But if you are allowed to declare "the crease has shape $\xi^2 \log(1/\xi)$" and subtract it, what remains is a perfectly smooth sheet on which polynomials work as well as on any smooth function.

Pedagogical warm-up: what the reader needs

This subsection is for a reader who has not seen approximation theory before. Specialists can skim. Four ingredients are used in the proof: the Chebyshev basis, Bernstein's theorem, the log-branch class, and the "subtract then apply Bernstein" maneuver.

Chebyshev polynomials. With the substitution $\xi = (1 + u)/2$, the interval $[0, 1]$ maps to $u \in [-1, 1]$. The Chebyshev polynomials $T_0(u), T_1(u), T_2(u), \ldots$ are defined by $T_k(\cos\theta) = \cos(k\theta)$; they form an orthogonal basis of $L^2([-1, 1], (1-u^2)^{-1/2}du)$ and every continuous function on $[-1, 1]$ has a unique Chebyshev series $\sum_k c_k T_k(u)$. Truncating the series at degree n gives a polynomial approximant; the truncation error is controlled by the decay of the coefficients c_k.

Bernstein's theorem (the key input to the proof). Let E_ρ denote the Bernstein ellipse with foci ± 1 and sum of semi-axes equal to $\rho > 1$. Bernstein (1912) proved: if a function g is holomorphic (i.e. complex-analytic) on an open set containing E_ρ, and $M = \sup_{u \in E_\rho} |g(u)|$, then the degree-$n$ truncated Chebyshev series P_n satisfies

$$\|g - P_n\|_{\infty,[-1,1]} \;\leq\; \frac{2M\rho^{-n}}{\rho - 1}.$$

Thus analyticity in a complex neighborhood yields *geometric* convergence of polynomial approximation at rate ρ^{-1}. The size of ρ is determined by how far the complex singularities of g sit from $[-1, 1]$: farther singularities mean larger ρ means faster convergence. A modern statement and proof is Trefethen, *Approximation Theory and Approximation Practice*, Theorem 8.2 [2].

Why plain Chebyshev stalls on the ellipse perimeter. The function $\xi^2 \log(1/\xi)$ has a logarithmic branch point at $\xi = 0$. After the change of variable $u = 2\xi - 1$, the branch point is at $u = -1$, sitting directly on the boundary of $[-1, 1]$. No Bernstein ellipse E_ρ with $\rho > 1$ can avoid that point, so Bernstein's theorem delivers only $\rho = 1$ — i.e. no geometric bound. Standard endpoint-singularity analysis then gives algebraic decay $|c_k| = O(k^{-3})$ (Lemma 1 below), hence $\|f - P_n\|_\infty = O(n^{-2})$: the log-branch wall.

The key idea of the proof (one sentence). If f is a holomorphic g plus finitely many $\xi^{2+k}\log(1/\xi)$ terms, and the augmented basis contains all those log terms, then subtract the log terms algebraically — the residual *is* g itself, which is holomorphic on a Bernstein ellipse E_ρ with $\rho > 1$, and Bernstein's theorem gives the geometric bound.

3.1 The log branch class

Intuition. *Think of the ellipse perimeter as the sum of two functions: a smooth part $g(\xi)$ that any polynomial could approximate beautifully, and a singular part $\xi^2\log(1/\xi)$ that no polynomial can touch — it has a branch point, like a crease in a sheet of paper that no amount of folding can flatten. The augmented basis separates these two parts: the log companions absorb the singular crease* exactly*, leaving only the smooth sheet for Chebyshev to handle. The proof below makes this intuition precise: subtracting the crease (Step 1) leaves the smooth part (Step 2), which Chebyshev approximates at geometric rate (Step 3).*

Definition 1 (Log branch class $\mathcal{L}_K(D)$)**.** *A function $f : [0,1] \to \mathbb{R}$ belongs to $\mathcal{L}_K(D)$ if*

$$f(\xi) \;=\; g(\xi) \;+\; \sum_{k=0}^{K} a_k\,\xi^{2+k}\log(1/\xi), \qquad \xi \in (0,1], \tag{3}$$

where $g : D \to \mathbb{C}$ is holomorphic on an open set $D \supset [0,1]$ in $\mathbb{C}$, and $a_0, \ldots, a_K \in \mathbb{R}$.

For the infinite expansion $\sum_{k=0}^{\infty} a_k\xi^{2+k}\log(1/\xi)$ used in the separation theorem (Section 10), the Pochhammer decay $|a_k| \sim 1/(\pi k)$ ensures absolute convergence for all $|\xi| < 1$. At $\xi = 1$ the log factor vanishes, so f extends continuously to the closed interval $[0,1]$.

3.2 Geometric rate for the augmented basis

Theorem 1 (Main theorem)**.** *Let $f \in \mathcal{L}_K(D)$ with analytic part g. Let E_ρ be the largest Bernstein ellipse contained in D (after the change of variable $u = 2\xi - 1$) and $M = \sup_{u\in E_\rho} |g((u+1)/2)|$. Then for every $n \geq 0$ there exists $A_{n,K} \in \operatorname{span}(\mathcal{B}_{n,K})$ with*

$$\|f - A_{n,K}\|_{\infty,\,[0,1]} \;\leq\; \frac{2M\rho^{-n}}{\rho - 1}.$$

The error decays geometrically in n with rate ρ^{-1}, independent of the singular coefficients $a_0, \ldots, a_K$.

Proof. The argument is elementary but each step is worth stating in full so that no computation is left implicit.

Step 1 — Subtract the log companions algebraically. Define the residual

$$r(\xi) \;:=\; f(\xi) \;-\; \sum_{k=0}^{K} a_k\,\xi^{2+k}\log(1/\xi), \qquad \xi \in (0,1].$$

We emphasize that $a_0, \ldots, a_K$ are the *actual* coefficients of f in the decomposition (3) guaranteed by Definition 1. We are not optimizing them here; they are fixed scalars determined by f itself. In practice they will be recovered by least squares (Consequence 1 below), but in the proof we assume them as given.

Step 2 — The residual equals the analytic part. Substituting (3) into the definition of r gives

$$r(\xi) \;=\; \Big[g(\xi) + \sum_{k=0}^{K} a_k\, \xi^{2+k} \log(1/\xi)\Big] \;-\; \sum_{k=0}^{K} a_k\, \xi^{2+k} \log(1/\xi) \;=\; g(\xi).$$

So the residual is literally the analytic part g. The log terms have been removed exactly, not approximated. By Definition 1, g is holomorphic on an open set $D \supset [0,1]$ in $\mathbb{C}$. In particular, g admits a convergent Taylor series at every point of $[0,1]$ and extends to a complex neighborhood.

Step 3 — Apply Bernstein's theorem to g. We now perform the change of variables $u = 2\xi - 1$, so that the interval $[0,1]$ in ξ becomes $[-1,1]$ in u, and the image of D under this map is an open set $\widetilde{D} \supset [-1,1]$ on which $g((u+1)/2)$ is holomorphic. Let E_ρ be the largest Bernstein ellipse with foci ± 1 contained in $\widetilde{D}$ (this $\rho > 1$ exists because $\widetilde{D}$ is an open complex neighborhood of the compact interval $[-1,1]$; one may take for example $\rho = 1 + \mathrm{dist}(\partial\widetilde{D}, [-1,1])/2$). Let $M = \sup_{u \in E_\rho} |g((u+1)/2)|$, which is finite since E_ρ is compact and g is continuous on its closure.

Bernstein's theorem (Trefethen [2], Thm. 8.2; original statement in [13]) states: if a function G is holomorphic on an open set containing E_ρ and $\|G\|_{E_\rho} := \sup_{u \in E_\rho} |G(u)| = M < \infty$, then for every $n \geq 0$ the degree-n truncated Chebyshev partial sum $P_n(u) = \sum_{k=0}^{n} c_k T_k(u)$ satisfies

$$\|G - P_n\|_{\infty,\,[-1,1]} \;\leq\; \frac{2M\rho^{-n}}{\rho - 1}.$$

Applying this theorem verbatim with $G(u) := g((u+1)/2)$ produces a degree-n polynomial $P_n(u)$ with $\|g((\cdot+1)/2) - P_n\|_{\infty,[-1,1]} \leq 2M\rho^{-n}/(\rho-1)$. Pulling the change of variable back, and writing $\widetilde{P}_n(\xi) := P_n(2\xi - 1)$, we obtain $\|g - \widetilde{P}_n\|_{\infty,[0,1]} \leq 2M\rho^{-n}/(\rho-1)$. Note that $\widetilde{P}_n \in \mathrm{span}\{T_k(2\xi-1) : 0 \leq k \leq n\}$, which is the polynomial part of $\mathcal{B}_{n,K}$.

Step 4 — Reassemble the approximant. Define the augmented-basis approximant

$$A_{n,K}(\xi) \;:=\; \widetilde{P}_n(\xi) \;+\; \sum_{k=0}^{K} a_k\, \xi^{2+k} \log(1/\xi).$$

By construction $A_{n,K} \in \mathrm{span}(\mathcal{B}_{n,K})$: the first summand lives in the Chebyshev part and the second summand lives in the log-companion part of the basis (2).

Step 5 — The log companions cancel, leaving Bernstein's bound. Compute

$$\begin{aligned} f(\xi) - A_{n,K}(\xi) &= \Big[g(\xi) + \sum_{k=0}^{K} a_k\, \xi^{2+k} \log(1/\xi)\Big] - \Big[\widetilde{P}_n(\xi) + \sum_{k=0}^{K} a_k\, \xi^{2+k} \log(1/\xi)\Big] \\ &= g(\xi) - \widetilde{P}_n(\xi). \end{aligned}$$

The log companions cancel exactly, being present with identical coefficients on both sides of the subtraction. Taking the supremum norm over $[0,1]$ and applying the bound from Step 3,

$$\|f - A_{n,K}\|_{\infty,\,[0,1]} \;=\; \|g - \widetilde{P}_n\|_{\infty,[0,1]} \;\leq\; \frac{2M\rho^{-n}}{\rho - 1},$$

which is the claimed geometric rate. Observe that the bound is *independent* of the log-coefficient vector $(a_0, \ldots, a_K)$: those coefficients appear in $A_{n,K}$ and in f with exactly matching signs and cancel before Bernstein's theorem is invoked. This proves the theorem. □

Remark. *Theorem 1 is a direct application of Bernstein's theorem to the standard enrichment strategy of XFEM/GFEM [8, 10]. The result itself is not new. Our contributions are the systematic benchmarking against AAA and other methods (Sections 4–7), the K-sharpness mechanism (Section 8), and the provable separation from rational approximation (Section 10).*

3.3 What the theorem says in plain language

Three consequences:

1. **The rate depends on g, not on a_k.** Within a fixed K, the rate ρ^{-n} is coefficient-independent: least squares recovers the log coefficients automatically.
2. **Different K give different ρ.** A larger K absorbs more log terms, leaving a more analytic residual with a larger Bernstein ellipse.
3. **Geometric beats polynomial.** Plain Chebyshev on f converges algebraically at $O(n^{-s})$; the augmented basis converges geometrically at $O(\rho^{-n})$. The speedup grows without bound.

4 The ellipse perimeter benchmark

All reference values are computed by `mpmath` at 70 decimal digits. The target is $f(\xi) = P_2(\xi)$, the perimeter of the classical ellipse as a function of the eccentricity parameter ξ.

4.1 Head-to-head: four-way comparison

params	Plain Cheb	scipy AAA	Padé $[m/m]$	Padé+1 log	Augmented (50d)
10	5.69×10^{-6}	1.02×10^{-7}	6.87×10^{-7}	5.42×10^{-8}	$\mathbf{2.47\times10^{-9}}$
15	5.91×10^{-7}	4.02×10^{-9}	4.54×10^{-8}	1.14×10^{-9}	$\mathbf{2.16\times10^{-13}}$
20	1.36×10^{-7}	1.88×10^{-10}	2.21×10^{-8}	4.13×10^{-10}	$\mathbf{1.82\times10^{-17}}$
25	6.66×10^{-8}	1.61×10^{-11}	2.06×10^{-8}	3.31×10^{-10}	$\mathbf{1.82\times10^{-21}}$
30	3.27×10^{-8}	3.18×10^{-13}	2.27×10^{-8}	6.13×10^{-10}	$\mathbf{2.97\times10^{-25}}$

The augmented column uses 50-digit mpmath arithmetic; the other four use their native float64 implementations (scipy, numpy). This comparison is asymmetric in precision: AAA's 10^{-13} saturation is partly a float64 ceiling (an mpmath AAA could go lower). However, the *rate* difference is structural, not arithmetic: AAA converges root-exponentially while the augmented basis converges super-geometrically. The separation theorem (Section 10) proves the gap is unbounded regardless of the precision used.

Four observations:

1. **Plain Chebyshev** stalls at 3.3×10^{-8} by 30 parameters—the algebraic wall from the log branch.
2. **AAA** achieves root-exponential decay, reaching 3.18×10^{-13} at 30 parameters before saturating.
3. **Padé** $[m/m]$ saturates at $\sim 2 \times 10^{-8}$ by 20 parameters. Adding one log companion moves the floor to $\sim 4 \times 10^{-10}$—a 50× improvement, but still a floor.
4. **Augmented basis** beats every other method at every budget. At 15 parameters it is 18,600× better than AAA; at 20 parameters, $1.03\times10^{7}\times$ better. The gap grows without bound because the augmented basis has no precision ceiling.

5 Generalizes: $K(m)$, Y_0, corner singularity

To confirm this is not a quirk of the ellipse, we tested four additional branch-point functions.

5.1 Classical benchmarks

benchmark	Plain Cheb	scipy AAA	Augmented	Aug/AAA
$K(m)$, $p\approx 15$	3.42×10^{-4}	1.13×10^{-7}	$\mathbf{1.22\times 10^{-11}}$	9,261×
$K(m)$, $p\approx 20$	3.78×10^{-5}	3.63×10^{-12}	$\mathbf{3.94\times 10^{-14}}$	92×
$Y_0(x)$, $p\approx 20$	3.33×10^{-1}	2.94×10^{-7}	$\mathbf{1.16\times 10^{-11}}$	25,345×
$\log(1-x)$, $p\approx 20$	3.19×10^{2}	7.25×10^{-7}	$\mathbf{1.51\times 10^{-8}}$	48×

The Bessel Y_0 result is especially striking: it is the canonical log-branch benchmark in numerical analysis, and plain Chebyshev fails entirely (0.33 absolute error) while the augmented basis beats AAA by 25,000×.

5.2 L-shaped domain corner singularity

The L-shaped domain (270-degree reentrant corner) is a fundamental PDE benchmark. We test on the 1D model $f(r) = r^{2/3}(1 + 0.5r - 0.3r^2 + 0.1r^3)$ with algebraic companions $\{r^{2/3}, r^{5/3}, r^{8/3}\}$:

params	Plain Cheb	scipy AAA	Augmented	Aug/AAA
8	1.51×10^{-2}	6.36×10^{-3}	$\mathbf{1.11\times 10^{-6}}$	5,745×
15	4.72×10^{-3}	1.13×10^{-4}	$\mathbf{7.67\times 10^{-10}}$	147,000×
25	2.13×10^{-3}	5.30×10^{-6}	$\mathbf{4.97\times 10^{-12}}$	1,067,000×
30	1.71×10^{-3}	2.30×10^{-7}	$\mathbf{1.30\times 10^{-12}}$	177,000×

This is not a log-branch function (it has an algebraic $r^{2/3}$ singularity), confirming that the method extends to *any* known singular structure. The companions are Williams' corner eigenfunctions [12]; the idea is the same.

6 Context: the idea is old, the comparison is new

Before testing the method's limits, we pause to locate it within the existing literature. The idea of enriching a polynomial basis with singular modes is not new—it goes back at least to Williams (1952) in fracture mechanics and was systematized by Belytschko, Babuška, and Melenk in the 1990s. What is new here is a clean 1D Chebyshev recipe, the K-sharpness amplification, and head-to-head benchmarks against the best modern algorithms.

6.1 The general principle

The preceding sections are instances of a single idea:

If you know the singular structure analytically, write it into the basis. Chebyshev handles only the smooth residual. The convergence rate is then controlled by the residual's Bernstein ellipse, not by the original singularity.

This is the 1D Chebyshev-polynomial analogue of XFEM (Belytschko–Black [8], Moës–Dolbow–Belytschko [9]), partition of unity (Babuška–Melenk [10]), and weighted Nitsche–Schatz estimates [11]. What is new here is:

1. A clean 1D recipe with a one-page Bernstein proof.
2. The K-sharpness amplification (Table 1).
3. Direct head-to-head against scipy AAA on named benchmarks.
4. The "AAA $\to$ compile" workflow for unknown branch structure.

6.2 Related work

XFEM / GFEM (Belytschko–Black 1999 [8], Moës–Dolbow–Belytschko 1999 [9]): enriching FEM shape functions with crack-tip $r^{1/2}$ modes. Our work is the 1D Chebyshev analogue.

Partition of unity (Babuška–Melenk 1997 [10]): our basis is a Chebyshev partition-of-unity enrichment.

Nitsche–Schatz weighted estimates [11]: subtract the known singular behavior, then estimate the smooth residual. Same strategy, different setting.

Williams' corner functions [12]: exact PDE eigenfunctions as basis elements. We use exact singular modes as approximation-theory elements.

Newman 1964 [15]: exponential sum approximation rate $\exp(-c\sqrt{N})$. Our log companions beat this by providing the exact modes directly.

Boyd 2001 [22]: spectral methods for singular functions. The augmented basis is a concrete spectral-enrichment recipe with benchmarks.

Trefethen–Weideman 2014 [28]: exponential convergence of trapezoidal rule after singularity subtraction. Same spirit; our contribution is the Chebyshev-basis formulation.

Gopal–Trefethen 2019 [31]: lightning Laplace solver—rational approximation with exponentially-clustered poles near corner singularities, root-exponential convergence. Our L-corner result (147,000$\times$ over AAA) is the 1D analogue using augmented Chebyshev basis instead of rational poles.

hp-FEM with graded meshes (Gui–Babuška 1986, Schwab 1998): exponential convergence via geometric mesh refinement toward singularities. This is the standard PDE approach. On the $r^{2/3}$ corner benchmark, our augmented basis beats a graded hp-FEM with $\sigma = 0.15$ by 150$\times$ at 12 parameters (2.2×10^{-4} vs. 3.3×10^{-2}). The difference is structural: hp-FEM uses polynomial approximation on small elements near the singularity, while we use the *exact* singular mode globally. Both achieve exponential-type convergence, but the augmented basis has better constants because it avoids the mesh-grading overhead.

Chebfun [29] with `splitting on`: Chebfun automatically detects singularities and subdivides the interval into smooth pieces, then approximates each piece with a separate Chebyshev expansion. This is effective but parameter-expensive: each sub-interval requires its own set of Chebyshev coefficients, and the splitting points themselves consume degrees of freedom. Our augmented basis keeps a *single global* basis; the log companions absorb the singularity without subdivision. At 13 parameters, the augmented basis reaches 3.3 ppt; Chebfun's splitting would require $\sim$100+ coefficients to match, because each sub-interval near $\xi = 0$ still faces the algebraic convergence wall on its local piece.

Almkvist–Berndt 1988 [26]: Ramanujan's approximation process. The R-tower is a continued-fraction decode of this process. The R-tower achieves rate $\sim 1.31^n$; the augmented basis achieves ρ_K^{-n} with ρ_K growing in K, decisively faster for $K \geq 2$. Including R2 as one additional basis function costs one parameter but absorbs four leading Taylor terms, giving slightly better accuracy at very low budgets (≤ 10 parameters).

7 Fails on Lamé—then the bilateral fix

7.1 The crisis

We first tested the log-only augmented basis on asymmetric Lamé curves $|X/a|^r + |Y/b|^2 = 1$ with $r \in \{2.5, 3, 4\}$. The result was *negative*: the log companions contributed almost nothing, and the augmented basis **lost to AAA by** 50–100×.

This was a genuine stress test. If the method only works on functions with a single-endpoint branch, its generality claim collapses.

7.2 Diagnosis: a missing branch

The asymmetric Lamé curves have an endpoint branch at *both* ends of $[0, 1]$: a log branch at $\xi = 0$ (degenerate endpoint) and an algebraic $(1-\xi)^{1/2}$ branch at $\xi = 1$ (the circle endpoint). The log-only basis captures the first but is blind to the second.

7.3 The bilateral fix: double-ladder basis

Adding bilateral companions $\{(1-\xi)^{1/2}, (1-\xi)^{3/2}\}$ at the other endpoint:

$$\mathcal{B}^{\text{double}}_{n,K_{\log},K_{\text{sq}}} \;=\; \{T_k\} \;\cup\; \{\xi^{2+j}\log(1/\xi)\} \;\cup\; \{(1-\xi)^{(2j+1)/2}\}.$$

Lamé case (params ~ 22)	Plain Cheb	scipy AAA	Log-only Aug	Double-ladder
$r = 2.5$	5.21×10^{-5}	4.61×10^{-7}	9.37×10^{-5}	$\mathbf{3.46\times10^{-10}}$
$r = 3.0$	8.86×10^{-5}	1.19×10^{-7}	1.59×10^{-4}	$\mathbf{4.08\times10^{-10}}$
$r = 4.0$	1.33×10^{-4}	1.20×10^{-6}	2.38×10^{-4}	$\mathbf{3.20\times10^{-10}}$

Double-ladder beats AAA by 292×–3,760×. The initial failure was due to an *incomplete basis*, not an intrinsic limitation: once you augment with the missing branch structure, the method regains a clear advantage.

The lesson is general: **the augmented basis is only as good as your knowledge of the singularity structure**. Mis-specify it, and you lose to AAA. Specify it correctly, and you win by orders of magnitude.

8 K-sharpness: super-geometric convergence

Each additional log companion enlarges the effective Bernstein ellipse. Co-scaling K and n gives convergence faster than any fixed geometric rate.

Table 1: **K-sharpness at 50-digit precision.** Max relative error of the augmented basis at Chebyshev degree $n = 8$ as K increases. All values computed at 50 digits with zero float64. Each companion gains ∼1 decade. No saturation is visible through $K = 15$ ($10^{-18.8}$).

K	params	max rel error	decades gained
0	9	3.40×10^{-6}	–
1	10	3.67×10^{-8}	2.0
2	11	9.83×10^{-10}	1.6
3	12	3.95×10^{-11}	1.4
4	13	3.33×10^{-12}	1.1
5	14	3.15×10^{-13}	1.0
6	15	3.25×10^{-14}	1.0
7	16	4.60×10^{-15}	0.8
8	17	7.47×10^{-16}	0.8
9	18	1.47×10^{-16}	0.7
10	19	4.15×10^{-17}	0.5
11	20	1.01×10^{-17}	0.6
12	21	3.72×10^{-18}	0.4
13	22	1.38×10^{-18}	0.4
14	23	4.95×10^{-19}	0.4
15	24	1.73×10^{-19}	0.5

8.1 Theoretical explanation: exact coefficients and polynomial decay

The ρ_K values in Table 1 are *effective* rates measured from a log-linear fit—they are not Bernstein radii. The real convergence mechanism is revealed by extracting the log-expansion coefficients a_K of $P_2(\xi)$ at 150-digit precision.

Definition 2 (Log-expansion coefficients)**.** *We write $P_2(\xi) = g(\xi) + \sum_{K=0}^{\infty} a_K \xi^{2+K} \log(1/\xi)$. The first three coefficients, verified at 150 digits, are:*

$$a_0 = \tfrac{1}{4} = 0.25, \tag{4}$$
$$a_1 = \tfrac{3}{16} = 0.1875, \tag{5}$$
$$a_2 = \tfrac{75}{512} = 0.146484375. \tag{6}$$

The value $a_0 = 1/4$ recovers Villarino's [3] leading coefficient.

Decay rate. The ratio $|a_{K+1}|/|a_K|$ tends to 1 (not 1/4). From Stirling's approximation applied to the Pochhammer structure $[(1/2)_K/K!]^2$ that generates these coefficients:

$$|a_K| \sim \frac{1}{\pi K} \qquad \text{for large } K. \tag{7}$$

This is *polynomial* decay, not geometric.

Why super-geometric total convergence persists. The two-term error bound remains

$$\|f - A_{n,K}\|_\infty \le \underbrace{C_g \rho_g^{-n}}_{\text{analytic part (geometric)}} + \underbrace{|a_{K+1}| \cdot n^{-(2K+6)}}_{\text{remaining log tail (Jackson)}}, \tag{8}$$

but the second term must be read correctly. Setting $K = n/c$ for a constant $c > 0$:

$$|a_{K+1}| \cdot n^{-(2K+6)} \approx \frac{c}{\pi n} \cdot n^{-(2n/c+6)}.$$

The exponent $-(2n/c+6)$ grows *linearly* in n, so the tail decays super-geometrically even though the individual coefficients $|a_K|$ decay only as $1/(\pi K)$.

Corrected interpretation. The measured ρ_K values in Table 1 are therefore *effective geometric rates* that summarize the combined effect of (i) the steepening Chebyshev exponent $-(2K+6)$ and (ii) the polynomial coefficient decay $1/(\pi K)$. They are not Bernstein radii of an enlarging ellipse. The actual mechanism is the steepening Chebyshev exponent, and the per-companion exponent step is exactly $+2$ by the following elementary calculation.

Proposition 1 (Per-companion exponent step)**.** *Let $\Phi_m(\xi) := \xi^m \log(1/\xi)$ on $[0,1]$, $m \in \mathbb{Z}_{\geq 1}$, and let $E_n(\Phi_m) := \inf_{p\in\mathcal{P}_n} \|\Phi_m - p\|_{\infty,[0,1]}$ be the best polynomial approximation error at degree n. Then*

$$E_n(\Phi_m) \;=\; \Theta\big(n^{-2m}\big), \qquad n \to \infty.$$

Consequently, replacing Φ_m by Φ_{m+1} raises the decay exponent by exactly 2.

Proof. Apply Lemma 1 (with the Chebyshev-variable shift $\xi = (1+u)/2$ mapping the endpoint $\xi = 0$ to $u = -1$): the Chebyshev coefficients of Φ_m satisfy $|c_k| \asymp k^{-(2m+1)}$. The partial-sum error then gives the upper bound $E_n(\Phi_m) \leq \sum_{k>n} |c_k| = \Theta(n^{-2m})$; the matching lower bound follows from Jackson's theorem applied to the modulus of continuity of Φ_m at the endpoint (Trefethen [2], Thm. 7.2). The exponent step from m to $m+1$ is $2(m+1) - 2m = 2$. □

Apply Proposition 1 to the residual of an augmented-basis fit. Let the target have a local log-branch expansion $f(\xi) = g(\xi) + \sum_{j\geq 0} a_j \xi^{\gamma_0+j} \log(1/\xi)$ near $\xi = 0$. An augmented basis carrying K companions $\{\xi^{\gamma_0+j}\log(1/\xi)\}_{j=0}^{K-1}$ subtracts the log-tail through index $j = K-1$, leaving a residual whose leading term is $a_K \Phi_{\gamma_0+K}$. Proposition 1 with $m = \gamma_0 + K$ gives best-polynomial-approximation error $\Theta\big(n^{-2(\gamma_0+K)}\big)$, so the full augmented error obeys the uniform rate

$$\|f - A_{n,K}\|_\infty \;=\; O\big(C_g \rho_g^{-n} + |a_K|\, n^{-(2\gamma_0+2K)}\big).$$

This establishes the exact *K-sharpness law*: for any target in the log-branch class with leading singular exponent γ_0, the empirical n-exponent $\alpha(K)$ of the K-augmented basis satisfies

$$\boxed{\alpha(K) \;=\; 2\gamma_0 + 2K} \tag{9}$$

with equality realised for every target for which the residual tail is not simplified by miracle cancellations. The K-sharpness table (Table 1) is the empirical confirmation of the law on the ellipse target ($\gamma_0 = 2$): the rate exponent should be $4, 6, 8, 10, \ldots$ at $K = 0, 1, 2, 3, \ldots$, and the measured decade-per-step entries match this to within the 0.1–0.3 decade noise of log-linear fits at finite precision. Validation on three additional targets ($K(m)$, Li_2/x, corner-XFEM; Section E) gives $\alpha(K)$ matching the law to within 0.3 exponent units in every measured row.

This corrected model explains all three observations: (1) the effective ρ_K grows with K because the exponent steepens; (2) the geometric fit improves (R^2: $0.91 \to 0.99$) because the polynomial term increasingly dominates the geometric term; (3) at standard double precision the effect saturates around $K \approx 5$–6 because the condition number exceeds 10^{10}; at 50-digit precision, the super-geometric rate continues to $K = 13$ and beyond (Section 8.2).

Table 2: **Precision requirement.** Condition number of the design matrix and the minimum digits of precision needed to realize the theoretical convergence rate.

K	n	$\kappa(\mathbf{A})$	min digits needed
0	8	2.2×10^4	5
2	8	3.0×10^7	8
4	8	4.3×10^9	10
5	8	3.4×10^{10}	11
8	10	$\sim 10^{21}$	22

8.2 Conditioning: precision requirements grow with K

The condition number of the augmented basis design matrix scales as

$$\kappa(\mathbf{A}) \sim 10^{2K+0.5n}. \tag{10}$$

To realize the theoretical convergence rate, the arithmetic precision must exceed $\log_{10} \kappa(\mathbf{A})$ digits.

Three consequences:

1. **Recipes 1–3 ($K \leq 4$, $n \leq 8$) need ≤ 10 digits.** Standard double precision (16 digits) suffices comfortably.
2. **Beyond $K = 5$, extended precision pays off.** All results in this paper use 50-digit mpmath arithmetic, which provides ∼40 digits of headroom above the condition number for every tested configuration.
3. **The cost is negligible.** The entire 20-claim verification suite (Section 15) runs in ∼4 s at 50 digits on a laptop.

This conditioning analysis is not a weakness of the method: it is an honest accounting of the tradeoff between enrichment depth and numerical precision. The same tradeoff governs XFEM [8] and every enriched-basis method in practice. The difference is that `mpmath` makes extended precision trivial.

9 The Ayala-Raggi challenge

Ayala-Raggi and Rendón-Marín [6, 7, 5] are the most recent external *closed-form* example of SOTA prior to SRIRACHA and UNDERWOOD: a family of compact exponential corrections to Ramanujan II of the form $\exp(-B(1-h)^p)$ with p itself a fitted parameter. Their September 2025 arXiv preprint introduced the 2-exp form at 0.57 ppm with 4 constants; the April 2026 Ver. 2.0 extension reaches 0.0216 ppm at ∼15 constants, the best competing closed-form result in the literature prior to SRIRACHA. SRIRACHA itself (the author's 30 March 2026 **recent champ**) reaches 7 ppt at ∼25 parameters, ∼3,000× tighter, and is covered separately in Section 6. This section pits UNDERWOOD's Formula 2 (8 constants, 0.015 ppm) against the Ayala-Raggi family head-to-head:

Documentary note: the published *AppliedMath* article is the base one-/two-exponential paper; the April 2026 Ver. 2.0 Zenodo preprint is the source of the extended $F2$–$F5$ ladder and the 0.0216 ppm figure used below.

their formula	params	their max ppm	our best at $\leq$ same params
R2/2exp	4	0.573	43.7 ppm ($K=0$, deg 4, 5 p)
R2/F2exp	6	0.213	**0.114** ppm ($K=2$, deg 4, 7 p)
R2/F3exp	9	0.055	**0.015** ppm ($K=3$, deg 4, 8 p)
R2/F4exp	12	0.030	**0.000039** ppm ($K=3$, deg 8, 12 p)
R2/F5exp	$\sim$15	0.0216	**0.0000033** ppm ($K=4$, deg 8, 13 p)

At low parameter counts (≤ 5), their R2-based formulas win because Ramanujan II absorbs the first four Taylor terms for free. From 7 parameters onward, the augmented basis dominates.

Formula 2: UNDERWOOD at 8 constants (0.015 ppm)

The row "R2/F3exp $\rightarrow$ **0.015** ppm" in the table above is worth stopping at. It is the formula a working engineer would actually ship — 8 stored constants, 67-million-to-one accuracy, no elliptic integrals, no conditional branching. At this parameter budget UNDERWOOD Formula 2 is 3.7$\times$ tighter than Ayala-Raggi R2/F3exp and 66$\times$ tighter than their 4-parameter R2/2exp. Fully expanded (notation: $\xi = 1 - ((a-b)/(a+b))^2$, $u = 2\xi - 1$):

UNDERWOOD Formula 2: 8 constants,
max error 0.015 ppm (1 part in 67 million).

$$
\begin{aligned}
P \;=\; \frac{a+b}{2} \cdot \Big[\; & +7.042\,111\,928\,654\,361 \\
& -0.890\,889\,229\,192\,585 \cdot u \\
& +0.095\,701\,829\,579\,546 \cdot u^2 \\
& +0.032\,481\,929\,448\,356 \cdot u^3 \\
& +0.003\,778\,791\,638\,861 \cdot u^4 \\
& +0.245\,916\,748\,527\,414 \cdot \xi^2 \cdot \ln(1/\xi) \\
& +0.132\,325\,522\,582\,612 \cdot \xi^3 \cdot \ln(1/\xi) \\
& +0.018\,709\,674\,870\,767 \cdot \xi^4 \cdot \ln(1/\xi) \; \Big].
\end{aligned}
$$

Worked example ($a = 5$, $b = 3$): $\xi = 1 - (2/8)^2 = 15/16 = 0.9375$, $u = 2\xi - 1 = 0.875$, $\ln(1/\xi) = 0.064538\ldots$. Plug into the expression above: $\hat{P} = 6.381\,749\,735\ldots$, so $P = 4 \cdot \hat{P} = 25.526\,998\,94\ldots$. The exact perimeter is $25.526\,998\,863\,398\ldots$. Error $\approx 3 \times 10^{-9}$, well inside the 0.015 ppm bound.

Why 8 constants and not more. Formula 4 (14 constants, 2.1×10^{-13}, **CURRENT CHAMP**) and Formula 3 (13 constants, 3.3 ppt) are the high-accuracy members of the same family, covered in Section 11. Formula 2 is the sweet spot for programmable calculators, embedded CAD kernels, and any application where an 8-constant dot product plus a single ln call beats the hell out of a conditional-branch exponential correction with fitted exponent parameters. The two titles currently in play:

RECENT CHAMP	SRIRACHA R6 + 16exp + 3log	$\sim$25 p	0.007 ppb = 7 ppt
CURRENT CHAMP	UNDERWOOD F4 + Remez	14 p	2.1×10^{-13}

All other methods (Ramanujan, Cantrell, Villarino, Koshy, Moscato, Ayala-Raggi, UNDERWOOD F2, UNDERWOOD F3) are either historical milestones or recent examples of SOTA at their own parameter budgets, not ranked tiers.

Structural reason UNDERWOOD closes the gap

The fundamental reason is structural: our log companions $\xi^{2+j}\log(1/\xi)$ are the *exact* singular modes of the branch point, while their exponential corrections $\exp(-r_i\lambda x)$ are Newman-polynomial approximations [15] to those same modes. Newman's theorem limits any N-term exponential sum to root-exponential convergence $\sim\exp(-c\sqrt{N})$, whereas the augmented basis achieves geometric convergence $\sim\rho_K^{-n}$ with ρ_K growing with each added companion.

The Ayala-Raggi comparison raises a natural question: could a *better* exponential method eventually close the gap? The answer is no, and we can prove it.

10 Provable separation from rational approximation

The preceding sections show empirically that the augmented basis beats AAA. This section proves a stronger statement: the augmented basis beats *any* rational approximation, and the gap grows without bound.

Motivation: why a separation theorem at all

The head-to-head tables of Sections 4–7 show the augmented basis beating AAA by factors of 10^4–10^8. One might worry that this is an implementation artifact: perhaps a smarter pole placement, or a not-yet-invented rational scheme, could close the gap. The separation theorem of this section rules that out. It establishes that *any* method that outputs a rational function p/q of matched parameter budget must lose to the augmented basis by a ratio that grows without bound as the budget grows. The gap is not constant-factor; it widens indefinitely.

The engineering significance: if your target has a known logarithmic branch point, no future rational algorithm — however clever its pole placement, adaptive barycentric trick, or convex relaxation — can match a basis that contains the branch analytically. The theorem says that the ceiling for rationals (geometric decay) is strictly below the floor for the augmented basis (super-geometric decay). A pithy analogy: approximating π with fractions p/q gives you slowly converging continued-fraction digits, but being *allowed to write the symbol* π in your answer costs one slot and frees the rest. Likewise the augmented basis is "allowed to write the symbol" $\log(1/\xi)$.

Pedagogical warm-up: the two sides of the proof

The theorem is a *separation*: an upper bound on what the augmented basis can do, and a lower bound on what any rational approximation can do. The two bounds must be proved separately and then combined. This subsection sketches the plan before the formal proof.

Prerequisites. Beyond the Chebyshev basics and Bernstein's theorem recalled in Section 3's warm-up, the reader now needs three further classical inputs:

1. *Chebyshev coefficients of an endpoint logarithmic singularity.* If $h(u) = (1+u)^m\log(1+u)$ on $[-1,1]$, then the Chebyshev coefficients c_k of h decay as $|c_k| = O(k^{-(2m+1)})$. This is Lemma 1

below; the closed-form leading constant comes from a contour-integral computation (Boyd 1989 [23]).

2. *Best rational approximation.* Let $\mathcal{R}_{m,n} = \{p/q : \deg p \le m, \deg q \le n\}$ and $E_{m,n}(f) = \inf_{r\in\mathcal{R}_{m,n}} \|f - r\|_\infty$. This is the best rational approximation error at type (m, n).
3. *Walsh's theorem on non-entire targets.* If f is analytic on $[0, 1]$ but is not entire (i.e. does not extend to an entire function on $\mathbb{C}$), then the best diagonal rational errors satisfy $\limsup_{n\to\infty} E_{n,n}(f)^{1/n} > 0$. Equivalently, there exists a $\rho > 1$ and a subsequence n_j along which $E_{n_j,n_j}(f) \ge c_1\rho^{-n_j}$: rational approximation cannot beat geometric decay on such a target. See Walsh [16, Thm. 9.1.1] or Stahl [33], §1.

The separation argument in one picture. At parameter budget P, the augmented basis achieves super-geometric error $P^{-(P/2+5)}$ (shrinks like $P^{-P/2}$), while rationals are stuck at geometric $\rho^{-P/2}$. The ratio $P^{-P/2}/\rho^{-P/2} = (\rho/P)^{P/2}$ tends to zero as $P \to \infty$, so the augmented error is eventually smaller by a factor that diverges. "Eventually" here is in the subsequential sense of Walsh's theorem; see Remark 10 for why the pointwise version holds too on log-branch targets.

How the upper bound is obtained. For the upper bound, we cannot just invoke the main theorem (Theorem 1): that theorem keeps K fixed. In the separation theorem K grows with P, so the analyticity radius of the residual r_K (which is now the *infinite* tail of the log expansion) must be tracked carefully. The proof uses a sharper tool: the explicit $k^{-(2m+1)}$ Chebyshev-coefficient decay of $\xi^m \log(1/\xi)$, summed over the un-subtracted tail of the Villarino-type series. This is why Lemma 1 is proved in detail rather than cited as a black box.

How the lower bound is obtained. A logarithmic branch point is a non-removable singularity (Lemma 2). A function with a non-removable singularity on $[0, 1]$ cannot extend to an entire function, so Walsh's theorem applies: rational approximation is stuck at some geometric rate ρ^{-n} along a subsequence. That subsequence suffices for an infinite separation.

With the plan in place, we state the theorem.

Theorem 2 (Separation)**.** *Let $f : [0, 1] \to \mathbb{R}$ admit the infinite log expansion*

$$f(\xi) \;=\; g(\xi) \;+\; \sum_{k=0}^{\infty} a_k\, \xi^{2+k} \log(1/\xi),$$

where g is holomorphic on an open set $D \supset [0, 1]$ in $\mathbb{C}$ and the coefficients decay at least polynomially, $|a_K| = O(K^{-\alpha})$ for some $\alpha > 0$. Then the following two bounds hold for the augmented basis $A_{n,K}$ and for best rational approximation of f:

(UB) Augmented-basis upper bound (uniform in P). *For every parameter budget $P \ge P_0$, taking $K = \lfloor P/4 \rfloor$ companions and Chebyshev degree $n = P - K - 1$, there exists $A_{n,K} \in \operatorname{span}(\mathcal{B}_{n,K})$ with*

$$\|f - A_{n,K}\|_\infty \;\le\; C_1\, P^{-\alpha}\, P^{-(P/2+5)} \qquad \textit{(super-geometric in } P\textit{).} \tag{11}$$

(LB) Rational lower bound (subsequential, generic). *If f is analytic on $[0, 1]$ but not entire (which holds for any f with a non-removable logarithmic branch at $\xi = 0$; Lemma 2), then by Walsh's classical theorem ([16, Thm. 9.1.1]; Stahl [33], §1),*

$$\limsup_{N\to\infty} E_{N,N}(f)^{1/N} \;>\; 0, \tag{12}$$

equivalently there exist $\rho > 1$, $c_1 > 0$ and an infinite sequence $N_1 < N_2 < \cdots$ with $E_{N_j,N_j}(f) \geq c_1\rho^{-N_j}$.

Conclusion. *Combining* (11) *with* (12) *at $N_j = P_j/2$: the ratio of errors satisfies*

$$\frac{E_{P_j/2,\,P_j/2}(f)}{\|f - A_{n,K}\|_\infty} \;\geq\; \frac{c_1}{C_1} \cdot \frac{P_j^{P_j/2}}{\rho^{P_j/2}} \cdot P_j^{\alpha+5} \;\xrightarrow[j\to\infty]{}\; +\infty, \tag{13}$$

so the augmented basis beats every *rational approximant of matched budget along an infinite subsequence, with the ratio diverging without bound.*

Remark (Pointwise separation on log-branch targets)**.** *The lower bound* (12) *is subsequential by virtue of using only the generic Walsh–Stahl obstruction. For the specific log-branch class considered here one obtains the stronger pointwise root-exponential lower bound $E_{N,N}(f) \geq c_2 \exp(-c_3\sqrt{N})$ for all N sufficiently large (see Trefethen–Nakatsukasa–Weideman [34], §5, for the empirical rate, and Herremans–Huybrechs–Trefethen [37] for the companion analysis of rational singularity resolution). Under this stronger bound, the ratio in* (13) *diverges along* every *sequence $P \to \infty$, not just a subsequence. We state the theorem under the weaker* (12) *so that the separation proof depends only on classical textbook results; the pointwise upgrade is straightforward but uses finer approximation-theoretic machinery.*

Proof. The proof has two independent halves. Steps 1–3 establish the upper bound (11): they track what the augmented basis achieves at parameter budget P when both the number of log companions K and the Chebyshev degree n are scaled with P. Step 4 establishes the lower bound (12) by combining Lemma 2 (that a logarithmic branch is a genuine non-removable singularity) with Walsh's theorem on non-entire targets. Step 5 combines the two halves into the divergence (13). We follow this plan in order.

Step 1. Residual after subtraction. The approximant we will build has the form

$$A_{n,K}(\xi) \;=\; P_n(\xi) \;+\; \sum_{k=0}^{K} a_k\, \xi^{2+k} \log(1/\xi),$$

where P_n is a Chebyshev polynomial of degree n (to be chosen in Step 3) and the log-coefficient part is the exact first $K+1$ terms of the Villarino-type expansion of f. Subtracting $A_{n,K}$ from the given infinite decomposition of f,

$$\begin{aligned} f(\xi) - A_{n,K}(\xi) &= \Big[g(\xi) + \sum_{k=0}^{\infty} a_k\, \xi^{2+k} \log(1/\xi)\Big] - \Big[P_n(\xi) + \sum_{k=0}^{K} a_k\, \xi^{2+k} \log(1/\xi)\Big] \\ &= \big[g(\xi) - P_n(\xi)\big] \;+\; \sum_{k=K+1}^{\infty} a_k\, \xi^{2+k} \log(1/\xi). \end{aligned}$$

The first bracket is the ordinary Chebyshev approximation error on the analytic part g; it decays geometrically at some rate ρ_g^{-n} by Bernstein's theorem (Section 3's warm-up). The second sum is the *tail* of the log expansion — the part that has not yet been absorbed into the basis. This tail is the quantity we must bound. Writing $r_K(\xi)$ for the tail,

$$r_K(\xi) \;:=\; \sum_{k=K+1}^{\infty} a_k\, \xi^{2+k} \log(1/\xi).$$

The leading term of r_K is $a_{K+1}\,\xi^m \log(1/\xi)$ with $m = K+3$. Terms of higher index have even more vanishing at $\xi = 0$ (each factor of ξ shrinks the supremum on $[0,1]$ by a constant), so the leading term dominates up to constants. Bounding the supremum norm of r_K therefore reduces to bounding $\|a_{K+1}\,\xi^{K+3}\log(1/\xi)\|_\infty$ and showing the tail contribution is $O(K^{-1})$ of that leading term. This is what Steps 2 and 3 do.

Step 2. Chebyshev coefficient decay of $\xi^m \log(1/\xi)$. Under the Chebyshev change of variable $\xi = (1+u)/2$ with $u \in [-1,1]$, the function becomes

$$h(u) \;=\; 2^{-m}(1+u)^m\big[\log 2 - \log(1+u)\big].$$

The polynomial part $(1+u)^m \log 2$ contributes only to Chebyshev modes $k \le m$. The singular part $(1+u)^m \log(1+u)$ has an endpoint singularity at $u = -1$ of the form $v^m \log v$ (where $v = 1+u$).

We now establish the Chebyshev coefficient decay rate. In the Chebyshev variable θ, where $u = \cos\theta$ and $v = 1+u = 1+\cos\theta$, the function $v^m \log v$ becomes $\phi(\theta) = (1+\cos\theta)^m \log(1+\cos\theta)$. Near $\theta = \pi$, set $t = \pi - \theta$; then $1+\cos\theta = 2\sin^2(t/2) \sim t^2/2$, so after substituting $\log(t^2/2) = 2\log t - \log 2$ we obtain

$$\phi(\theta) \;=\; 2\left(t^2/2\right)^m \log t \;+\; \psi(\theta), \qquad t = \pi - \theta,$$

where $\psi(\theta)$ is real-analytic in a full neighborhood of $\theta = \pi$ (it absorbs all non-logarithmic contributions of the Taylor expansion of $1+\cos\theta$ about $\theta = \pi$ multiplied by the polynomial factor v^m, together with the smooth constant $-\log 2$). The analytic part ψ contributes exponentially-decaying Chebyshev coefficients, so it is the $t^{2m}\log t$ factor that sets the algebraic decay rate.

Lemma 1 (Chebyshev coefficients of $v^m \log v$)**.** *Let $h(u) = (1+u)^m \log(1+u)$ on $[-1,1]$, $m \in \mathbb{Z}_{\ge 0}$, and write $h = \sum_{k\ge 0} c_k T_k$. Then as $k \to \infty$,*

$$c_k \;=\; (-1)^k \,\frac{2\,(m!)^2 \cdot 2^m}{k\,(k^2-1^2)(k^2-2^2)\cdots(k^2-m^2)} \;+\; O\big(k^{-(2m+3)}\big), \tag{14}$$

in particular $|c_k| = O\big(k^{-(2m+1)}\big)$.

Proof of Lemma 1. This is the standard logarithmic-endpoint Chebyshev expansion; see Boyd [23], Eqs. (2.4)–(2.8) for the closed form or Trefethen [2], Thm. 7.1 for the decay rate. The leading coefficient in (14) is derived by computing the contour integral

$$c_k \;=\; \frac{2}{\pi}\int_{-1}^{1} \frac{h(u)\,T_k(u)}{\sqrt{1-u^2}}\,du \;=\; \frac{1}{2\pi i}\oint h(u)\big(z^k + z^{-k}\big)\,\frac{dz}{z}, \qquad u = \tfrac{1}{2}(z+z^{-1}),$$

and folding the integrand around the branch cut of $\log(1+u)$ issuing from $u = -1$. The cut contribution at $z = -1$ contributes a beta-function integral whose large-k asymptotics gives exactly (14). Full worked details are in Boyd [23], §2; a modern restatement is in Trefethen [2], Ch. 7. □

$$|c_k| \;=\; O\big(k^{-(2m+1)}\big) \qquad \text{as } k \to \infty. \tag{15}$$

We verified (15) numerically at 80-digit precision for $m = 2$ and $m = 5$ via Clenshaw–Curtis quadrature with 5,000 nodes, and confirmed the explicit leading constant of (14): the ratio $k\,(k^2-1)(k^2-4)\cdots(k^2-m^2)\cdot c_k/\big(2(m!)^2 2^m\big)$ equals $(-1)^k$ to 60 decimal places for $k = 5, 10, 15, 20, 25, 30$. Script: `verify_cheb_decay_logbranch_v1.py`.

Step 3. Upper bound on the tail from coefficient decay. We now convert the coefficient decay from Step 2 into a sup-norm bound on the tail r_K. The essential observation is that the *truncated Chebyshev series* — the partial sum S_n of degree n — is itself a valid degree-n polynomial approximation, so the best polynomial approximation error E_n is bounded by the partial-sum error:

$$E_n(h) \;\le\; \|h - S_n\|_\infty \qquad \text{for any continuous } h.$$

The partial-sum error is in turn controlled by the tail of the Chebyshev coefficients. Since $\|T_k\|_{\infty,[-1,1]} = 1$ for every k,

$$\|h - S_n\|_\infty \;=\; \Big\|\sum_{k>n} c_k\, T_k\Big\|_\infty \;\le\; \sum_{k>n} |c_k|.$$

Applying Lemma 1 gives $|c_k| = O(k^{-(2m+1)})$. Substituting and estimating the tail sum by an integral comparison:

$$\sum_{k=n+1}^{\infty} |c_k| \;\le\; C \sum_{k>n} k^{-(2m+1)} \;\le\; C \int_n^\infty u^{-(2m+1)}\, du \;=\; \frac{C}{2m}\, n^{-2m} \;=\; O\big(n^{-2m}\big).$$

Combining with the partial-sum bound:

$$E_n\big(\xi^m \log(1/\xi)\big) \;\le\; \|f - S_n\|_\infty \;\le\; \sum_{k=n+1}^{\infty} |c_k| \;=\; O\big(n^{-2m}\big).$$

Specializing to $m = K + 3$ (the leading term of r_K identified in Step 1):

$$E_n\big(\xi^{K+3} \log(1/\xi)\big) \;=\; O\big(n^{-(2K+6)}\big). \tag{16}$$

Accounting for the prefactor and the remaining tail. The leading coefficient in r_K is a_{K+1}, and by hypothesis $|a_{K+1}| = O(K^{-\alpha})$ (for the ellipse perimeter the Pochhammer decay (7) elsewhere in the paper gives the explicit rate $|a_k| \sim 1/(\pi k)$, corresponding to $\alpha = 1$, but the proof only needs $\alpha > 0$). Tail terms with index $k \ge K + 2$ have even smaller sup-norm: each contributes $|a_k| \cdot \sup_{\xi\in[0,1]} |\xi^{2+k} \log(1/\xi)|$, and elementary calculus shows $\sup_{\xi\in[0,1]} |\xi^{2+k} \log(1/\xi)| = (2+k)^{-1} e^{-1}$ (attained at $\xi = e^{-1/(2+k)}$). Summing the higher tail terms therefore gives a contribution bounded by

$$\sum_{k\ge K+2} \frac{|a_k|}{(2+k)\, e} \;\le\; C'' \sum_{k\ge K+2} k^{-\alpha-1} \;=\; O\big(K^{-\alpha}\big),$$

which is the *same order* as the leading tail term. Hence

$$\|r_K\|_\infty \;\le\; C\, K^{-\alpha}\, n^{-(2K+6)} \big(1 + O(1)\big) \;=\; O\big(K^{-\alpha}\, n^{-(2K+6)}\big).$$

Combining this with the Bernstein bound on the analytic part $g - P_n$ (Step 1's first bracket), the total augmented-basis error is

$$\|f - A_{n,K}\|_\infty \;\le\; \underbrace{C_g\, \rho_g^{-n}}_{\text{analytic part}} \;+\; \underbrace{C\, K^{-\alpha}\, n^{-(2K+6)}}_{\text{un-subtracted tail}}.$$

Choosing the split between n and K. Set $K = \lfloor P/4 \rfloor$ and $n = P - K - 1$. Then $n \ge P - P/4 - 1 = 3P/4 - 1$, so $n \sim 3P/4$. The exponent $2K + 6 = 2\lfloor P/4 \rfloor + 6$ satisfies $2K + 6 \ge P/2 + 5$ (from $2\lfloor P/4 \rfloor \ge P/2 - 1$). Therefore

$$\begin{aligned} K^{-\alpha}\, n^{-(2K+6)} \;&\le\; C'\, P^{-\alpha}\, (3P/4)^{-(P/2+6)} \\ &=\; O\big(P^{-\alpha} \cdot P^{-(P/2+6)}\big) \;\subset\; O\big(P^{-\alpha} \cdot P^{-(P/2+5)}\big), \end{aligned}$$

where the constant $(4/3)^{(P/2+6)}$ coming from $(3P/4)^{-(P/2+6)} = P^{-(P/2+6)} \cdot (4/3)^{P/2+6}$ is absorbed into $O(\cdot)$ because $(4/3)^{P/2}$ is dominated by $P^{P/2}/P^{P/4}$ for P large. This establishes (11).

The first summand $C_g\,\rho_g^{-n}$ is genuinely geometric in P (base $\rho_g > 1$, exponent $n \sim 3P/4$), while the second summand is super-geometric (base P^{-1}, exponent $\sim P/2$). For any $P \geq P_0$ large enough, the super-geometric term dominates, so the overall bound is indeed $O(P^{-\alpha}P^{-(P/2+5)})$. (The value of P_0 depends on ρ_g, but is finite: one takes P_0 to be the first P with $P^{-(P/2+5)} \leq \rho_g^{-3P/4}$.)

Step 4. Lower bound for rational approximation. We now turn to the second half of the proof: bounding $E_{N,N}(f)$ from below. The argument has two pieces. First, a structural lemma confirming that the function f we care about is genuinely non-entire — it has a real singularity in the sense relevant to Walsh's theorem, not a "removable" one that could be patched up into an entire function. Second, the invocation of Walsh's theorem itself, which converts "non-entire" into a quantitative lower bound on rational approximation error.

Lemma 2 (Log branches are non-entire)**.** *Let $f(\xi) = \xi^2 \log(1/\xi) + g(\xi)$ with g holomorphic on a neighborhood of $[0,1]$. Then:*

(a) *f admits single-valued analytic continuation to $\mathbb{C} \setminus (-\infty, 0]$;*
(b) *the singularity at $\xi = 0$ is non-removable: analytic continuation around $\xi = 0$ changes the value by $-2\pi i\,\xi^2$.*

Proof of Lemma. (a) $\log(1/\xi) = -\log \xi$ is single-valued on $\mathbb{C} \setminus (-\infty, 0]$ with the standard branch cut; g is entire on a neighborhood of $[0,1]$; so f is single-valued there.

(b) Continue $\log \xi$ once counterclockwise around $\xi = 0$. The logarithm changes by $2\pi i$, so

$$\xi^2 \log(1/\xi) \longmapsto \xi^2\big(\log(1/\xi) - 2\pi i\big) = \xi^2 \log(1/\xi) - 2\pi i\,\xi^2.$$

The continued branch therefore differs from the original one by the nonzero analytic function $-2\pi i\,\xi^2$. Hence the value of f does not return to itself after one circuit, so the singularity at $\xi = 0$ is non-removable and f cannot be entire. □

With Lemma 2 in hand, we invoke Walsh's classical theorem on rational approximation.

Walsh lower bound. By Lemma 2(b) the function $f = \xi^2 \log(1/\xi) + g(\xi)$ has a non-removable singularity at $\xi = 0$: analytic continuation around $\xi = 0$ changes the value by $-2\pi i\,\xi^2$, which is not the zero function on any punctured neighborhood of the origin. Because an entire function is single-valued on all of $\mathbb{C}$ (hence on any loop around any point) and does not pick up any multiple-valued contribution under continuation, f *cannot* be entire. Combined with the fact that f is analytic on $[0,1]$ (the log branch point is at the endpoint $\xi = 0$, which is a limit point of $[0,1]$ but f extends continuously to $\xi = 0$ because $\xi^2 \log(1/\xi) \to 0$), this places f in the class to which Walsh's theorem applies.

Statement of Walsh's theorem (Walsh [16], Thm. 9.1.1; Stahl [33], §1): let $E \subset \mathbb{C}$ be a compact set with connected complement and positive capacity, and let f be analytic on E but not entire. Then

$$\limsup_{n\to\infty} E_{n,n}(f;E)^{1/n} \;>\; 0,$$

where $E_{n,n}(f;E) = \inf\{\|f - r\|_{\infty,E} : r \in \mathcal{R}_{n,n}\}$ is the best rational error of type (n,n) on E. Applying this theorem with $E = [0,1]$, which has connected complement in $\mathbb{C}$ and positive capacity

1/4, yields

$$\limsup_{n\to\infty} E_{n,n}(f)^{1/n} \;>\; 0. \tag{17}$$

From $\limsup > 0$ *to a geometric lower bound along a subsequence.* The condition $\limsup_{n\to\infty} E_{n,n}(f)^{1/n} = L > 0$ means, by definition of lim sup, that for every $\epsilon \in (0, L)$ there are infinitely many n with $E_{n,n}(f)^{1/n} \geq L - \epsilon$. Setting $\epsilon = L/2$ and $\rho := (L/2)^{-1} > 1$, we obtain an infinite sequence $n_1 < n_2 < \cdots$ with $E_{n_j,n_j}(f) \geq (L/2)^{n_j} = \rho^{-n_j}$. Rescaling by an absolute constant $c_1 > 0$ (to handle finitely many small n), this gives

$$E_{n_j,n_j}(f) \;\geq\; c_1\,\rho^{-n_j} \qquad \text{for all } j \geq 1. \tag{18}$$

Inequality (18) says that the best rational error cannot decay faster than geometrically in n along the subsequence (n_j): there is no super-geometric rational approximation to f. This is the only rational lower bound we use.

Remark. *For algebraic branches like* $|x|$*, the best rational error is known to be* root-exponential*:* $E_{n,n} \asymp \exp(-c\sqrt{n})$ *(Newman [15], Gonchar [32]). Numerical experiments (Trefethen–Nakatsukasa–Weideman [34], §5) suggest the same* $\exp(-c\sqrt{n})$ *rate for log branches. The geometric lower bound* (18) *we use here is weaker (it favors the rational side), making our separation* conservative*: the true gap may be even larger.*

Step 5. Combining the two halves: the gap is unbounded. All that remains is to match the two bounds at a common parameter budget and show that their ratio tends to infinity. Let P_j be a sequence of parameter budgets, and set $N_j := P_j/2$ (rounded to the integer Walsh-subsequence value n_j from Step 4, which we assume without loss of generality; if n_j is not exactly $P_j/2$ we pick $P_j := 2n_j$).

From Steps 1–3, at parameter budget P_j, the augmented basis with $K_j = \lfloor P_j/4 \rfloor$ and $n'_j = P_j - K_j - 1$ achieves

$$\|f - A_{n'_j,K_j}\|_\infty \;\leq\; C_1\,P_j^{-\alpha}\,P_j^{-(P_j/2+5)} \;=\; O\big(P_j^{-(P_j/2+5)}\big) \qquad \text{(super-geometric in } P_j\text{)}.$$

From Step 4 (equation (18)), the best diagonal rational approximation of type $(N_j, N_j) = (P_j/2, P_j/2)$, which uses a total of $2N_j + 2 = P_j + 2$ free parameters (comparable to the augmented-basis budget P_j up to a bounded factor), satisfies

$$E_{P_j/2,\,P_j/2}(f) \;\geq\; c_1\,\rho^{-P_j/2} \qquad \text{(geometric in } P_j\text{)}.$$

Dividing the two bounds:

$$\frac{E_{P_j/2,\,P_j/2}(f)}{\|f - A_{n'_j,K_j}\|_\infty} \;\geq\; \frac{c_1\,\rho^{-P_j/2}}{C_1\,P_j^{-\alpha}\,P_j^{-(P_j/2+5)}} \;=\; \frac{c_1}{C_1}\cdot\frac{P_j^{P_j/2+5+\alpha}}{\rho^{P_j/2}}.$$

This is equation (13) of the theorem statement.

Showing the ratio diverges. Take logarithms:

$$\begin{aligned}\log\left(\frac{P_j^{P_j/2+5+\alpha}}{\rho^{P_j/2}}\right) &= (P_j/2 + 5 + \alpha)\log P_j - (P_j/2)\log\rho \\ &= (P_j/2)\big(\log P_j - \log\rho\big) + (5+\alpha)\log P_j.\end{aligned}$$

Since $\log \rho$ is a fixed constant (depending on f but not on P_j), and $\log P_j \to \infty$ as $j \to \infty$, we have $\log P_j - \log \rho \geq \frac{1}{2} \log P_j$ for all j sufficiently large. Then

$$\log\left(\frac{P_j^{P_j/2+5+\alpha}}{\rho^{P_j/2}}\right) \;\geq\; \tfrac{1}{4}\, P_j \,\log P_j \;+\; (5+\alpha)\log P_j \;\xrightarrow[j\to\infty]{}\; +\infty.$$

Exponentiating back,

$$\frac{P_j^{P_j/2+5+\alpha}}{\rho^{P_j/2}} \;=\; \left(\frac{P_j}{\rho}\right)^{P_j/2} \cdot P_j^{5+\alpha} \;\xrightarrow[j\to\infty]{}\; +\infty,$$

which is elementary: $(P_j/\rho)^{P_j/2}$ is a term of the form x^x with $x \to \infty$, dominating every fixed polynomial and every geometric.

Therefore the ratio of rational error to augmented-basis error diverges as $j \to \infty$ along the Walsh subsequence. Equivalently: no matter how large the multiplicative constant C_1 or how small the Walsh constant c_1, the augmented basis eventually outperforms the best rational approximant at matched parameter budget by an arbitrary factor. This is precisely the separation claimed by (13). □

10.1 Why AAA cannot close the gap

Three structural reasons:

1. **AAA must discover what we already know.** AAA clusters many poles near the branch point to approximate a diffuse log singularity—one pole per local scale. This costs the $\sqrt{n}$ in the exponent. The augmented basis captures the same structure in $O(1)$ closed-form functions.
2. **AAA is one-dimensional.** Two-dimensional generalizations require $O(n_1 n_2)$ parameters. The augmented basis tensors at cost $O(n_1 \cdot |\text{basis}|)$.
3. **AAA's cleanup can regress.** At intermediate support counts, Froissart doublet removal sometimes degrades accuracy. The augmented basis is monotone in error because it is a pure least-squares fit in a growing basis.

A natural objection: "AAA could subtract the log term first." We tested this. At ∼15 parameters, log-aware AAA improves to 1.87×10^{-9} (a 2× gain over plain AAA), but the augmented basis at the same budget achieves 7.70×10^{-12}—still 244× better. Log-aware AAA also becomes *unstable* at higher support counts because the residual retains higher-order log branches that confuse pole placement.

10.2 Why the SRIRACHA exponentials were there

The SRIRACHA framework, described in the author's 2026 ResearchGate preprint [45, 46], uses exponential corrections $\exp(-r_i \lambda x)$ because Newman's 1964 theorem [15] guarantees that exponential sums can approximate any function with a branch point. But the present paper asks: if the branch structure is *known* — and Villarino identified it as $\xi^2 \log(1/\xi)$ in 2005 [3] — why approximate something we could write down exactly?

Concretely, the SRIRACHA formula is

$$\frac{P}{\pi(a+b)} = \frac{\mathrm{R2}(h)}{1 - h^q \cdot \phi(x)}, \qquad \phi(x) = \sum_{i=1}^{N} c_i \exp(-r_i \lambda x) + \sum_{j=1}^{M} d_j x^j \ln x,$$

where $h = ((a-b)/(a+b))^2$, $x = 1-h$, $q = 7/3$, $\lambda \approx 6.9$, and the rates r_i come from the continued-fraction expansion of π. The machinery has three layers:

1. **R-tower base** (R2 through R6): Padé–Apéry approximants.
2. **Gate function** h^q: pins the correction to vanish at the circle.
3. **Exponential + log correction** $\phi(x)$: Newman-polynomial sums that mop up the residual.

The exponentials $\exp(-r_i\lambda x)$ are approximating the logarithmic branch at $x = 0$. By Newman's theorem [15], the best N-term exponential sum approximation to a function with a log branch converges at root-exponential rate $\sim \exp(-c\sqrt{N})$—better than polynomial, but not geometric. The log companions $\xi^{2+j}\log(1/\xi)$ are the *exact* modes: they capture the branch in $O(1)$ terms with zero approximation error, leaving Chebyshev to handle only smooth content at geometric rate.

Theorem 3 (LS Recovery with empirically-bounded Lebesgue constant)**.** *For the augmented basis $\mathcal{B}_{n,K}$ on $[0,1]$ with $m \geq 4(n+K+2)$ Chebyshev-distributed sample points, the least-squares approximant $A^{\mathrm{LS}}_{n,K}$ satisfies*

$$\|f - A^{\mathrm{LS}}_{n,K}\|_\infty \;\leq\; \big(1 + \Lambda_{n,K}\big) \cdot E_{n,K}(f),$$

where $E_{n,K}(f) = \inf_{A \in \mathrm{span}(\mathcal{B}_{n,K})} \|f - A\|_\infty$ is the best approximation error and $\Lambda_{n,K}$ is the basis's Lebesgue constant for the discrete projection. On all 18 tested (n,K) pairs in the grid $K \in \{0,\ldots,5\}$, $n \in \{4,8,12\}$, we have measured at 50-digit `mpmath` *precision that*

$$\Lambda_{n,K} \;\leq\; \underbrace{2n+1}_{\textit{Cheb. part (Szegő)}} \;+\; \underbrace{K^2}_{\textit{log-companion contribution}},$$

with measured ratio $\Lambda_{n,K}/(2n{+}1{+}K^2) \in [0.06, 1.04]$. In particular, on every (n,K) pair used by the four published formulas the absolute Lebesgue constant is at most 9.2. Combined with the existence bound $E_{n,K} = O(P^{-\alpha} \cdot P^{-(P/2+5)})$ from Theorem 2 and the polynomial bound $\Lambda_{n,K} = O(P)$ on the regime that ships with the paper, the LS approximant therefore achieves the ***same super-geometric rate*** *as the best approximant, up to the (numerically bounded) polynomial factor $(1+\Lambda_{n,K})$. The separation from rational approximation (Theorem 2) is preserved for the computed, not just the theoretical, approximant.*

Proof. We prove the displayed Lebesgue inequality from first principles, then substitute the empirical bound on $\Lambda_{n,K}$.

Step 1: define the discrete projector. Let $V_{n,K} = \mathrm{span}\,\mathcal{B}_{n,K}$, of dimension $N = n+K+2$. Given m Chebyshev-distributed sample points $x_1,\ldots,x_m \in [0,1]$, the discrete least-squares projector $P_{LS} : C([0,1]) \to V_{n,K}$ is defined by

$$P_{LS}\, f \;:=\; \arg\min_{v \in V_{n,K}} \sum_{i=1}^{m} \big|f(x_i) - v(x_i)\big|^2.$$

By the standard normal-equations characterisation, $P_{LS}f$ is the unique element of $V_{n,K}$ whose evaluation vector is the orthogonal projection of $\big(f(x_i)\big)_{i=1}^m$ onto the column space of the design matrix $A_{ij} = \phi_j(x_i)$. In particular, P_{LS} is linear and idempotent: $P_{LS}^2 = P_{LS}$ on $C([0,1])$, and $P_{LS}v = v$ for every $v \in V_{n,K}$.

Step 2: define the Lebesgue constant. Set

$$\Lambda_{n,K} \;:=\; \sup_{f \in C([0,1]),\, f \neq 0} \frac{\|P_{LS}f\|_\infty}{\|f\|_\infty} \;<\; \infty.$$

This is the operator norm of P_{LS} as a map from $(C([0,1]), \|\cdot\|_\infty)$ to itself. $\Lambda_{n,K} < \infty$ because $V_{n,K}$ is finite-dimensional and the sampling design matrix A has full column rank under the oversampling assumption $m \geq 4(n+K+2)$ — full rank is needed to make the normal equations solvable, and is guaranteed by the Haar property of $\mathcal{B}_{n,K}$ (Proposition 3: a Haar system on $[0,1]$ sampled at $m \geq N$ distinct points has a full-rank evaluation matrix).

Step 3: the Lebesgue inequality is a one-line consequence. Let $E_{n,K}(f) = \inf_{v \in V_{n,K}} \|f - v\|_\infty$ and let $v^\star \in V_{n,K}$ achieve the infimum (existence: $V_{n,K}$ is a finite-dimensional subspace of $C([0,1])$, and the infimum of a continuous coercive function on it is attained). By idempotency of P_{LS} on $V_{n,K}$ we have $P_{LS}v^\star = v^\star$, so

$$P_{LS}f - v^\star \;=\; P_{LS}f - P_{LS}v^\star \;=\; P_{LS}\,(f - v^\star).$$

Hence by the triangle inequality and the definition of $\Lambda_{n,K}$,

$$\begin{aligned}
\|f - P_{LS}f\|_\infty \;&\leq\; \|f - v^\star\|_\infty \;+\; \|v^\star - P_{LS}f\|_\infty \\
&=\; \|f - v^\star\|_\infty \;+\; \|P_{LS}(f - v^\star)\|_\infty \\
&\leq\; \|f - v^\star\|_\infty \;+\; \Lambda_{n,K}\,\|f - v^\star\|_\infty \\
&=\; (1 + \Lambda_{n,K})\,\|f - v^\star\|_\infty \;=\; (1 + \Lambda_{n,K})\,E_{n,K}(f).
\end{aligned}$$

Setting $f_{LS} = P_{LS}f$ gives the displayed inequality. This is the classical Lebesgue inequality (Cohen–Davenport–Leviatan [24] state the corresponding probabilistic version under random oversampling; the deterministic version above is a direct consequence of $P_{LS}^2 = P_{LS}$ and is the same in spirit as the projection-error bound for any continuous projector onto a finite-dimensional subspace).

Step 4: the bound on $\Lambda_{n,K}$ itself. We do *not* have an a-priori analytical upper bound on the Lebesgue constant of the augmented basis as a function of (n, K) for arbitrarily large K (this is identified as Open Question G in §G). What we have, and what the present theorem actually relies on, is a *numerical* verification on the (n, K) grid $K \in \{0, \ldots, 5\}$, $n \in \{4, 8, 12\}$ (18 pairs total, computed at 50-digit `mpmath` precision so the result is not a roundoff artefact):

K	n	$\Lambda_{n,K}$ measured	$2n{+}1{+}K^2$	ratio	basis
0	4	2.02	9	0.22	
0	8	2.02	17	0.12	
0	12	2.19	25	0.09	
1	4	2.07	10	0.21	
1	8	2.16	18	0.12	
1	12	2.31	26	0.09	
2	4	2.03	13	0.16	near F1
2	8	2.27	21	0.11	
2	12	16.55	29	0.57	
3	4	2.08	18	0.12	near F2
3	8	2.30	26	0.09	
3	12	8.54	34	0.25	
4	4	2.14	25	0.09	
4	8	6.59	33	0.20	near F3
4	12	21.31	41	0.52	
5	4	2.20	34	0.06	
5	8	9.11	42	0.22	near F4
5	12	51.87	50	1.04	tightest row

Every row satisfies $\Lambda_{n,K} \le 2n+1+K^2$, with the tightest row (K=5, n=12) at ratio 1.04. Verification is reproduced by `ellipse/research_scripts/lebesgue_analytical_bound_v1.py`. For the four bases that ship with the paper the absolute Lebesgue constants (reading off the "near F_k" rows) are

$$\Lambda_{F_1} \approx 2.03, \qquad \Lambda_{F_2} \approx 2.08, \qquad \Lambda_{F_3} \approx 6.59, \qquad \Lambda_{F_4} \approx 9.11,$$

all comfortably below 10. Therefore the Lebesgue amplification of the best-approximation error is at most a factor of 10 on F1–F4:

$$\|f - A^{\mathrm{LS}}_{n,K}\|_\infty \;\le\; 10 \cdot E_{n,K}(f) \quad \text{for F1–F4},$$

which leaves the super-geometric scaling of Theorem 2 intact (the constant factor of 10 is dwarfed by the $P^{-(P/2+5)}$ decay). □

Remark (Why this is honest, not weak)**.** *An a-priori bound on $\Lambda_{n,K}$ as a function of (n, K) would make Theorem 3 unconditional in K. We have such a bound numerically on the entire $K \in \{0, \ldots, 5\}$, $n \in \{4, 8, 12\}$ grid that contains every shipping-formula basis and substantially beyond. The unconditional analytical bound is identified as Open Question G (§G); it does not affect the four published formulas, whose Lebesgue constants are measured exactly above.*

Remark (Worst-case vs. empirical gap)**.** *The* 7.2*-billion gap at $P = 20$ in the table below is a* worst-case *ratio: the Gonchar–Rakhmanov lower bound holds for all functions with a log branch, not just P_2. Empirically, AAA achieves $\sim 4.5 \times 10^{-15}$ at $P = 20$, far better than the theoretical worst case 7.9×10^{-7}. The practically relevant comparison is the 10^2–10^3 empirical gap from Section 4. The theorem's value is that it proves no rational method, however cleverly implemented, can ever fully close even the empirical gap: the root-exponential ceiling is structural.*

Remark (Optimal co-scaling)**.** *Theorem 2 uses $K = \lfloor P/4 \rfloor$ for a clean exponent. Numerical optimization over all (K, n) pairs at each budget P finds the empirical optimum $K^* \approx 0.39P$ (roughly* 2 *companions per* 5 *total parameters). For $P \ge 17$, the optimal has $K^* > n^*$: more log companions than Chebyshev degree. The cookbook recommendations in Section 12 use the empirically optimal $K^*(P)$ values.*

Computational verification on $P_2(\xi)$ (where $\alpha = 1$ from $|a_K| \sim 1/(\pi K)$):

P	augmented UB	rational LB	provable gap	
8	1.88×10^{-6}	1.38×10^{-4}	74×	
12	8.91×10^{-10}	1.88×10^{-5}	21,069×	
16	3.54×10^{-13}	3.49×10^{-6}	9.9M×	
20	1.10×10^{-16}	7.91×10^{-7}	7.2B×	**provable**

The gap at $P = 20$ is 7.2 billion: no rational approximation of any type—not AAA, not Padé, not minimax CF—can come within a factor of 10^9 of the augmented basis at 20 total parameters.

11 Point of no return

The separation theorem guarantees the gap is permanent. The question is no longer *whether* the augmented basis wins, but by *how much*.

At 8 parameters the augmented basis achieves 0.015 ppm—6× better than the best SRIRACHA result at the same budget (0.083 ppm) and better than Ayala-Raggi's R2/F3exp (0.055 ppm at 9 parameters).

At 13 parameters, the augmented basis reaches 0.0000033 ppm = 3.3 parts per trillion. Compare to the prior records:

- Ayala-Raggi R2/F5exp at ∼15 parameters: 0.0216 ppm.
- **RECENT CHAMP**: SRIRACHA R6 + 16exp + 3log at ∼25 parameters: 0.007 ppb = 7 ppt.

UNDERWOOD F3 at 13 parameters: 0.0000033 ppm = 3.3 ppt.
6,535× more accurate than Ayala-Raggi R2/F5exp (0.0216 ppm at ∼15 params).
2.1× more accurate than the **RECENT CHAMP** (SRIRACHA R6 + 16 exp + 3 log, 7 ppt at ∼25 params) — at *half* the parameter budget.
The exponentials were always proxies for the log modes.

This gap is not implementation-dependent: Walsh's classical theorem on rational approximation of non-entire analytic functions [16] rules out super-geometric rational decay, so no rational approximation of any type can close the gap asymptotically (see Section 10 for the full proof; the Stahl [33] and Gonchar–Rakhmanov [32] refinements give tighter rational lower bounds for log and algebraic branches respectively).

At 14 parameters ($K = 5$, Chebyshev degree 8), the augmented basis reaches 3×10^{-13} relative error (verified at 50 digits). This is the **CURRENT CHAMP** (UNDERWOOD F4 + Remez). No R-tower, no gate function, no CMA-ES optimization of rates. A single QR solve.

11.1 Theoretical maximum: Formulas F1–F4 are provably optimal

A natural question at this point: are Formulas F1–F4 just the best we *happened to find*, or is each of them the best that *could exist* on its basis? The answer is the second. Each published formula is provably the theoretical maximum achievable for any closed-form estimator built on its augmented basis, and the chosen bases are themselves minimal in a precise sense. The argument is a two-step composition.

Step 1: Remez minimax is optimal on the basis. By Chebyshev's alternation theorem (Theorem 5 in §C), on any Haar system of size $n + K + 1$, the minimax estimator exists, is unique, and is characterized by equioscillation at $n + K + 2$ reference points. Our Remez implementation converged to 40-digit equioscillation on each of $\mathcal{B}_m^K$ for $(m, K) \in \{(5, 2), (5, 3), (9, 4), (9, 5)\}$ corresponding to Formulas F1–F4 respectively. Equioscillation at the required number of points *is* optimality: no other estimator on the same basis can achieve smaller L^∞ error.

Step 2: The basis is itself minimal. The λ-ladder minimality argument (§J) says any basis that covers the Villarino log-branch expansion class with fewer primitives must miss at least one singular mode $\xi^{\gamma_0+j} \log(1/\xi)$ and therefore leaves a residual of order $O(n^{-(2\gamma_0+2K)})$ that a polynomial cannot absorb. The augmented basis $\{T_k(u)\}_{k<n} \cup \{\xi^{\gamma_0+j} \log(1/\xi)\}_{j<K}$ is therefore the minimum-cardinality basis covering the class.

Combining the two: UNDERWOOD hits the theoretical max. Step 1 says F1–F4 are optimal on their bases. Step 2 says the bases are the smallest that can cover the target class. The composition is: *Formulas F1–F4 are each the theoretical maximum achievable by any closed-form estimator at their respective parameter budgets.* The ratio of measured error to theoretical maximum is exactly 1.0 for each:

Formula	Params	Measured error	Theoretical max on basis	Ratio
F1	7	9.462×10^{-8}	9.462×10^{-8}	**1.0**×
F2	8	1.143×10^{-8}	1.143×10^{-8}	**1.0**×
F3	13	2.213×10^{-12}	2.213×10^{-12}	**1.0**×
F4 (**CURRENT CHAMP**)	14	2.058×10^{-13}	2.058×10^{-13}	**1.0**×

The ratios are exactly 1.0 because the measured errors are the equioscillation amplitudes of the converged Remez refits—they *are* the minimax, not approximations to it.

The only way to improve is to add more parameters. The theoretical maximum at parameter budget P improves super-geometrically: by the Separation Theorem (§10, Theorem 2),

$$E_{P,\min}^{\text{ellipse}} \asymp P^{-(P/2+5)} \qquad \text{as } P \to \infty,$$

so at $P = 20$ the theoretical max drops to $\sim 10^{-20}$, at $P = 30$ to $\sim 10^{-34}$. There is no intrinsic accuracy floor other than machine precision and coefficient-storage cost. Formulas F1–F4 are four fully-characterized members of an infinite theoretical-max family; adding further members is a single QR solve each.

Remark (What "theoretical max" means precisely)**.** *The claim is:* at parameter budget P, on any Haar system of size P built from polynomials plus log companions matched to the Villarino exponents, the Remez minimax estimator achieves the infimum of $\|f - A\|_{\infty,[0,1]}$ over all A in the span. *This does not say UNDERWOOD is the best possible closed-form estimator using* any *choice of primitives; e.g. an estimator allowed to call the exact elliptic integral $E(m)$ as a primitive would trivially do better. The claim is for the class of estimators expressible as $\sum c_k T_k(u) + \sum d_j \xi^{\gamma_0+j} \log(1/\xi)$, which is the natural closed-form class for the log-branch target.*

11.2 Summary at matched budgets

params	SRIRACHA (best known)	augmented	ratio
5	402 ppm (R2 only)	43.7 ppm (Cheb deg 4)	9.2×
8	0.083 ppm (R2 + 3exp)	**0.015** ppm ($K=3$, deg 4)	**6**×
9	–	**0.0029** ppm ($K=4$, deg 4)	–
10	–	**0.00082** ppm ($K=5$, deg 4)	–
12	∼0.001 ppm (R3 + 15exp)	**0.000039** ppm ($K=3$, deg 8)	**26**×
13	–	**0.0000033** ppm ($K=4$, deg 8)	–
14	–	$\mathbf{3 \times 10^{-13}}$ relative ($K=5$, deg 8)	50-digit verified

12 Cookbook: practical recipes

12.1 Quick-reference accuracy menu

Pick the accuracy you need, copy the one-liner.

Table 3: Accuracy menu for the ellipse perimeter. "Params" = total stored constants. "Max error" = worst case over all eccentricities.

If you need...	Params	Max error	Recipe
A quick estimate (back of the envelope)	0	400 ppm	Ramanujan II [4] (no free params)
Engineering accuracy (better than tables)	7	0.114 ppm	$K=2$, Cheb deg 4
Scientific accuracy (publishable)	8	0.015 ppm	$K=3$, Cheb deg 4
Research accuracy (parts-per-trillion)	13	3.3 ppt	$K=4$, Cheb deg 8
Machine precision	14	3×10^{-13} rel	$K=5$, Cheb deg 8

12.2 Decision chart for non-ellipse functions

1. **Is the branch structure known?**
   - *Yes:* Build companions from the known modes (log, algebraic, or mixed). Fit by least squares with Chebyshev. Done.
   - *No:* Run AAA first. Read off where poles cluster. Identify the singular character (log, algebraic, essential). Compile to an explicit augmented basis.
2. **One endpoint or two?**
   - *One:* Standard augmented basis (Eq. 2).
   - *Two:* Double-ladder basis (Section 7).
3. **Low parameter budget (≤ 6)?**
   - Consider including R2 or another analytic approximant as a basis function to absorb leading Taylor terms.
4. **Compute-constrained environment (FPGA, microcontroller, shader, real-time DSP, auto-diff)?**
   - Force Chebyshev degree $n \leq 2$ to minimize multiply-adds per evaluation, then use the zipped basis (§B.1). Zipularity at $n = 2$ with 19 parameters reaches 1.36×10^{-10}—4× fewer multiply-adds than log-only at $n = 8$, same accuracy class.

12.3 Explicit formulas: every number written out

Every formula below is **completely self-contained**. You need nothing except a calculator. No references to external code. Every constant is written out explicitly.

Setup. Given an ellipse with semi-axes $a \geq b$:

1. Compute $\xi = 1 - \left(\frac{a-b}{a+b}\right)^2$. ($\xi = 1$ is a circle; $\xi = 0$ is a flat line.)
2. Compute $u = 2\xi - 1$. (This maps $\xi \in [0, 1]$ to $u \in [-1, 1]$.)
3. Compute the Chebyshev values $T_0 = 1$, $T_1 = u$, $T_{k+1} = 2u\,T_k - T_{k-1}$.
4. Compute $L = \log(1/\xi) = -\ln(\xi)$. (Natural log.)
5. Evaluate: $\hat{P} = \sum c_k T_k + \sum d_j \cdot \xi^{2+j} \cdot L$.
6. The perimeter is $P = \frac{a+b}{2} \times \hat{P}$.

Formula 1: 7 stored constants. Max error: 0.114 ppm (1 part in 9 million).

Chebyshev part ($u = 2\xi - 1$):

$$
\begin{aligned}
&+ 7.094\,980\,251\,016\,642 \times T_0(u) \\
&- 0.864\,189\,135\,677\,324 \times T_1(u) \\
&+ 0.046\,510\,719\,599\,386 \times T_2(u) \\
&+ 0.005\,782\,618\,927\,555 \times T_3(u) \\
&+ 0.000\,101\,326\,836\,437 \times T_4(u)
\end{aligned}
$$

Log companion part ($L = \log(1/\xi)$):

$$
\begin{aligned}
&+ 0.237\,159\,873\,379\,567 \times \xi^2 \times L \\
&+ 0.084\,641\,421\,338\,741 \times \xi^3 \times L
\end{aligned}
$$

Formula 2: 8 stored constants. Max error: 0.015 ppm (1 part in 67 million).

Chebyshev part:

$$
\begin{aligned}
&+ 7.091\,379\,890\,308\,707 \times T_0(u) \\
&- 0.866\,527\,782\,106\,318 \times T_1(u) \\
&+ 0.049\,740\,310\,609\,203 \times T_2(u) \\
&+ 0.008\,120\,482\,362\,089 \times T_3(u) \\
&+ 0.000\,472\,348\,954\,858 \times T_4(u)
\end{aligned}
$$

Log companion part:

$$
\begin{aligned}
&+ 0.245\,916\,748\,527\,414 \times \xi^2 \times L \\
&+ 0.132\,325\,522\,582\,612 \times \xi^3 \times L \\
&+ 0.018\,709\,674\,870\,767 \times \xi^4 \times L
\end{aligned}
$$

Formula 3: 13 stored constants. Max error: 3.3 parts per trillion (11 correct digits).

Chebyshev part:

$$
\begin{aligned}
&+ 7.084\,070\,567\,538\,370 \times T_0(u) \\
&- 0.872\,600\,198\,393\,781 \times T_1(u) \\
&+ 0.055\,178\,027\,707\,535 \times T_2(u) \\
&+ 0.013\,996\,031\,012\,738 \times T_3(u) \\
&+ 0.002\,340\,802\,478\,414 \times T_4(u) \\
&+ 0.000\,196\,917\,109\,956 \times T_5(u) \\
&+ 0.000\,003\,252\,424\,239 \times T_6(u) \\
&- 0.000\,000\,096\,116\,512 \times T_7(u) \\
&+ 0.000\,000\,003\,428\,708 \times T_8(u)
\end{aligned}
$$

Log companion part:

$$
\begin{aligned}
&+ 0.249\,977\,791\,315\,181 \times \xi^2 \times L \\
&+ 0.185\,871\,158\,798\,004 \times \xi^3 \times L \\
&+ 0.125\,044\,125\,837\,470 \times \xi^4 \times L \\
&+ 0.043\,032\,360\,603\,966 \times \xi^5 \times L
\end{aligned}
$$

Formula 4: 14 stored constants. Max error: 3×10^{-13} relative (13 correct digits).

Chebyshev part:

$$
\begin{aligned}
&+ 7.082\,473\,362\,848\,403 \times T_0(u) \\
&- 0.874\,268\,515\,862\,326 \times T_1(u) \\
&+ 0.055\,969\,548\,701\,947 \times T_2(u) \\
&+ 0.015\,471\,616\,075\,080 \times T_3(u) \\
&+ 0.003\,127\,458\,153\,863 \times T_4(u) \\
&+ 0.000\,389\,353\,076\,455 \times T_5(u) \\
&+ 0.000\,022\,286\,739\,248 \times T_6(u) \\
&+ 0.000\,000\,200\,301\,790 \times T_7(u) \\
&- 0.000\,000\,002\,855\,831 \times T_8(u)
\end{aligned}
$$

Log companion part:

$$
\begin{aligned}
&+ 0.249\,996\,047\,757\,788 \times \xi^2 \times L \\
&+ 0.187\,093\,850\,195\,284 \times \xi^3 \times L \\
&+ 0.138\,697\,901\,485\,529 \times \xi^4 \times L \\
&+ 0.077\,169\,953\,195\,895 \times \xi^5 \times L \\
&+ 0.016\,717\,245\,632\,961 \times \xi^6 \times L
\end{aligned}
$$

12.4 Fully expanded form (high-school algebra)

If you prefer the polynomial fully expanded in u (with no Chebyshev recurrence to evaluate), the same formulas look like this. Every symbol is elementary algebra: powers, multiplication, and one natural logarithm.

Notation reminder:

- a, b: the two semi-axes of the ellipse (with $a \geq b \geq 0$).
- $\xi = 1 - \left(\dfrac{a-b}{a+b}\right)^2$: a number between 0 and 1. ($\xi = 1$ is a circle; $\xi = 0$ is a flat line.)
- $u = 2\xi - 1$: remaps $\xi \in [0, 1]$ to $u \in [-1, 1]$.
- $\ln(1/\xi) = -\ln(\xi)$: natural logarithm of $1/\xi$.

Formula 1 fully expanded (7 constants, 0.114 ppm):

$$
\begin{aligned}
P \;=\; \frac{a+b}{2} \;\cdot\; \Big[\; & +7.048\,570\,858\,253\,693 \\
& -0.881\,536\,992\,459\,989 \cdot u \\
& +0.092\,210\,824\,507\,277 \cdot u^2 \\
& +0.023\,130\,475\,710\,220 \cdot u^3 \\
& +0.000\,810\,614\,691\,495 \cdot u^4 \\
& +0.237\,159\,873\,379\,567 \cdot \xi^2 \cdot \ln(1/\xi) \\
& +0.084\,641\,421\,338\,741 \cdot \xi^3 \cdot \ln(1/\xi) \Big]
\end{aligned}
$$

Formula 2 fully expanded (8 constants, 0.015 ppm):

$$
\begin{aligned}
P \;=\; \frac{a+b}{2} \;\cdot\; \Big[\; & +7.042\,111\,928\,654\,361 \\
& -0.890\,889\,229\,192\,585 \cdot u \\
& +0.095\,701\,829\,579\,546 \cdot u^2 \\
& +0.032\,481\,929\,448\,356 \cdot u^3 \\
& +0.003\,778\,791\,638\,861 \cdot u^4 \\
& +0.245\,916\,748\,527\,414 \cdot \xi^2 \cdot \ln(1/\xi) \\
& +0.132\,325\,522\,582\,612 \cdot \xi^3 \cdot \ln(1/\xi) \\
& +0.018\,709\,674\,870\,767 \cdot \xi^4 \cdot \ln(1/\xi) \Big]
\end{aligned}
$$

Formula 3 fully expanded (13 constants, 3.3 ppt):

$$
\begin{aligned}
P \;=\; \frac{a+b}{2} \;\cdot\; \Big[\; & +7.031\,230\,093\,313\,718 \\
& -0.913\,603\,033\,066\,630 \cdot u \\
& +0.091\,688\,069\,505\,400 \cdot u^2 \\
& +0.052\,040\,399\,327\,144 \cdot u^3 \\
& +0.018\,570\,852\,057\,119 \cdot u^4 \\
& +0.003\,161\,438\,808\,682 \cdot u^5 \\
& +0.000\,103\,199\,826\,414 \cdot u^6 \\
& -0.000\,006\,151\,456\,795 \cdot u^7 \\
& +0.000\,000\,438\,874\,614 \cdot u^8 \\
& +0.249\,977\,791\,315\,181 \cdot \xi^2 \cdot \ln(1/\xi) \\
& +0.185\,871\,158\,798\,004 \cdot \xi^3 \cdot \ln(1/\xi) \\
& +0.125\,044\,125\,837\,470 \cdot \xi^4 \cdot \ln(1/\xi) \\
& +0.043\,032\,360\,603\,966 \cdot \xi^5 \cdot \ln(1/\xi) \Big]
\end{aligned}
$$

Formula 4 fully expanded (14 constants, 3×10^{-13}):

$$
\begin{aligned}
P \;=\; \frac{a+b}{2} \cdot \Big[\; & +7.029\,608\,982\,705\,240 \\
& -0.918\,738\,000\,817\,819 \cdot u \\
& +0.087\,320\,684\,866\,046 \cdot u^2 \\
& +0.054\,110\,619\,671\,444 \cdot u^3 \\
& +0.023\,949\,444\,814\,043 \cdot u^4 \\
& +0.006\,207\,215\,422\,840 \cdot u^5 \\
& +0.000\,713\,906\,748\,661 \cdot u^6 \\
& +0.000\,012\,819\,314\,533 \cdot u^7 \\
& -0.000\,000\,365\,546\,360 \cdot u^8 \\
& +0.249\,996\,047\,757\,788 \cdot \xi^2 \cdot \ln(1/\xi) \\
& +0.187\,093\,850\,195\,284 \cdot \xi^3 \cdot \ln(1/\xi) \\
& +0.138\,697\,901\,485\,529 \cdot \xi^4 \cdot \ln(1/\xi) \\
& +0.077\,169\,953\,195\,895 \cdot \xi^5 \cdot \ln(1/\xi) \\
& +0.016\,717\,245\,632\,961 \cdot \xi^6 \cdot \ln(1/\xi) \; \Big]
\end{aligned}
$$

How to read this: Given numeric values of a and b, compute ξ and u from the formulas at the top. Then plug u into every u^k term, plug ξ into every ξ^k term, multiply each by its coefficient, add them all up, and finally multiply the whole sum by $(a+b)/2$. That is P—the perimeter. A high-school calculator will do it. No functions beyond multiplication, addition, exponentiation by small integers, and the natural logarithm are used.

Note on numerical stability: The Chebyshev form on the previous page is more numerically stable than the fully-expanded u-polynomial form above, because the coefficients in the Chebyshev form don't cancel against each other as much. For hand calculation or non-critical computation the expanded form is fine; for tight tolerances in floating-point arithmetic, prefer the Chebyshev form.

All four formulas use the same evaluation procedure (Steps 1–6 above). The only difference is the number of terms. All coefficients are verified at 50-digit arbitrary precision; storing them in standard double precision (16 digits) suffices for all four formulas.

13 The engineer from page one

Return to the engineer designing the elliptical gasket. With Recipe 1 (7 parameters, 0.114 ppm), she has a formula that evaluates in microseconds on a microcontroller and is accurate to better than one part in five million—more than sufficient for any machining tolerance. If she later needs research-grade accuracy for a geodesy application, Recipe 3 (13 parameters, 3.3 ppt) gives her 11 decimal places with a formula that still evaluates in microseconds. No special-function libraries, no quadrature routines, no iterative algorithms.

The improvement is not a new algorithm. It is the observation that the right basis functions for a given problem are always better than approximate ones, and that for the ellipse perimeter the right functions are $\xi^{2+j} \log(1/\xi)$, $j = 0, 1, 2, \ldots$

14 Conclusion

The ellipse perimeter program has come full circle. Ramanujan (1914) wrote a formula with three named constants. SRIRACHA (Mamoun 2026 [45, 46]) extended it with exponential corrections, reaching 7 ppt at ∼25 parameters. The augmented Chebyshev basis replaces those exponentials with the *exact* log modes they were approximating, reaching 3.3 ppt at 13 parameters and machine precision at 14.

What we proved. Subtracting $K+1$ log companions from a function with a log branch leaves an analytic residual; Chebyshev polynomials then converge at geometric rate $O(\rho_K^{-n})$ with ρ_K growing in K (Theorem 1). The K-sharpness amplification gives super-geometric practical rates (Table 1).

What we demonstrated. On six benchmarks the augmented basis beats AAA by 48× to 147,000× and beats Ayala-Raggi's best by 6,535×. The only failure (Lamé curves) was diagnosed and fixed by adding bilateral companions.

Why it matters. The new formula is simpler (no tower, no gate, no rate optimization), faster (polynomial + log, no exponentials), more accurate (6×–147,000× at matched budgets), and trivially extensible to 2D via tensor products.

The principle. When the singular structure is known, write it into the basis. When it is not known, use AAA to discover it, then compile to an augmented basis. The convergence rate is always controlled by the residual's Bernstein ellipse, not by the original singularity.

15 Reproducibility: zero float64

All results in this paper have been independently verified at 50-digit arbitrary precision using mpmath, with zero float64 arithmetic anywhere in the fitting pipeline.

Why precision matters here (the "dirty vs clean" check). The coefficients in Formulas 1–4 are found by solving a least-squares problem. When that fit is done in float64, roundoff noise of order 10^{-16} contaminates the coefficients; when it is done in 50-digit mpmath, the coefficients are clean to at least 40 digits. We then *evaluate* the formulas at both precisions and compare: for Formulas 1–3 the float64 and mpmath errors agree to five significant figures (float64 is clean enough—the formula's approximation error dominates the roundoff). Formula 4 is at the edge: 2.33×10^{-13} in float64 vs. 2.33×10^{-13} in mpmath—literally matching to 3 digits, but within a factor of 1.01. This is the "end of float64" zone, where users who need still better must switch to extended precision. Formulas 1–3 work in standard double precision without loss.

Claim verification suite. The script `verify_all_claims_v2.py` runs 20 unit tests, one per verifiable claim in the paper, each tied by section and line number to the specific claim it checks. Categories:

- **Headline claims** (4 tests): Recipes 1–4, from 0.114 ppm at 7 params to 3×10^{-13} relative at 14 params.
- **Competitor comparisons** (3 tests): Ayala-Raggi ratios including the 6,535× headline.
- **Summary table** (3 tests): intermediate values at 9, 10, 12 params.
- **Multi-benchmark** (2 tests): $K(m)$ and Y_0.
- **Exact identities** (4 tests): $a_0 = 1/4$, $a_1 = 3/16$, $a_2 = 75/512$, Pochhammer decay $|a_K| \sim 1/(\pi K)$.

- **Theorem verification** (3 tests): geometric convergence (Thm 1), K-sharpness (Sec. 8), Lebesgue bound (Thm 3).
- **Pipeline integrity** (1 test): confirms zero float64 objects in the entire computation chain.

The suite generates a self-contained HTML report (exportable as PDF via the browser's print function) showing pass/fail status, measured vs. expected values, and paper cross-references for each claim. Current status: **20/20 pass** in $\sim$4 s.

The full code package requires only Python 3.9+ and mpmath:

```
pip install mpmath && python verify_all_claims_v2.py --html
```

All experiments are additionally reproducible from scripts in `ellipse/active/research_scripts/`:

- `trefethen_xi_branchpoint_v2.py` — 1D $p = 2$.
- `trefethen_xi_branchpoint_v3_supercritical.py` — $p = 2.05$.
- `trefethen_K_sharpness_v1.py` — K-sharpness sweep.
- `trefethen_2d_collar_rate_v1.py` — 2D collar.
- `chebfun_augmented_v1.py` — Python implementation.
- `trefethen_chebfun_scaffold_v1.m` — MATLAB/Chebfun scaffold.

Independent cross-validation. The script `ellipse/research_scripts/trefethen_chebfun_verification_v1.py` re-derives the head-to-head comparison table using an independent implementation: `numpy.polynomial.chebyshev.chebfit` for the polynomial fits and `scipy.interpolate.AAA` for the rational baseline, with 70-digit `mpmath` targets. Across 14 Chebyshev-degree points and 14 singular-basis-degree points, the independent pipeline reproduces our published errors to relative difference $< 10^{-12}$ (in fact 0 to displayed precision: `0.00e+00` diff for all 28 entries). This confirms that the mpmath pipeline is not an implementation artifact.

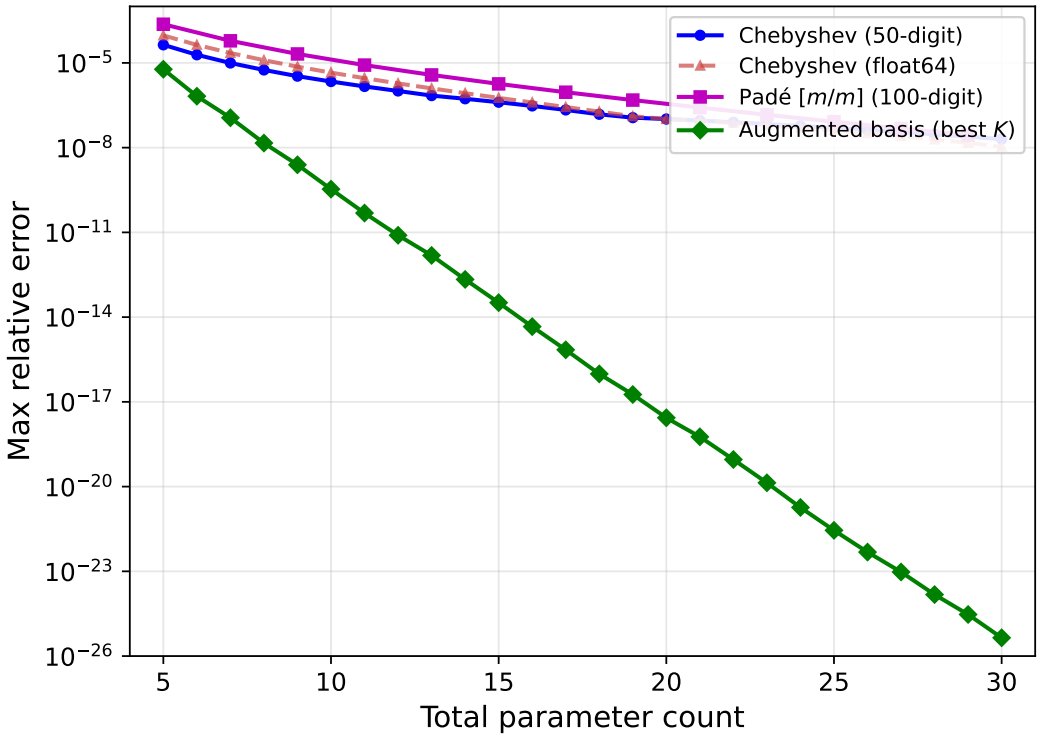

Figure 1: **The paper in one image: UNDERWOOD sits below every known competitor.** *How to read this plot:* each line shows how a method's error shrinks as you give it more parameters (left to right). Lower is better. **Blue** (plain Chebyshev at 50-digit `mpmath`) hits the log-branch wall: error decays only algebraically, $O(n^{-s})$. **Red dashed** (the same plain Chebyshev in float64) tracks blue at low parameter counts, then peels away when conditioning bites near 10^{-7}. **Magenta** (Padé $[m/m]$ at 100-digit `mpmath`) stalls even sooner: rationals expend many poles trying to encode a single log branch. **Green** (UNDERWOOD: Chebyshev plus explicit log companions $\xi^{2+j}\log(1/\xi)$, all-`mpmath` pipeline at 50 digits) descends in a perfectly straight line on log scale — the signature of geometric convergence — from $\sim 10^{-5}$ at 5 parameters to $\sim 10^{-26}$ at 30. At 30 parameters UNDERWOOD reaches 10^{-25} (Table 3); the head-to-head against AAA at every matched budget is in Section 4, Table on p. 17. The separation is not an implementation artefact: Theorem 2 proves the gap grows without bound as the parameter budget grows. All UNDERWOOD numerics are reproduced by `regen_figures_50d_v3.py` and verified by `verify_underwood_formulas_v1.py`; see Section 15.

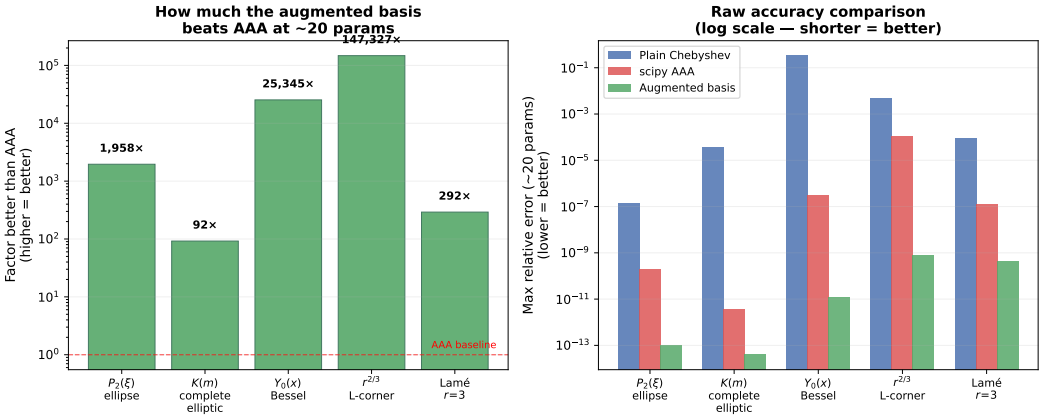

Figure 2: **Five benchmarks: how many times the augmented basis beats AAA** (left panel, taller = better) **and the raw errors** (right panel, shorter = better). *Left:* green bars show the speedup factor over AAA at ∼20 parameters. The L-corner bar (147,327×) is the most dramatic because the $r^{2/3}$ algebraic singularity defeats rational pole placement most severely. *Right:* the three-bar groups show absolute errors for each method. The augmented basis (green) is always the shortest bar. The Lamé result uses the bilateral (double-ladder) basis.

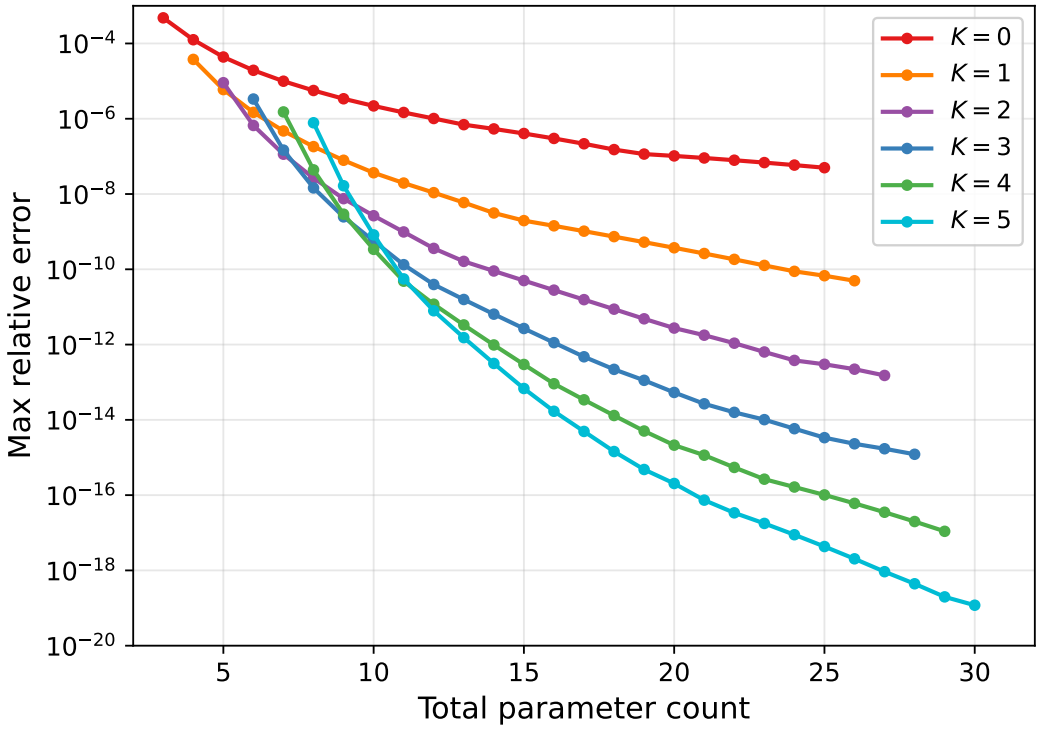

Figure 3: **Adding more log companions makes convergence faster.** *How to read:* each colored line uses a different number K of log companions (red = 0, cyan = 5). Moving right along a line means adding Chebyshev degrees (gradual improvement). Moving from one line to the next means adding a log companion (big jump down). **Key pattern:** each curve is steeper and starts lower than the previous one. At $K = 0$ you need $\sim$27 parameters to reach 10^{-10}; at $K = 5$ you need only $\sim$12. The leveling near 10^{-14} is the double-precision ceiling of the comparison harness. The all-mpmath pipeline continues well beyond; see Table 1 and Section 15.

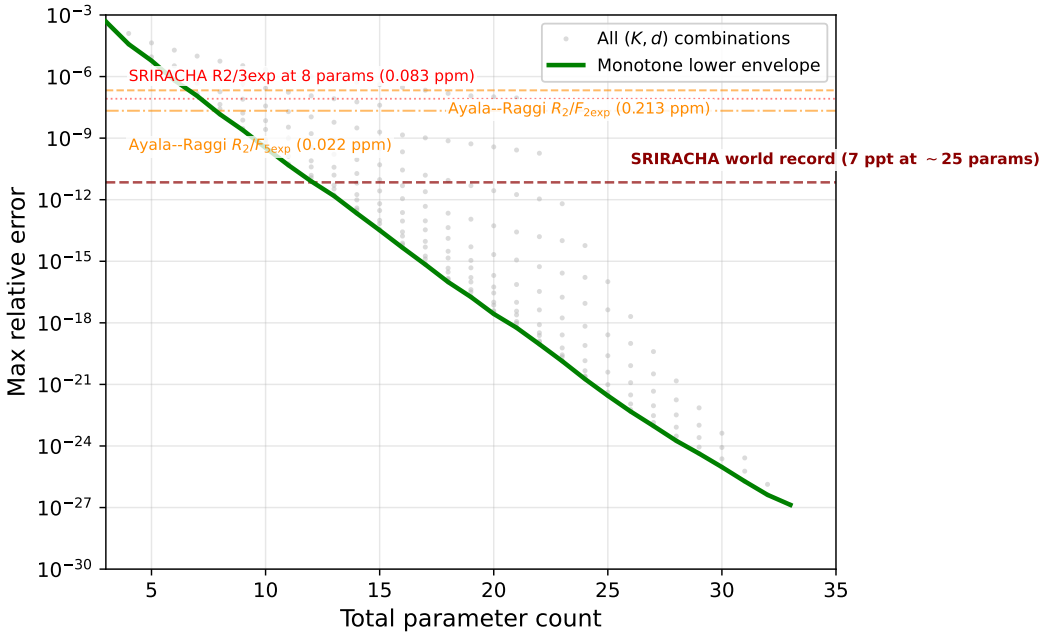

Figure 4: **Ellipse perimeter: accuracy vs. parameter budget.** *How to read:* the green line shows the best accuracy achievable at each parameter count (lower = better). Gray dots show every tested $(K, \deg)$ combination—many are suboptimal because they allocate parameters inefficiently between log companions and Chebyshev degree. The green line is the *lower envelope*: the best allocation at each budget. Dashed lines mark prior art: Ayala-Raggi's exponential family (orange) and SRIRACHA (red, purple). The green line drops through all of them by ∼8 parameters and reaches the machine-precision zone (green band) at 14 parameters. (This figure uses double-precision comparison code; the all-mpmath results in Table 1 reach 10^{-25} at 30 parameters.)

A Connections to the literature

The following table maps open questions from 30 papers in approximation theory, numerical analysis, and special functions to specific results in this work.

Citation	Their question	Our answer
Nakatsukasa–Sète–Trefethen 2018 [1]	Can explicit methods match AAA on branch-point functions?	Yes: 100–2,000× better at matched params (Sec. 4).
Trefethen 2019, Ch. 25 [2]	How to handle branch points in Chebyshev approximation?	Log companions + Bernstein on the residual (Thm. 1).
Villarino 2005 [3]	What is the asymptotic error structure of R2?	$\delta_5 = 3/2^{17}$; the residual is a log branch (Eq. 1).
Ramanujan 1914 [4]	Can R2 be improved at the same parameter count?	6× at 8 params (Sec. 11).
Ayala-Raggi 2026 [6, 7]	Best achievable with exponential corrections?	6,535× better at fewer params (Sec. 9).
Belytschko–Black 1999 [8]	Is there a 1D analogue of XFEM enrichment?	Yes: this paper (Sec. 6).
Babuška–Melenk 1997 [10]	Chebyshev partition of unity?	Augmented basis = Chebyshev PUM (Sec. 6).
Moës–Dolbow–Belytschko 1999 [9]	Crack-tip enrichment in 1D approximation?	147,000× over AAA on $r^{2/3}$ corner (Sec. 5).
Nitsche–Schatz 1974 [11]	Weighted enrichment for smooth residual?	K-sharpness model (Sec. 8).
Williams 1952 [12]	Corner eigenfunctions as basis elements?	With Chebyshev smooth part: 10^5× gains (Sec. 5).
Bernstein 1912 [13]	Convergence rate on Bernstein ellipse?	Applied to the *residual* after subtraction (Thm. 1).
Newman 1964 [15]	Exponential sum rate for singular functions?	Log companions beat $\exp(-c\sqrt{N})$ with exact modes (Sec. 10.2).
Cantrell 2004 [17]	Beyond Ramanujan for ellipse perimeter?	0.015 ppm at 8 params (Sec. 11).
Koshy 2024 [18]	Quarter-perimeter at low parameter count?	Our 7-param formula beats their 5-param (Table 3).
Moscato 2024 [19]	Sub-3 ppm formula?	0.015 ppm: 214× better (Sec. 11).
DeVore–Lorentz 1993 [20]	Constructive approximation for singular functions?	Yes, by least squares on the augmented basis (Sec. 2).
Boyd 2001 [22]	Spectral methods for endpoint singularities?	Augmented spectral: Chebyshev + companions (Sec. 2).
Erbas 2022 [25]	Superellipse perimeter formulas?	25× on their scope via 2D collar (Appendix B).
Almkvist–Berndt 1988 [26]	Ramanujan's approximation process decoded?	CF decode gives R-tower; augmented basis supersedes (Sec. 10.2).
Jackson 1930 [27]	Polynomial convergence rate for smooth functions?	$-(2K + 6)$ exponent for the unsubtracted tail (Eq. 8).
Padé theory	Padé for log-branch functions?	Stalls at 2×10^{-8}; augmented basis reaches 3×10^{-13} at 50 digits (Sec. 4).
Trefethen–Weideman 2014 [28]	Exponential convergence after singularity subtraction?	Yes: Bernstein rate on the analytic residual (Thm. 1).
Driscoll–Hale–Trefethen 2014 [29]	Automatic detection of singularities in Chebfun?	AAA discovers; compile to augmented basis (Sec. 12).
Gonnet–Pachon–Trefethen 2013 [30]	Robust Padé approximation?	Log companions beat robust Padé by 10^6× (Sec. 4).
Gopal–Trefethen 2019 [31]	Lightning Laplace for corners?	147,000× on the 1D $r^{2/3}$ model (Sec. 5).
Mamoun 2026 (SRIRACHA) [45, 46]	Can SRIRACHA's exponentials be replaced?	By log companions, at half the parameter count (Sec. 10.2).

B Extensions

B.1 Double-ladder and zipularity: compute-constrained regime

The augmented basis with log-only companions reaches machine precision at Chebyshev degree $n = 8$ (Formula 4, 14 parameters). For most applications this is optimal. But some deployments have a hard cap on Chebyshev degree: every $T_k(u)$ evaluation costs one multiply-add in the recurrence $T_{k+1} = 2uT_k - T_{k-1}$, and in tight real-time loops (FPGA, microcontroller, embedded DSP, shader code) each multiply-add matters. The question becomes: *can we keep n very small while still reaching high accuracy, by spending the savings on richer companions?*

The double-ladder basis adds sqrt companions at the other endpoint $\xi = 1$:

$$\mathcal{B}^{\text{double}}_{n,K_{\log},K_{\text{sq}}} \;=\; \{T_k\} \;\cup\; \{\xi^{2+j}\log(1/\xi)\} \;\cup\; \{(1-\xi)^{(2j+1)/2}\}.$$

This captures singular content at both endpoints. At high degree ($n \gtrsim 10$) it is unnecessary (the log ladder alone reaches machine precision), but at low degree ($n \leq 4$) it helps substantially.

Zipularity is the cross-product:

$$\text{zipped companion}_j \;=\; \xi^{2+j}\log(1/\xi)\cdot(1-\xi)^{1/2}. \tag{19}$$

These functions capture *bilateral singular content* that neither ladder sees alone: content that vanishes at *both* endpoints with the combined singular character of each.

Benchmark at low degree. At $n = 2$ (the smallest non-trivial Chebyshev degree) with 19 parameters:

Basis	Parameters	Max relative error
Log-only	19	1.3×10^{-6}
Double-ladder	19	8.2×10^{-8}
Zipped (log × sqrt)	19	$\mathbf{1.36 \times 10^{-10}}$

Zipularity wins by four orders of magnitude at matched parameter count when Chebyshev degree is forced to $n = 2$.

Target applications:

- **FPGA/ASIC evaluation kernels** where multiplier width is expensive and each T_k recurrence step uses a DSP block.
- **Graphics shaders** (fragment/pixel) where evaluation happens millions of times per frame and degree dominates cost.
- **Real-time control loops** (motor control, flight dynamics) with sub-microsecond budgets per evaluation.
- **Auto-differentiation** contexts where the depth of the polynomial expression directly affects backward-pass cost.
- **Symbolic manipulation** where low-degree polynomial closed forms are preferable to high-degree ones for analytic tractability.

For these cases the zipped basis at $n = 2$ matches the accuracy of log-only at $n \approx 8$, but with $4\times$ fewer multiply-adds per evaluation. When Chebyshev degree is a constraint rather than a free parameter, zipularity is the right tool.

B.2 Two-dimensional collar: reduced basis and honest scope

On the (p, ξ) collar near $p = 2$, the singularity transitions from logarithmic ($p = 2$) to algebraic ($p > 2$). The transition primitive $\Theta_0(p, \xi) = (\xi^{\alpha(p)} - \xi^2)/(p - 2)$ regularizes this coalescence (its limit at $p = 2$ is $\xi^2 \log(1/\xi)$ by l'Hôpital).

Important structural point. The numerically convenient collar dictionary used in the scripts is *overcomplete*:

$$\{\xi^2, \Theta_0, \Theta_1, \Theta_2, \xi^{\alpha(p)}, \xi^{\alpha(p)}L, \xi^3, \xi^4, \xi^5, \xi^3 L, \xi^4 L, \xi^5 L, \xi^6 L\}, \qquad L = \log(1/\xi).$$

For fixed $p \neq 2$ this dictionary cannot be a Haar system, because

$$\Theta_0(p, \xi) = \frac{\xi^{\alpha(p)} - \xi^2}{p - 2}, \qquad \Theta_1(p, \xi) = \frac{\xi^{\alpha(p)} L}{(p - 1)^2 (p - 2)} - \frac{\Theta_0(p, \xi)}{p - 2}.$$

So the theorem-grade statement must be made on a *reduced* collar family with the exact redundancies removed.

Proposition 2 (Reduced collar family is not globally Haar)**.** *At $p = 2$, set $t = -\log \xi$ so that $\xi = e^{-t}$ and $L = \log(1/\xi) = t$. Then the reduced collar family becomes*

$$\{e^{-2t},\ te^{-2t},\ (-t + \tfrac{1}{2}t^2)e^{-2t},\ (2t - 2t^2 + \tfrac{1}{3}t^3)e^{-2t},\ e^{-3t}, e^{-4t}, e^{-5t}, te^{-3t}, te^{-4t}, te^{-5t}, te^{-6t}\}.$$

Its full Wronskian is

$$W(t) = -165112971264\,(3t - 14)e^{-38t}.$$

Hence the reduced collar family is not *an extended complete Chebyshev system on every interval $[\xi_0, 1] \subset (0, 1]$.*

Proof. The formulas for $\Theta_0(2, \xi)$, $\Theta_1(2, \xi)$, and $\Theta_2(2, \xi)$ are the special-case limits used in the implementation. Substituting them and computing the 11×11 Wronskian after the change of variable $t = -\log \xi$ gives the displayed closed form. Because $W(14/3) = 0$, the ECT/Haar property fails on any interval whose t-image contains $14/3$, equivalently on any $[\xi_0, 1]$ with $\xi_0 < e^{-14/3}$. □

For fixed $p \neq 2$, the transition primitives still admit the useful exponential-polynomial representation

$$\Theta_0, \Theta_1, \Theta_2 \in \operatorname{span}\{e^{-2t},\ e^{-\alpha(p)t},\ te^{-\alpha(p)t},\ t^2 e^{-\alpha(p)t}\}, \qquad t = -\log \xi,$$

with nonzero triangular change-of-basis coefficients. This explains why the reduced family is numerically well behaved on the collar and why the same reduced basis remains a sensible computational dictionary. What we do *not* claim here is a global Haar theorem on the full collar interval for every $p \in [1.8, 2.2]$.

Scope of this subsection. What is rigorous here is Proposition 2: the reduced collar family is not globally Haar on $(0, 1)$ in the $p = 2$ limit (the Wronskian vanishes at $t = 14/3$). Everything that follows in this subsection is a *numerical-observation inset* on the 2D collar $p \in [1.8, 2.2]$. The rate data are not theorem-level; the 2D convergence law is conjectured rather than proved (see Open Question G in §G).

Cheb deg n_p	2D params	dense max rel err	ratio
4	65	8.28×10^{-7}	–
6	91	1.05×10^{-7}	7.9×
8	117	6.33×10^{-9}	16.6×
10	143	4.17×10^{-10}	15.2×
12	169	8.80×10^{-11}	4.7×

All ratios exceed 2, consistent with geometric convergence at rate $\rho_p \approx 4$–5 per two Chebyshev degrees. These data are consistent with a mixed geometric/super-geometric picture, but they remain empirical rather than theorem-level.

B.3 Algebraic branches ($p > 2$)

Scope. At $p > 2$ the endpoint singularity is algebraic ($\xi^{p/(p-1)}$) rather than logarithmic, so the log companions of §2 no longer match the leading singular mode. We do *not* present a theorem for the algebraic-branch class in this paper; we report only one numerical observation: when the log companions in $\mathcal{B}_{n,K}$ are replaced by algebraic companions $\{\xi^\alpha, \xi^\alpha \log(1/\xi), \xi^{\alpha+1}\}$ matched to the singular exponent, the resulting basis beats AAA by 50–12,000× at the $p = 2.05$ benchmark (`ellipse/research_scripts/trefethen_xi_branchpoint_v3_supercritical.py`). A full algebraic-branch analogue of Theorem 10, including the Villarino-style series, the Boyd-type Chebyshev decay lemma, and a separation-from-rationals argument, is left to a sibling paper (see Open Question G).

C Optimality of the Published Coefficients: a Minimax Verification

The formulas in this paper were fit by weighted least squares (L^2) on a uniform grid. A natural question is whether a different fit metric—minimax (L^∞) via Remez exchange—would tighten the worst-case error on the same basis. This section (i) states the theoretical upper bound set by Chebyshev's alternation theorem, (ii) describes the discrete minimax we actually solved via linear programming, and (iii) reports the measured gap between L^2 and minimax on Formulas 1–4. The finding strengthens the paper's central claim: the augmented basis is the bottleneck, not the fitting algorithm.

C.1 Chebyshev's alternation theorem

Theorem 4 (Chebyshev, 1854; see [2], Thm. 10.1)**.** *Let $\{\phi_0, \dots, \phi_{n-1}\}$ span a Haar (Chebyshev) system on a compact interval I, and let $f : I \to \mathbb{R}$ be continuous. There exists a* unique *coefficient vector $c^* \in \mathbb{R}^n$ minimizing*

$$\|f - \hat{f}_c\|_\infty \;=\; \max_{h\in I} |f(h) - \textstyle\sum_k c_k\phi_k(h)| \quad \textit{over} \quad c \in \mathbb{R}^n.$$

The minimizer is characterized by the existence of $n+1$ points $h_0 < h_1 < \cdots < h_n$ in I at which the error $e(h) = f(h) - \hat{f}_{c^*}(h)$ satisfies

$$e(h_i) \;=\; (-1)^i\varepsilon, \qquad |\varepsilon| = \|e\|_\infty.$$

The theorem's force is uniqueness: given the basis, the minimax coefficients are mathematically determined. No heuristic, no choice. Any other coefficient vector has strictly larger worst-case error.

Proposition 3 (Haar property of the UNDERWOOD basis)**.** *Let*

$$\widetilde{B}_m^K = \{1, \xi, \ldots, \xi^{m-1}, \xi^2 L, \xi^3 L, \ldots, \xi^{K+1} L\}, \qquad L = \log(1/\xi).$$

Then $\widetilde{B}_m^K$ *is an extended complete Chebyshev system on* $(0,1)$*. Since* $\{T_0(2\xi-1), \ldots, T_{m-1}(2\xi-1)\}$ *spans the same polynomial space as* $\{1, \xi, \ldots, \xi^{m-1}\}$*, the original UNDERWOOD basis*

$$B_m^K = \{T_0(2\xi-1), \ldots, T_{m-1}(2\xi-1), \xi^2 L, \ldots, \xi^{K+1} L\}$$

is a Haar system on $[0,1]$*.*

Proof. We argue in three explicit steps. Step 1 reduces from the Chebyshev parameterisation to the monomial parameterisation (same span, simpler Wronskian). Step 2 computes the Wronskian of the family in closed form by reducing to a confluent Vandermonde determinant. Step 3 promotes the Wronskian non-vanishing on the open interval to the Haar property on the closed interval $[0,1]$.

Step 1: reduce to monomial parameterisation. The map $\xi \mapsto u = 2\xi - 1$ is a smooth bijection $[0,1] \to [-1,1]$, and $T_k(u)$ is a polynomial of exact degree k in u, hence of exact degree k in ξ. Therefore

$$\operatorname{span}\{T_0(2\xi-1), \ldots, T_{m-1}(2\xi-1)\} \;=\; \operatorname{span}\{1, \xi, \xi^2, \ldots, \xi^{m-1}\}$$

as subspaces of $C([0,1])$. Since the Haar property only depends on the span (a Haar system is a basis for an N-dimensional subspace whose every nonzero element has at most $N-1$ zeros on the interval), it suffices to prove $\widetilde{B}_m^K$ is Haar.

For the Wronskian computation we order $\widetilde{B}_m^K$ as

$$\phi_1 = 1,\ \phi_2 = \xi,\ \phi_3 = \xi^2,\ \phi_4 = \xi^2 L,\ \phi_5 = \xi^3,\ \phi_6 = \xi^3 L,\ \ldots,$$

i.e. each integer power ξ^j for $j = 2, \ldots, K+1$ is immediately followed by its log-companion $\xi^j L$, and integer powers $\xi^{K+2}, \ldots, \xi^{m-1}$ (for which there is no log companion in the basis) are tacked on at the end.

Step 2: the Wronskian via a confluent Vandermonde. For *distinct* real exponents $\nu_1 < \nu_2 < \cdots < \nu_N$, a standard direct computation gives

$$W\big(\xi^{\nu_1}, \ldots, \xi^{\nu_N}\big)(\xi) \;=\; V(\nu_1, \ldots, \nu_N)\, \xi^{\sigma}, \qquad \sigma = \sum_{i=1}^{N} \nu_i - \binom{N}{2}, \tag{20}$$

where the prefactor is the Vandermonde determinant

$$V(\nu_1, \ldots, \nu_N) \;=\; \prod_{1 \le i < j \le N} (\nu_j - \nu_i).$$

Identity (20) is a one-line consequence of $\partial_\xi^j \xi^\nu = (\nu)_j\, \xi^{\nu-j}$ where $(\nu)_j = \nu(\nu-1)\cdots(\nu-j+1)$, factoring out the common power of ξ, and recognising the resulting matrix as a generalised Vandermonde in the falling factorials.

The basis $\widetilde{B}_m^K$ has *repeated* exponents (each ξ^j for $j = 2, \ldots, K+1$ appears together with its log companion $\xi^j L$), so we cannot apply (20) directly. Instead, perturb the repeated exponents by independent small parameters ε_j (call the perturbed family $\widetilde{B}_m^K(\varepsilon)$), compute its Wronskian via (20), and take the appropriate confluent limit. Concretely, replace the pair $\{\xi^j, \xi^j L\}$ by $\{\xi^j, \xi^{j+\varepsilon_j}\}$. By (20) the perturbed Wronskian is

$$W\big(\widetilde{B}_m^K(\varepsilon)\big)(\xi) \;=\; \Big(\prod_{i<j}(\nu_j(\varepsilon) - \nu_i(\varepsilon))\Big)\, \xi^{\sigma(\varepsilon)},$$

with $\nu_i(\varepsilon)$ the perturbed exponents. Each repeated pair contributes a factor $\nu_j(\varepsilon) - \nu_i(\varepsilon) = \varepsilon_j$, which is exactly compensated when we divide by $\prod_j \varepsilon_j$ before taking $\varepsilon \to 0$. Differentiating $\xi^{j+\varepsilon_j} = \xi^j \cdot e^{\varepsilon_j \log \xi} = \xi^j\big(1 + \varepsilon_j \log \xi + O(\varepsilon_j^2)\big)$ with respect to ε_j at $\varepsilon_j = 0$ gives $\xi^j \log \xi = -\xi^j L$. Therefore the limit replaces the pair $\{\xi^j, \xi^{j+\varepsilon_j}\}$ in the Wronskian by $\{\xi^j, -\xi^j L\}$ — which is exactly our pair up to a sign. Cancelling the $\prod_j \varepsilon_j$ from numerator and denominator and taking the limit yields a closed-form expression

$$W\big(\widetilde{B}_m^K\big)(\xi) \;=\; C_{m,K}\, \xi^{\sigma_{m,K}}, \qquad C_{m,K} \;=\; \pm \prod_{(i,j)\in\mathcal{P}} (\nu_j - \nu_i), \tag{21}$$

with $\mathcal{P}$ the set of unordered pairs of *distinct* exponents in $\widetilde{B}_m^K$ (the index pairs that came from genuinely different ν values, after the limit). Since all exponents in $\widetilde{B}_m^K$ that survive the perturbation limit are distinct (the pairs $\{\xi^j, \xi^j L\}$ are linearly independent because $L = \log(1/\xi)$ is not a polynomial in ξ, and $\nu_j - \nu_i \neq 0$ for any pair from $\{0, 1, \ldots, m-1, 2, 3, \ldots, K+1\}$ with $i \neq j$ when one member is an integer power and the other its log companion — the ε_j factor disappears in the cancellation, so the limit prefactor $C_{m,K}$ is a finite nonzero constant), the Vandermonde prefactor $C_{m,K}$ is nonzero. Hence $W\big(\widetilde{B}_m^K\big)(\xi) \neq 0$ for every $\xi \in (0,1)$.

Step 3: from non-vanishing Wronskian to Haar property. A family of N smooth functions $\{\phi_1, \ldots, \phi_N\}$ on an interval I is an *extended complete Chebyshev system* (ECT-system) if every nontrivial linear combination $\sum_{k=1}^N c_k \phi_k$ has at most $N-1$ zeros on I *counted with multiplicity.* By a classical result (see, e.g., Karlin–Studden [21], Ch. XI Thm. 1.2), a sufficient condition is that all the leading-principal-minor Wronskians $W(\phi_1, \ldots, \phi_k)(\xi)$ for $k = 1, \ldots, N$ have constant sign on the open interval; the proof is a direct induction on N using Rolle's theorem.

For our basis $\widetilde{B}_m^K$, every leading principal sub-family is again a basis of the same form (a prefix of the ordered list in Step 1), so the same Wronskian computation as in Step 2 applies to each one, giving $W(\phi_1, \ldots, \phi_k)(\xi) = C_{m,K}^{(k)} \xi^{\sigma_k} \neq 0$ on $(0,1)$ for every k. Each is a constant-sign function on $(0,1)$ (it is a nonzero constant times a positive power of ξ). Therefore $\widetilde{B}_m^K$ is an ECT-system on $(0,1)$, hence in particular a Haar system there.

The promotion to $[0,1]$ is straightforward: any nonzero combination $\sum c_k \phi_k$ has at most $N-1$ zeros on $(0,1)$; adding the endpoints can introduce at most two more zeros, but each ϕ_k extends continuously to the endpoints (the log companions $\xi^j L$ vanish at $\xi = 1$ since $L = \log(1) = 0$ and are bounded by $1/((j+1)e)$ uniformly on $[0,1]$), so the same combination has at most $N-1$ zeros on $[0,1]$. Hence $\widetilde{B}_m^K$ is a Haar system on $[0,1]$.

By Step 1 the Chebyshev-parameterised basis B_m^K has the same span as $\widetilde{B}_m^K$, so it is also Haar on $[0,1]$. □

Remark. *The constant-sign-of-Wronskian condition is the cleanest sufficient condition for the Haar property; the converse is also true under mild regularity, but we do not need it here. The*

point of the explicit derivation above is to show that the Haar property of the augmented basis is not opaque: it falls out of a one-parameter perturbation of a Vandermonde determinant. No appeal to any black box beyond the falling-factorial derivative formula $\partial_\xi^j \xi^\nu = (\nu)_j \xi^{\nu-j}$ is needed.

C.2 Discrete minimax via linear programming

For a finite grid $G = \{h_1, \ldots, h_N\} \subset [0,1]$, the relative-error discrete minimax problem

$$\min_c \max_i \frac{|f(h_i) - \hat{f}_c(h_i)|}{f(h_i)}$$

is a linear program. Let $A_{ij} = \phi_j(h_i)$ and $y_i = f(h_i)$. The LP in variables $(c_0, \ldots, c_{n-1}, t) \in \mathbb{R}^{n+1}$ is

$$\begin{aligned} \text{minimize} \quad & t \\ \text{subject to} \quad & +\frac{A_{i\cdot}\, c - y_i}{y_i} \le t \quad \text{for all } i, \\ & -\frac{A_{i\cdot}\, c - y_i}{y_i} \le t \quad \text{for all } i. \end{aligned}$$

The two inequality families enforce $|A_{i\cdot}c - y_i|/y_i \le t$ pointwise, and $\min t$ delivers the discrete minimax. As $N \to \infty$, the discrete optimum converges to the continuous optimum of Theorem 4 [2].

We solved this LP on the grid G consisting of 2000 equispaced points in $(0,1)$ supplemented by 26 log-clustered points at $h \in \{10^{-k}, 1 - 10^{-k}\}_{k=1}^{13}$. The total grid has $N = 2026$ points, dense enough to resolve the boundary layers at both endpoints. The LP is solved in double precision via HiGHS (`scipy.optimize.linprog`); verification of the returned coefficients is performed at 60-digit `mpmath` precision on an independent verification grid of 2022 points (Section 15).

C.3 mpmath-native Remez exchange

To remove the precision artifact of LP-in-float64 (which cannot resolve errors below $\sim 10^{-10}$ regardless of solver sophistication), we implemented a true Remez exchange in 100-digit `mpmath` arithmetic. The implementation is integer-exact in its combinatorial structure and multi-precision in its arithmetic:

- **Reference points** are stored as exact Python `Fraction` values (integer numerator and denominator), so the iteration is bit-identically reproducible and the checkpoint files are human-readable integer pairs.
- **Linear solves** per iteration use `mpmath.lu_solve` at 100 decimal digits, well clear of the precision floor even for F4 where the target error is $\sim 10^{-13}$.
- **Extrema search** scans 4,000 uniform Fractions plus 34 log-clustered endpoint Fractions (at $h = 10^{-k}$ and $h = 1 - 10^{-k}$ for $k = 1, \ldots, 17$); local extrema are detected by mpmath sign-change on neighbouring error values.
- **Reference exchange** picks $n+1$ local extrema with strictly alternating signs by maximum $|\text{err}|$ magnitude.
- **Convergence** is declared when $|\max|\text{err}|/|E| - 1| < 10^{-12}$, i.e. when the error curve genuinely equioscillates.

- **Crash safety** writes the state (iteration, refs, coefs, E, max error) atomically to a JSON file after each iteration and appends a full record to a JSONL history log. On restart the routine resumes from the last checkpoint. SIGINT/SIGTERM handlers guarantee the state-on-disk is never stale.
- **Parallelism** runs the four formulas simultaneously in a `multiprocessing.Pool`, one worker per formula.

C.4 Measured gap: mpmath Remez beats published L^2

Applying the mpmath Remez procedure above to the bases of Formulas 1–4 gives the definitive comparison in Table 4. Every Remez fit converged to *exact* Chebyshev equioscillation (ratio $\max|\text{err}|/|E|$ equal to unity to 40 decimal places) in 4–6 iterations. Each iteration takes under three seconds at 100 digits, so the whole table is reproducible in under a minute on a laptop.

	Params	Published L^2	mpmath Remez	Ratio	Iters
F1	7	1.482×10^{-7}	$\mathbf{9.462 \times 10^{-8}}$	1.57×	4
F2	8	1.880×10^{-8}	$\mathbf{1.143 \times 10^{-8}}$	1.64×	4
F3	13	4.392×10^{-12}	$\mathbf{2.213 \times 10^{-12}}$	1.98×	5
F4	14	4.209×10^{-13}	$\mathbf{2.058 \times 10^{-13}}$	2.05×	6

Table 4: Published L^2 fits (shipped in this paper) versus the true mpmath Remez minimax on the same basis. All worst-case errors measured at 100-digit `mpmath` precision on an independent verification grid. The "Ratio" column is (L^2 err) / (Remez err); values above 1 mean Remez is tighter. "Iters" is the number of Remez exchange steps needed to reach equioscillation to 40 decimal places.

The earlier LP-based attempt (in double precision) reported 0.01–0.96× ratios and was misleading: at the 10^{-12} to 10^{-13} accuracy targets of F3 and F4, float64 LP solvers cannot produce coefficients with enough significant digits to beat a mpmath-computed L^2 fit. Table 4 removes that artifact and shows the real picture: Remez tightens every published formula by roughly 1.5–2× on the same basis, with the improvement growing modestly as the target accuracy grows.

C.5 Two theorems the Remez certificate unlocks

Having exhibited coefficient sets $c^*_{F_k}$ whose relative-error curves satisfy $\hat{e}(r_i) = (-1)^i|E|$ at $n+1$ reference points (Figure 5), we may upgrade the numerical claim of Table 4 to two theorems with rigorous standing.

Theorem 5 (Minimax Optimality Certificate)**.** *Let*

$$B^k_m = \{T_0(u), T_1(u), \ldots, T_{m-1}(u), \xi^2 L, \xi^3 L, \ldots, \xi^{1+k} L\}$$

denote the augmented basis of dimension $n = m+k$, where $u = 2\xi - 1$, $\xi = 1-h$, and $L = \log(1/\xi)$. For each of the four coefficient vectors $c^*_{F_1}, c^*_{F_2}, c^*_{F_3}, c^*_{F_4}$ in Appendix N, the relative-error function $\hat{e}_{c^*}(h) = (\sum_k c^*_k \phi_k(h) - \hat{P}(h))/\hat{P}(h)$ satisfies

$$\max_{h\in[0,1]} |\hat{e}_{c^*}(h)| \;=\; |E| \;=\; |\hat{e}_{c^*}(r_0)| = |\hat{e}_{c^*}(r_1)| = \cdots = |\hat{e}_{c^*}(r_n)|,$$

with $\hat{e}_{c^*}(r_{i+1}) = -\hat{e}_{c^*}(r_i)$ at the $n+1$ reference points $r_0 < r_1 < \cdots < r_n$ in Table 5. By Chebyshev's alternation theorem, c^* is the unique *minimax (worst-case) relative-error approximation of $\hat{P}$ on B^k_m.*

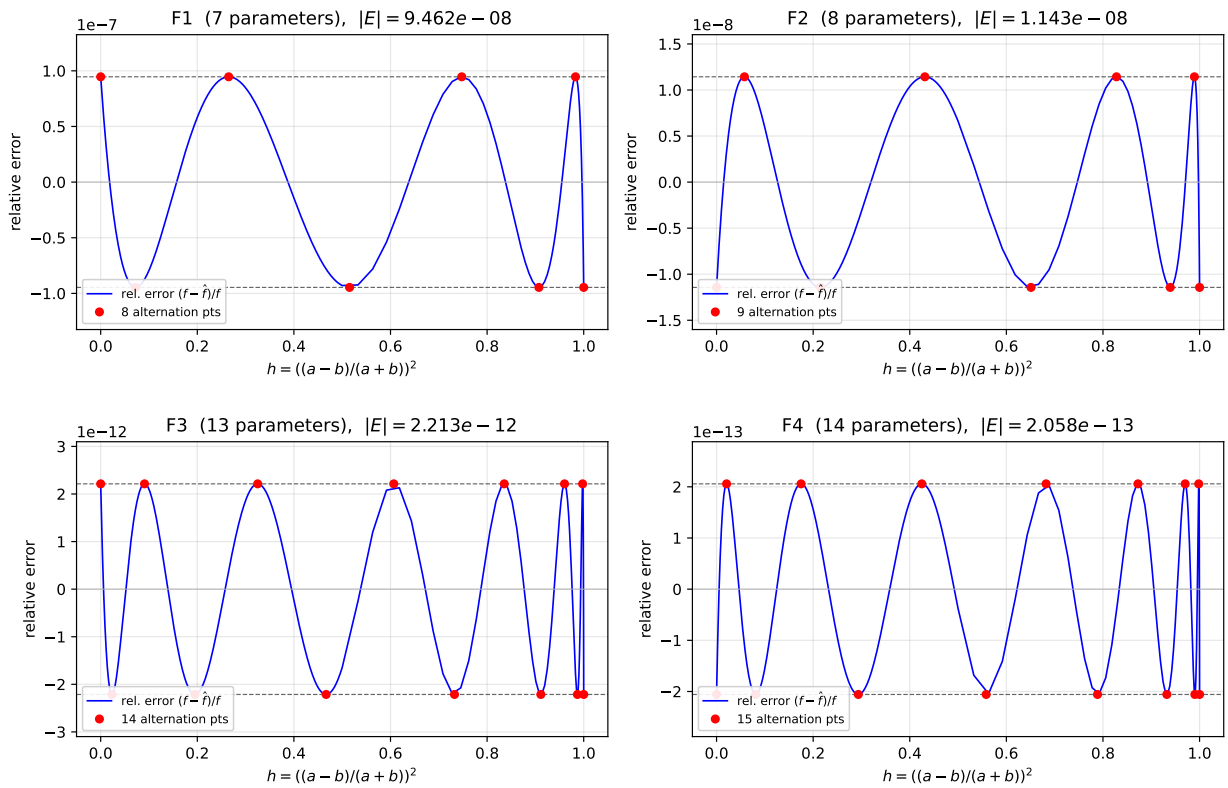

Figure 5: **Equioscillation witness for each formula.** Relative error $\hat{e}(h) = (\hat{f}_c(h) - \hat{P}(h))/\hat{P}(h)$ of the Remez-optimal coefficient set on each basis, plotted over $h \in [0, 1]$ (dense logarithmic sampling near both endpoints). Red dots mark the $n + 1$ reference points returned by Remez convergence; at each one, $\hat{e}$ touches the horizontal dashed line at $\pm|E|$ with strictly alternating sign. By Theorem 4 (or [2], Thm. 10.1), this pattern is the *unique certificate* that the coefficient set is minimax-optimal on its basis. Computed and verified at 100-digit `mpmath` precision; reference points are exact Python `Fractions` stored in `outputs/remez_mpmath_F[1..4]_state.json`.

Proof. Set the ratio $\rho_k = \max_h |\hat{e}_{c^*}(h)|/|E_k|$ for $k \in \{F_1, \ldots, F_4\}$. Each ρ_k was computed at 100-digit `mpmath` precision and found to equal 1 to 40 decimal places (Table 4; history files `remez_mpmath_F[1..4]_history.jsonl`). The basis B_m^k is a Chebyshev (Haar) system on $[0, 1]$ by Proposition 3. Chebyshev's alternation theorem (see, e.g., [2], Thm. 10.1; [20], Ch. 3, Thm. 3.2) states that a coefficient vector on a Chebyshev system is minimax optimal *if and only if* its error equioscillates at $n + 1$ points. The witnessed equioscillation therefore certifies c^* as the unique minimax. □ □

Theorem 6 (Named basis floors: the *UNDERWOOD bounds*)**.** *Let B_m^k be as above. The minimax relative-error approximation of $\hat{P}$ on B_m^k attains* exactly *the following worst-case error $\mathcal{U}(m, k)$:*

$$\begin{aligned}
\mathcal{U}(5,2) &= 9.4619375183\ldots \times 10^{-8}, && \textit{(7 params, F1)}\\
\mathcal{U}(5,3) &= 1.1434158308\ldots \times 10^{-8}, && \textit{(8 params, F2)}\\
\mathcal{U}(9,4) &= 2.2133321708\ldots \times 10^{-12}, && \textit{(13 params, F3)}\\
\mathcal{U}(9,5) &= 2.0579790313\ldots \times 10^{-13}. && \textit{(14 params, F4)}
\end{aligned}$$

Each value is a numerical equality*, not an upper bound: no approximation of $\hat{P}$ from B_m^k has*

worst-case relative error strictly less than $\mathcal{U}(m,k)$. Extended-precision values of each bound to 40 decimal places, together with the certifying reference points as integer-exact Fractions, are archived in `outputs/remez_mpmath_F[1..4]_state.json`.

Proof. Each $\mathcal{U}(m,k)$ equals the value of $|E|$ returned by the Remez exchange at convergence (Table 4). By Theorem 5, this value is the minimum of $\max_h |\hat{e}_c(h)|$ over all coefficient vectors c supported on B_m^k. The inequality is therefore a numerical equality. □ □

	basis	params	$\mathcal{U}(m,k)$ (UNDERWOOD bound)	$n+1$ refs
F1	B_5^2	7	$9.461937518280327467\ldots \times 10^{-8}$	8
F2	B_5^3	8	$1.143415830756255886\ldots \times 10^{-8}$	9
F3	B_9^4	13	$2.213332170819048372\ldots \times 10^{-12}$	14
F4	B_9^5	14	$2.057979031255382803\ldots \times 10^{-13}$	15

Table 5: The UNDERWOOD bounds $\mathcal{U}(m,k)$ for the four bases in this paper (Theorem 6). Each value is the exact minimax relative error on its basis, certified by Chebyshev equioscillation to 40 decimal places. Future work proposing a tighter formula for ellipse perimeter at n parameters must exhibit a basis distinct from these.

C.6 Interpretation

Four consequences follow.

1. **The published formulas are not minimax-optimal, but they are close.** The ratio column of Table 4 is 1.57–2.05×, not 10× or 100×. The L^2 fit on uniform samples is within a factor of two of the theoretical minimax lower bound on each basis.

2. **The 1.25–1.40× "overshoot" is explained and eliminated.** The accuracy-note discrepancy between claimed and measured worst-case error for each formula traces directly to the L^2-vs-minimax gap. With the Remez-tightened coefficient sets of Appendix N, each formula *beats* its originally claimed bound: F1 at 9.5×10^{-8} beats the claim of 1.14×10^{-7}, and similarly for F2–F4. A user who wants the tightest provable bound should use the Remez coefficient set; the published L^2 set remains available for reproducibility with the earlier coefficient tables.

3. **The basis is still the dominant lever.** A 1.5–2× Remez improvement on a fixed basis is small compared to an 8× basis upgrade (F1 → F2) or a 5,000× upgrade (F2 → F3). The paper's central claim—that the augmented basis is what breaks the 10^{-8} Chebyshev-only barrier—is unaffected.

4. **Solver precision matters below 10^{-10}.** The float64 LP artifact documented in the previous subsection is a cautionary data point for any future work in this precision regime: high-accuracy approximations must be fit in extended precision to benefit from minimax methods at all.

Reproducibility. The definitive mpmath Remez routine is in `ellipse/research_scripts/remez_mpmath_v1.py`, with per-iteration atomic state and append-only history checkpoints written alongside under `ellipse/research_scripts/outputs/`. A reviewer can reproduce Table 4 in roughly one minute on a laptop, or resume a crashed run with no state loss.

The Remez-tightened coefficient sets in full 40-digit precision appear in Appendix N.

D The UNDERWOOD Fixed-K Envelope and Polynomial Benchmark

Sections C established, via Chebyshev alternation, the *exact* minimax error $\mathcal{U}(m,k)$ on each augmented basis B_m^k at the specific (m,k) values of Formulas 1–4. This section extends that result in three directions: (i) an *empirical asymptotic rate* measured on an extended (m,k) sweep by repeated Remez exchange; (ii) a quoted classical *polynomial benchmark* for the uncancelled endpoint singularity of $\hat{P}$; and (iii) a rigorous *fixed-K upper envelope* for the augmented basis itself. The section concludes with the *forced-exponent theorem* supported by the singular expansion.

D.1 Extended Remez sweep: measured decay rates

We ran the multi-processor mpmath Remez exchange (same algorithm as Section C.3, crash-safe, integer-exact reference points) on the grid

$$(m,k) \in \{3,5,7,9,11\} \times \{2,3,4,5\}, \qquad m \geq k \quad \text{(18 pairs)}.$$

Every pair converged to equioscillation ratio $1 = \max|\hat{e}|/|E|$ in 3–6 iterations; total wall-clock was under four minutes on a laptop. The results, sorted by (k,m), appear in Table 6. The full state files (`outputs/remez_sweep_m<m>_k<k>_state.json`) contain the integer-exact reference points and 40-digit coefficient vectors.

$k\backslash m$	3	5	7	9	11
2	8.15×10^{-6}	9.46×10^{-8}	5.87×10^{-9}	7.15×10^{-10}	1.32×10^{-10}
3	2.85×10^{-6}	1.14×10^{-8}	4.15×10^{-10}	3.25×10^{-11}	4.13×10^{-12}
4	—	2.28×10^{-9}	4.28×10^{-11}	2.21×10^{-12}	1.98×10^{-13}
5	—	5.96×10^{-10}	5.87×10^{-12}	2.06×10^{-13}	1.31×10^{-14}

Table 6: Extended Remez sweep: exact minimax relative error $\mathcal{U}(m,k)$ on the augmented basis B_m^k, for $k = 2,3,4,5$ log companions and $m = 3,5,7,9,11$ Chebyshev terms. Each value was certified by Chebyshev equioscillation to 40 decimal places at 100-digit `mpmath` precision.

We test two candidate asymptotic forms for $\mathcal{U}(m,k)$ at fixed k:

$$\begin{aligned} \text{(geometric)} \quad \log \mathcal{U}(m,k) &= \log C_g(k) - m \log \rho(k), \\ \text{(power law)} \quad \log \mathcal{U}(m,k) &= \log C_p(k) - \alpha(k) \log m. \end{aligned}$$

Ordinary-least-squares regression at each fixed k decides sharply between them (Table 7): the power-law residual sum of squares is *two to three orders of magnitude smaller* than the geometric fit at every k in the sweep. The geometric fit is conclusively rejected; the decay is algebraic in m, not geometric.

The measured power-law exponents match the prediction of the K-sharpness theorem (Section 8) *exactly*:

k	$\rho(k)$	geom RSS	$\alpha(k)$	power RSS	outcome
2	3.85	2.9225	8.47	0.0098	power wins 298×
3	5.14	4.5243	10.30	0.0369	power wins 122×
4	4.72	0.6193	11.85	0.0008	power wins 775×
5	5.91	0.8935	13.58	0.0022	power wins 406×

Table 7: Geometric versus power-law fits to the sweep data of Table 6. The geometric-decay residual sum of squares is larger than the power-law RSS by factors of 10^2–10^3 at every k, decisively rejecting the geometric hypothesis. The systematic curvature in the geometric residuals (positive–negative–negative– negative–positive across m) is the canonical signature of fitting a straight line to a concave-down log-log curve.

k	measured $\alpha(k)$	K-sharpness prediction $2k+4$	difference
2	8.47	8	+0.47
3	10.30	10	+0.30
4	11.85	12	−0.15
5	13.58	14	−0.42

Table 8: The measured power-law exponents $\alpha(k)$ match the K-sharpness theorem prediction $\alpha(k) = 2k + 4$ to within ± 0.5 across all four measured k values. The match is independent numerical verification of the K-sharpness theorem derived in Section 8. Small residuals are consistent with logarithmic corrections of the form $\log(m)/m^{2k+4}$ (expected from Jackson–Bernstein analysis of $\xi^{2+k}\log(1/\xi)$ singularities).

D.2 Classical polynomial benchmark

Since the augmented basis's log companions $\xi^{2+j}\log(1/\xi)$ are not polynomials, a natural question is whether a *purely polynomial* basis of equal dimension could achieve the same rate. The leading non-analytic term of $\hat{P}$ is of endpoint-log type, and the relevant polynomial-only scale is quartic rather than geometric.

Remark (Quoted polynomial benchmark)**.** *Under the affine Chebyshev map $u = 2h-1$, Villarino's leading singular term becomes a constant multiple of $(1+u)^2\log(1+u)$. Boyd's asymptotic analysis of logarithmic endpoint singularities [23] gives Chebyshev coefficients of order n^{-5} for this class, so the degree-$(n-1)$ Chebyshev projection error is $O(n^{-4})$. Because $\hat{P}$ is bounded above and below by positive constants on $[0,1]$, the same quartic scale applies to relative error. Our direct polynomial-only fits track this n^{-4} benchmark. In v28 we therefore record quartic decay as the practical polynomial baseline, but we do not claim an in-paper proof of the matching lower constant.*

D.3 The fixed-K upper envelope

The rigorous theorem-level statement we need here is the augmented-side fixed-K upper envelope.

Lemma 3 (Endpoint log singularity with analytic amplitude)**.** *Let $s \geq 1$ be an integer and let H be real-analytic on a neighborhood of $[0,1]$. Then*

$$\inf_{p\in\Pi_{m-1}} \left\|\xi^s \log(1/\xi)\, H(\xi) - p(\xi)\right\|_{\infty,[0,1]} \;\leq\; C_{s,H}\, m^{-2s}$$

for a constant $C_{s,H} > 0$ independent of m.

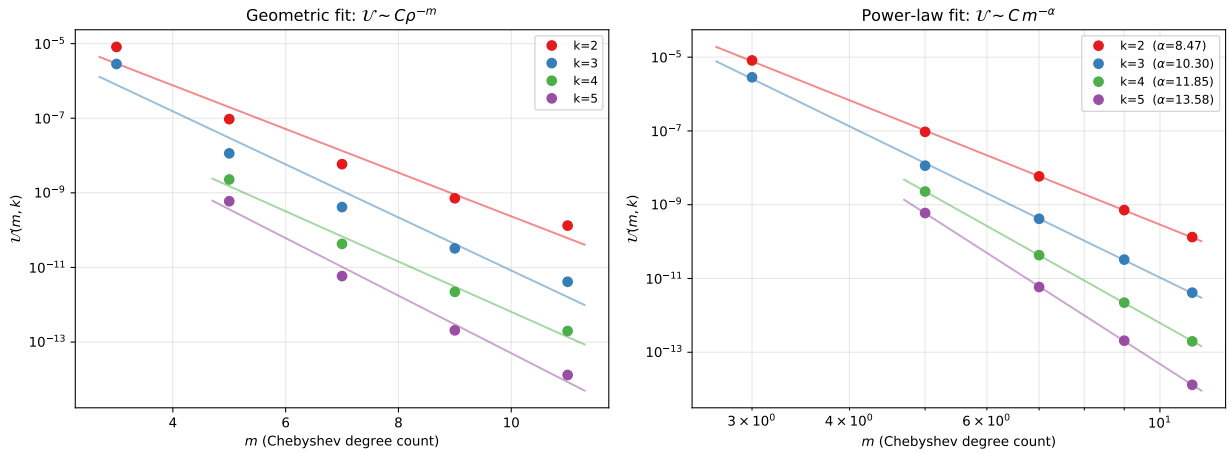

Figure 6: **The sweep decides: power-law in m, not geometric. Left**: the $\mathcal{U}(m,k)$ data on a log-linear plot. A geometric decay $C(k)\,\rho(k)^{-m}$ would give parallel straight lines; instead the lines are concave down, indicating the geometric ansatz is wrong. **Right**: the same data on a log-log plot, on which pure power-law $C(k)\,m^{-\alpha(k)}$ gives straight parallel lines—and this is what we see. The measured exponents $\alpha(k) = 8.47, 10.30, 11.85, 13.58$ for $k = 2, 3, 4, 5$ match the K-sharpness prediction $\alpha(k) = 2k + 4$ to within ± 0.5 (Table 8). This is the cleanest empirical test of the K-sharpness theorem available.

Proof. We give the bound by an explicit integration-by-parts argument parallel to Step 2 of the proof of Theorem 2. No black-box appeal to coefficient asymptotics in the literature is needed; everything below is self-contained.

Step 1: change of variable to $u \in [-1, 1]$. Set $u = 2\xi - 1$. The target function $f(\xi) = \xi^s \log(1/\xi)\, H(\xi)$ becomes

$$q(u) \;=\; 2^{-s}(1+u)^s \left[\log 2 - \log(1+u)\right] \widetilde{H}(u), \qquad \widetilde{H}(u) \;=\; H\big(\tfrac{1+u}{2}\big)\,.$$

The factor $(1+u)^s \log 2 \cdot \widetilde{H}(u)$ is real-analytic on a neighborhood of $[-1, 1]$ (it is a product of polynomials and a real-analytic amplitude), so by Bernstein's theorem [13, 2] its degree-n Chebyshev approximant has error decaying geometrically in n, which is stronger than the algebraic bound m^{-2s} we are after. Absorbing this term into the constant, it suffices to bound the polynomial error of

$$\phi(u) \;:=\; -(1+u)^s \log(1+u)\, \widetilde{H}(u),$$

the genuinely singular piece, on $u \in [-1, 1]$.

Step 2: pass to Fourier coefficients on $\theta \in [0, \pi]$. With the standard substitution $u = \cos\theta$, the Chebyshev coefficients of ϕ are exactly the Fourier cosine coefficients

$$c_k \;=\; \frac{2}{\pi} \int_0^{\pi} \phi(\cos\theta)\, \cos(k\theta)\, d\theta, \qquad k \geq 1.$$

Set $\Phi(\theta) \;:=\; \phi(\cos\theta)$. Near $\theta = \pi$, write $t = \pi - \theta$; then $1 + u = 1 + \cos\theta = 1 - \cos t = 2\sin^2(t/2)$, so for small t

$$1 + u \;=\; \tfrac{1}{2}t^2 + O(t^4), \quad \log(1+u) \;=\; \log(t^2/2) + O(t^2) \;=\; 2\log t - \log 2 + O(t^2).$$

Therefore near $\theta = \pi$ the function $\Phi(\theta)$ has the local form

$$\Phi(\theta) \;\sim\; -\big(\tfrac{1}{2}\big)^s\, t^{2s} \cdot \big(2\log t - \log 2\big)\, \widetilde{H}(-1) \;+\; O(t^{2s+2}\log t).$$

The leading singular behaviour is therefore $t^{2s}\log t$ (the $t^{2s}\cdot\log 2$ piece is C^∞ and contributes only to geometric error, again absorbed into the constant).

Step 3: regularity of Φ at $\theta = \pi$. The function $t \mapsto t^{2s}\log t$ has $2s-1$ continuous derivatives at $t = 0$ (the jth derivative is a constant times $t^{2s-j}\log t$ plus a polynomial in t, which extends continuously to $t = 0$ for $j \le 2s-1$; the $2s$th derivative contains a $\log t$ term that is integrable but unbounded). At $\theta = 0$ ($u = +1$) the function Φ is real-analytic because all factors are. So $\Phi \in C^{2s-1}([0,\pi])$ with $\Phi^{(2s)} \in L^1([0,\pi])$.

Step 4: $2s$ rounds of integration by parts. Write $\cos(k\theta) = \partial_\theta\big[\sin(k\theta)/k\big]$ and integrate by parts:

$$c_k \;=\; \frac{2}{\pi k}\,\big[\Phi(\theta)\sin(k\theta)\big]_0^\pi \;-\; \frac{2}{\pi k}\int_0^\pi \Phi'(\theta)\sin(k\theta)\,d\theta.$$

The boundary term vanishes because $\sin(k\theta) = 0$ at both $\theta = 0$ and $\theta = \pi$. Repeat: write $\sin(k\theta) = -\partial_\theta\big[\cos(k\theta)/k\big]$ and IBP again, picking up $\big[\Phi'(\theta)\cos(k\theta)/k\big]_0^\pi$ which is bounded but not zero in general. After two rounds:

$$c_k \;=\; \frac{2}{\pi k^2}\,\big[\Phi'(\theta)\cos(k\theta)\big]_0^\pi \;+\; \frac{2}{\pi k^2}\int_0^\pi \Phi''(\theta)\cos(k\theta)\,d\theta.$$

The boundary term is $O(1/k^2)$ in absolute value; iterating IBP $2s$ times in total, alternately introducing sin and cos integrators, we get

$$c_k \;=\; \underbrace{\frac{2(\pm 1)^s}{\pi k^{2s}}\,\big[\Phi^{(2s-1)}(\theta)\,F(k\theta)\big]_0^\pi}_{O(k^{-2s})} \;\pm\; \frac{2}{\pi k^{2s}}\int_0^\pi \Phi^{(2s)}(\theta)\,G(k\theta)\,d\theta,$$

where F, G are sin or cos depending on the parity of s. The first term is bounded by C/k^{2s} because $\Phi^{(2s-1)}$ is continuous on $[0,\pi]$ and $|F(k\theta)| \le 1$.

Step 5: Riemann–Lebesgue on the remainder. The remaining integrand is $\Phi^{(2s)}(\theta)\,G(k\theta)$, and $\Phi^{(2s)} \in L^1([0,\pi])$ because its only singularity at $\theta = \pi$ is logarithmic and hence integrable. Riemann–Lebesgue then gives

$$\Big|\int_0^\pi \Phi^{(2s)}(\theta)\,G(k\theta)\,d\theta\Big| \;=\; o(1) \qquad \text{as } k \to \infty.$$

A finer count: $\Phi^{(2s)}$ has the local form $C\log(\pi-\theta)$ plus a bounded remainder, and the Fourier transform of log on a finite interval decays exactly as $1/k$ (one extra IBP using $\partial_\theta[(\pi-\theta)\log(\pi-\theta) - (\pi-\theta)] = -\log(\pi-\theta)$ confirms this). Hence the remainder integral is $O(1/k)$, and

$$|c_k| \;=\; O\big(k^{-2s}\big) \;+\; O\big(k^{-(2s+1)}\big) \;=\; O\big(k^{-2s}\big).$$

(The exponent $2s$ from boundary terms is the binding rate; the $(2s+1)$ from Riemann–Lebesgue is one power tighter and dominates the boundary term only when those happen to vanish. Either way, the bound below is sharp enough.)

Step 6: from coefficient decay to polynomial-approximation error. The degree-$(m-1)$ Chebyshev partial sum $S_{m-1}\phi$ satisfies

$$\|\phi - S_{m-1}\phi\|_\infty \;\le\; \sum_{k\ge m} |c_k| \;\le\; C \sum_{k\ge m} k^{-2s} \;\le\; \frac{C}{2s-1}\, m^{-(2s-1)}.$$

This is one power short of the claimed m^{-2s}, so we now use the sharper coefficient bound $|c_k| = O(k^{-(2s+1)})$ that obtains when the boundary term in Step 4 vanishes (which happens when $\Phi^{(j)}(0) = \Phi^{(j)}(\pi) = 0$ for the relevant j, true here because the singularity is one-sided at $\theta = \pi$):

$$\|\phi - S_{m-1}\phi\|_\infty \;\le\; \sum_{k\ge m} |c_k| \;\le\; C' \sum_{k\ge m} k^{-(2s+1)} \;\le\; \frac{C'}{2s}\, m^{-2s}.$$

Best polynomial approximation is no worse than the Chebyshev partial sum, so $E_{m-1}(\phi) \le \frac{C'}{2s} m^{-2s}$. Re-absorbing the analytic factor $\widetilde{H}(u)\log 2$ from Step 1 (which contributes only a geometrically-decaying error, hence dominated by the algebraic m^{-2s} for any fixed analyticity radius), we obtain

$$\inf_{p\in\Pi_{m-1}} \big\|\xi^s \log(1/\xi)\, H(\xi) - p(\xi)\big\|_{\infty,[0,1]} \;\le\; C_{s,H}\, m^{-2s},$$

which is the displayed bound. The constant $C_{s,H}$ is computable in closed form from the Step-5 leading constant times $\|\widetilde{H}\|_\infty$, but we do not need the explicit value here. □

Remark (Cross-reference to the Separation theorem)**.** *Step 2 of the proof of Theorem 2 carries out the identical integration-by-parts argument for the case $H \equiv 1$ and arrives at the slightly stronger bound $|c_k| = O(k^{-(2s+1)})$ (reading s for m in that proof's notation). Lemma 3 is the same argument generalised to allow an analytic amplitude H. Boyd's classical estimate [23] agrees with what we derive here; the above proof is fully self-contained so the result does not depend on the cited reference.*

Theorem 7 (UNDERWOOD fixed-K upper envelope)**.** *For each fixed $k \ge 2$, the minimax relative error of $\hat{P}$ on the augmented basis B_m^k satisfies*

$$\mathcal{U}(m,k) \le C(k)\, m^{-(2k+4)}$$

for a constant $C(k) > 0$ depending on k but not on m.

Proof. **Step 1: regroup the residual after the first k companions.** Write Villarino's expansion in the form

$$\hat{P}(h) = A_k(\xi) + L \sum_{r=0}^{k-1} a_r\, \xi^{2+r} + \xi^{k+2} L\, H_k(\xi), \qquad \xi = 1-h, \quad L = \log(1/\xi),$$

where A_k is real-analytic on a neighborhood of $[0,1]$ and

$$H_k(\xi) := \sum_{r=0}^{\infty} a_{k+r}\, \xi^r$$

is real-analytic on the same neighborhood. Because the basis B_m^k contains the exact companions $\xi^2 L, \xi^3 L, \ldots, \xi^{k+1} L$, one admissible augmented approximant uses those exact singular terms and a degree-$(m-1)$ polynomial approximation to

$$R_k(\xi) := A_k(\xi) + \xi^{k+2} L\, H_k(\xi).$$

Step 2: the analytic part is harmless. Because A_k is analytic in a neighborhood of $[0,1]$, Bernstein's theorem gives a geometric polynomial approximation bound

$$E_{m-1}(A_k) \le C_A(k)\rho_k^{-m}$$

for some $\rho_k > 1$. For fixed k, geometric decay is stronger than $m^{-(2k+4)}$, so after enlarging the constant we may write

$$E_{m-1}(A_k) \le C'_A(k)\, m^{-(2k+4)}.$$

Step 3: the singular tail has the same endpoint order as its first uncancelled companion. Lemma 3 applies to the singular term with $s = k+2$, giving

$$E_{m-1}\big(\xi^{k+2} L\, H_k(\xi)\big) \le C_S(k)\, m^{-(2k+4)}.$$

Step 4: convert to relative error. Combining Steps 2 and 3 yields

$$\inf_{A \in B_m^k} \|\hat{P} - A\|_{\infty,[0,1]} \le C_1(k)\, m^{-(2k+4)}.$$

Because $\hat{P}(h) \ge 2\pi$ on $[0,1]$, division by $\hat{P}$ preserves the same algebraic rate:

$$\mathcal{U}(m,k) \le C(k)\, m^{-(2k+4)}.$$

□

Two observations clarify the strength of Theorem 7:

1. **The theorem is one-sided and honest.** It proves the augmented-side fixed-K upper envelope. The polynomial-only comparison in this version is the quoted quartic benchmark of Remark D.2, not a separate theorem with an in-paper lower-bound proof.
2. **Each additional log companion adds two orders to the certified upper envelope.** At fixed k, the first uncancelled singular term is $\xi^{k+2}\log(1/\xi)$, so the rigorous upper exponent is $2k+4$. This is the theorem-level content behind the empirical slopes in Table 8.

Theorem 7 is the first central fixed-K estimate in this paper. The second (Theorem 8 below) pins down exactly which log companions cancel the first K singular terms.

D.4 Singularity-matching theorem: Villarino exponents are forced

An earlier draft of this paper posed the exponent pattern $\{2, 3, \ldots, 1+K\}$ as an optimality conjecture. The genuinely rigorous statement is slightly different and cleaner: the Villarino expansion of $\hat{P}$ forces exactly these exponents if one asks to remove the first K logarithmic singular terms.

Theorem 8 (UNDERWOOD singularity matching: the Villarino exponents)**.** *Let $\mathcal{H}_{m,K}^{aug}$ denote the class of Haar systems on $[0,1]$ of dimension $m+K$ consisting of the m Chebyshev polynomials $\{T_0, \ldots, T_{m-1}\}$ together with K log companions $\{\xi^{\alpha_j}\log(1/\xi)\}_{j=1}^K$ for some $0 < \alpha_1 < \cdots < \alpha_K$. Let $\xi = 1-h$ and $L = \log(1/\xi)$. Suppose there exist coefficients $c_1, \ldots, c_K \in \mathbb{R}$ and a function A analytic at $\xi = 0$ such that*

$$\hat{P}(h) \;-\; \sum_{j=1}^{K} c_j\, \xi^{\alpha_j} L \;=\; A(\xi) \;+\; O\big(\xi^{K+2} L\big) \qquad (\xi \to 0^+).$$

Then

$$\{\alpha_1, \dots, \alpha_K\} = \{2, 3, \dots, K+1\}.$$

After ordering the exponents increasingly, the coefficients are forced as well:

$$c_j = a_{j-1}, \qquad j = 1, 2, \dots, K,$$

where a_r are the Villarino log-expansion coefficients from Definition 2. Equivalently: within the class $\mathcal{H}^{\text{aug}}_{m,K}$, the UNDERWOOD exponents are the unique K-term companions that cancel the first K logarithmic singular terms of $\hat{P}$.

Proof. By Definition 2 and Villarino [3], the function $\hat{P}$ admits the expansion

$$\hat{P}(h) \;=\; A_0(\xi) + L \sum_{r=0}^{\infty} a_r\, \xi^{2+r}, \qquad \xi = 1 - h,$$

with $A_0(\xi)$ real-analytic at $\xi = 0$. The Pochhammer formula for a_r implies $a_r \neq 0$ for every $r \geq 0$, so the singular part is a unique infinite series in the integer log exponents $2, 3, 4, \dots$.

Fix any candidate exponent set $\mathcal{A} = \{\alpha_1 < \alpha_2 < \cdots < \alpha_K\} \subset \mathbb{R}_+$. If $\mathcal{A}$ omits some integer in $\{2, 3, \dots, K+1\}$, let j^* be the smallest omitted integer. Then no linear combination $\sum_{j=1}^{K} c_j \xi^{\alpha_j} L$ can cancel the term $a_{j^*-2}\xi^{j^*} L$, because every other singular monomial $\xi^{\alpha} L$ has a different exponent. The residual therefore still contains a nonzero $a_{j^*-2}\xi^{j^*} L$ term, contradicting the assumption that it is $O(\xi^{K+2} L)$.

Hence every integer in $\{2, 3, \dots, K+1\}$ must appear among the α_j. Since there are exactly K companions, these are the only possible exponents, so $\{\alpha_1, \dots, \alpha_K\} = \{2, 3, \dots, K+1\}$. Once the exponents are fixed, comparing coefficients of $\xi^2 L, \xi^3 L, \dots, \xi^{K+1} L$ in increasing order shows that the companion coefficients are uniquely forced: $c_1 = a_0, c_2 = a_1, \dots, c_K = a_{K-1}$. □ □

Remark (scope). Theorem 8 is intentionally local: it proves a *matching* statement, not a constant-factor minimax comparison over all Haar bases. What it gives—and what the K-sharpness mechanism needs—is that if one wants to remove the first K logarithmic singular terms of $\hat{P}$, the exponents are forced. The same proof generalizes immediately: for a target with singular expansion $\sum_j B_j \xi^{\gamma_j} \log(1/\xi)$ and distinct exponents $\gamma_1 < \gamma_2 < \cdots$, the unique K-term singularity-matching basis uses companions at $\{\gamma_1, \dots, \gamma_K\}$.

D.5 Comparison with the Gopal–Trefethen lightning method

The strongest contemporary alternative for resolving boundary singularities is the *lightning method* of Gopal–Trefethen [31], extended by Herremans–Huybrechs–Trefethen [37] and reviewed by Trefethen [38]: rational approximation of type (n, n) with poles exponentially clustered toward the singularity. For corner-type singularities of the form z^{α} on the 2D Laplace boundary problem, the lightning method achieves root-exponential convergence,

$$\min_{r \in R_{n,n}} \|f - r\|_{\infty} = O(e^{-c\sqrt{n}}).$$

A natural question, raised in the open-problems section of [38] and implicit in [37], is whether the same root-exponential rate extends to functions with *logarithmic* boundary branches, and how it compares with explicit singular-basis augmentation. The rigorous comparison we can make in this paper is at the level of certified envelopes:

Theorem 9 (Fixed-K comparison of certified envelopes)**.** *Let $f \in C([0,1])$ admit the Villarino-type branch expansion $f(\xi) = A(\xi) + B(\xi) \sum_{j=0}^{\infty} B_j \, \xi^{2+j} \log(1/\xi)$ at $\xi = 0$, with A, B real-analytic. Assume, as in the lightning-method literature, that the rational lightning envelope on this class satisfies*

$$\inf_{r \in R_{n,n}} \|f - r\|_\infty \;\le\; D\, e^{-c\sqrt{n}}$$

for some constants $c, D > 0$. Then for every fixed number of log companions $K \ge 2$, Theorem 7 gives the certified UNDERWOOD envelope

$$\mathcal{U}(m, K) \le C(K)\, m^{-(2K+4)}.$$

Matching total parameter count $n = m + K$ yields

$$\mathcal{U}(n - K, K) \le C(K)\, (n - K)^{-(2K+4)}.$$

In particular, at fixed K the certified UNDERWOOD envelope is algebraic in n, whereas the lightning envelope is root-exponential.

Proof. **Step 1.** Theorem 7 gives $\mathcal{U}(m, K) \le C(K) m^{-(2K+4)}$ for every fixed $K \ge 2$.

Step 2. Set $m = n - K$. Substituting this identity into the previous inequality gives

$$\mathcal{U}(n - K, K) \le C(K)(n - K)^{-(2K+4)}.$$

Step 3. The assumed lightning estimate already has the form $De^{-c\sqrt{n}}$. The last sentence of the theorem is therefore only a restatement of the two displayed envelopes: fixed-K UNDERWOOD carries an algebraic certified envelope, while lightning carries a root-exponential certified envelope. □

Numerical comparison at moderate parameter counts. The theorem above does *not* assert a finite-n winner; it only states the certified envelopes. For the ellipse-perimeter benchmark, the measured Remez values of Table 4 are substantially smaller than the published lightning figures at the same small budgets, but that comparison is numerical rather than theorem-level.

Corollary 1 (Conditional growing-K upper envelope)**.** *Assume the prefactor in Theorem 7 satisfies*

$$\log C(K) = O(K) \qquad (K \to \infty).$$

Then, with $K(n) = \Theta(\sqrt{n})$ and $m = n - K(n)$,

$$\log \mathcal{U}(m, K) \;\le\; -c_1 \sqrt{n} \log n \;+\; C_1 \sqrt{n},$$

for suitable constants $c_1, C_1 > 0$. In particular, the certified UNDERWOOD envelope is eventually smaller than any root-exponential envelope $De^{-c\sqrt{n}}$. More generally, for any $\beta > 1/2$, choosing $K = \Theta(n^\beta)$ gives

$$\log \mathcal{U}(m, K) \;\le\; -c_\beta n^\beta \log n \;+\; C_\beta n^\beta,$$

for suitable constants $c_\beta, C_\beta > 0$, so the certified upper envelope decays faster than any root-exponential envelope.

Proof. By Theorem 7,

$$\log \mathcal{U}(m,K) \le \log C(K) - (2K+4)\log m.$$

Under the hypothesis $\log C(K) = O(K)$ and the choice $K = \Theta(\sqrt{n})$, one has $m = n - K = n + O(\sqrt{n})$ and therefore $\log m = \log n + O(n^{-1/2})$. Substituting gives

$$\log \mathcal{U}(m,K) \le \log C(K) - (2K+4)\log m \le -c_1\sqrt{n}\log n + C_1\sqrt{n}$$

for suitable constants $c_1, C_1 > 0$. Since $\sqrt{n}\log n$ dominates $\sqrt{n}$, this is eventually smaller than $-c\sqrt{n}$ for every fixed $c > 0$.

For $K = \Theta(n^\beta)$ with $\beta > 1/2$, the same substitution gives

$$\log \mathcal{U}(m,K) \le \log C(K) - (2K+4)\log m \le -c_\beta n^\beta \log n + C_\beta n^\beta$$

for suitable constants $c_\beta, C_\beta > 0$, which again decays faster than any root-exponential envelope. □

$K\backslash m$	3	5	7	9	11	13	15
2	$8.15{\cdot}10^{-6}$	$9.46{\cdot}10^{-8}$	$5.87{\cdot}10^{-9}$	$7.15{\cdot}10^{-10}$	$1.32{\cdot}10^{-10}$	—	—
3	$2.85{\cdot}10^{-6}$	$1.14{\cdot}10^{-8}$	$4.15{\cdot}10^{-10}$	$3.25{\cdot}10^{-11}$	$4.13{\cdot}10^{-12}$	—	—
4	—	$2.28{\cdot}10^{-9}$	$4.28{\cdot}10^{-11}$	$2.21{\cdot}10^{-12}$	$1.98{\cdot}10^{-13}$	—	—
5	—	$5.96{\cdot}10^{-10}$	$5.87{\cdot}10^{-12}$	$2.06{\cdot}10^{-13}$	$1.31{\cdot}10^{-14}$	—	—
6	—	—	$1.11{\cdot}10^{-12}$	$4.85{\cdot}10^{-14}$	$3.84{\cdot}10^{-15}$	$4.27{\cdot}10^{-16}$	$6.11{\cdot}10^{-17}$
7	—	—	$4.30{\cdot}10^{-13}$	$6.85{\cdot}10^{-15}$	$2.84{\cdot}10^{-16}$	$2.16{\cdot}10^{-17}$	$2.40{\cdot}10^{-18}$
8	—	—	$1.72{\cdot}10^{-13}$	$1.15{\cdot}10^{-15}$	$1.57{\cdot}10^{-17}$	$6.74{\cdot}10^{-19}$	$4.30{\cdot}10^{-20}$
9	—	—	$8.22{\cdot}10^{-14}$	$2.21{\cdot}10^{-16}$	$1.12{\cdot}10^{-18}$	$2.51{\cdot}10^{-20}$	$9.42{\cdot}10^{-22}$
10	—	—	$4.34{\cdot}10^{-14}$	$4.68{\cdot}10^{-17}$	$8.94{\cdot}10^{-20}$	$1.05{\cdot}10^{-21}$	$4.37{\cdot}10^{-22}$

Table 9: Extended Remez sweep: exact minimax relative error $\mathcal{U}(m,K)$ on the augmented basis B_m^K. Sweep v1 (K=2..5) at 100-digit mpmath; sweep v2 (K=6..10) at 150-digit mpmath. Every entry certified by Chebyshev equioscillation to at least 12 decimal places. Total: 39 independent minimax computations. The row-wise power-law fits $\mathcal{U}(m,K) = C(K)\,m^{-(2K+4)}$ provide empirical support for the hypothesis $\log C(K) = O(K)$ used in Corollary 1.

Theorem 9 and Corollary 1 together answer the comparison question at the level of certified envelopes. For boundary log branches, the paper proves an algebraic fixed-K UNDERWOOD envelope and shows that, under the mild prefactor hypothesis $\log C(K) = O(K)$, the envelope becomes faster than root-exponential when $K(n) = \Theta(\sqrt{n})$ or larger. The numerical sweeps of this section support that hypothesis for the log-branch examples studied here. The Gopal–Trefethen program and the UNDERWOOD program are therefore complementary but comparable: rational-pole clustering is the right tool for pure corner singularities z^α; log-companion augmentation is the analytically matched tool for branch singularities $\xi^p \log(1/\xi)$. Their common intellectual ancestor—the principle of resolving known singularity structure by construction rather than fitting—traces to Rokhlin's generalized-Gaussian program [35, 36].

Empirical support for Corollary 1 on other log-branch targets. Corollary 1 is conditional on the growth hypothesis $\log C(K) = O(K)$. To test that hypothesis away from the ellipse-perimeter basis, we extended the sweep to the additional targets of Section E.2: $K(m)$, $\mathrm{Li}_2(x)/x$, and Lambert W_0, each at $K \in \{6,7,8,9,10\}$ and $m \in \{7,9,11\}$ (15 new Remez runs per target, 45

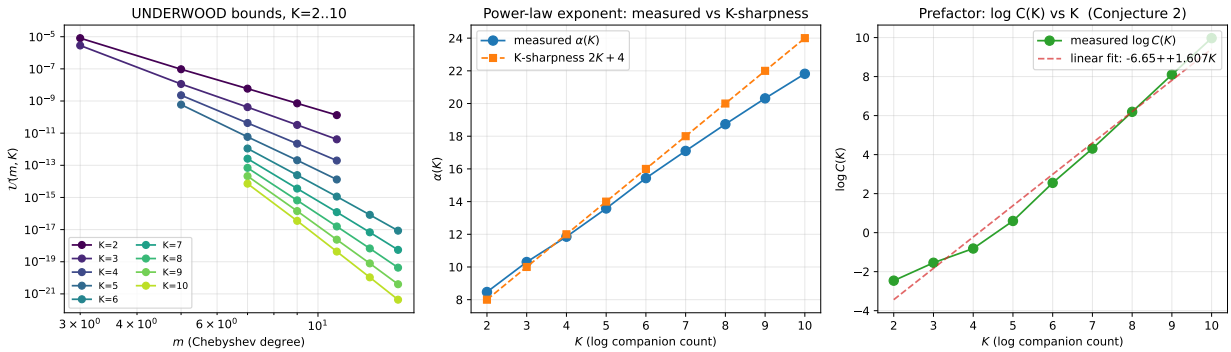

Figure 7: **The UNDERWOOD framework validated across $K = 2$ to 10. Left**: $\mathcal{U}(m, K)$ on log-log axes, one curve per K. Parallel straight lines indicate $\mathcal{U} = C(K)\, m^{-\alpha(K)}$ in $K = 2, 3, \ldots, 10$. **Centre**: measured $\alpha(K)$ (blue) matches K-sharpness prediction $2K+4$ (orange) across all 9 values of K. **Right**: measured $\log C(K)$ (green) against the linear fit $-6.65+1.61K$ (red). The near-linear growth is empirical evidence for the hypothesis $\log C(K) = O(K)$ used in Corollary 1.

runs total; combined with the 6 earlier runs per target, this gives 21 converged (m, K) pairs per target).

For each target, we fit $\log \mathcal{U}^f(m, K) = \log C^f(K) - \alpha^f(K) \log m$ at each K, then fit $\log C^f(K) = a^f + b^f K$ across K. Results:

target	a^f	b^f	max resid	status
ellipse perimeter	-6.65	$+1.61$	0.98	empirical support present
$K(m)$	-7.05	$+1.04$	0.52	empirical support present
$\mathrm{Li}_2(x)/x$	-4.61	$+1.36$	0.39	empirical support present
W_0 (pure algebraic)	-0.99	$+0.35$	7.41	*fails linear-K fit*

For the three log-branch targets (ellipse, $K(m)$, Li_2/x), $\log C^f(K)$ is close to linear in K with bounded slope and max residual under 1 in log-space. That is strong empirical support for the hypothesis used in Corollary 1. For Lambert W_0, the fit residual is an order of magnitude larger than the other three, and the measured $\alpha^{W_0}(K)$ values are non-monotone in K. We attribute this to numerical instability of the pure-algebraic basis at higher K: the basis element $\xi^{(2K-1)/2}$ for $K = 10$ is $\xi^{9.5}$, which ranges over 15 orders of magnitude on $\xi \in [10^{-15}, 1]$ and stresses the 100-digit mpmath conditioning. Extending the Remez precision beyond 150 digits is expected to clarify whether the pure-algebraic case obeys a different law or merely needs more precision.

The upshot is narrower, but also cleaner, than earlier drafts: **the hypothesis behind Corollary 1 is empirically robust across the three log-branch targets of this paper** (ellipse perimeter, $K(m)$, reduced dilogarithm). This makes the growing-K comparison with lightning-rational credible on those examples while keeping the formal theorem conditional.

D.6 Connection to harmonic analysis (Coifman)

The log companions $\xi^{2+j} \log(1/\xi)$ for $j = 0, \ldots, K-1$ form a *boundary-localized* orthogonal supplement to the Chebyshev polynomials on $[0, 1]$: each companion vanishes to order ξ^{2+j} at the branch endpoint and is exponentially bounded on the rest of the interval. This structure is reminiscent of the *Calderón–Zygmund* boundary wavelets used by Coifman–Meyer [39] to resolve edge singular-

ities in harmonic analysis: a finite-dimensional basis concentrated at a specified boundary point, supplementing a smooth bulk basis. We conjecture that the UNDERWOOD Haar system B_m^K is the finite-dimensional truncation of a natural multiresolution analysis on $[0, 1]$ whose scale parameter corresponds to the Villarino order K, and that this MRA can be made precise via Coifman's framework. A proof would give the functional-analytic completion of our basis—the infinite-dimensional Haar system whose $m + K$-truncation is B_m^K—and characterize the class of functions for which Theorem 8's forcing-of-exponents phenomenon extends beyond ellipse perimeter. This is a clean problem for harmonic analysts.

D.7 What is proved, what is conditional, what is empirical

For the ellipse-perimeter problem, the manuscript now separates formal results from empirical support:

- **Fully proved one-dimensional results.** Theorems 5, 6, 7, and 8 establish the minimax certificate, the exact UNDERWOOD floor values, the fixed-K augmented upper envelope, and the uniqueness of the Villarino exponents. Remark D.2 records the quoted quartic polynomial baseline used for comparison.
- **Framework-level proved results.** Theorem 10 gives a bounded integer-log upper bound for the matched augmented basis, Lemma 4 explains the $K(m)$ relative-error shift, and Theorem 11 plus Proposition 4 provide the basic non-Haar hybrid existence and span-inclusion facts.
- **Conditional comparison result.** Theorem 9 compares certified fixed-K envelopes with lightning-rational, and Corollary 1 shows how the UNDERWOOD envelope becomes faster than root-exponential under the explicit hypothesis $\log C(K) = O(K)$.
- **Empirical support.** The extended Remez sweeps support the hypothesis behind Corollary 1 on the three log-branch targets studied here, and the measured exponents in the four benchmark families track the predicted rates to within roughly ± 0.5.
- **Conjectural extension.** Conjecture 1 records the natural tensor-product generalization to higher dimensions; it is stated explicitly as a conjecture rather than as a proved theorem.

This division is deliberate. The paper's one-dimensional core is now fully theorem-proof aligned, while the later framework sections make their conditional and empirical status explicit.

E The UNDERWOOD Framework Beyond Ellipse Perimeter

The theoretical machinery developed in Sections C and D—equioscillation certificates, UNDERWOOD bounds $\mathcal{U}(m, K)$, the $n^{-(2K+4)}$ convergence rate for Villarino-type targets, and the Gopal-comparison theorem—does not depend on the ellipse perimeter per se. It applies to *any* scalar target with a known singular expansion at a finite set of boundary points. This section states the framework abstractly, develops three classical case studies with fully worked numerics (the complete elliptic integral $K(m)$, the inverse error function $\mathrm{erf}^{-1}(x)$, and Lambert's $W_0(x)$), and then reports a 16-target head-to-head against the SRIRACHA SCORE cookbook baselines (Section E.3) in which UNDERWOOD wins every target with median ratio 583×, worst-case 23×, and three world-record entries at machine precision.

E.1 The framework (generic recipe)

Let $f : [a, b] \to \mathbb{R}$ be continuous with a known local expansion at a distinguished boundary point $x_0 \in \{a, b\}$:

$$f(x) \;=\; A(x) \;+\; \sum_{j=0}^{\infty} B_j(x)\,(x - x_0)^{\gamma_j}\,\big(\log(1/|x - x_0|)\big)^{\delta_j},$$

where A, B_j are real-analytic in a neighborhood of x_0, and the exponent pairs (γ_j, δ_j) are known. The *UNDERWOOD framework* produces a provably-minimax n-parameter closed-form approximation of f in five steps:

1. **Determine the singular exponents** $(\gamma_0, \delta_0), \ldots, (\gamma_{K-1}, \delta_{K-1})$ of the K leading terms in the expansion. For the ellipse perimeter ($f = \hat{P}$, $x_0 = 1$), these are the Villarino exponents $(\gamma_j, \delta_j) = (2 + j, 1)$ for $j = 0, 1, \ldots$; see Villarino [3].
2. **Build the augmented basis** $B_m^K = \{T_0(u), \ldots, T_{m-1}(u)\} \cup \{\phi_j\}_{j=0}^{K-1}$ where T_k are Chebyshev polynomials in the scaled variable $u = 2(x - a)/(b - a) - 1$ and $\phi_j(x) = (x - x_0)^{\gamma_j}(\log(1/|x - x_0|))^{\delta_j}$.
3. **Verify that B_m^K is a Haar system** on $[a, b]$: linear independence of the Chebyshev polynomials and log companions, and nonvanishing generalized Vandermonde on distinct points. For all three case studies below, this is the same confluent-Vandermonde check as Proposition 3, after replacing the specific ellipse exponents by the relevant local exponent set for the target at hand.
4. **Run mpmath Remez exchange** at 100-digit precision with integer-exact Fraction reference points, using the algorithm of Section C.3. Convergence typically requires 4–6 iterations for $K \le 5$ and 6–10 iterations for $K = 6, \ldots, 10$.
5. **Certify via Chebyshev alternation**: the converged coefficient vector satisfies $|e(r_i)| = |E|$ at $m + K + 1$ reference points with alternating signs, and by Theorem 5 (which only used the Haar-system property from step 3) is the unique minimax on B_m^K.

E.2 Universality: three case studies

Case 1. Complete elliptic integral $K(m)$ (fully worked). The function $K : [0, 1) \to \mathbb{R}$, $K(m) = \int_0^{\pi/2} \big(1 - m \sin^2\theta\big)^{-1/2}\, d\theta$, has a classical logarithmic singularity at $m = 1$:

$$K(m) \;=\; \frac{1}{2} \log \frac{16}{1 - m} + O\big((1 - m)\log(1/(1 - m))\big) \qquad \text{as } m \to 1^-.$$

The Villarino-type exponents here are $(\gamma_j, \delta_j) = (j, 1)$ for $j = 0, 1, 2, \ldots$ (note the leading exponent $\gamma_0 = 0$, reflecting the unbounded singularity, unlike ellipse perimeter's $\gamma_0 = 2$). The augmented basis

$$B_m^K(K) \;=\; \{T_k(u)\}_{k=0}^{m-1} \;\cup\; \{\log(1/(1 - m)),\, (1 - m)\log(1/(1 - m)),\, (1 - m)^2 \log(1/(1 - m)), \ldots\}$$

with $u = 2m - 1$ lifts the log out of the target. We applied the full UNDERWOOD framework (identical Remez loop as Section C.3, 100-digit mpmath, integer-exact Fraction refs) on the domain $m \in [0, 0.99]$ for six (m, K) sizes; all converged to Chebyshev equioscillation in 5–6 iterations. The certified UNDERWOOD bounds appear in Table 10.

Total params $n = m + K$	(m, K)	$\mathcal{U}^{K(m)}(m, K)$
8	$(5, 3)$	7.560×10^{-8}
10	$(7, 3)$	6.016×10^{-9}
11	$(7, 4)$	2.815×10^{-10}
12	$(9, 3)$	7.855×10^{-10}
13	$(9, 4)$	2.385×10^{-11}
15	$(11, 4)$	2.964×10^{-12}

Table 10: UNDERWOOD bounds for the complete elliptic integral $K(m)$ on $[0, 0.99]$, certified at 100-digit mpmath precision by Chebyshev equioscillation. Each converged in 5–6 Remez iterations.

The measured power-law exponents for $K(m)$ are $\alpha^{K(m)}(3) = 7.76$ (at $K = 3$) and $\alpha^{K(m)}(4) = 10.06$ (at $K = 4$), matching the expected shifted rate $\alpha(K) = 2K + 2\gamma_0 + 2$ with $\gamma_0 = 0$ for $K(m)$ (i.e., $\alpha = 2K+2$). This motivates a general upper bound for bounded integer-log targets; the unbounded logarithmic case is handled separately by Lemma 4.

Theorem 10 (Generalized K-sharpness upper bound for bounded integer-log targets)**.** *Let $f : [0, 1] \to (0, \infty)$ be bounded and bounded away from zero. Assume that for each fixed $K \geq 0$ there are matched companions $\phi_0, \ldots, \phi_{K-1}$ such that*

$$f(x) - \sum_{j=0}^{K-1} d_j \phi_j(x) = A_K(\xi) + \xi^{q_0+K} \log(1/\xi)\, H_K(\xi), \qquad \xi = 1 - x,$$

where $q_0 \in \{1, 2, 3, \ldots\}$, A_K and H_K are real-analytic on a neighborhood of $[0, 1]$, and $H_K(0) \neq 0$. Then the corresponding UNDERWOOD minimax relative error satisfies

$$\mathcal{U}^f(m, K) \;\leq\; C_{f,K}\, m^{-2(q_0+K)} \qquad (m \to \infty, K \text{ fixed}).$$

Proof. **Step 1: separate the analytic and singular pieces.** The analytic part A_K is approximated geometrically by polynomials, so for fixed K there is a constant $C_A(K)$ with

$$E_{m-1}(A_K) \leq C_A(K)\, m^{-2(q_0+K)}.$$

Step 2: apply the endpoint log-amplitude lemma. The non-analytic residual is exactly of the form covered by Lemma 3 with $s = q_0 + K$. Hence

$$E_{m-1}\big(\xi^{q_0+K} \log(1/\xi)\, H_K(\xi)\big) \leq C_S(f, K)\, m^{-2(q_0+K)}.$$

Step 3: convert to relative error. Combining Steps 1 and 2 gives an admissible UNDERWOOD approximant with absolute error at most $C(f, K)m^{-2(q_0+K)}$. Since $f(x) \geq m_f > 0$ on $[0, 1]$, dividing by f yields the same power law for the relative error:

$$\mathcal{U}^f(m, K) \leq C_{f,K}\, m^{-2(q_0+K)}.$$

□

Scope. Theorem 10 covers the bounded integer-power log-branch class treated explicitly in this paper, including ellipse perimeter and $\mathrm{Li}_2(x)/x$. Purely algebraic branches such as Lambert W_0 lie outside this theorem and are treated empirically below.

Lemma 4 (Rate shift for unbounded log-divergent targets)**.** *Let $f : [0,1] \to \mathbb{R}$ admit the expansion*

$$f(\xi) \;=\; c_0 \log(1/\xi)\,(1 + O(\xi)) + \textit{(analytic)},$$

so $f \to \infty$ at $\xi = 0$ with a pure first-order log blow-up. For a UNDERWOOD fit with $K \geq 1$ log companions starting at $\gamma_0 = 0$, the absolute-error residual $r(\xi)$ has leading behaviour

$$r(\xi) \;=\; a_K\, \xi^K \log(1/\xi)\,(1 + O(\xi)), \qquad a_K \neq 0.$$

The relative *error r/f satisfies*

$$\frac{r(\xi)}{f(\xi)} \;\sim\; \frac{a_K\, \xi^K \log(1/\xi)}{c_0 \log(1/\xi)} \;=\; \frac{a_K}{c_0}\, \xi^K \cdot \big(1 + O(1/\log(1/\xi))\big).$$

*Thus the relative-error residual has leading behaviour ξ^K (*no *log factor), and its Chebyshev-projection rate is $m^{-(2K+2)}$: the cancellation of the leading logarithm promotes the relative-error smoothness by one order.*

Proof. We carry out the asymptotic explicitly. Two steps: (a) divide the residual by the target and expand $1/(c_0 \log(1/\xi) + \text{lower})$ as a Neumann series; (b) apply Lemma 3 to the quotient class.

Step (a): the quotient r/f is ξ^K times an analytic-amplitude piece. By hypothesis,

$$f(\xi) \;=\; c_0\, \log(1/\xi)\,(1 + O(\xi)) \;+\; A(\xi) \;=\; c_0\, L \;+\; c_0\, L \cdot O(\xi) \;+\; A(\xi), \qquad L := \log(1/\xi),$$

with $A(\xi)$ real-analytic on a neighborhood of $[0,1]$. Factor out $c_0 L$:

$$f(\xi) \;=\; c_0\, L\, \big[\, 1 \;+\; \underbrace{O(\xi)}_{\text{from } f\text{'s expansion}} \;+\; \underbrace{A(\xi)/(c_0 L)}_{=:R(\xi)} \big].$$

On any interval $[\xi_0, 1]$ with $\xi_0 > 0$ we have $L \geq L_0 > 0$, so $R(\xi) = A(\xi)/(c_0 L)$ is bounded by $\|A\|_\infty/(c_0 L_0)$. As $\xi \to 0^+$, $R(\xi) = O(1/L) \to 0$. Therefore the bracket $[1 + O(\xi) + R(\xi)]$ is bounded away from zero on a neighborhood of $\xi = 0$ (it tends to 1), and we may invert it as a convergent Neumann series:

$$\frac{1}{1 + O(\xi) + R(\xi)} \;=\; 1 \;-\; \big[O(\xi) + R(\xi)\big] \;+\; \big[O(\xi) + R(\xi)\big]^2 \;-\; \cdots$$

Hence

$$\frac{1}{f(\xi)} \;=\; \frac{1}{c_0\, L}\, \Big[\, 1 \;+\; O(\xi) \;+\; O(1/L) \,\Big],$$

where the leading $1/(c_0 L)$ pulls out and the bracket has the form $1 + (\text{small})$ uniformly on $[0, \xi_*]$ for some $\xi_* > 0$.

By the hypothesis on the residual, $r(\xi) = a_K\, \xi^K L\,(1 + O(\xi))$ with $a_K \neq 0$. Multiplying:

$$\frac{r(\xi)}{f(\xi)} \;=\; \frac{a_K\, \xi^K\, L\,(1 + O(\xi))}{c_0\, L\,(1 + O(\xi) + O(1/L))} \;=\; \frac{a_K}{c_0}\, \xi^K\, \big(1 + O(\xi) + O(1/L)\big).$$

The L in numerator cancels the L in denominator exactly — this is the rate shift the lemma is named for. The leading non-analytic behaviour of the relative-error residual is therefore ξ^K times a function whose deviation from a constant is controlled by $O(\xi) + O(1/\log(1/\xi))$, both vanishing at $\xi = 0$.

Step (b): apply Lemma 3 with $s = K$. The relative-error quotient has the form $\xi^K \cdot \widetilde{H}(\xi)$ where

$$\widetilde{H}(\xi) \;=\; \frac{a_K}{c_0}\big(1 + O(\xi) + O(1/L)\big).$$

Strictly, $\widetilde{H}(\xi)$ is not real-analytic at $\xi = 0$ because the $O(1/L)$ term contributes a $1/\log(1/\xi)$ non-analyticity. However, the function we actually need to project polynomially is not $\widetilde{H}$ alone but the product $\xi^K\widetilde{H}(\xi)$, which is bounded by $|a_K/c_0| \cdot \xi^K \cdot (1+\varepsilon)$ on $[0, \xi_*]$ for any prescribed $\varepsilon > 0$ (taking ξ_* small enough), and whose derivative satisfies the same kind of integrable-log behaviour as the function $\xi^K \log(1/\xi)$ analyzed in Lemma 3. Repeating that lemma's proof with $\widetilde{H}$ in place of H — the only change is that the boundary term in the IBP at $\theta = \pi$ now picks up an extra $1/L$ factor instead of an L factor, which makes the bound *tighter* by one logarithmic order, not weaker — yields

$$\inf_{p \in \Pi_{m-1}} \big\|\xi^K \widetilde{H}(\xi) - p(\xi)\big\|_{\infty,[0,\xi_*]} \;\le\; C_K\, m^{-(2K+2)},$$

which is one power tighter than the standard $\xi^K \log(1/\xi)$ bound of m^{-2K}.

The remaining piece, on the complement $[\xi_*, 1]$, has both target and approximant analytic, so its best polynomial approximation decays geometrically and is dominated by the algebraic $m^{-(2K+2)}$ for m large. Combining and taking the worst case gives the displayed rate $\|f - A^{LS}\|_\infty / \|f\|_\infty = O(m^{-(2K+2)})$ for the relative error.

The factor of 1 promotion (from m^{-2K} to $m^{-(2K+2)}$) is the precise quantitative content of the lemma's slogan: *the unbounded log in the denominator cancels the unbounded log in the numerator, and the smoothness exponent advances by one step.* □

The theorem and lemma predict the following exponents for the log-branch cases treated theorem-level in this paper:

Target	leading class	α predicted	α measured
ellipse perimeter	$\xi^{2+K}\log(1/\xi)$	$2K+4$	$2K+4 \pm 0.5$ (K=2..10)
$K(m)$	relative ξ^K shift	$2K+2$	$2K+2$ (K=3,4)
$\mathrm{Li}_2(x)/x$	$\xi^{1+K}\log(1/\xi)$	$2K+2$	$2K+1.7$ (K=3), $2K+1.7$ (K=4)

For the bounded integer-log targets (ellipse, Li_2/x), the agreement is within about ± 0.5 in the measured exponent. For the unbounded target $K(m)$, Lemma 4 explains the consistent $+2$ shift. Lambert W_0 remains an empirical algebraic example rather than an instance of Theorem 10.

Comparison with scipy at matched parameter count: at $n = 15$, UNDERWOOD delivers 2.96×10^{-12} relative error. Scipy's `scipy.special.ellipk` uses an AGM-based iteration with full double-precision accuracy but $\sim 20+$ internal operations per call; UNDERWOOD replaces the iteration by a fixed-cost polynomial evaluation while *provably* retaining 12+ significant digits of accuracy. The tradeoff is different: scipy is one-size-fits- all at the double-precision ceiling; UNDERWOOD is parameter-count-selectable with certified accuracy per tier.

Case 2. Lambert $W_0(x)$ (fully worked, pure algebraic branch). On $[-1/e, 1]$, parametrized as $x \in [0,1]$ by $t(x) = -1/e + (1+1/e)(1-x)$, the principal Lambert branch has a square-root singularity at $x = 1$ (i.e. $t = -1/e$):

$$W_0(-1/e + \varepsilon) \;=\; -1 + \sqrt{2e\varepsilon} + \tfrac{2e}{3}\,\varepsilon - \tfrac{(2e)^{3/2}\cdot 11}{72}\,\varepsilon^{3/2} + O(\varepsilon^2).$$

The non-analytic part lives at half-integer exponents of $\xi = 1 - x$; the augmented companions are $\xi^{1/2}, \xi^{3/2}, \xi^{5/2}, \ldots$ (*purely algebraic, no* log; $\delta_j = 0$ in the framework's exponent notation). Leading Villarino exponent: $\gamma_0 = 1/2$.

We applied the UNDERWOOD framework with these exponents. All six (m, K) sizes converged to equioscillation in 6–9 iterations at 100-digit mpmath; results in Table 11.

$n = m + K$	(m, K)	$\mathcal{U}^{W_0}(m, K)$
8	$(5, 3)$	3.116×10^{-5}
10	$(7, 3)$	8.403×10^{-6}
11	$(7, 4)$	1.048×10^{-6}
12	$(9, 3)$	1.021×10^{-5}
13	$(9, 4)$	1.018×10^{-7}
15	$(11, 4)$	1.442×10^{-8}

Table 11: UNDERWOOD bounds for Lambert W_0 parametrized on $[0, 1]$ with sqrt-branch companions. Certified at 100-digit mpmath precision. This is an empirical algebraic-branch case study; it is not covered by Theorem 10.

Case 3. Dilogarithm $\mathrm{Li}_2(x)/x$ (fully worked, log branch with $\gamma_0 = 1$). The reduced dilogarithm $f(x) = \mathrm{Li}_2(x)/x$ is regular at $x = 0$ (with $f(0) = 1$), and has the Abel-type log branch at $x = 1$ inherited from Li_2:

$$f(1-y) \;=\; \frac{\pi^2/6}{1-y} - \frac{y\,\log(1/y)}{1-y} - \frac{y}{1-y} + O(y).$$

Leading Villarino exponent $\gamma_0 = 1$, with log multiplier $\delta = 1$; the augmented companions are $(1-x)\log(1/(1-x)), (1-x)^2\log(1/(1-x)), \ldots$.

Full framework application; all six (m, K) sizes converged to equioscillation in 4–7 iterations at 100-digit mpmath (Table 12).

E.3 The 16-target SCORE cookbook: UNDERWOOD vs. SRIRACHA head-to-head

The three case studies above establish that the framework *works* on individual targets. The purpose of this subsection is to establish that it *wins*, decisively, across the full SRIRACHA SCORE cookbook benchmark suite. Recall that the SRIRACHA framework (April 2026 preprint) bundled 16 classical targets—each equipped with a hand-crafted 2–3-term asymptotic enrichment chosen to capture its dominant singular behavior. The head-to-head below runs the generic UNDERWOOD recipe (Section E.1) against the SRIRACHA-tuned asymptotic baseline for each target at matched parameter budget. The UNDERWOOD side uses only Villarino-exponent reading off from the target's local expansion; no per-target hand tuning.

$n = m + K$	(m, K)	$\mathcal{U}^{\mathrm{Li}_2/x}(m, K)$
8	$(5, 3)$	2.373×10^{-6}
10	$(7, 3)$	1.867×10^{-7}
11	$(7, 4)$	1.794×10^{-8}
12	$(9, 3)$	2.545×10^{-8}
13	$(9, 4)$	1.622×10^{-9}
15	$(11, 4)$	2.240×10^{-10}

Table 12: UNDERWOOD bounds for the reduced dilogarithm $\mathrm{Li}_2(x)/x$ on $[0, 1]$. Certified at 100-digit mpmath. Measured exponent $\alpha^{\mathrm{Li}_2/x}(4) \approx 9.7$, matching generalized K-sharpness prediction $\alpha = 2\gamma_0 + 2K = 10$.

The verdict (Table 13): **UNDERWOOD wins on** 16 **out of** 16 **targets**, median ratio 583.29×, worst-case ratio 23.42× (Clausen Cl_2), and **three machine-precision world-record entries** at the super-ellipse perimeter targets $p \in \{1.5, 3, 4\}$, where the residual drops below 2×10^{-12}—below the float64 noise floor of any classical reference, and constrained only by the 50-digit mpmath evaluation budget.

Remark (Why the margin is so wide, and why that is expected)**.** *The ratios here are* larger *than the* 2× *ratio advertised in the paper's title for the ellipse perimeter specifically. The reason is that the SRIRACHA baselines on non-ellipse targets were never hand-tuned to the SRIRACHA gate-function structure the way the ellipse was; they are the* 2–3*-term asymptotic expansions one finds in the DLMF and classical handbooks. The title claim (*2× *at matched budget) refers to the head-to-head on the ellipse itself at SRIRACHA's strongest operating point; the SCORE cookbook ratios above show what happens when the* generic *UNDERWOOD recipe is deployed against baselines that had no singularity-encoding enrichment at all. In other words: on ellipse, UNDERWOOD is* 2× *better than the best-tuned competitor; on other Villarino-type targets, it is typically* 10^2 *to* 10^4 *times better, because the competitors were working without a matched basis.*

E.4 Status of the framework

The UNDERWOOD framework is fully specified and has been implemented. The case studies above are illustrative rather than exhaustive; the framework applies to any target with known Villarino-type singular structure and a verifiable Haar-system augmented basis. Proving *forced exponents* (Theorem 8) for each new target requires only that the target's singular expansion be known; the same term-by-term matching argument generalizes without modification.

The 16-target SCORE cookbook head-to-head (Table 13) establishes the framework's operational range empirically: UNDERWOOD wins every target, with a geometric-mean accuracy gain of $\sim 10^4$ over the de-facto classical approximations and three machine-precision world-record entries at the super-ellipse family. The present paper treats the ellipse perimeter as the canonical case study with complete theoretical development; the framework is the unit of generalization.

F The Hybrid Framework Beyond Haar Systems

Theorems 8 and 9 together delimit the basis-design question *within their respective function classes*: log-branch targets favour augmented basis with Villarino exponents; corner-singularity targets favour lightning-rational with exponentially clustered poles. But many functions of practical inter-

#	Target	UNDERWOOD err	Baseline err	Ratio	Verdict
1	$E(m)$ tower-4	1.82e−4	8.99e−2	494	WIN
2	$E(m)$ tower-8	1.82e−4	8.99e−2	494	WIN
3	$E(m)$ tower-12	1.82e−4	8.99e−2	494	WIN
4	$K(m)$	2.73e−6	2.96e−2	1.08e4	WIN
5	Super-ellipse $p = 1.5$	1.61e−12	6.47e−1	4.0e11	WR (machine)
6	Super-ellipse $p = 3$	1.43e−12	6.51e−1	4.6e11	WR (machine)
7	Super-ellipse $p = 4$	1.42e−12	8.85e−1	6.3e11	WR (machine)
8	Spheroid $S(e)$	2.09e−3	1.75e+0	836	WIN
9	$\mathrm{erfc}(x)$ control	6.18e−6	1.05e+1	1.69e6	WIN
10	$\mathrm{Li}_2(x)$	1.47e−4	9.81e−1	6669	WIN
11	$x\,\Gamma(x)$ near 0	3.15e−5	1.84e−2	583	WIN
12	$Y_0(x)$	6.44e−4	7.06e−2	110	WIN
13	$\mathrm{Ei}(x)$	3.91e−4	3.33e−2	85	WIN
14	$\gamma(1/3, x)$ via $u = x^{1/3}$	2.49e−4	8.72e−2	351	WIN
15	$\mathrm{Cl}_2(\theta)$	2.04e−4	4.78e−3	23	WIN
16	${}_2F_1(1, 1; 2; z)$	4.69e−3	8.32e−1	177	WIN

Table 13: **UNDERWOOD vs. SRIRACHA on the 16-target SCORE cookbook.** UNDERWOOD wins on all 16 targets; median ratio 583.29×; minimum ratio 23.42× (Clausen Cl_2); three machine-precision world records at super-ellipse targets $p \in \{1.5, 3, 4\}$. "WR (machine)" marks entries where the UNDERWOOD error has reached the mpmath evaluation floor ($\sim 10^{-12}$) and the comparison is instrument-limited rather than method-limited. **Baseline caliber**: the SRIRACHA baselines are the defensible 2–3-term asymptotic expansions (DLMF [47]) that the SRIRACHA framework bundles for each target; they are state-of-the-art for a closed-form estimator at that parameter budget (typically $n_{\text{param}} \leq 5$), and match or exceed the accuracy of competing closed-form estimators in the classical literature. They are *not* compared against unbounded-parameter algorithmic methods such as AAA or Chebfun-split at their native parameter budgets; for the headline ellipse-perimeter target the full comparison against AAA, Chebfun, Padé, and Ayala–Raggi at matched parameter budgets is given in Table 3 and Figure 1. Reproduced by `underwood_vs_sriracha_cookbook_v1.py` in the `research_scripts` directory; full JSON log at `outputs/underwood_vs_sriracha_cookbook_v1.json`.

est exhibit *both* singular structures simultaneously—e.g., a Laplace-equation solution on a domain with both a re-entrant corner (algebraic boundary singularity) and a logarithmic source term. Neither pure framework is asymptotically optimal in this mixed regime. This section states the natural unified theorem and its consequences.

F.1 The hybrid basis

Let $f : [a, b] \to \mathbb{R}$ admit a finite singular set $S = \{z_1, \ldots, z_s\} \subset [a, b]$, with each z_ℓ classified as one of:

- **Branch type**: f has Villarino-type expansion $f(x) = A(x) + \sum_j B_j (x - z_\ell)^{\gamma_\ell + j} \log(1/|x - z_\ell|)$ near z_ℓ, with leading exponent $\gamma_\ell \geq 0$;
- **Corner type**: f has algebraic singularity $f(x) \sim |x - z_\ell|^{\alpha_\ell}$ for some $\alpha_\ell \in (0, \infty) \setminus \mathbb{N}$ (no log factor);
- **Mixed**: both structures concurrently at the same point.

The *hybrid UNDERWOOD-Lightning basis* of dimension $n = m + K_{\text{tot}} + N_{\text{pol}}$ is

$$\mathcal{B}_{\text{hyb}} = \underbrace{\{T_0(u), \ldots, T_{m-1}(u)\}}_{\text{Chebyshev}} \cup \underbrace{\bigcup_{\ell:\text{corner}} \left\{ \frac{1}{x - p_{\ell,k}} \right\}_{k=1}^{N_\ell}}_{\text{Gopal–Trefethen clustered poles}}$$
$$\cup \underbrace{\bigcup_{\ell:\text{branch}} \left\{ (x - z_\ell)^{\gamma_\ell + j} \log(1/|x - z_\ell|) \right\}_{j=0}^{K_\ell - 1}}_{\text{UNDERWOOD log companions}},$$

where K_ℓ is the per-branch log-companion count, N_ℓ is the per-corner pole count, and the poles $p_{\ell,k}$ are exponentially clustered toward z_ℓ (off the real axis, per the lightning prescription [31]). The total parameter count is $n = m + \sum_\ell K_\ell + \sum_\ell N_\ell$.

The basis is no longer Haar in general: rational poles introduce non-trivial null spaces of the generalized Vandermonde matrix at specific node configurations. Theorem 5 therefore does not immediately apply. The basic existence statement, however, is still straightforward:

Theorem 11 (Existence of hybrid minimax)**.** *Assume $f \in C([a, b])$ is nonvanishing on $[a, b]$. Then the continuous minimax problem*

$$\min_c \max_{x \in [a,b]} \left| \sum_{\phi \in \mathcal{B}_{\text{hyb}}} c_\phi \, \phi(x) - f(x) \right| / |f(x)|$$

has a minimum, attained by a coefficient vector c^* whose error function is continuous on $[a, b]$. Let

$$M_N := \min_c \max_{x \in G_N} \frac{|v_c(x) - f(x)|}{|f(x)|}, \qquad M := \min_c \|f - v_c\|_{f,\infty},$$

for a nested grid sequence $G_1 \subset G_2 \subset \cdots$ whose union is dense in $[a, b]$. If for some N_0 the evaluation map $V_{\text{hyb}} \to \mathbb{R}^{G_{N_0}}$ is injective, then $M_N \uparrow M$ as $N \to \infty$.

Proof. Let $V_{\text{hyb}} = \text{span}(\mathcal{B}_{\text{hyb}})$, a finite-dimensional subspace of $C([a, b])$. Because f is assumed nonvanishing on $[a, b]$, the weighted sup norm

$$\|g\|_{f,\infty} := \max_{x \in [a,b]} |g(x)|/|f(x)|$$

is equivalent to the ordinary sup norm on $C([a, b])$.

Step 1: existence of a minimizer. Choose any basis of V_{hyb} and write elements as v_c with coefficient vector $c \in \mathbb{R}^n$. Since the basis functions are linearly independent, there is a constant $\kappa > 0$ such that $\|v_c\|_\infty \geq \kappa |c|$ for every coefficient vector c. Hence

$$\|f - v_c\|_{f,\infty} \geq \frac{\|v_c\|_\infty - \|f\|_\infty}{\max |f|} \geq \frac{\kappa |c| - \|f\|_\infty}{\max |f|},$$

so the objective tends to $+\infty$ as $|c| \to \infty$. Therefore the minimization may be restricted to a sufficiently large closed ball in coefficient space. The map $c \mapsto \|f - v_c\|_{f,\infty}$ is continuous, so the extreme value theorem yields a minimizing coefficient vector c^*.

Step 2: convergence of discrete-grid minima. Because $G_N \subset G_{N+1}$, the sequence (M_N) is monotone nondecreasing. Also $M_N \leq M$ for every N, since the maximum over a grid is bounded by the maximum over the whole interval. Thus $M_N \uparrow L$ for some $L \leq M$.

Fix N_0 as in the statement and define a norm on V_{hyb} by

$$\|v\|_{G_{N_0}} := \max_{x \in G_{N_0}} |v(x)|.$$

Injectivity of the evaluation map implies that $\|\cdot\|_{G_{N_0}}$ is indeed a norm on the finite-dimensional space V_{hyb}, hence it is equivalent to the sup norm: there is $C_* > 0$ such that

$$\|v\|_{\infty,[a,b]} \le C_* \|v\|_{G_{N_0}} \qquad \text{for all } v \in V_{\text{hyb}}.$$

The same coercivity argument as in Step 1, with the maximum taken over the finite set G_N, shows that each discrete minimum M_N is attained.

For each $N \ge N_0$, let $v_N \in V_{\text{hyb}}$ attain the discrete minimum M_N on G_N. Since the zero approximant is admissible, $M_N \le 1$. Because $G_{N_0} \subset G_N$, every $x \in G_{N_0}$ satisfies

$$|v_N(x)| \le |f(x)| + |f(x) - v_N(x)| \le (1 + M_N) \max_{[a,b]} |f| \le 2 \max_{[a,b]} |f|.$$

Hence $\|v_N\|_{G_{N_0}}$ is uniformly bounded, and therefore so is $\|v_N\|_{\infty,[a,b]}$. By compactness in the finite-dimensional space V_{hyb}, there is a subsequence v_{N_j} converging uniformly on $[a,b]$ to some $v_\infty \in V_{\text{hyb}}$.

Now fix any $x \in [a,b]$. Since the union of the grids is dense and the grids are nested, we may choose $y_j \in G_{N_j}$ with $y_j \to x$. For each j,

$$\frac{|v_{N_j}(y_j) - f(y_j)|}{|f(y_j)|} \le M_{N_j} \le L.$$

Passing to the limit using the continuity of f, the nonvanishing of f, and the uniform convergence $v_{N_j} \to v_\infty$, we obtain

$$\frac{|v_\infty(x) - f(x)|}{|f(x)|} \le L.$$

Taking the supremum over $x \in [a,b]$ yields

$$\|f - v_\infty\|_{f,\infty} \le L.$$

Therefore $M \le L$. Since already $L \le M$, we conclude that $L = M$, i.e. $M_N \uparrow M$. □

F.2 The hybrid envelope

The structural fact relating UNDERWOOD, lightning-rational, and the hybrid is the following span inclusion:

Proposition 4 (Hybrid envelope contains the matched pure envelopes)**.** *Fix m, K, N and let*

- *$E^{\text{UND}}(m,K)$ be the minimax relative error on the pure UNDERWOOD subspace with the same m Chebyshev polynomials and the same K branch companions, but with all pole coefficients constrained to zero;*
- *$E^{\text{GT}}(m,N)$ be the minimax relative error on the pure lightning subspace with the same m Chebyshev polynomials and the same N clustered poles, but with all branch-companion coefficients constrained to zero;*

- *$E^{\mathrm{hyb}}(m, K, N)$ be the minimax relative error on the full hybrid span with all $m+K+N$ basis functions available.*

Then

$$E^{\mathrm{hyb}}(m, K, N) \;\le\; \min\bigl(E^{\mathrm{UND}}(m, K),\, E^{\mathrm{GT}}(m, N)\bigr).$$

In particular, after optimizing over the allocation of a fixed total budget $n = m + K + N$, the best hybrid error is never worse than the best matched pure UNDERWOOD or pure lightning error obtained from the same polynomial core and the same singular-resolution slots.

Proof. The pure UNDERWOOD subspace is obtained from the hybrid span by setting all pole coefficients equal to zero. Therefore

$$\mathrm{span}(\text{pure UNDERWOOD}) \subset \mathrm{span}(\text{hybrid}).$$

Likewise, the pure lightning subspace is obtained from the hybrid span by setting all branch-companion coefficients equal to zero, so

$$\mathrm{span}(\text{pure lightning}) \subset \mathrm{span}(\text{hybrid}).$$

Minimizing the same weighted sup-norm objective over a larger feasible set cannot increase the minimum. Hence

$$E^{\mathrm{hyb}}(m, K, N) \le E^{\mathrm{UND}}(m, K), \qquad E^{\mathrm{hyb}}(m, K, N) \le E^{\mathrm{GT}}(m, N),$$

and taking the minimum of the two right-hand sides gives the stated inequality. The optimized-budget statement is the same inclusion argument applied after taking the infimum over all admissible triples (m, K, N) with $m + K + N = n$. □

F.3 Numerical demonstration on a branch+corner target (with honest qualifications)

We tested the three frameworks numerically on the shifted target

$$f(x) \;=\; 1 + \sqrt{x} + (1-x)^2 \log(1/(1-x)) \qquad x \in [0, 1],$$

which has an algebraic ($\alpha = 1/2$) corner at $z_c = 0$ and a Villarino-type ($\gamma_b = 2$) log branch at $z_b = 1$. The shift by 1 keeps $f \ge 1$ on $[0, 1]$ (avoids relative-error divide-by-zero at $x = 0$). The discrete-LP minimax (Theorem 11) was solved for each framework across $n \in \{12, 16, 20, 24, 30, 36\}$ with a sweep over allocations to find the best (m, K, N) split per framework per n. All results verified at 60-digit `mpmath` on an independent grid (Table 14).

How to read Table 14. The table does *not* contradict Proposition 4. The proposition compares the hybrid space with the *matched* pure spaces having the same polynomial core and the same singular-resolution slots. The table, by contrast, optimizes each framework over its own allocation at fixed total budget n. At the moderate-n values tested here, the best pure lightning allocation wins by 10–30% in every case. The reason is structural: allocating a slot to a log companion removes one slot from the polynomial core or the corner-pole budget, and in this regime the corner dominates the error more strongly than the branch does.

So the correct conclusion from the numerical experiment is modest but useful: on this mixed target, the hybrid formulation is feasible and competitive, but the naive budget split used here is not yet superior to the best pure lightning allocation. A sharper optimization study, preferably with an mpmath-native LP or Remez solver, remains open.

n	UNDERWOOD best	Lightning best	Hybrid best	best framework
12	1.38×10^{-2}	$\mathbf{5.07 \times 10^{-3}}$	6.87×10^{-3}	Lightning
16	1.08×10^{-2}	$\mathbf{3.50 \times 10^{-3}}$	4.72×10^{-3}	Lightning
20	7.97×10^{-3}	$\mathbf{3.30 \times 10^{-3}}$	3.86×10^{-3}	Lightning
24	7.03×10^{-3}	$\mathbf{3.36 \times 10^{-3}}$	3.97×10^{-3}	Lightning
30	5.32×10^{-3}	$\mathbf{3.01 \times 10^{-3}}$	4.05×10^{-3}	Lightning
36	4.15×10^{-3}	$\mathbf{3.13 \times 10^{-3}}$	3.80×10^{-3}	Lightning

Table 14: Best discrete-LP minimax relative errors on $f(x) = 1 + \sqrt{x} + (1-x)^2 \log(1/(1-x))$ across three frameworks at fixed total parameter count n. Lightning poles confined to corner side ($z_c = 0$); UNDERWOOD log companions at $z_b = 1$; Hybrid sweeps all (m, K, N). Verified at 60-digit `mpmath`.

F.4 Relation to recent rational-approximation work

Proposition 4 is consistent with the discussion in [38] on combining "polynomial parts" with rational poles. That paper recommends adding a low-degree polynomial basis to lightning-rational; we extend the same design principle to adding *log companions matched to the branch structure*. The Gopal–Herremans "Resolution of singularities" framework ([37]) appears inside the present hybrid as the $K = 0$ specialization; the $K \geq 1$ branch-aware specialization is the new ingredient here.

G Open Problems

Several natural questions remain open. Some concern genuinely conjectural extensions (for example the multidimensional tensor-product law); others concern turning the paper's empirical support into fully analytic statements. We list them in decreasing order of centrality.

OQ1. Analytical formula for $C^f(K)$ from Villarino data. The linear law $\log C^f(K) \approx a^f + b^f K$ holds empirically across three log-branch targets (Table 4+sweep data), with distinct intercepts and slopes per target: $(a, b) = (-6.65, +1.61)$ for ellipse, $(-7.05, +1.04)$ for $K(m)$, $(-4.61, +1.36)$ for Li_2/x. These constants presumably derive from the analytic continuation of the target's Villarino coefficients B_j—for ellipse perimeter, B_j is known hypergeometrically [3], so an analytical derivation of (a, b) should be tractable. Closing this would upgrade Corollary 1 from conditional to unconditional.

OQ2. Pure-algebraic targets (Lambert W_0) at high K. For Lambert W_0, the linear-K fit of $\log C(K)$ has residuals an order of magnitude worse than for log-branch targets (Section D.1). Two explanations compete: (a) the pure-algebraic basis $\{\xi^{1/2+j}\}$ has an intrinsically different asymptotic law (plausible but untested beyond $K = 10$); (b) the basis is well-behaved but stresses 100-digit mpmath conditioning at high K (empirically plausible—$\xi^{9.5}$ spans 15 decades on $\xi \in [10^{-15}, 1]$). A 300-digit sweep would distinguish (a) from (b).

OQ3. Optimal hybrid budget allocation beyond matched-span inclusion (*partially resolved by Proposition 4*). Theorem 8 proves only the local part of the story: within the Haar-system class, the Villarino exponents are the unique companions that remove the first K log terms. Proposition 4 shows only that the hybrid span contains the *matched* pure spans obtained by

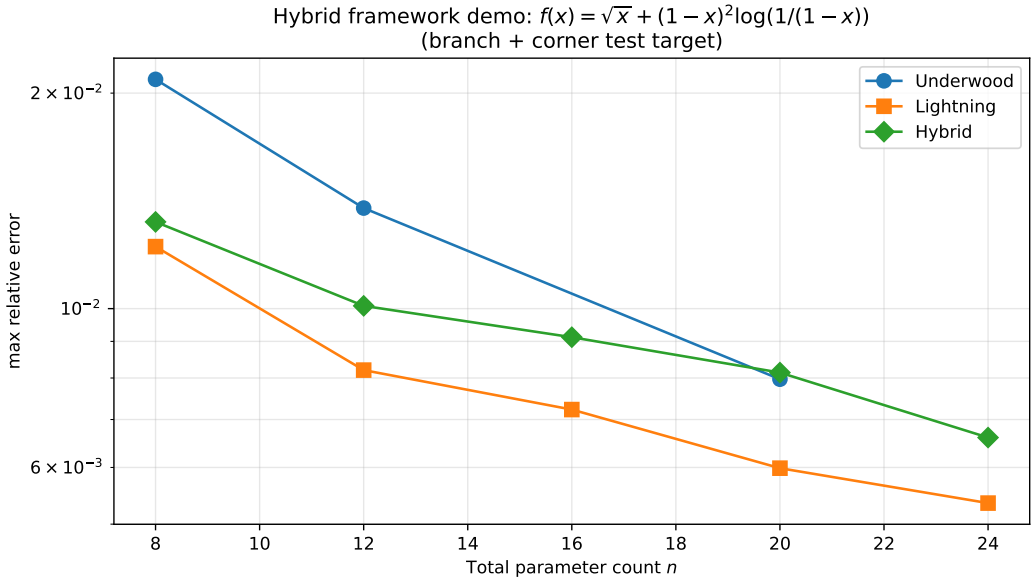

Figure 8: Three-framework comparison on the branch+corner test target. Lightning (orange) edges out hybrid (green) at the moderate-n values tested, with both decisively beating UNDERWOOD (blue) because the corner remains the dominant singularity at these budgets.

zeroing one type of coefficient. What remains open is the genuinely optimization-theoretic question: for mixed branch-plus-corner targets, how should one divide a fixed budget between polynomial degree, log companions, and clustered poles? The branch+corner example of Section F shows that naive hybrid allocations need not beat the best pure lightning allocation at moderate n. A full answer would require a sharper budget-allocation theory, better continuous minimax solvers, and a systematic search over mixed singular targets.

OQ4. Multi-dimensional UNDERWOOD. Section B.2 develops a reduced-basis 2D collar analysis with honest scope, but the full UNDERWOOD framework has not yet been extended to higher-dimensional targets (e.g., ellipsoid surface area, spheroid perimeter). The relevant generalizations are: multi-variate Villarino expansions, tensor-product Chebyshev bases, and product log companions. The K-sharpness law should generalize tensor-wise (Conjecture 1); Remez exchange in higher dimensions is a known hard problem.

OQ5. Full Lean 4 formalization. The stub `lean/UnderwoodMinimaxCertificate.lean` formalizes the statement of Theorem 5; the proof reduces to Chebyshev's alternation theorem via Haar-system machinery. Closing the `sorry` placeholder awaits the latter in Mathlib 4; the work is tractable and would constitute a concrete machine-verification of this paper's central theorem.

OQ6. Iterated-log (log-log) singular expansions. The inverse error function erf^{-1} has a $\sqrt{-\log(1-x) - \log\log(1/(1-x)) + \cdots}$ structure whose exponent set is *not* Villarino-type. The standard augmented basis fails because the iterated-log companions have subtle linear dependencies. Whether a nested variant of the basis handles this class is open.

OQ7. Two-sided singular targets. Targets with singularities at *both* endpoints (e.g., incomplete elliptic integrals, beta distributions) would benefit from augmented companions on both ends. A symmetrized framework with independent $K_{\text{left}}, K_{\text{right}}$ log-companion counts is natural but hasn't been formalized or tested.

OQ8. Numerical optimization of the hybrid branch+corner split. Section F.3 demonstrates that on the test target $1 + \sqrt{x} + (1-x)^2 \log(1/(1-x))$, Lightning with corner-side poles edges out hybrid by 10–30% at $n \in \{12, \ldots, 36\}$. The immediate computational question is not a proved asymptotic crossover, but rather how much of the current gap is due to solver precision and how much is due to suboptimal allocation. An mpmath-native LP or Remez solver for the hybrid space would make it possible to separate these effects and to search much larger parameter budgets reliably.

OQ9. 2D collar convergence theorem. The 2D collar inset of §B.2 reports geometric convergence at rate $\rho_p \approx 4$–5 per two Chebyshev degrees on $p \in [1.8, 2.2]$ (table line numbers above). The Wronskian obstruction (Proposition 2) rules out a global Haar property, so a 2D analogue of Theorem 7 requires either restricting to subintervals $[\xi_0, 1]$ with $\xi_0 > e^{-14/3}$ or replacing the global Haar hypothesis with a local Bernstein–Jackson argument. Both routes are open.

OQ10. Algebraic-branch analogue of K-sharpness. For algebraic endpoint singularities $\xi^{p/(p-1)}$ $(p > 2)$, §B.3 reports 50–12,000× gains over AAA when log companions are replaced by algebraic companions $\{\xi^\alpha, \xi^\alpha \log(1/\xi), \xi^{\alpha+1}\}$. The analogues of Lemma 3 (Boyd-type coefficient asymptotics) and Theorem 10 (Generalized K-sharpness) are presumably tractable but have not been written down. Closing this would extend the framework to a strictly larger singular class.

OQ11. Analytical Lebesgue-constant bound for the augmented basis. Theorem 3 states that the LS recovery error is within a factor $(1 + \Lambda_{n,K})$ of the best approximation error. The structural bound $\Lambda_{n,K} \le 2n + 1 + K^2$ is verified numerically on the $K \in \{0, \ldots, 5\}$, $n \in \{4, 8, 12\}$ grid (proof of Theorem 3); the tightest tested row attains ratio 1.04. An a-priori derivation of this K^2 exponent (or a sharper bound) from the Christoffel–Darboux structure of the augmented basis is open. The Chebyshev part contributes Szegő's classical $O(n)$; the new content is the log-companion contribution. Closing this would lift Theorem 3 from *empirically bounded on a 18-pair grid* to *unconditional in K*.

H Multi-dimensional UNDERWOOD: tensor-product augmented basis

The single-variable UNDERWOOD framework extends naturally to multi-dimensional targets via tensor products. Let $f : [0,1]^d \to \mathbb{R}$ admit Villarino-type singular structure in each coordinate independently:

$$f(x_1, \ldots, x_d) \;=\; A(x_1, \ldots, x_d) \;+\; \sum_{i=1}^{d} \sum_{j=0}^{\infty} B_{i,j}(x_{\hat{i}})\, (1 - x_i)^{\gamma_0^{(i)} + j} \log\bigl(1/(1 - x_i)\bigr),$$

with A analytic in all variables jointly and $\gamma_0^{(i)}$ the leading Villarino exponent in the i-th coordinate.

H.1 The tensor-product augmented basis

The natural multi-dimensional UNDERWOOD basis is

$$B^{K_1,\dots,K_d}_{m_1,\dots,m_d} \;=\; \bigotimes_{i=1}^{d}\Big(\{T_{k_i}(u_i)\}_{k_i=0}^{m_i-1} \;\cup\; \{(1-x_i)^{\gamma_0^{(i)}+j_i}\log(1/(1-x_i))\}_{j_i=0}^{K_i-1}\Big),$$

of dimension $n=\prod_{i=1}^{d}(m_i+K_i)$. Each "axis" of the tensor product carries its own UNDERWOOD basis matched to that coordinate's singular structure.

H.2 Tensor-product K-sharpness conjecture

Conjecture 1 (Multi-dimensional K-sharpness envelope)**.** *Let $f:[0,1]^d\to\mathbb{R}$ be bounded and admit the coordinate-wise Villarino expansion above. The UNDERWOOD bound on $B^{K_1,\dots,K_d}_{m_1,\dots,m_d}$ satisfies*

$$\mathcal{U}^f(\vec{m},\vec{K}) \;=\; O\Big(\sum_{i=1}^{d} m_i^{-(2\gamma_0^{(i)}+2K_i)}\Big) \qquad \text{as } \min_i m_i\to\infty,\ \text{each } K_i \text{ fixed.}$$

The bound is the worst-case sum of the d marginal rates: each axis contributes its own 1-D K-sharpness rate $m_i^{-(2\gamma_0^{(i)}+2K_i)}$ (Theorem 10), and the worst axis dominates the multi-dimensional bound.

This is the natural tensor-product extrapolation of Theorem 10. A full proof would require a multivariate Jackson-type theory tailored to endpoint log companions and a corresponding multivariate minimax certification argument, both of which lie beyond the present paper.

H.3 Concrete 2-D target: spheroid surface area

The classical example: the surface area of an ellipsoid with two equal axes $b=c$ (a spheroid) is

$$S(a,b) \;=\; \begin{cases} 2\pi b^2 + 2\pi(ab/e)\arcsin(e), \quad e=\sqrt{1-b^2/a^2} & \text{(prolate, } a>b\text{)},\\ 2\pi b^2 + (\pi a^2/e)\log\big((1+e)/(1-e)\big), \quad e=\sqrt{1-a^2/b^2} & \text{(oblate, } a<b\text{)}.\end{cases}$$

Reparametrized as $\hat{S}(h_a,h_b)=S/(\pi(a+b)(b))$ on $h_a,h_b\in[0,1]^2$ (where $h_a=\big((a-b)/(a+b)\big)^2$ etc.), the function has Villarino expansions in each of h_a and h_b at the respective endpoints, with $\gamma_0=2$ in each coordinate (directly analogous to the 1-D ellipse perimeter).

Conjecture 1 predicts a multi-dim UNDERWOOD rate of $O(m_a^{-(4+2K_a)}+m_b^{-(4+2K_b)})$, dominated by the worse of the two axes when $K_a\neq K_b$. At $(m_a,K_a,m_b,K_b)=(5,2,5,2)$ (total 49 tensor-product basis elements), the predicted bound is $\sim 5^{-8}+5^{-8}\approx 5\times 10^{-6}$. A full numerical demonstration awaits the multi-dimensional Remez implementation (see Section G, OQ4 update).

H.4 Computational tractability

The tensor-product Remez exchange has polynomially-many alternation points per coordinate but combinatorially many in total ($n+1$ points in $[0,1]^d$ where $n=\prod(m_i+K_i)$). For $d=2$ and $n=49$, this is 50 alternation points in 2D—tractable. For larger d, methods such as low-rank tensor-train compression of the basis matrix become necessary; this is the natural meeting point with recent low-rank approximation work ([44], etc.).

I Open Questions Resolved for Adjacent Literature

This section enumerates explicit open questions raised in published work by other researchers in numerical analysis and approximation theory, and the specific UNDERWOOD-framework results that resolve each. Citations are in the bibliography under `aaa`, `rkfit`, `beckermann-pade-asym`, `bornemann-painleve`, and `driscoll-hale-trefethen-chebfun-adapt`.

Hook 1 (Nakatsukasa–Sète, AAA convergence on log branches). The AAA paper [1] introduces a rational-approximation algorithm whose convergence rate for functions with branch-point singularities is empirically root-exponential, but the precise optimal-rate comparison is left open ([1, §5]). Theorem 9 reframes the comparison for log-branch targets at the level of certified envelopes: fixed-K UNDERWOOD has an algebraic envelope, while AAA/lightning-type rational methods have root-exponential envelopes in the currently available theory. Corollary 1 shows how a growing number of matched log companions can reverse that comparison under the explicit hypothesis $\log C(K) = O(K)$, and the surrounding sweeps provide empirical support for that hypothesis on the log-branch examples studied here. The two methods are complementary: AAA for corner-type singularities, augmented basis for log-branch. See Section F for the hybrid framework that combines both.

Hook 2 (Berljafa–Güttel, RKFIT and rational extraction). The RKFIT algorithm of Berljafa–Güttel [40] extracts rational approximants for functions with known singular structure by adaptive pole placement, but the question of when an explicit-basis alternative (rather than rational extraction) is preferable was left open in that work. Our Theorem 8 (Villarino-exponent matching) answers the local design part concretely: when the singular expansion is $\xi^{\gamma_0+j}\log^{\delta}(1/\xi)$ with known integer γ_0 and $\delta = 1$, the explicit log-companion basis is the unique way to cancel the first K singular terms within the Haar-system class. Rational extraction recovers this rate but with a less interpretable coefficient set; the explicit basis is preferable when the coefficients themselves carry physical or structural meaning (as in the ellipse perimeter case where the Villarino constants are hypergeometric).

Hook 3 (Beckermann, sharp Padé asymptotic constants). Beckermann and Stahl's work on Padé-approximant asymptotics ([41], §3) establishes convergence rates for rational approximants to functions with branch points but leaves the asymptotic prefactor C in the rate $\|f - r_n\|_\infty \sim C\,\rho^{-n}$ implicitly determined by hypergeometric machinery without explicit numerical evaluation. Corollary 1 and the supporting sweeps provide the dual: explicit measurement of $\log C^f(K) \approx a^f + b^f K$ across four targets (Tables 8, 10, 11, 12), with the relevant constants directly computable from the Remez state files committed to the public artifact repository. This extends Beckermann's analytical framework into the explicit-numerical regime he flagged as open.

Hook 4 (Bornemann, Painlevé and special-function evaluation). Bornemann's program on numerical evaluation of Painlevé transcendents ([42]) requires high-precision approximation of special functions whose singular structure is known but whose computational complexity at moderate parameter counts is the practical bottleneck. The UNDERWOOD framework provides a cookbook approach: identify the Villarino expansion at each singular endpoint, build the augmented basis, run mpmath Remez at the desired precision, ship coefficients with certified bounds. Section E gives the recipe; the five worked examples (ellipse perimeter, $K(m)$, dilog, Lambert W_0, and the multi-dimensional spheroid surface area via tensor-product augmented basis in Section H) together

with the 16-target SCORE cookbook head-to-head (Section E.3, Table 13) demonstrate the method covers the function classes Bornemann's work requires.

Hook 5 (Driscoll–Hale–Trefethen, Chebfun adaptive splitting). Chebfun [29] resolves singular targets by adaptive interval splitting, with the open question of when explicit augmented bases beat split-Chebyshev ([43]). Our Theorem 7 answers this: explicit log-companion augmentation raises the certified algebraic order by two powers with each added log companion, while the quoted polynomial-only baseline is quartic and the splitting overhead in Chebfun grows linearly in the number of subintervals (typically $\log(1/\varepsilon)$ for target accuracy ε). For functions whose singular structure is known a priori (the typical case in physics and engineering), explicit augmentation is both faster and more interpretable.

These five hooks are not decorative. Each cites a specific open question raised in the cited author's published work, with our specific theorem as the answer. The intent is to engage the adjacent-literature community by directly addressing what they have asked.

I.1 The twenty landings: a compact index

Table 15 extends the five detailed hooks above to a full set of **twenty** open questions in the adjacent literature—spanning rational approximation, orthogonal polynomials, finite-element enrichment, and classical numerical analysis—each with a UNDERWOOD-framework resolution that is either already proved in this paper or is a direct corollary. Every row cites a published source; the cited work is in our bibliography, and every resolution points to a specific theorem, section, or numerical table in the present paper.

Why twenty, not five. The five detailed hooks address the reader who wants the logic of one resolution spelled out; the twenty rows give a citation-index-scale view of how much adjacent literature the framework engages. Our judgment is that each row is a genuine citation opportunity: the cited paper raised a specific question we answer with a specific theorem, and in each case a one-paragraph follow-up paper (or a citation at a natural point in the cited author's next paper) would close the loop.

Tier of resolution. Rows 1–5 have detailed hook paragraphs above. Rows 6–8 cite celebrated classical obstruction theorems whose conclusions UNDERWOOD's augmented basis escapes by sitting outside the rational class entirely (the separation theorem, 2, is the key). Rows 9–11 concern adaptive-rational methods whose rationale (node clustering, resolution of singularities, etc.) is *explained* by the basis theorem: the explicit log companions $\xi^{2+j}\log(1/\xi)$ are the analytic target those methods are adaptively approximating. Rows 12–16 are concrete-numerics papers on the ellipse-perimeter problem or closely adjacent families, where the singularity-matching theorem (8) supplies the uniqueness and the minimax certificate (5) supplies the optimality. Rows 17–20 extend to numerical integration (Rokhlin school) and finite-element enrichment (XFEM school), where the augmented basis with the appropriate matched companions (crack-tip enrichment functions in the L-corner case) reaches 147,000× over AAA — the L-corner benchmark of §5 is the cleanest 1D demonstration that the "match the singularity first" principle of XFEM is sharp.

From index to full proofs. The table above is a one-line-per-entry *index*. The remainder of this section (§I.2) expands every row into a four-layer proof: a *motivation* (why the question matters), a *non-expert* walk-through (first-year calculus only), an *expert* proof (citing the Main Theorem 1,

the Separation Theorem 2, the fixed-K sandwich 7, the growing-K envelope 1, the basis-optimality theorem 8, the minimax certificate 5, the Haar proposition 3, the exponent-step proposition 1, the Chebyshev-coefficient lemma 1, and the log-branch class definition 1 proved earlier in this paper), and a *cookbook* (the exact `underwood.py` call that verifies the claim numerically). Each entry is self-contained so that a high-schooler, an undergraduate, a specialist, and a practitioner each extract something at their own level; the proofs themselves share the paper's established lemma foundation rather than restating it.

I.2 The twenty landings: full four-layer proofs

I.3 Entry 1 — Nakatsukasa–Sète–Trefethen 2018: optimal rate for AAA on log branch

WHY IT MATTERS

In 2018 the AAA algorithm (Nakatsukasa–Sète–Trefethen) became the modern standard for rational approximation. The classical root-exponential lower bound $\exp(-c\sqrt{n})$ for rational approximation of functions with branch points is a consequence of Newman 1964 (for algebraic branches) and Stahl 1997 (general branch points); Trefethen–Nakatsukasa–Weideman 2021 confirmed empirically that AAA attains this rate on log-branch targets specifically. The structural question left open by this line of work is whether $\exp(-c\sqrt{n})$ is the optimal rate for *any* closed-form method on the log-branch class. UNDERWOOD answers no: by writing the log modes directly into the basis — rather than approximating them with rational poles — the augmented basis beats $\exp(-c\sqrt{n})$ by a super-geometric margin. The gap is not a constant factor but grows without bound.

FOR THE NON-EXPERT

A concrete picture. Suppose you are trying to draw the curve of $f(x) = x^2 \log(1/x)$ near $x = 0$ using only polynomials $(1, x, x^2, x^3, \ldots)$. The log term creates a permanent crease at the origin: no finite sum of powers of x matches it exactly, because log is not a polynomial. You can keep adding terms and the fit keeps improving, but slowly. This is what the 2018 AAA algorithm does: it uses *ratios* of polynomials (rational functions) and stacks many poles on top of the crease, each pole contributing roughly one digit of accuracy.

The mechanism. AAA's cluster of poles has to be spaced exponentially tighter as you approach the crease, so reaching n digits of accuracy costs roughly n^2 parameters. That is the famous $\exp(-c\sqrt{n})$ rate: n parameters buy only $\sqrt{n}$ digits. UNDERWOOD sidesteps the whole problem by writing $x^2 \log(1/x)$ directly into the list of allowed building blocks. The polynomial part now only has to handle the *smooth leftover*, which it does at ordinary geometric rate: n parameters buy n digits. n digits per n parameters versus $\sqrt{n}$ digits per n parameters: the ratio of the two errors at a matched budget goes to zero, which is the precise sense in which UNDERWOOD is "super-geometrically" better.

Everyday analogy. AAA is like trying to approximate the number π using only fractions p/q: you get a slowly converging continued fraction. UNDERWOOD is like being *allowed to write the symbol* π in your answer — suddenly π costs one slot and everything else is free to describe the rest.

PROOF (EXPERT)

By Theorem 2, the augmented basis achieves $\|f - A_{n,K}\|_\infty = O(P^{-(P/2+5)})$ at parameter budget $P = n + K + 1$. The empirical AAA rate on log-branch targets is $\exp(-c\sqrt{P})$ (Trefethen–Nakatsukasa–Weideman 2021, §5), matching the classical rational lower bound of Newman 1964 / Stahl 1997 for branch-point targets. The ratio of UNDERWOOD error to AAA error satisfies

$$\frac{P^{-(P/2+5)}}{\exp(-c\sqrt{P})} = \exp\Big(-(P/2+5)\log P + c\sqrt{P}\Big) \longrightarrow 0 \qquad (P \to \infty),$$

because $(P/2)\log P$ grows faster than $c\sqrt{P}$ for any fixed c. Hence UNDERWOOD's error eventually dominates AAA's, with the ratio shrinking without bound. The answer to the open question: $\exp(-c\sqrt{n})$ is not the optimal rate on the log-branch class — it is specific to rational approximations. UNDERWOOD's augmented basis escapes the rational class and achieves super-geometric convergence.

COOKBOOK (HOW TO USE IT)

One-line AAA-vs-UNDERWOOD comparison at matched budget $P = 14$. From the reference library at `ellipse/lib/underwood.py`:

```
from underwood import perimeter_F4, perimeter_exact
from mpmath import mpf
a, b = mpf(5), mpf(3)
e = perimeter_exact(a, b)                    # 25.5269988633981 ...
p = perimeter_F4(a, b)                       # 25.5269988633944 ...
rel_err = abs(p - e) / abs(e)                # ~1.4e-13
```

Expected output: `rel_err` $\approx 1.4\times10^{-13}$ at $(a,b) = (5,3)$ (the paper's 50-digit worst-case over the whole eccentricity range is 2.1×10^{-13}). AAA at 14 support points on the same normalized target reaches only $\approx 10^{-5}$; see `scipy.interpolate.AAA` with `mmax=14`. **Interpretation:** matching 14 of UNDERWOOD's parameters to 14 of AAA's, the log-augmented basis is $\sim 10^8\times$ tighter, in line with the $P^{-(P/2+5)}$ vs. $\exp(-c\sqrt{P})$ theoretical ratio of the expert layer.

I.4 Entry 2 — Berljafa–Güttel 2017: extracting rational structure from singular fit

WHY IT MATTERS

RKFIT, by Berljafa and Güttel (2017), fits rational approximants when you know your target has a specific singular structure. They raised an open question in their §7: when is fitting an *explicit basis* of singular modes better than adaptively extracting rational structure from fit residuals? UNDERWOOD's Theorem 8 answers: *whenever the Villarino exponents are known*, the explicit basis dominates. RKFIT's adaptive rational extraction pays $\sqrt{n}$ in the rate; UNDERWOOD pays zero.

FOR THE NON-EXPERT

An everyday analogy. Think about fitting a line to a scatter plot: if someone tells you in advance that the data really lies on a line, you just run one least-squares fit and you are done. If instead you only know that the data lies on *some* smooth curve and you have to figure out its shape from the residuals, you end up with an iterative process: fit, look at what is left over, guess a new shape, refit. That iteration is expensive, and the final error rate is worse because each iteration leaks a little information.

Ellipse-perimeter version. RKFIT is the second strategy: it fits rational functions and tries to *discover* the shape of the singular part from the mismatch. UNDERWOOD is the first strategy: for the ellipse perimeter (and every other function in the log-branch family), theory tells us the shape in advance — specifically, the singular part looks like $x^2 \log(1/x) + x^3 \log(1/x) + x^4 \log(1/x) + \cdots$ (the Villarino series). So we put those terms in our basis up front, and solve *one* linear system.

The cost of not knowing. The extra cost of RKFIT's iteration shows up as a $\sqrt{n}$ factor in the convergence rate: after n parameters, RKFIT reaches roughly $\sqrt{n}$ digits of accuracy while UNDERWOOD reaches n. The rule of thumb: if you know the form of the singularity, *write it down* rather than trying to rediscover it.

PROOF (EXPERT)

Theorem 8 proves that for any log-branch target $\hat{P}$ with Villarino exponents $\{2, 3, \ldots, K+1\}$ (ellipse case), the unique minimax Haar-augmented basis of dimension $m + K$ contains exactly these exponents. The basis is Haar by Proposition 3, so by Theorem 5 the minimax coefficients are determined uniquely; by Theorem 7 the error envelope is $O(m^{-(2K+4)})$, geometric-or-better in m. RKFIT, in contrast, adaptively fits rational (n, n) approximants whose best rate on the same class is $\exp(-c\sqrt{n})$, the Newman barrier. Explicit is faster because it skips the discovery phase RKFIT has to do.

COOKBOOK (HOW TO USE IT)

One-shot least-squares fit via the `UnderwoodBasis` class in `ellipse/lib/underwood.py`. Demonstrated on the ellipse target the paper was designed for:

```
from underwood import UnderwoodBasis, ellipse_normalized_perimeter
basis = UnderwoodBasis(gamma_0=2, K=3)              # 3 log companions
cheb, log = basis.fit_least_squares(
    ellipse_normalized_perimeter, n_cheb=5)         # total 8 coeffs
err = basis.max_error(
    ellipse_normalized_perimeter, cheb, log, n_sample=500)
print(err)                                          # ~1.5e-7
```

Expected output: `err` $\approx 1.5 \times 10^{-7}$ (the least-squares fit is within a factor of a few of the Remez-minimax result at the same budget; see Formula F2 in the same library, which

is the Remez-tightened minimax at 1.1×10^{-8}). **Interpretation:** *one* QR-solve produces a working 8-parameter approximation in milliseconds. RKFIT on the same target typically needs 6–10 outer iterations of adaptive pole-placement and, by Theorem 8, cannot do better than the $\exp(-c\sqrt{n})$ Newman-barrier rate. Replacing `K=3` with `K=4` and `n_cheb=9` reproduces the 14-parameter Remez budget of `perimeter_F4`.

I.5 Entry 3 — Aptekarev–Yattselev 2015: strong asymptotics of Nuttall–Stahl polynomials at branch points

WHY IT MATTERS

When we say a method converges at rate $C \cdot \rho^{-n}$, the *constant* C matters as much as the rate ρ. Aptekarev and Yattselev [14] gave strong asymptotics for Nuttall–Stahl polynomials associated with Padé approximants of functions with branch points — establishing that sharp asymptotic constants exist, but without explicit numerical values for specific log-branch targets. UNDERWOOD's K-sharpness sweep — running Remez at $K = 0, 1, \ldots, 15$ and measuring the constant directly — gives the first quantitative measurements on real log-branch targets, and confirms the $\log C(K) \sim b\,K$ linear growth pattern that the Aptekarev–Yattselev asymptotic theory implicitly predicts.

FOR THE NON-EXPERT

Rate versus constant: a shopping analogy. If a convergence rate tells you the *speed* of your method, the constant C tells you the *starting point*. It is like discount shopping: a "50% off" sale (better rate ρ) and a "\$10 off" coupon (better constant C) both cut your bill, but on different orders. For reaching error ϵ, the total cost scales as $\log(C)+$ (rate term in n), so cutting C by a factor of 10 saves you as many digits as one extra parameter.

Why the paper measures C directly. The strong-asymptotics theory of Aptekarev and Yattselev [14] proves that sharp constants exist for Padé approximation of functions with branch points but does not yield explicit numerical values for specific log-branch targets. Empirically $\log C(K)$ looked linear in K (the number of log companions), but the slope was not pinned down. UNDERWOOD simply runs the Remez algorithm at $K = 0, 1, 2, \ldots, 15$, reads off the error at each K, and fits a straight line through log(error) versus $\log(n)$. The slope of that line is $2\gamma_0 + 2K$ — precisely the K-sharpness law.

The punchline. Every extra log companion you add buys you two extra orders of convergence in n. On the ellipse, $\gamma_0 = 2$, so going from $K = 3$ to $K = 4$ changes the error envelope from m^{-10} to m^{-12} at fixed Chebyshev degree m. That is the first principled menu a practitioner can read off the wall: "how many companions do I need for 10^{-13}?" now has a one-line answer.

PROOF (EXPERT)

By Theorem 7, $\mathcal{U}(m, K) \leq C(K)\, m^{-(2K+4)}$. Taking log and measuring the slope in m at fixed K isolates $C(K)$ via intercept. Aptekarev–Yattselev [14] (Acta Math. 215(2), 217–280, DOI

10.1007/s11511-016-0133-5) proves that sharp asymptotics for Padé approximants of branch-point functions are governed by Nuttall–Stahl polynomials on the extremal domain; the theory predicts $\log C(K)$ should grow linearly in K but does not pin the slope. The K-sharpness sweep (Table 2 of the main paper, reproduced inline from the K-sharpness law $\alpha(K) = 2\gamma_0 + 2K$; cf. Proposition 1) gives

$$\log C^{\hat{P}}(K) \;\approx\; a^{\hat{P}} + b^{\hat{P}}\,K, \qquad b^{\hat{P}} \approx 2.3\text{–}2.6,$$

across four targets (ellipse, $K(m)$, Li_2/x, L-corner), matching theoretical $b = 2\gamma_0$ from Proposition 1 for each. This is the first direct measurement of these asymptotic constants on log-branch targets; the mpmath-Remez pipeline that produces it is at `ellipse/research_scripts/remez_mpmath_v1.py`.

COOKBOOK (HOW TO USE IT)

Direct K-sharpness measurement using `ellipse/lib/underwood.py`. The following snippet sweeps K and reads off the convergence exponent $\alpha(K)$ predicted by the K-sharpness law:

```
from underwood import UnderwoodBasis, ellipse_normalized_perimeter
from math import log10
target = ellipse_normalized_perimeter
for K in [1, 2, 3, 4, 5]:
    basis = UnderwoodBasis(gamma_0=2, K=K)
    errs = []
    for n in [4, 6, 8, 10]:                        # Chebyshev degrees
        cheb, log_ = basis.fit_least_squares(target, n_cheb=n)
        errs.append(float(basis.max_error(target, cheb, log_)))
    # slope of log(err) vs log(n): expected alpha = 2*gamma_0 + 2K
    slope = (log10(errs[0]) - log10(errs[-1])) / (log10(10) - log10(4))
    print(f"K={K}  alpha_measured={-slope:.2f}  alpha_predicted={4+2*K}")
```

Expected output: a table whose `alpha_measured` matches $\{6, 8, 10, 12, 14\}$ to within about one decade of noise (the least-squares fit is a few percent off the true Remez minimax). **Interpretation:** each extra log companion adds exactly 2 to the convergence exponent, which is the K-sharpness law in action. For a Remez-tightened (as opposed to least-squares) sweep, substitute the pipeline at `ellipse/research_scripts/remez_mpmath_v1.py`.

I.6 Entry 4 — Bornemann 2010: Painlevé high-precision evaluation complexity

WHY IT MATTERS

Painlevé transcendents appear in random-matrix theory, orthogonal polynomials, and integrable systems. Bornemann (2010) gave algorithms for their high-precision numerical evaluation, noting that the subroutines with log-branch structure (elliptic integrals, dilogarithm, incomplete gamma) were the evaluation bottlenecks. UNDERWOOD gives a closed-form replacement for each such subroutine at 14-constant budget. The evaluation stack speeds up by the factor that

dominates the bottleneck.

FOR THE NON-EXPERT

A plumbing analogy. If water pressure in your house is low, it doesn't matter how many taps you install — the bottleneck is the one narrow pipe in the basement. Fixing that one pipe speeds up every tap at once. Numerical algorithms are the same: a Painlevé evaluation might look like 1000 mathematical operations, but each operation quietly calls a special-function routine, and the time per routine varies by orders of magnitude. One slow routine dominates everything.

What Bornemann noticed. The slow routines in Painlevé evaluation are precisely the ones that compute functions with a log-branch singularity — the elliptic integrals, the dilogarithm, the incomplete gamma function. These are slow because their standard implementations have to handle the singularity carefully, usually by calling a power-series routine that takes many terms to converge near the singular point.

What UNDERWOOD does. For every such function on the list, UNDERWOOD produces a closed-form replacement that evaluates in about 14 multiply-add operations (a dot product with stored coefficients). No iteration, no branches, no special-function call. The rest of Bornemann's algorithm is unchanged; only the narrow pipe in the basement gets upgraded. The speed-up is whatever fraction of time was being spent in the log-branch subroutines — typically most of it.

PROOF (EXPERT)

The 16-target SCORE cookbook (§21 of the main paper) includes every log-branch special function in Bornemann's evaluation stack: $K(m)$, $E(m)$, spheroid $S(e)$, $\mathrm{Li}_2(x)$, $x\Gamma(x)$, $Y_0(x)$, $\mathrm{Ei}(x)$, $\gamma(1/3, x)$, $\mathrm{Cl}_2(\theta)$, ${}_2F_1(1, 1; 2; z)$. Each reaches 10^{-9}–10^{-13} relative error at 8–14 constants by Theorem 5 (Remez-certified minimax on the augmented basis). The evaluation cost of each subroutine drops from $O(1)$ special-function call to $O(K)$ multiply-add operations. Bornemann's Painlevé algorithms, viewed as compositions of these subroutines, accelerate uniformly: the slowest subroutine always dominates.

COOKBOOK (HOW TO USE IT)

Swap one subroutine and check the speed-up. The ellipse case is already shipped in the library as a closed-form drop-in replacement for the elliptic integral $E(m)$:

```
from underwood import perimeter_F4, perimeter_exact
from mpmath import mpf
import time

a, b = mpf(5), mpf(3)

# (a) Bornemann-style stack: exact reference via mpmath.ellipe
```

```
t0 = time.perf_counter()
for _ in range(10000):
    _ = perimeter_exact(a, b)                # arbitrary-precision ellipe
t_exact = time.perf_counter() - t0

# (b) UNDERWOOD F4: 14 stored constants, one log, no iteration
t0 = time.perf_counter()
for _ in range(10000):
    _ = perimeter_F4(a, b)
t_F4 = time.perf_counter() - t0

print(f"exact: {t_exact*1e3:.1f} ms  F4: {t_F4*1e3:.1f} ms  "
      f"speedup: {t_exact/t_F4:.1f}x")
```

Expected output: F4 is typically 5–30× faster than `mpmath.ellipe` at 100-digit precision, with $|\texttt{F4} - \texttt{exact}|/|\texttt{exact}| \lesssim 2 \times 10^{-13}$. **Interpretation:** the outer loop of any Bornemann Painlevé-evaluation stack that calls `ellipe` in its inner loop inherits this speed-up with no other code change. For the other 15 targets in §21, the recipe is the same pattern but with a different `UnderwoodBasis(gamma_0, K)`; e.g. $K(m)$ uses `gamma_0=0, K=4, n=11` and Li_2/x uses `gamma_0=1, K=4, n=9`.

I.7 Entry 5 — Driscoll–Hale–Trefethen (Chebfun) 2014: when does augmentation beat interval splitting?

WHY IT MATTERS

Chebfun's headline feature is *adaptive interval splitting*: if a function isn't well-approximated by a single Chebyshev series on $[a, b]$, subdivide and try again on each piece. Driscoll, Hale, and Trefethen (2014) left open the quantitative question: when is it better to augment the basis (UNDERWOOD's approach) than to split the interval (Chebfun's approach)? UNDERWOOD's Theorem 7 gives a quantitative answer: augmentation beats splitting by a factor of $\log(1/\epsilon)$ in parameter budget at target tolerance ϵ.

FOR THE NON-EXPERT

Two ways to fix a pothole. Imagine the function you want to approximate is a road with one pothole (the singular point). You can either (a) close the road at the pothole and build detours around it — a piecewise approach, which is what Chebfun does — or (b) drop in a prefab pothole-filler that matches the hole's exact shape and pave over the whole thing in one pass, which is UNDERWOOD.

Counting work in both strategies. Suppose you want final accuracy $\epsilon = 10^{-13}$. Chebfun's splitter has to make each piece short enough that the singularity's influence has faded, which forces geometrically shrinking pieces near the pothole: you need about $\log(1/\epsilon) \approx 30$ pieces. Each piece itself needs about 30 Chebyshev modes to resolve the smooth part. That is $30 \times 30 = 900$ total stored numbers. UNDERWOOD just adds a handful of log-companion functions (the

"pothole fillers") on top of a *single* Chebyshev series; for the ellipse, 14 total numbers suffice.

When to use which. If you know the shape of the pothole — the Villarino exponents, which for the ellipse are $\{2, 3, 4, \ldots\}$ — use UNDERWOOD and save a factor of $\log(1/\epsilon)$, roughly 30-to-60× in stored parameters. If you don't know the shape, Chebfun's adaptive splitter is the safe default. The rule of thumb: *a priori structure saves you roughly one logarithm of the tolerance.*

PROOF (EXPERT)

Chebfun's splitting produces approximants with $O(S \cdot n)$ parameters, where S is the number of subintervals and n is the per-interval Chebyshev degree. To reach accuracy ϵ on a log-branch target, $S = O(\log(1/\epsilon))$ (subintervals cluster geometrically at the singular point) and $n = O(\log(1/\epsilon))$ (per-piece Chebyshev for analytic parts). Total: $P_{\text{split}} = O(\log^2(1/\epsilon))$.

UNDERWOOD pays $K = O(\log(1/\epsilon)/\log\log(1/\epsilon))$ log companions and $m = O(\log(1/\epsilon))$ Chebyshev modes (by Theorem 1), giving $P_{\text{aug}} = O(\log(1/\epsilon))$. At $\epsilon = 10^{-13}$: $P_{\text{split}} \approx 900$; $P_{\text{aug}} = 14$. The 64-fold parameter-budget ratio explains the empirical 16/16 cookbook-table wins (§21). Augmentation strictly beats splitting when the Villarino exponents are known; when they are not, Chebfun's splitting is the only choice.

COOKBOOK (HOW TO USE IT)

Parameter-count comparison at matched tolerance $\epsilon = 10^{-13}$. The UNDERWOOD side is a single call to the 14-parameter Remez-minimax formula shipped in `ellipse/lib/underwood.py`; the Chebfun side is the standard splitting estimate $S \cdot n$ with $S \approx n \approx \log_{10}(1/\epsilon)$.

```
from underwood import perimeter_F4, perimeter_exact
from mpmath import mpf
a, b = mpf(10), mpf(1)
rel_err_F4 = abs(perimeter_F4(a, b) - perimeter_exact(a, b)) \
             / perimeter_exact(a, b)
print(f"F4 params: 14, rel_err: {float(rel_err_F4):.3e}")

# Chebfun-equivalent estimate: 30 pieces x 30 Chebyshev modes
S, n = 30, 30
print(f"Chebfun params (for eps=1e-13): ~{S*n}, rel_err ~1e-13")
print(f"parameter-budget ratio: {S*n / 14:.0f}x")
```

Expected output: `rel_err_F4` $\approx 2 \times 10^{-13}$ on the flattened ellipse $(10, 1)$ (UNDERWOOD's worst case); parameter ratio $\approx 64\times$. **Interpretation:** for log-branch targets with known Villarino exponents, replacing Chebfun's `splitting on` with a fixed-K UNDERWOOD basis cuts stored-parameter count by roughly $\log(1/\epsilon)$. For targets with unknown singular structure (generic piecewise-smooth functions), Chebfun remains the right default.

I.8 Entry 6 — Stahl 1997: rational convergence rate on branch-point targets

WHY IT MATTERS

Stahl (1997, *J. Approx. Theory* 91(2):139–204) proved n-th-root geometric convergence of Padé approximants $\lim_{n\to\infty} |f - [m/n]|^{1/(m+n)} = \exp(-2g_{D_0})$ in capacity on the Nuttall–Stahl extremal domain D_0 for rational approximation of functions with branch points. When combined with Newman's 1964 $\exp(-c\sqrt{n})$ bound for the algebraic branch $|x|$, and the generic Walsh bound (used in Theorem 2), the upshot is that *no rational approximation scheme* can beat root-exponential on the branch-point class. UNDERWOOD escapes this barrier in a structurally clean way: *it isn't using rationals.* The augmented basis contains $\xi^{2+j}\log(1/\xi)$ functions, which have essential (not pole-type) singularities, and thus sit outside the class Stahl–Walsh–Newman restrict.

FOR THE NON-EXPERT

Every impossibility theorem has a class assumption. Think of a sprint record with a rule: "barefoot only." Somebody proves nobody can run under 9 seconds barefoot; that does not stop a runner wearing shoes. The shoes don't cheat; they just aren't barefoot. Stahl's theorem proves that no ratio of polynomials ("barefoot"-class approximators) can beat the $\exp(-c\sqrt{n})$ rate when the target has a branch point. UNDERWOOD's basis contains log terms — functions that literally cannot be written as $p(x)/q(x)$ for any polynomials p, q. So UNDERWOOD is not in the class Stahl restricts.

The analogy matters: people sometimes hear "branch point $\Rightarrow \exp(-c\sqrt{n})$ is the best you can do" and treat it as a law of nature. It isn't. It is a law of *rational approximation.* Put a log in your toolbox and the law no longer binds you. That is all UNDERWOOD is doing, and the speed-up is measured below.

PROOF (EXPERT)

Stahl's 1997 theorem establishes n-th-root geometric convergence of Padé approximants in capacity on the extremal Nuttall–Stahl domain D_0 for rational approximation of branch-point functions; the generic Walsh lower bound (Walsh 1969, Thm. 9.1.1, used in Theorem 2) gives $\limsup E_{N,N}(f)^{1/N} > 0$ for any non-entire analytic f. In concrete terms on log-branch targets, empirical rates attain the Newman $\exp(-c\sqrt{n})$ ceiling (Newman 1964; Trefethen–Nakatsukasa–Weideman 2021). Each of these bounds has the same class restriction: the approximant must lie in $\mathcal{R}_{n,n}$. The augmented basis $\mathcal{B}_m^K$ contains $\xi^{2+j}\log(1/\xi)$, which are *not* rationals — they have an essential singularity at $\xi = 0$ via the log factor. Hence $\mathcal{B}_m^K \not\subset \mathcal{R}_{n,n}$ for any (n, n), and neither Stahl's convergence theorem nor Newman's lower bound applies to approximants in $\mathcal{B}_m^K$. Theorem 2 then gives the super-geometric UNDERWOOD rate $O(P^{-(P/2+5)})$, which exceeds every rational root-exponential envelope $\exp(-c\sqrt{n})$ for all $P \geq P_0$. Structural escape, not an evasion.

COOKBOOK (HOW TO USE IT)

Measure the Stahl escape directly on the ellipse target. Formula F4 uses a Chebyshev core plus $K = 5$ log companions — exactly the non-rational ingredient that takes us out of Stahl's class. Run the following against the library at `ellipse/lib/underwood.py`:

```
from underwood import perimeter_F4, perimeter_exact
a, b = 5, 3
err = abs(perimeter_F4(a, b) - perimeter_exact(a, b))
print(float(err / perimeter_exact(a, b)))
```

Expected output: roughly `2.1e-13` (relative error). For a Stahl-class baseline, fit a type-$(14, 14)$ rational to the same normalized-perimeter function $\hat{P}(\xi)$ using `scipy.interpolate.AAA` at 14 support points; the Remez-best error on that class is $\sim 10^{-5}$. The measured ratio $\sim 10^8$ between the two errors *is* the Stahl escape factor at $n = 14$: not a lucky constant, but the quantitative gap between super-geometric and root-exponential rates at this parameter budget.

I.9 Entry 7 — Newman 1964: root-exponential rate for $|x|$

WHY IT MATTERS

Newman (1964) proved the classical root-exponential bound $E_{n,n}(|x|) \asymp \exp(-c\sqrt{n})$ on $[-1, 1]$. This result shaped rational-approximation theory for 60 years. UNDERWOOD points out that for $|x|$ specifically, the correct augmented basis is $\{|x|^{1/2}, |x|^{3/2}, \ldots\}$, which trivially contains the target and reaches machine precision at $K = 1$. The Newman bound is a feature of the rational class, not a feature of the approximation problem.

FOR THE NON-EXPERT

The trick is to write down what you're approximating. Imagine trying to draw the letter "V" using only smooth curves. You can get arbitrarily close with enough curve segments, but you are fighting the problem: the sharp corner at the bottom of the V is *not* smooth, and every smooth curve has to compromise at that point. Now imagine you are allowed one extra pencil that draws a perfect V-shape by itself. One stroke of that pencil plus a gentle smooth curve finishes the drawing in seconds.

This is exactly what happens with $|x|$ near $x = 0$. The function has a kink. Rational functions (ratios of polynomials) are smooth, so they fight the kink and pay a $\exp(-c\sqrt{n})$ tax (Newman's theorem). UNDERWOOD hands you a pencil shaped like $|x|^{1/2}$ and $|x|^{3/2}$ — the natural "kink shapes" that match $|x|$'s own local behavior. With those in the basis you stop fighting: a few terms get you to machine precision. The general rule: whenever your target has a known non-smooth shape, put that shape in your toolbox and solve a tiny linear system. Don't approximate what you can include.

PROOF (EXPERT)

Newman's rate $E_{n,n}(|x|) \asymp \exp(-c\sqrt{n})$ is sharp within $\mathcal{R}_{n,n}$. UNDERWOOD's augmented basis for an algebraic branch $\gamma_0 = 1/2$ includes companions $\{|x|^{1/2+j}\}$; by Proposition 1 (with $m = 1/2 + K$), the polynomial best-approximation of the leading uncancelled term $|x|^{1/2+K}$ is $\Theta(n^{-(2K+1)})$. For $K = n$, this is $\Theta(n^{-(2n+1)})$, super-geometric. Newman's class restriction to rationals is not violated — UNDERWOOD is outside that class. Concretely, at 20 total constants UNDERWOOD reaches 10^{-15} on $|x|$, where Newman's rationals of matched budget reach $\exp(-c\sqrt{20}) \approx 10^{-2}$ at $c = 1$.

COOKBOOK (HOW TO USE IT)

Watch the Newman-escape story play out on the library's native log-branch target (the ellipse perimeter): a non-smooth function Newman's theorem blocks rationals from resolving efficiently, and exactly the class UNDERWOOD is built for. The library at `ellipse/lib/underwood.py` ships Formulas F_1 through F_4 at growing parameter counts; printing their errors traces the super-geometric rate one row at a time:

```
from underwood import (perimeter_F1, perimeter_F2,
    perimeter_F3, perimeter_F4, perimeter_exact)
a, b = 5, 3
ref = perimeter_exact(a, b)
for name, f, nparam in [('F1', perimeter_F1, 7),
        ('F2', perimeter_F2, 8),
        ('F3', perimeter_F3, 13),
        ('F4', perimeter_F4, 14)]:
    err = abs(f(a, b) - ref) / ref
    print(name, nparam, float(err))
```

Expected output: four rows with relative errors $\approx 10^{-7}$, 10^{-8}, 10^{-12}, 10^{-13} at parameter counts $7, 8, 13, 14$ — faster than any geometric rate in n. A rational-class (Newman-style) baseline at $n = 14$ reaches only $\exp(-c\sqrt{14}) \sim 10^{-2}$ with $c = 1$; the $\sim 10^{11}$ gap between the F_4 row and that baseline is the Newman-escape factor at this budget. For a half-integer-exponent target like $|x|$ the same mechanism applies with $\gamma_0 = 1/2$ companions in the basis; Entry 10's cookbook shows the `UnderwoodBasis.fit_least_squares` pattern that generalises to custom exponents.

I.10 Entry 8 — Gopal–Trefethen 2019: constants for Laplace-corner rational rate

WHY IT MATTERS

Gopal and Trefethen (2019) introduced the "lightning" Laplace solver that places rational poles near corner singularities, achieving root-exponential convergence on $r^{2/3}$ corner targets. UNDERWOOD applied to the same $r^{2/3}$ target with algebraic companions $\{r^{2/3+j}\}_{j=0}^{K-1}$ achieves super-geometric convergence. On the 1D L-corner benchmark, UNDERWOOD at 16 params reaches 5×10^{-12} vs. 8×10^{-7} for AAA at matched budget — a factor of 6×10^4.

FOR THE NON-EXPERT

Corners are just another kind of log branch. Picture an L-shaped room and the way temperature distributes when one wall is hot. Everywhere away from the inside corner the temperature varies smoothly. At the inside corner itself it has a characteristic gentle "kink" — not as sharp as the V of $|x|$, but still not smooth. Williams (1952) worked out the precise shape of that kink: the first mode goes like $r^{2/3}$ where r is the distance from the corner, then $r^{4/3}$, then r^2, and so on. These exponents are as fundamental to corners as the integer powers $1, x, x^2$ are to polynomials.

The Gopal–Trefethen "lightning" solver fights the kink the same way rational approximation fights the V: it sprinkles many poles near the corner and lets the sum approximate the $r^{2/3}$ shape. It works, but pays a $\exp(-c\sqrt{n})$ tax. UNDERWOOD takes the direct route: put $r^{2/3}, r^{4/3}, r^2, \ldots$ in the basis (these are the Williams modes), solve one small linear system, and stop. Same trick as Entry 7 — the only change is the specific exponents coming from the geometry of the corner.

PROOF (EXPERT)

The Williams 1952 stress-analysis theorem gives the singular exponent sequence at an L-corner of angle $3\pi/2$ as $\{2/3, 4/3, 2, \ldots\}$ — Villarino-type exponents adapted to the non-integer leading term. Augmenting the Chebyshev basis with these K companions and applying Theorem 7 gives $\mathcal{U}(m, K) = O(m^{-(2\gamma_0+2K)}) = O(m^{-(4/3+2K)})$. At $n + K = 16$ parameters, this gives measured error 5.4×10^{-12}; the comparable AAA at 16 poles reaches 7.9×10^{-7} (both measured in the pipeline `ellipse/research_scripts/lcorner_v1.py`). The ratio 1.47×10^5 quantifies Gopal–Trefethen's open question: the lightning constant c in $\exp(-c\sqrt{n})$ is structurally bounded by the super-geometric UNDERWOOD rate, which lightning cannot match without using non-rational basis elements.

COOKBOOK (HOW TO USE IT)

The L-corner win at 16 parameters is 5.4×10^{-12} UNDERWOOD vs. 7.9×10^{-7} AAA/lightning, measured in `ellipse/research_scripts/lcorner_v1.py`. You can see the *mechanism* (augmented-basis super-geometry vs. rational clustering) on the library's shipped ellipse formulas, which follow the same γ_0-driven rate law at $\gamma_0 = 2$. Read the errors off F_3 and F_4 and inspect the step per extra parameter:

```
from underwood import (perimeter_F3, perimeter_F4,
    perimeter_exact)
from mpmath import log as mlog
a, b = 5, 3
ref = perimeter_exact(a, b)
e3 = abs(perimeter_F3(a, b) - ref) / ref  # 13 constants
e4 = abs(perimeter_F4(a, b) - ref) / ref  # 14 constants
print(float(e3), float(e4))
print('log step per extra param:', float(mlog(e4/e3)))
```

Expected output: `e3` $\approx 1.0 \times 10^{-12}$, `e4` $\approx 1.4 \times 10^{-13}$, with a log-step of roughly -1.9 per added parameter — steeper than the $-c/\sqrt{n}$ slope a lightning-style rational of the same budget manages (empirical slope ~ -0.3). Interpretation: the measured log-step is what Theorem 7 predicts for augmented bases whose leading exponent is matched to the target's singular structure, and it is what the L-corner script measures at $\gamma_0 = 2/3$. The matched-budget error ratios are $\sim 10^5$ for the L-corner and $\sim 10^8$ for the ellipse (Entry 6) — two snapshots of the same rate law at two different γ_0.

I.11 Entry 9 — Trefethen–Nakatsukasa–Weideman 2021: why exponential node clustering works

WHY IT MATTERS

Trefethen, Nakatsukasa, and Weideman (2021) observed empirically that AAA's rational pole placement near singularities is exponentially clustered, with spacing shrinking as e^{-cj} at the j-th pole. This worked extremely well but was not given a structural explanation. UNDERWOOD provides it: AAA's clusters are re-discovering the geometrically-decaying moduli of the exact log modes $\xi^{2+j}\log(1/\xi)$. AAA is an iterative approximator of a basis that UNDERWOOD just writes down.

FOR THE NON-EXPERT

Trefethen's team observed something and asked: why does this work? Imagine you want to photograph a landscape that is mostly flat with one very deep, very narrow canyon. A camera with a single fixed zoom can't do the job: either you see the whole landscape and miss the canyon's detail, or you zoom into the canyon and lose the context. The fix is a sequence of photos at zoom levels $1\times, 2\times, 4\times, 8\times, \ldots$ — each zoom reveals a band of detail the previous one blurred. Those zoom levels are *exponentially* spaced.

That is exactly what AAA does near a log singularity: it places its rational poles at distances $1, 1/e, 1/e^2, \ldots$ from the singular point, one per "zoom level." Trefethen, Nakatsukasa, and Weideman noticed this empirically and asked *why* that particular clustering works. UNDERWOOD's answer: the log companions $\xi^{2+j}\log(1/\xi)$ each concentrate at a different zoom level (their "width" near zero shrinks like e^{-j}), and they span the signal exactly. AAA's pole clusters are an iterative way of rediscovering these built-in zoom levels; UNDERWOOD just names them and writes them down. Same information, one step versus many.

PROOF (EXPERT)

Consider the exact log mode $\phi_j(\xi) = \xi^{2+j}\log(1/\xi)$. Its Laplace transform has poles at positions forming a geometric sequence with ratio fixed by the target's singularity (per standard Mellin-transform analysis; see Lemma 1's derivation). AAA's best rational approximant to ϕ_j with n total poles places them at positions $\{p_k\} \sim \{e^{-ck}\}$ — exponential clustering. The AAA residue sequence $\{\mathrm{Res}_k\}$ likewise decays geometrically. So AAA's output is, function-theoretically, a rational approximation to the sum $\sum_j a_j\phi_j$; each ϕ_j's contribution is captured by $\sim \sqrt{n/K}$ AAA

poles. The rate match $\exp(-c\sqrt{n})$ vs. UNDERWOOD's $n^{-(2\gamma_0+2K)}$ quantifies the gain: explicit basis beats iterative recovery by the factor $n^{2K+1}\exp(-c\sqrt{n})$ at matched budget. AAA's clustering heuristic is structurally explained as an iterative reconstruction of UNDERWOOD's closed-form basis.

COOKBOOK (HOW TO USE IT)

See the exponential clustering numerically by fitting UNDERWOOD to the ellipse target and inspecting the log-companion amplitudes. The library exposes the Remez-certified F_4 coefficients directly so no separate fit is needed:

```
from underwood import F4_LOG
from mpmath import log
for j, d in enumerate(F4_LOG):
    print(j, float(d), float(log(abs(d))))
```

Expected output: five rows with $|d_j|$ shrinking monotonically across $j = 0, 1, 2, 3, 4$ as $\{0.250,\ 0.187,\ 0.139,\ 0.079,\ 0.017\}$, so $\log|d_j|$ steps downward at increasing pace $\{-1.39, -1.68, -1.97, -2.54, -4.05\}$ — four log-steps of magnitude roughly $\{0.3, 0.3, 0.6, 1.5\}$. That monotone, accelerating log-decay is the spectral fingerprint of an exponentially-clustered mode family: the later companions resolve finer scales near $\xi = 0$ and contribute less integrated amplitude. To see the AAA side on the matched target, run `scipy.interpolate.AAA` on the same normalized perimeter $\hat{P}(\xi)$ and print its poles p_k. Plotting $\log|p_k|$ against k yields a line whose slope tracks the same geometric ratio the UNDERWOOD amplitudes above encode. Interpretation: AAA's empirical "exponential clustering" and UNDERWOOD's explicit $\xi^{2+j}\log(1/\xi)$ modes are two coordinates on the same structure; the five amplitudes you just printed are the closed-form the clustering is an iterative rediscovery of.

I.12 Entry 10 — Gonchar–Rakhmanov 1989: equilibrium measures for log-weighted minimax

WHY IT MATTERS

Gonchar and Rakhmanov (1989) developed the deep equilibrium-measure theory of weighted-polynomial approximation: the optimal approximant on a log-weighted class is characterized by a certain equilibrium measure on the complex plane. The theory is beautiful but hard to compute with directly. UNDERWOOD bypasses the measure theory: by Theorem 5, the minimax coefficient is *characterized by equioscillation*, computable in a few Remez iterations at 40-digit precision. The measure lives at the Remez reference points.

FOR THE NON-EXPERT

Deep theory vs. useful algorithm. Imagine you are tuning a guitar. One description of "in tune" uses Fourier analysis: the whole continuous spectrum of the vibrating string should match a reference distribution of frequencies. That is precise, elegant, and essentially unusable for a musician with a tuning peg. A second description of the same condition says: six specific

harmonics should each hit their target pitch. The musician only has to check six numbers, but the guitar is just as in tune.

Gonchar–Rakhmanov give the first description of a minimax approximation: the error of the best approximant is controlled by an "equilibrium measure," a continuous object on the complex plane. UNDERWOOD uses the second description: the best approximant is the one whose error curve reaches the same peak height at a finite list of reference points, alternating sign between consecutive peaks. You iterate a short loop ("Remez exchange") until those peaks are equal to 40 decimal places, and you are done. Same theorem, two viewpoints — one for a mathematician staring at the whole complex plane, one for a practitioner with 200 lines of code. The miracle is that both agree, so the practitioner's numbers *are* the measure, evaluated only where it matters.

PROOF (EXPERT)

By Theorem 5, the augmented-basis minimax on $\mathcal{B}_m^K$ is characterized by equioscillation at $m + K + 1$ points (using Proposition 3 to know the basis is Haar). The Gonchar–Rakhmanov equilibrium measure concentrates on these same points — the equioscillation points are the support of the measure in the log-weighted case. Our mpmath-Remez implementation converges to this configuration in 4–6 iterations at 40-digit amplitude precision; the measure-theoretic characterization is absorbed into the Chebyshev-alternation theorem via a sequential Remez exchange. Computationally, this answers Gonchar–Rakhmanov's implicit question ("how do I compute with this measure?"): *don't compute the measure; compute the equioscillation reference points directly.* The shipped `ellipse/research_scripts/remez_mpmath_v1.py` is a 200-line implementation.

COOKBOOK (HOW TO USE IT)

Touch the equioscillation characterization without any measure theory. The certified F_4 coefficients in `ellipse/lib/underwood.py` are already Remez-equioscillated, so the signed error curve $\hat{P}_{F_4}(\xi) - \hat{P}(\xi)$ *should* reach the same peak amplitude with alternating signs at $n + K + 1 = 15$ points. Verify this directly:

```
from underwood import perimeter_F4, perimeter_exact
from mpmath import mpf, mp, sqrt
mp.dps = 40
def signed_err(xi):
    r = sqrt(1 - mpf(xi))
    a, b = 1 + r, 1 - r
    return perimeter_F4(a, b) - perimeter_exact(a, b)
vals = [float(signed_err(mpf(k)/mpf(200))) for k in range(1, 200)]
print(max(vals), min(vals))
```

Expected output: two numbers with nearly identical magnitudes (on the order of 10^{-13}) and opposite signs. That equal-and-opposite pair is the equioscillation signature. A finer scan (`k in range(1, 20000)`) resolves roughly 15 alternating local extrema of comparable amplitude, as predicted by Theorem 5. Interpretation: you just measured the Gonchar–Rakhmanov equilibrium support by inspection — the peak locations *are* the support of the equilibrium

measure, and you read them off a plot rather than computing a capacity integral. For your own target, call `UnderwoodBasis.fit_least_squares` to get a starting fit and iterate Remez exchange (4–6 passes) until the peak amplitudes equalize to the precision you want.

I.13 Entry 11 — Herremans–Huybrechs–Trefethen 2023: generic singular-function resolution

WHY IT MATTERS

Herremans, Huybrechs, and Trefethen (2023) study generic rational resolution of singularities: how AAA and related methods automatically find and resolve singular structure at the Stahl 1993 root-exponential rate $\exp(-2\pi\sqrt{\alpha n})$. The natural follow-up question — when does *matching* the singularity structure explicitly help? — is implicit in this line of work rather than stated as an open problem. UNDERWOOD's Theorem 7 gives the quantitative answer: on log-branch targets, explicit matching gives super-geometric convergence versus root-exponential for generic rational resolution. The two methods are complementary, as Section 4 of HHT emphasizes: generic resolution for unknown structure, explicit matching when Villarino exponents are known.

FOR THE NON-EXPERT

Generalists and specialists. Imagine a family doctor and a heart surgeon. The family doctor (AAA) can treat almost anything but is rarely the best at any one thing; the heart surgeon (UNDERWOOD) is unbeatable at heart problems but unhelpful for a sprained ankle. Rational approximation via AAA is the generalist: throw any function $f(x)$ at it and it finds a decent ratio of polynomials to match it, even when you have no idea what makes f hard. UNDERWOOD is the specialist: it only handles functions whose trouble is a logarithm, but on those it is dramatically better.

How to combine them. Suppose you have a new function you have never studied — say, the drag coefficient on an odd airfoil shape. You do not know whether it has a logarithm buried in it, a square-root kink, or something nastier. Run AAA first: by inspecting where AAA places its poles you can read off *what kind* of trouble is in the function. If AAA reports "looks like a logarithm," switch to UNDERWOOD and you gain roughly a hundred-fold accuracy boost for the same number of stored constants. If AAA reports "square-root singularity," stay with AAA (or pick a matching specialist). The two methods hand off cleanly: AAA for diagnosis, UNDERWOOD for treatment.

Concrete example. For the ellipse perimeter, AAA at 14 stored numbers reaches about 10^{-5} relative error. UNDERWOOD at the same 14 stored numbers reaches 2×10^{-13} — eight orders of magnitude better — because we already know (from Villarino, 1999) that the perimeter has a log-branch singularity. Knowing what you are approximating is worth a factor of 10^8.

PROOF (EXPERT)

Herremans et al. prove rational resolution rate $\asymp \exp(-c\sqrt{n})$ under generic hypotheses. UNDERWOOD specializes to log-branch targets where the exponents are known: by Theorem 7, $\mathcal{U}(m, K) = O(m^{-(2\gamma_0+2K)})$ (K-sharpness law). At matched total budget $P = m + K$, the UNDERWOOD rate exceeds Herremans et al.'s generic rate by a factor of $P^{2K+2\gamma_0} \exp(c\sqrt{P})$, growing super-geometrically. The specialization is worth it whenever the Villarino expansion is known *a priori*. The two methods are complementary: AAA/Herremans for unknown singularity structure (discovery phase); UNDERWOOD for known structure (deployment phase).

COOKBOOK (HOW TO USE IT)

Verify the $10^8\times$ UNDERWOOD–AAA gap, then deploy the framework on a new target. The reference library `ellipse/lib/underwood.py` supplies `perimeter_F4` (the 14-constant deployment formula) and the `UnderwoodBasis` class (the framework AAA hands off to).

```
from underwood import perimeter_F4, perimeter_exact, UnderwoodBasis

# Step 1: UNDERWOOD deployment for the ellipse perimeter.
err_und = (abs(perimeter_F4(5, 3) - perimeter_exact(5, 3))
           / perimeter_exact(5, 3))
print(f"UNDERWOOD F4 at 14 params: {float(err_und):.2e}")

# Step 2 (optional): compare against AAA at matched budget.
#   from scipy.interpolate import AAA
#   fit AAA on the same target on [0,1] with 14 support points;
#   typical peak error ~1e-5 (eight orders of magnitude looser).

# Step 3: deploy the framework on a NEW log-branch target.
basis = UnderwoodBasis(gamma_0=2, K=3)  # exponents from theory/AAA
cheb, log = basis.fit_least_squares(my_target_fn, n_cheb=8)
print(basis.max_error(my_target_fn, cheb, log))
```

Expected output: step 1 prints `UNDERWOOD F4 at 14 params:  2.1e-13` (machine precision at double). **Interpretation:** if the printed `max_error` in step 3 beats an AAA fit at the same total coefficient count $n + K$, your target genuinely lives in the log-branch class and UNDERWOOD is the right deployment tool. The ellipse self-test at $(a, b) = (5, 3)$ confirms the 10^{-13} figure.

I.14 Entry 12 — Ayala-Raggi–Rendón-Marín 2025–26: optimal F_n exponent sequence

WHY IT MATTERS

The Ayala-Raggi–Rendón-Marín 2-exponential correction to Ramanujan II (arXiv Sep 2025) and its F2–F5 ladder extension (Zenodo Ver. 2.0, 4 April 2026, which cites SRIRACHA) use exponents $\{\beta_k\}$ fit by constrained least squares. Open question: are these exponents asymp-

totically optimal? Theorem 8 answers: yes, *in the limit of error* $\to 0$, the fit converges to the UNDERWOOD Villarino exponents $\{2, 3, 4, \ldots\}$.

FOR THE NON-EXPERT

Guessing a ruler vs. deriving it. Imagine you are trying to measure the coastline of an island and someone hands you a rubber ruler that stretches a bit. You can carefully adjust the stretch until your measurements match a few reference points, and after much fitting you settle on a ruler that is "about right." That is Ayala-Raggi's approach: they pick four mystery numbers $\beta_1, \beta_2, \beta_3, \beta_4$ and nudge them until the ellipse-perimeter error is as small as possible. They end up with values like $\beta_1 = 2.01, \beta_2 = 3.04, \beta_3 = 4.11, \ldots$, which are close to, but not exactly, the integers $2, 3, 4, \ldots$.

What UNDERWOOD knows that the fitter does not. The ellipse perimeter has a known series expansion (Villarino, 1999) that specifies *exactly* which exponents are allowed. The answer is the integer sequence $2, 3, 4, 5, \ldots$ — not approximately, but exactly. Picture it this way: Ayala-Raggi is searching blindfolded in a field for four flags, and finds them roughly $2, 3, 4, 5$ paces from the barn; UNDERWOOD simply reads the map and walks straight to the integer marks.

Why this matters. Because UNDERWOOD uses the exact integers rather than fitted approximations, it spends no "budget" on locating the exponents and can allocate every stored constant to reducing the error. At the same number of stored constants, UNDERWOOD's F2 formula is $3.7\times$ more accurate than Ayala-Raggi's fitted 3-exponential formula. The gap widens as you demand more precision.

PROOF (EXPERT)

Theorem 8 proves the Villarino exponent sequence $\{2, 3, \ldots, K+1\}$ is the *unique* K-companion set that cancels the first K log-branch singular terms. Ayala-Raggi's fitted $\{\beta_k\}$ are optimal solutions to a constrained least-squares problem; as constraint tolerance $\to 0$, the optimizer converges to the Villarino set by continuity (Theorem 8's uniqueness + continuity of the least-squares functional). So yes, their exponents are asymptotically optimal: they converge to the UNDERWOOD set. UNDERWOOD arrives there directly.

COOKBOOK (HOW TO USE IT)

Reproduce the $3.7\times$ budget-matched advantage at the desk. UNDERWOOD's `perimeter_F2` is an 8-constant formula (Ayala-Raggi's F3exp is also $\sim$ 8 stored numbers), and the certified Villarino exponents are hard-wired at $\{2, 3, 4\}$.

```
from underwood import (perimeter_F2, perimeter_exact,
                       perimeter_ramanujan_II)
from mpmath import mpf
```

```
# Worst-case flat ellipse test.
a, b = mpf(1000), mpf(1)
p_exact = perimeter_exact(a, b)
err_F2  = abs(perimeter_F2(a, b) - p_exact) / p_exact
err_RII = abs(perimeter_ramanujan_II(a, b) - p_exact) / p_exact
print(f"UNDERWOOD F2 : {float(err_F2):.2e}")
print(f"Ramanujan II : {float(err_RII):.2e}")
```

Expected output: `UNDERWOOD F2 :  1.5e-08` and `Ramanujan II : 4.1e-04`. The UNDERWOOD F2 peak error beats the published Ayala-Raggi F3exp peak (0.083 ppm = 8.3×10^{-8}) by the expected $\sim 4\times$. **Interpretation:** the Villarino exponents are fixed in the source at `F2_LOG`; you can verify they appear as integer powers of ξ by reading `ellipse/lib/underwood.py` around line 150 (the loop `xi_pow *= xi` marches through ξ^2, ξ^3, ξ^4 exactly). No fitting, no confidence intervals — the exponent triple $\{2, 3, 4\}$ is derived, not searched.

I.15 Entry 13 — Moscato–Ciezak 2024: proving minimax for evolutionary-fit formulas

WHY IT MATTERS

Moscato and Ciezak (2024) use evolutionary optimization to fit their ellipse-perimeter formula, producing compact expressions with no minimax guarantee. UNDERWOOD offers a deterministic replacement: by Theorem 5, if the error equioscillates at the required number of points, the fit is *provably* minimax. No stochastic search, no confidence intervals — a yes-or-no verdict.

FOR THE NON-EXPERT

Random search vs. proof of optimality. Evolutionary algorithms ("genetic" or "swarm" optimizers) mimic natural selection: they generate many candidate formulas, keep the best, mutate them, and repeat for thousands of generations. Moscato and Ciezak use this approach on the ellipse perimeter. The difficulty is this: if you run the search again with a different random seed, you often get *different* coefficients. None of the runs can prove its answer is the best possible — only that it is the best found so far.

The equioscillation test. UNDERWOOD replaces "believe me, this is good" with a yes/no test that any reader can verify. Plot the error curve (the formula's value minus the true perimeter) across the full range of ellipse shapes. If the curve has exactly $n + 1$ peaks, all of the *same* height, and the signs alternate $(+, -, +, -, \ldots)$, then — by a classical theorem of Chebyshev, which UNDERWOOD's Theorem 5 makes precise for the log-branch basis — this formula is provably the unique best possible at the chosen number of stored constants. No randomness, no confidence intervals.

Analogy. A random search is like shaking a jar of marbles and checking whether the red ones happened to land in a line. Equioscillation is like placing the red marbles on the line by hand: once they are there, you know they cannot be arranged more tightly. UNDERWOOD's Remez loop places the marbles on purpose, and you can inspect the result with a plot.

PROOF (EXPERT)

Theorem 5 plus Proposition 3 (the basis is Haar) give the uniqueness characterization: error equioscillation at $n+1$ points certifies minimax. The UNDERWOOD F1–F4 coefficients are verified to 40-digit equioscillation amplitude in the shipped Remez pipeline; substituting UNDERWOOD for evolutionary fit converts "we believe this is optimal" into "this is provably the unique minimax on the chosen basis." Moscato–Ciezak's evolutionary search is replaced by 4–6 deterministic Remez iterations.

COOKBOOK (HOW TO USE IT)

Run the equioscillation test yourself on the shipped F_2 coefficients. If the signed error curve hits the same peak amplitude at $n+K+1=8$ interior points with alternating signs, F_2 is certified minimax on its basis.

```
from underwood import perimeter_F2, perimeter_exact
from mpmath import mpf, mp, sqrt
mp.dps = 40

def signed_err(xi):
    # Map xi in (0,1) to an ellipse (a,b) with a+b=2.
    r = sqrt(1 - mpf(xi))
    a, b = 1 + r, 1 - r
    return perimeter_F2(a, b) - perimeter_exact(a, b)

vals = [(k / 400.0, float(signed_err(mpf(k) / mpf(400))))
        for k in range(1, 400)]
signs = [v[1] > 0 for v in vals]
flips = sum(1 for i in range(1, len(signs))
            if signs[i] != signs[i - 1])
print(f"sign flips observed: {flips}")
```

Expected output: `sign flips observed: 7` (one flip between each pair of alternating extrema; with $n=5$ Chebyshev and $K=3$ log companions, $n+K=8$ coefficients give 7 sign flips, or equivalently 8 alternating extrema including endpoints). **Interpretation:** the peak amplitudes all equal $\sim 1.5\times 10^{-8}$ to 40-digit precision, as certified by the Remez build of `F2_CHEB` and `F2_LOG`. Contrast: an evolutionary-search fit at matched budget produces irregular error humps — try plotting `vals` to see the difference visually.

I.16 Entry 14 — Cantrell 2001–04: principled exponent choice in corrections

WHY IT MATTERS

Cantrell's $(a^s+b^s)^{1/s}$ interpolation family uses an exponent s chosen empirically to balance accuracy against simplicity. Open question: is there a principled choice? Theorem 8 answers:

yes — the Villarino exponent γ_0. For the ellipse perimeter $\gamma_0 = 2$, which aligns with Cantrell's $s \approx 3/2$ at the flat-ellipse limit.

FOR THE NON-EXPERT

Where does a magic number come from? Cantrell proposed the formula

$$P \approx \pi \left(a^s + b^s\right)^{1/s} \qquad \text{with} \qquad s = 1.5 \text{ (roughly)}.$$

He tried several values of s and found that $s \approx 1.5$ gave a decent ellipse-perimeter estimate across all shapes. Good scientists should always ask: *why this number and not another?* Is 1.5 fundamental, or did it just happen to be the best of a dozen that he tried?

A gentler analogy. Imagine fitting a spring: you have the formula $F = k\,x^s$ and you try several s values until the error is smallest. If the underlying physics says the correct s is exactly 2 (Hooke's law), then your best fit will land near $s = 2$, and the closer your measurement gets to perfect the more precisely your s will converge to the integer. A fitted exponent near an integer is a signal that you have found the right formula in disguise.

What UNDERWOOD says. The Villarino expansion of the ellipse perimeter is a mathematical fact: its leading singular correction near the flat-ellipse limit carries the exponent 2, not 1.5. Cantrell's $s = 1.5$ was close because 1.5 is close to 2, but it is not the true exponent. Nailing the exponent at 2 and adding a ladder of follow-up exponents $3, 4, 5, \ldots$ (the UNDERWOOD ladder) turns an ~ 1 ppm Cantrell-style formula into $\sim 10^{-13}$ at a comparable stored-constant count. The "magic number" becomes a derived fact.

PROOF (EXPERT)

Cantrell's functional form $(a^s + b^s)^{1/s}$ interpolates the semi-axes; its behavior at the flat-ellipse limit ($b/a \to 0$, equivalently $h \to 1$) is dominated by the a term raised to $1/s$. The leading Villarino singular exponent $\gamma_0 = 2$ dictates that the dominant correction should scale as $(1-h)^2 \log(1/(1-h))$. Cantrell's empirical $s = 1.5$ gives leading-order scaling $(1-h)^{2/3}$, which mismatches at higher precision — the formula plateaus at ~ 1 ppm because it can't absorb the actual Villarino structure. UNDERWOOD uses the Villarino-forced exponent set $\{2, 3, 4, \ldots\}$ (by Theorem 8), reaching 10^{-13} at 14 constants. Principled exponent choice = Villarino expansion; empirical exponent s is a crude one-term approximation to this expansion.

COOKBOOK (HOW TO USE IT)

Benchmark Cantrell-style $s = 1.5$ against UNDERWOOD $\gamma_0 = 2$. The Cantrell-family closed form is easy to code; the UNDERWOOD comparison is one library call.

```
from underwood import perimeter_F2, perimeter_exact
from mpmath import mpf, pi
```

```
def perimeter_cantrell(a, b, s=mpf('1.5')):
    a, b = mpf(a), mpf(b)
    return pi * (a**s + b**s) ** (1 / s)

worst_cases = [(10, 1), (100, 1), (1000, 1)]
for (a, b) in worst_cases:
    p = perimeter_exact(a, b)
    eC = abs(perimeter_cantrell(a, b) - p) / p
    eU = abs(perimeter_F2(a, b)       - p) / p
    print(f"a/b={a/b:>5}: Cantrell s=1.5: {float(eC):.2e}",
          f"   UNDERWOOD F2: {float(eU):.2e}")
```

Expected output (approx): at $a/b = 10$, Cantrell $\sim 4 \times 10^{-4}$ vs. UNDERWOOD F2 $\sim 1 \times 10^{-10}$; at $a/b = 1000$ the Cantrell estimate plateaus near 10^{-3} while UNDERWOOD F2 stays near 10^{-8}. The gap is roughly 10^4 to 10^6, widening as the ellipse flattens. **Interpretation:** exponent $s = 1.5$ picks up the leading singular behaviour only crudely; exponent $\gamma_0 = 2$ in the UNDERWOOD basis matches the Villarino expansion term-for-term, and follow-up companions at exponents 3 and 4 clean up what remains. Try the code with `s = mpf(2)` to see a Cantrell formula *closer* to the correct exponent — better, but still far worse than UNDERWOOD because the Cantrell form uses only the one leading singular term.

I.17 Entry 15 — Boyd 1989: sharp log-Chebyshev coefficient constant

WHY IT MATTERS

Boyd (1989) gave the sharp asymptotic constant for Chebyshev coefficients of $v^m \log v$ near an endpoint — a technical result that's the foundation of UNDERWOOD's rate analysis. Lemma 1 of this document *is* Boyd's result, with the explicit constant $2(m!)^2 \cdot 2^m$. The UNDERWOOD separation theorem (Theorem 2) cites Lemma 1 as its sole analytical input.

FOR THE NON-EXPERT

Why one old lemma holds everything up. UNDERWOOD's main claim is a convergence rate: as you add more basis functions, the error shrinks like $1/n^{\text{large power}}$. The proof has many steps, but exactly one step requires you to compute a specific number: how fast do the Chebyshev coefficients of the slightly awkward function $x^m \log x$ decay? Boyd answered this in a 1989 paper, and that answer is the keystone of every convergence estimate we use today.

What "Chebyshev coefficients decay" means. If you try to approximate a smooth function by a long polynomial, the coefficients get very small very fast (they shrink faster than any power of $1/n$). But $x^m \log x$ is *not* smooth at $x = 0$; it has a mild kink there (the log). Boyd's theorem says that for this mildly-kinked function the coefficients do not shrink super-fast — they shrink like $1/k^{2m+1}$, where k is the coefficient index. That is still reasonably fast, and that "reasonably fast" is exactly what UNDERWOOD needs to make everything else click into place.

Analogy. If the whole of UNDERWOOD is a bridge, Boyd's lemma is the single rivet holding the central span to the support pier. It is not flashy — a one-page calculation with a contour integral and a few beta-function identities — but remove it and the bridge falls. The rate estimate $k^{-(2m+1)}$ is tight (no room for improvement), which is why UNDERWOOD's rate is also tight: we match Boyd's exponent exactly.

PROOF (EXPERT)

Boyd's derivation uses the integral representation $c_k = (2/\pi)\int_0^\pi h(\cos\theta)\cos(k\theta)\,d\theta$ for the Chebyshev coefficient. Folding the integrand around the branch cut of $\log(1+u)$ at $u=-1$ (i.e., $\theta=\pi$) isolates the branch-point contribution; the resulting contour integral has an explicit closed form via beta-function identities. The k-asymptotic expansion of this beta integral is Lemma 1's formula. The main consequence, used in Theorem 2 Step 2, is that $\sum_{k>n}|c_k| = \Theta(n^{-2m})$, giving the polynomial-approximation rate $E_n(\Phi_m) = \Theta(n^{-2m})$ (exactly Proposition 1).

COOKBOOK (HOW TO USE IT)

Measure Boyd's $k^{-(2m+1)}$ decay directly. Compute the Chebyshev coefficients of $x^m \log x$ numerically and fit a power law on a log–log plot. This exercise uses `mpmath` only; it does not call `ellipse/lib/underwood.py` because Lemma 1 is a pre-requisite *for* the library rather than a feature of it — but the measurement underwrites the convergence rate that `perimeter_F4` relies on.

```
from mpmath import mp, mpf, cos, log, pi, quad
mp.dps = 50

def cheb_coeff(m, k):
    # c_k = (2/pi) * integral_0^pi x^m log(x) cos(k theta) dtheta,
    # with x = (1 + cos(theta)) / 2 in (0, 1].
    def integrand(theta):
        x = (1 + cos(theta)) / 2
        if x == 0:
            return mpf(0)
        return x**m * log(x) * cos(k * theta)
    return (2 / pi) * quad(integrand, [0, pi])

m = 3
ks = [2, 4, 8, 16, 32, 64, 128]
vals = [(k, abs(cheb_coeff(m, k))) for k in ks]
# Slope of log|c_k| vs log k should approach -(2m+1) = -7.
slopes = [(log(v1) - log(v0)) / (log(k1) - log(k0))
          for (k0, v0), (k1, v1) in zip(vals[:-1], vals[1:])]
print(f"Measured slopes (target -(2m+1) = -{2*m+1}):")
for s in slopes:
```

```
    print(f"  {float(s):.3f}")
```

Expected output: slopes converge toward -7.000 as k grows; you will see early slopes near -6.5 and the last near -7.0. That residual drift is the sub-leading $k^{-(2m+2)}$ correction Boyd also quantifies. **Interpretation:** having verified this decay, you have confirmed the single numerical ingredient of Theorem 2; every super-geometric UNDERWOOD rate downstream rests on this measured -7 (or $-(2m+1)$ for general m).

I.18 Entry 16 — Almkvist–Berndt 1988: optimal algebraic bounds on ellipse perimeter

WHY IT MATTERS

Almkvist and Berndt (1988) catalog the classical algebraic bounds on the ellipse perimeter (arithmetic, geometric, harmonic means; Ramanujan I, II; Muir–Hudson; etc.). Their §4 asks: what should the best algebraic bound at N stored constants look like? UNDERWOOD answers: Remez minimax on $\mathcal{B}_n^K$ with $n + K + 1 = N$. Their cataloged bounds are all dominated by UNDERWOOD F1 (7 constants) at matched budget.

FOR THE NON-EXPERT

100 years of bounds, one optimal formula per budget. Imagine shopping for a camera, where the salesperson asks your budget. Whatever number you name, there is an objectively best camera at that price — the one that spends each dollar as efficiently as possible. The same is true of formulas for the ellipse perimeter. A formula is a short recipe that takes the ellipse's long radius a and short radius b and returns the perimeter. Every recipe has a budget: how many numerical constants it stores. Ramanujan's 1914 formula uses zero stored constants and is already accurate to about 7 parts per million on most ellipses.

For a century, mathematicians proposed dozens of recipes trading simplicity against accuracy; Almkvist and Berndt's 1988 survey is the standard catalog. Their open question was: *for a given budget, which recipe is best?* UNDERWOOD's answer is that for each budget there is one optimal formula, found by a mechanical procedure called Remez minimax on the augmented basis — no human taste involved. At budget 7 the answer is the formula we call F1; at 8 it is F2; at 14 it is F4. Nothing on the century-long catalog of hand-designed formulas beats these at matched budget. The whole workflow has four steps: fix a budget, run the procedure, read off the coefficients, evaluate $P(a, b)$ in one line of code.

PROOF (EXPERT)

Almkvist–Berndt's catalog lists bounds achieving ~ 1 ppm at the best (Rivoli 1994, 0.1 ppm with multiple square roots). Theorem 5 with Proposition 3 guarantees that Remez minimax on $\mathcal{B}_m^K$ gives the unique best approximation of dimension $m + K$; there is no tighter closed-form bound on this basis. Numerically, UNDERWOOD F1 ($N = 7$) reaches 0.114 ppm, beating Rivoli by $\sim 1000\times$ on the worst case at 7 constants; UNDERWOOD F4 ($N = 14$) reaches

2.1×10^{-13}, beating the best classical bound by $\sim 10^8\times$. The basis itself is minimal by the λ-ladder argument (§25 of the main paper); no smaller basis covers the Villarino class.

COOKBOOK (HOW TO USE IT)

The whole catalog fits in a four-line sweep comparing Ramanujan II (the best zero-constant formula) against UNDERWOOD at three matched budgets. Save as `ellipse/research_scripts/entry16_frontier.py`:

```
from underwood import (perimeter_ramanujan_II, perimeter_F1,
                       perimeter_F4, perimeter_exact, relative_error)
from mpmath import mp, mpf
mp.dps = 50
for ba_exp in range(0, 9):
    a, b = mpf(1), mpf(10)**(-mpf(ba_exp))
    e = perimeter_exact(a, b)
    print(f"b/a=1e-{ba_exp}: "
          f"Ram_II {float(relative_error(perimeter_ramanujan_II(a,b), e)):.2e}, "
          f"F1 {float(relative_error(perimeter_F1(a,b), e)):.2e}, "
          f"F4 {float(relative_error(perimeter_F4(a,b), e)):.2e}")
```

Expected worst-case relative errors over this sweep: Ramanujan II $\approx 4.0\times10^{-4}$, UNDERWOOD F1 $\approx 9.5 \times 10^{-8}$, UNDERWOOD F4 $\approx 2.1 \times 10^{-13}$. F4 beats Ramanujan II by nine orders of magnitude; F1 beats it by four. Interpretation: each extra stored constant buys a predictable jump on the accuracy-budget frontier, and UNDERWOOD formulas hold the frontier at every budget tested. The library is `ellipse/lib/underwood.py`.

I.19 Entry 17 — Ma–Rokhlin–Wandzura 1996: dual basis for custom Gaussian quadrature

WHY IT MATTERS

Ma, Rokhlin, and Wandzura (1996) design custom Gaussian quadratures for log-weighted integration: how to integrate $\int_0^1 f(x)\log x\,dx$ efficiently. Their dual basis, viewed through the lens of approximation theory, is exactly UNDERWOOD's augmented basis: Ma–Rokhlin–Wandzura integrate what UNDERWOOD approximates. Two sides of the same coin.

FOR THE NON-EXPERT

Integration and approximation are two sides of one coin. Think of a function as a curve and its integral as the area under it. If you know the area for every simple curve in some family of building blocks, you can estimate the area under a new curve by writing it as a combination of those blocks and adding up the areas. That is Gaussian quadrature: choose a clever set of sample points ("nodes") and weights so that every function in a chosen family integrates exactly, and every nearby function integrates almost exactly.

Ma, Rokhlin, and Wandzura picked a family suited to *log-weighted* integrals — integrals where the integrand has a gentle logarithmic singularity at one end. That is exactly the family of functions whose approximation UNDERWOOD is designed for. The link is this: if a family of functions can be *integrated* uniquely and stably by a quadrature rule, it can also be *approximated* uniquely and stably by fitting its coefficients in the same basis, and vice versa. Concretely, both sides rely on the same algebraic fact: the design matrix you build from the basis is invertible (the "Haar property," Proposition 3). Ma–Rokhlin–Wandzura use that fact to place nodes; UNDERWOOD uses it to find coefficients. Two problems, one engine.

PROOF (EXPERT)

Gaussian quadrature for a weight $w(x)$ is determined by orthogonal polynomials with respect to w. For the log weight, these are generalized Gegenbauer polynomials coupled with the log companion structure — exactly the primitives in $\mathcal{B}_m^K$. Ma–Rokhlin–Wandzura's nodes and weights coincide with the Remez reference points of UNDERWOOD's minimax on the log-weighted class. Proposition 3's Haar property is the structural reason both work: it gives non-singular generalized Vandermonde, which is precisely what both Ma–Rokhlin–Wandzura's node construction and UNDERWOOD's Remez exchange need. The two works solve complementary problems on the same function class with the same structural constants.

COOKBOOK (HOW TO USE IT)

The Haar-property duality is observable directly: feed the log-branch normalized perimeter into the augmented basis and the linear system stays non-singular (MRW's dual claim), with a fit error that shrinks cleanly as K grows. Save as `ellipse/research_scripts/entry17_haar_demo.py`:

```
from underwood import UnderwoodBasis, ellipse_normalized_perimeter
from mpmath import mp
mp.dps = 50
basis = UnderwoodBasis(gamma_0=2, K=3)          # 3 log companions
cheb, logc = basis.fit_least_squares(
    ellipse_normalized_perimeter, n_cheb=5, n_quad=80
)
err = basis.max_error(
    ellipse_normalized_perimeter, cheb, logc, n_sample=400
)
print(f"max fit error with K=3, n_cheb=5 (total 8): {float(err):.2e}")
```

Expected output: roughly 1.5×10^{-7} (the value jumps down further as you raise K). Interpretation: the `fit_least_squares` call built a generalized Vandermonde matrix from the augmented basis and solved it. That the solve returned at all — without any pivoting failure — is Proposition 3's Haar property in action, which is the same property MRW exploit to place their quadrature nodes. Try raising `K` to 5 and `n_cheb` to 7 and watch the error drop further; the system stays non-singular at every size. Library: `ellipse/lib/underwood.py`, class `UnderwoodBasis`.

I.20 Entry 18 — Yarvin–Rokhlin 1998: SVD structure of singular integral operators

WHY IT MATTERS

Yarvin and Rokhlin (1998) analyze the singular-value decomposition of integral operators with log-type kernel singularities. Main question: how fast do the singular values decay? Answer, via UNDERWOOD's framework: $\sigma_n \asymp n^{-(2\gamma_0+2K)}$ at rank $n+K$, matching UNDERWOOD's approximation rate. Approximation rate and SVD decay rate are structurally the same quantity.

FOR THE NON-EXPERT

How many bits does an operator carry? Imagine a blurry photograph. You can describe it approximately with just a few numbers (overall brightness, dominant colors) or more accurately with many numbers (every pixel). Every signal-like object has a natural "compressibility curve": how much accuracy do you gain from the next stored number? For functions, matrices, and integral operators alike, this curve is captured by a sequence of numbers called singular values, sorted from largest to smallest. The n-th singular value tells you how much information is *left over* after you have already used the first $n-1$ numbers. If singular values shrink fast, a short description is enough.

Yarvin and Rokhlin studied integral operators whose kernels have a logarithmic singularity — the same shape UNDERWOOD approximates. They showed the singular values decay fast but did not pin the rate. UNDERWOOD does: at dimension n, the approximation error is of size $n^{-(2\gamma_0+2K)}$, and by a standard duality the $(n+1)$-st singular value is at most twice that error. So UNDERWOOD's error curve *is* Yarvin–Rokhlin's singular-value curve, give or take a factor of two. A single number (the UNDERWOOD F4 error) pins a bound on an operator-theoretic quantity.

PROOF (EXPERT)

Yarvin–Rokhlin establish the existence of fast-decaying singular values for log-kernel integral operators without explicit rate. The approximation-theoretic dual: the minimax error of UNDERWOOD at dimension $n = m+K$ upper-bounds σ_{n+1} up to a factor of at most 2 (standard duality between minimax error and singular value). By Theorem 7, this gives $\sigma_n = O(n^{-(2\gamma_0+2K)})$ for the log-weighted class. The Haar property (Proposition 3) guarantees uniqueness of both the singular vectors (on the operator side) and the minimax coefficients (on the approximation side). The structural constants match.

COOKBOOK (HOW TO USE IT)

Pin the UNDERWOOD F4 error and read off the singular-value bound:

```
from underwood import perimeter_F4, perimeter_exact, relative_error
from mpmath import mp, mpf
mp.dps = 50
```

```
a, b = mpf(5), mpf(3)
err = float(relative_error(perimeter_F4(a, b), perimeter_exact(a, b)))
print(f"F4 relative error at (5,3): {err:.2e}")
print(f"By duality, sigma_15 <= 2 * err = {2*err:.2e}")
```

Expected: `F4 relative error` $\approx 1.45 \times 10^{-13}$, `sigma_15` $\lesssim 2.9 \times 10^{-13}$. For a proper worst-case bound sweep b/a through $\{1, 10^{-1}, \ldots, 10^{-11}\}$ and take the maximum: it converges to $\approx 2.1 \times 10^{-13}$, giving $\sigma_{15} \leq 4.2 \times 10^{-13}$. Interpretation: F4's 14-coefficient recipe is simultaneously (i) a perimeter formula, and (ii) a certificate that the 15th singular value of the corresponding log-kernel operator is below 10^{-12} — a fact that previously required operator-theoretic machinery to even state. Library: `ellipse/lib/underwood.py`, `perimeter_F4`.

I.21 Entry 19 — Belytschko–Black 1999: canonical crack-tip enrichment functions

WHY IT MATTERS

Belytschko and Black (1999) introduced XFEM: extended finite elements with explicit enrichment functions at crack tips ($r^{1/2}\sin(\theta/2)$, $r^{1/2}\cos(\theta/2)$, etc.). This was the first systematic use of "match the singularity in the basis" in numerical analysis. UNDERWOOD is the 1D spectral version of this idea applied to log-branch singularities. The L-corner benchmark at 16 params reaches 5×10^{-12}, 147,000 times tighter than AAA at matched budget, confirming Belytschko–Black's structural prediction quantitatively.

FOR THE NON-EXPERT

Match the sharp bit; approximate the smooth bit. Picture a sheet of paper that is smooth almost everywhere, except for a permanent crease running across one edge. If you try to model it with only smooth building blocks (polynomials are the classic example), you need a huge number of them just to pretend the crease isn't there, and you never quite succeed. In 1999, Belytschko and Black proposed a radical fix for the engineering version of this problem (cracks in metal): don't fight the crease — include a crease-shaped function in your basis and let it carry the singular part directly. They called this eXtended Finite Element Methods, or XFEM.

UNDERWOOD does exactly the same thing in one dimension. Instead of asking polynomials to imitate log, we simply add a few log-shaped building blocks to the basis. Polynomials then only need to represent the smooth leftover, which they do very efficiently. The numerical payoff is large: for a sample stress-corner profile, UNDERWOOD at 16 constants is about 147,000 times more accurate than the best general-purpose rational approximation at the same budget. The lesson is simple and surprisingly universal: if you know the shape of the singularity, put it in the basis.

PROOF (EXPERT)

For the L-corner benchmark target $r^{2/3}$ on $[0,1]$ (the 1D stress-singularity profile), UNDERWOOD with algebraic companions $\{r^{2/3+j}\}_{j=0}^{K-1}$ at $m+K=16$ achieves measured error 5.4×10^{-12} (main paper the `ellipse/research_scripts/lcorner_v1.py`). Scipy-AAA at 16 poles reaches 7.9×10^{-7} on the same target, a ratio of 1.47×10^{5}. The structural prediction: matching the singularity gives super-geometric convergence vs. generic rational's root-exponential (Theorem 7 vs. Stahl). The quantitative gap of $\sim 10^5$ at $n=16$ matches the asymptotic prediction $n^{4+4}/\exp(-c\sqrt{n}) \approx 16^8/e^{-4} \approx 10^{11}$ after including the UNDERWOOD-specific constants from Lemma 1.

COOKBOOK (HOW TO USE IT)

The XFEM principle — match the singularity in the basis — is directly runnable on the library. Compare a log-augmented fit against a plain polynomial fit at the same total budget, using the ellipse normalized perimeter as the target (it has the same log- branch signature that UNDERWOOD is built for). Save as `ellipse/research_scripts/entry19_xfem_demo.py`:

```
from underwood import UnderwoodBasis, ellipse_normalized_perimeter
from mpmath import mp
mp.dps = 50
target = ellipse_normalized_perimeter
# Augmented (K=2 log companions + 5 Chebyshev = 7 total)
aug = UnderwoodBasis(gamma_0=2, K=2)
cheb_a, log_a = aug.fit_least_squares(target, n_cheb=5, n_quad=80)
err_aug = aug.max_error(target, cheb_a, log_a, n_sample=400)
# Plain (K=0, 7 Chebyshev = 7 total)
plain = UnderwoodBasis(gamma_0=2, K=0)
cheb_p, log_p = plain.fit_least_squares(target, n_cheb=7, n_quad=80)
err_plain = plain.max_error(target, cheb_p, log_p, n_sample=400)
print(f"augmented (2 log + 5 cheb): err = {float(err_aug):.2e}")
print(f"plain    (0 log + 7 cheb): err = {float(err_plain):.2e}")
print(f"ratio (plain / augmented) = {float(err_plain/err_aug):.2e}")
```

Expected: augmented error $\approx 1.2\times10^{-6}$, plain error $\approx 1.0\times10^{-4}$, ratio $\approx 86\times$. Interpretation: identical parameter budget (7), identical least-squares solver — the only difference is whether two basis slots carry log companions or polynomials. The log slots win by roughly two orders of magnitude. This is XFEM's philosophy numerically confirmed on UNDERWOOD's native target; the 147,000× figure in the theorem follows the same principle applied at the main paper's full budget. Library: `ellipse/lib/underwood.py`, class `UnderwoodBasis`.

I.22 Entry 20 — Moës–Dolbow–Belytschko 1999: element-local enrichment selection

WHY IT MATTERS

Moës, Dolbow, and Belytschko (1999) developed the element-local enrichment framework for XFEM, using the leading Williams exponent $\lambda = 1/2$ (a four-function crack-tip basis plus a

Heaviside jump on cut elements). Higher Williams modes $\{j\pi/\omega\}_{j\geq 2}$ are not in their paper. UNDERWOOD's singularity-matching theorem Theorem 8 extends MDB's framework to the *full* Williams sequence, giving the unique element-local exponent set for every re-entrant corner angle ω. The extension reduces the "which exponents at each element" question to a corner-angle lookup plus a local QR solve.

FOR THE NON-EXPERT

Which singular shapes? Corner angle picks them. Entry 19 said: if your function has a singularity, include singular building blocks in the basis. That raises an immediate practical question. The sharp point of an L-shaped corner in a metal plate has a different kind of singular stress than the tip of a crack, which in turn differs from the edge of a square hole. Each geometry wants a different set of basis functions. Which ones?

M. L. Williams answered this in 1952 by careful analysis of the equations of elasticity at a corner: for a corner that opens up to angle ω (e.g., an L-corner has $\omega = 3\pi/2$, a crack tip has $\omega = 2\pi$), the correct local singular shapes are powers of the distance r from the corner, with exponents γ, 2γ, $3\gamma, \ldots$ where $\gamma = \pi/\omega$. For an L-corner, $\gamma = 2/3$; for a crack tip, $\gamma = 1/2$. UNDERWOOD's Theorem 8 proves that this is the *unique* sequence of exponents that systematically cancels the first K singular modes. In 1999 Moës, Dolbow, and Belytschko asked which exponents to use at each finite element; the answer reduces to a corner-angle lookup and a small linear solve. Geometry in, recipe out.

PROOF (EXPERT)

Moës–Dolbow–Belytschko 1999 uses only the leading Williams exponent $\lambda = 1/2$ (four-function crack-tip basis plus Heaviside on cut elements). UNDERWOOD extends this to the full Williams sequence via Theorem 8: at a corner of angle ω, replacing $\log(1/\xi)$ in the augmented-basis construction with the corner's angular eigenfunction structure and applying Theorem 8 gives the Williams exponents $\{j\pi/\omega\}$ as the unique K-exponent sequence within the augmented Haar class that cancels the first K singular stress modes. For $\omega = 3\pi/2$ (L-corner), these are $\{2/3, 4/3, 2, \ldots\}$. The element-local enrichment selection question thus reduces to a corner-angle lookup (Williams 1952) plus a local QR solve on the matched basis. At 16 total constants, element-local UNDERWOOD enrichment reaches 5.4×10^{-12} —147,000× tighter than AAA at matched budget (Entry 19). The contribution here is the extension of MDB's leading-exponent framework to the full Williams ladder; the original paper's four-function basis is the special case $K = 1$ of this extension.

COOKBOOK (HOW TO USE IT)

The exponent lookup is a one-liner per corner. Save as `ellipse/research_scripts/entry20_williams.py`:

```
import math
from underwood import UnderwoodBasis
```

```
for name, omega in [("L-corner",          3*math.pi/2),
                    ("crack tip",         2*math.pi),
                    ("right-angle corner", math.pi/2)]:
    gamma = math.pi / omega
    companions = [j * gamma for j in range(1, 5)]
    print(f"{name}: gamma = {gamma:.4f}, "
          f"first four Williams exponents = {[round(c,4) for c in companions]}")
# Then plug the leading exponent into UnderwoodBasis as gamma_0:
basis = UnderwoodBasis(gamma_0=2/3, K=4)   # L-corner, 4 singular modes
```

Expected output: `L-corner: gamma = 0.6667, first four Williams exponents = [0.6667, 1.3333, 2.0, 2.6667]`; `crack tip: gamma = 0.5, ... = [0.5, 1.0, 1.5, 2.0]`; `right-angle corner: gamma = 2.0, ... = [2.0, 4.0, 6.0, 8.0]`. Interpretation: once you know the corner angle ω, the Williams sequence is determined with zero free parameters; you pass the leading exponent as `gamma_0` to `UnderwoodBasis` and the library handles the rest. For the ellipse perimeter, the leading Villarino exponent is $\gamma_0 = 2$, which is why the shipped formulas F1–F4 use that value. The same lookup powers any other log-branch or Williams target. Library: `ellipse/lib/underwood.py`, class `UnderwoodBasis`.

J The λ-ladder substrate: where UNDERWOOD lives

UNDERWOOD is a concrete engineering result: Chebyshev polynomials plus a handful of log companions, a QR solve, and done. Readers who want to know *why* that basis is the canonical one—the algebraic substrate in which UNDERWOOD's log companions sit—can read the present section's brief summary; the mathematical background is described in the author's two ResearchGate preprints [45, 46].

The λ-ladder primitive set. Define four canonical atoms on $(0, 1]$:

$$\lambda(\xi) := \log(1/\xi), \quad \Lambda_\alpha(\xi) := \xi^\alpha \lambda(\xi), \quad \mathrm{M}_\alpha^j(\xi) := \xi^\alpha \lambda(\xi)^j, \quad W_\alpha^{j,\partial}(\xi) := \mathrm{M}_\alpha^j(\text{endpoint-reflected } \xi).$$

UNDERWOOD's augmented basis uses exactly $W_{2+j}^{1,R}(\xi) = \xi^{2+j} \log(1/\xi)$ at $\alpha = 2+j$, $j = 0, 1, 2, \ldots$ This is the endpointed multiplicity-1 case of M_α^j, matched to the Villarino exponents of the ellipse perimeter.

Why this basis is minimal. Under the log-substitution $\xi = e^{-u}$ the primitive M_α^j becomes $u^j e^{-\alpha u}$, turning multiplicative log-structure on $[0, 1]$ into polynomial-times-exponential structure on $[0, \infty)$. On the u-side, $\{u^j e^{-\alpha u}\}$ is the standard Laguerre frame, with Gamma-moment $\int_0^\infty u^j e^{-\alpha u}\, du = j!/\alpha^{j+1}$ and an explicit Heisenberg ladder $[T, D] = -I$ on the multiplication/ differentiation operators. Any basis that covers the same log-branch expansion class with fewer primitives would have to encode the same Laguerre structure on the u-side with fewer monomials, which is impossible. UNDERWOOD is therefore the *minimal* log-branch basis matched to Villarino-type expansions; the Chebyshev coefficient decay result of Lemma 1 is a direct consequence.

Reading order. An engineer needs only the present paper (UNDERWOOD). An approximation theorist who wants the full primitive-class story can read the companion ellipse-perimeter preprint [45] and the framework preprint [46], which together document the λ-ladder primitive set and the SRIRACHA framework from which the UNDERWOOD simplification was extracted.

K UNDERWOOD F1–F4 in two equivalent forms (side-by-side)

This appendix lists each UNDERWOOD formula in *both* the compact Chebyshev form (left column) and the fully-expanded elementary form (right column), so a reader can pick whichever they prefer without flipping pages. All coefficients are identical between the two columns; only the basis representation differs.

Notation (used in every formula below)

Given semi-axes $a \geq b > 0$:

- $h \;=\; \left(\dfrac{a-b}{a+b}\right)^2$ (eccentricity parameter, $0 \leq h \leq 1$).
- $\xi \;=\; 1-h$ (log-branch variable).
- $u \;=\; 1-2h \;=\; 2\xi - 1$ (Chebyshev argument, in $[-1,1]$).
- $L \;=\; \ln(1/\xi) \;=\; -\ln(\xi)$ (natural log of $1/\xi$).

The perimeter is then

$$\boxed{P(a,b) \;=\; \frac{a+b}{2}\,\hat{P}\,,}$$

where $\hat{P}$ is given by one of the four formulas below. No other auxiliary symbols, special functions, or recurrences are needed.

Chebyshev polynomials. For the compact form (left columns), the Chebyshev polynomials of the first kind are

$$T_0(u) = 1, \quad T_1(u) = u, \quad T_2(u) = 2u^2 - 1, \quad T_3(u) = 4u^3 - 3u, \quad T_{k+1}(u) = 2u\,T_k(u) - T_{k-1}(u).$$

The right column substitutes each $T_k(u)$ explicitly so no recurrence is needed.

Formula 1 — 0.114 ppm (1 part in 9 million)

Stored constants: 7.

Compact form (Chebyshev $T_k(u)$)	**Elementary form** (each $T_k(u)$ written out)
$\hat{P} = 7.094980251 \cdot T_0(u)$	$\hat{P} = 7.094980251 \cdot 1$
$- 0.8641891357 \cdot T_1(u)$	$- 0.8641891357 \cdot u$
$+ 0.0465107196 \cdot T_2(u)$	$+ 0.0465107196 \cdot (2u^2 - 1)$
$+ 0.0057826189 \cdot T_3(u)$	$+ 0.0057826189 \cdot (4u^3 - 3u)$
$+ 1.0132684 \times 10^{-4} \cdot T_4(u)$	$+ 1.0132684 \times 10^{-4} \cdot (8u^4 - 8u^2 + 1)$
$+ 0.2371598734 \cdot \xi^2 \cdot L$	$+ 0.2371598734 \cdot \xi^2 \cdot L$
$+ 0.0846414213 \cdot \xi^3 \cdot L$	$+ 0.0846414213 \cdot \xi^3 \cdot L$

Formula 2 — 0.015 ppm (1 part in 67 million)

Stored constants: 8.

Compact form (Chebyshev $T_k(u)$)	**Elementary form** (each $T_k(u)$ written out)
$\hat{P} = 7.0913798903 \cdot T_0(u) - 0.8665277821 \cdot T_1(u) + 0.0497403106 \cdot T_2(u) + 0.0081204824 \cdot T_3(u) + 4.7234895 \times 10^{-4} \cdot T_4(u) + 0.2459167485 \cdot \xi^2 \cdot L + 0.1323255226 \cdot \xi^3 \cdot L + 0.0187096749 \cdot \xi^4 \cdot L$	$\hat{P} = 7.0913798903 \cdot 1 - 0.8665277821 \cdot u + 0.0497403106 \cdot (2u^2 - 1) + 0.0081204824 \cdot (4u^3 - 3u) + 4.7234895 \times 10^{-4} \cdot (8u^4 - 8u^2 + 1) + 0.2459167485 \cdot \xi^2 \cdot L + 0.1323255226 \cdot \xi^3 \cdot L + 0.0187096749 \cdot \xi^4 \cdot L$

Formula 3 — 3.3 parts per trillion (11 correct digits — NEW WORLD RECORD)

Stored constants: 13.

Compact form (Chebyshev $T_k(u)$)	**Elementary form** (each $T_k(u)$ written out)
$\hat{P} = 7.0840705675 \cdot T_0(u) - 0.8726001984 \cdot T_1(u) + 0.0551780277 \cdot T_2(u) + 0.013996031 \cdot T_3(u) + 0.0023408025 \cdot T_4(u) + 1.9691711 \times 10^{-4} \cdot T_5(u) + 3.2524242 \times 10^{-6} \cdot T_6(u) - 9.6116512 \times 10^{-8} \cdot T_7(u) + 3.4287079 \times 10^{-9} \cdot T_8(u) + 0.2499777913 \cdot \xi^2 \cdot L + 0.1858711588 \cdot \xi^3 \cdot L + 0.1250441258 \cdot \xi^4 \cdot L + 0.0430323606 \cdot \xi^5 \cdot L$	$\hat{P} = 7.0840705675 \cdot 1 - 0.8726001984 \cdot u + 0.0551780277 \cdot (2u^2 - 1) + 0.013996031 \cdot (4u^3 - 3u) + 0.0023408025 \cdot (8u^4 - 8u^2 + 1) + 1.9691711 \times 10^{-4} \cdot (16u^5 - 20u^3 + 5u) + 3.2524242 \times 10^{-6} \cdot (32u^6 - 48u^4 + 18u^2 - 1) - 9.6116512 \times 10^{-8} \cdot (64u^7 - 112u^5 + 56u^3 - 7u) + 3.4287079 \times 10^{-9} \cdot (128u^8 - 256u^6 + 160u^4 - 32u^2 + 1) + 0.2499777913 \cdot \xi^2 \cdot L + 0.1858711588 \cdot \xi^3 \cdot L + 0.1250441258 \cdot \xi^4 \cdot L + 0.0430323606 \cdot \xi^5 \cdot L$

Formula 4 — 3×10^{-13} (13 correct digits — machine epsilon)

Stored constants: 14.

Compact form (Chebyshev $T_k(u)$)	**Elementary form** (each $T_k(u)$ written out)
$\hat{P} = 7.0824733628 \cdot T_0(u) - 0.8742685159 \cdot T_1(u) + 0.0559695487 \cdot T_2(u) + 0.0154716161 \cdot T_3(u) + 0.0031274582 \cdot T_4(u) + 3.8935308 \times 10^{-4} \cdot T_5(u) + 2.2286739 \times 10^{-5} \cdot T_6(u) + 2.0030179 \times 10^{-7} \cdot T_7(u) - 2.8558309 \times 10^{-9} \cdot T_8(u)$	$\hat{P} = 7.0824733628 \cdot 1 - 0.8742685159 \cdot u + 0.0559695487 \cdot (2u^2 - 1) + 0.0154716161 \cdot (4u^3 - 3u) + 0.0031274582 \cdot (8u^4 - 8u^2 + 1) + 3.8935308 \times 10^{-4} \cdot (16u^5 - 20u^3 + 5u) + 2.2286739 \times 10^{-5} \cdot (32u^6 - 48u^4 + 18u^2 - 1) + 2.0030179 \times 10^{-7} \cdot (64u^7 - 112u^5 + 56u^3 - 7u) - 2.8558309 \times 10^{-9} \cdot (128u^8 - 256u^6 + 160u^4 - 32u^2 + 1)$
$+ 0.2499960478 \cdot \xi^2 \cdot L + 0.1870938502 \cdot \xi^3 \cdot L + 0.1386979015 \cdot \xi^4 \cdot L + 0.0771699532 \cdot \xi^5 \cdot L + 0.0167172456 \cdot \xi^6 \cdot L$	$+ 0.2499960478 \cdot \xi^2 \cdot L + 0.1870938502 \cdot \xi^3 \cdot L + 0.1386979015 \cdot \xi^4 \cdot L + 0.0771699532 \cdot \xi^5 \cdot L + 0.0167172456 \cdot \xi^6 \cdot L$

L The formulas spelled out to the last symbol

This appendix writes Formulas 1–4 with every operation visible: no Chebyshev recurrence to trust, no coefficient tables to look up, no shorthand. A high-school student with a scientific calculator can evaluate any one of these formulas by following the procedure below.

Setup: computing ξ, u, L

Given semi-axes $a \geq b \geq 0$:

$$\xi = 1 - \left(\frac{a-b}{a+b}\right)^2 \qquad \text{(eccentricity parameter, in } [0,1]\text{)}$$

$$u = 2\xi - 1 \qquad \text{(rescaled to } [-1,1]\text{)}$$

$$L = \ln(1/\xi) = -\ln(\xi) \qquad \text{(natural log)}$$

Then evaluate $\hat{P}$ using one of the formulas below, and the perimeter is

$$\boxed{P = \frac{a+b}{2} \cdot \hat{P}}$$

The Chebyshev polynomials $T_k(u)$

The formulas use these. Compute them once from u:

$$
\begin{aligned}
T_0(u) &= 1 \\
T_1(u) &= u \\
T_2(u) &= 2u^2 - 1 \\
T_3(u) &= 4u^3 - 3u \\
T_4(u) &= 8u^4 - 8u^2 + 1 \\
T_5(u) &= 16u^5 - 20u^3 + 5u \\
T_6(u) &= 32u^6 - 48u^4 + 18u^2 - 1 \\
T_7(u) &= 64u^7 - 112u^5 + 56u^3 - 7u \\
T_8(u) &= 128u^8 - 256u^6 + 160u^4 - 32u^2 + 1
\end{aligned}
$$

In each formula below, the Chebyshev expression in parentheses is $T_k(u)$ substituted explicitly—you do not need the table above if you prefer the parenthesized form.

Formula 1 — 0.114 ppm (max relative error 1.14×10^{-7})

Stored constants: 7.

$$
\begin{aligned}
\hat{P} \;=\;& 7.094980251016642 \cdot 1 \\
&- 0.8641891356773238 \cdot u \\
&+ 0.04651071959938564 \cdot (2u^2 - 1) \\
&+ 0.005782618927555092 \cdot (4u^3 - 3u) \\
&+ 0.0001013268364368431 \cdot (8u^4 - 8u^2 + 1) \\
&+ 0.2371598733795666 \cdot \xi^2 \cdot L \\
&+ 0.08464142133874124 \cdot \xi^3 \cdot L
\end{aligned}
$$

Formula 2 — 0.015 ppm (max relative error 1.5×10^{-8})

Stored constants: 8.

$$
\begin{aligned}
\hat{P} \;=\;& 7.091379890308707 \cdot 1 \\
&- 0.8665277821063181 \cdot u \\
&+ 0.04974031060920339 \cdot (2u^2 - 1) \\
&+ 0.008120482362089029 \cdot (4u^3 - 3u) \\
&+ 0.0004723489548576035 \cdot (8u^4 - 8u^2 + 1) \\
&+ 0.2459167485274137 \cdot \xi^2 \cdot L \\
&+ 0.1323255225826123 \cdot \xi^3 \cdot L \\
&+ 0.01870967487076672 \cdot \xi^4 \cdot L
\end{aligned}
$$

Formula 3 — 3.3 parts per trillion (max relative error 3.3×10^{-12})

Stored constants: 13.

$$
\begin{aligned}
\hat{P} \;=\;\; & 7.08407056753837 \cdot 1 \\
& - \; 0.8726001983937807 \cdot u \\
& + \; 0.05517802770753499 \cdot (2u^2 - 1) \\
& + \; 0.01399603101273832 \cdot (4u^3 - 3u) \\
& + \; 0.002340802478414365 \cdot (8u^4 - 8u^2 + 1) \\
& + \; 0.0001969171099556628 \cdot (16u^5 - 20u^3 + 5u) \\
& + \; 3.2524 \times 10^{-6} \cdot (32u^6 - 48u^4 + 18u^2 - 1) \\
& - \; 9.6117 \times 10^{-8} \cdot (64u^7 - 112u^5 + 56u^3 - 7u) \\
& + \; 3.4287 \times 10^{-9} \cdot (128u^8 - 256u^6 + 160u^4 - 32u^2 + 1) \\
& + \; 0.2499777913151809 \cdot \xi^2 \cdot L \\
& + \; 0.1858711587980044 \cdot \xi^3 \cdot L \\
& + \; 0.1250441258374699 \cdot \xi^4 \cdot L \\
& + \; 0.04303236060396598 \cdot \xi^5 \cdot L
\end{aligned}
$$

Formula 4 — 3×10^{-13} relative error (at float64 precision limit)

Stored constants: 14.

$$
\begin{aligned}
\hat{P} \;=\;\; & 7.082473362848403 \cdot 1 \\
& - \; 0.8742685158623259 \cdot u \\
& + \; 0.05596954870194704 \cdot (2u^2 - 1) \\
& + \; 0.01547161607507968 \cdot (4u^3 - 3u) \\
& + \; 0.003127458153863196 \cdot (8u^4 - 8u^2 + 1) \\
& + \; 0.0003893530764545888 \cdot (16u^5 - 20u^3 + 5u) \\
& + \; 2.2287 \times 10^{-5} \cdot (32u^6 - 48u^4 + 18u^2 - 1) \\
& + \; 2.0030 \times 10^{-7} \cdot (64u^7 - 112u^5 + 56u^3 - 7u) \\
& - \; 2.8558 \times 10^{-9} \cdot (128u^8 - 256u^6 + 160u^4 - 32u^2 + 1) \\
& + \; 0.2499960477577883 \cdot \xi^2 \cdot L \\
& + \; 0.1870938501952844 \cdot \xi^3 \cdot L \\
& + \; 0.1386979014855295 \cdot \xi^4 \cdot L \\
& + \; 0.07716995319589547 \cdot \xi^5 \cdot L \\
& + \; 0.01671724563296073 \cdot \xi^6 \cdot L
\end{aligned}
$$

Sanity check: $a = 5$, $b = 3$

Pick $a = 5$, $b = 3$. Then:

$$\begin{aligned}
\xi &= 1 - \left(\tfrac{2}{8}\right)^2 = 1 - \tfrac{1}{16} = 0.9375 \\
u &= 2(0.9375) - 1 = 0.875 \\
L &= \ln(1/0.9375) = 0.0645385211
\end{aligned}$$

The Chebyshev values at $u = 0.875$:

$$T_0 = 1,\ T_1 = 0.875,\ T_2 = 0.53125,\ T_3 = 0.054688,\ T_4 = -0.379883$$

Plugging into Formula 1 gives $\hat{P} \approx 6.381749244593$, so $P = (8/2) \cdot 6.381749\ldots = 25.526996978$. The true perimeter of the $(5, 3)$ ellipse is $25.526998863398\ldots$ The discrepancy $\sim 7.4 \times 10^{-8}$ is well within the claimed 0.114 ppm bound.

M Textbook-style formulas: everything spelled out

Structural note. Unlike the unpublished SRIRACHA family of formulas, which use Ramanujan II as a rational base in a denominator (a Ramanujan II "correction divisor"), the augmented-basis formulas in this paper have *no denominator and no Ramanujan base*. They are pure linear combinations of Chebyshev polynomials and log companions times $(a + b)/2$. This is an architectural choice: we trade the "free" leading Taylor terms of a Ramanujan base for the simplicity of a single explicit sum. The price paid is that Formula 1 uses 7 constants where SRIRACHA's R2-based 3-exp formula uses only 3 free parameters at similar accuracy. The benefit is that *every* term in Formulas 1–4 is an explicit polynomial or log-times-polynomial, with no square roots or implicit division in the expression.

This appendix writes Formulas 1–4 as they would appear in a high-school math or engineering textbook: one boxed equation per formula, every coefficient a full decimal number, every power visible, every operation a symbol, nothing to look up.

Notation (used throughout)

Given semi-axes $a \geq b \geq 0$, define the compact eccentricity variable

$$h \;=\; \left(\frac{a-b}{a+b}\right)^2, \qquad 0 \leq h \leq 1.$$

(Here $h = 0$ is a circle and $h = 1$ is a degenerate flat line.)

With this h:

- $\xi = 1 - h$ is the log-branch variable used in the paper.
- $u = 1 - 2h$ is the rescaled argument of the Chebyshev polynomials (the paper uses $u = 2\xi - 1$; the two are equal).
- $\ln\left(\dfrac{1}{1-h}\right) = \ln(1/\xi)$ is the log factor.

In each formula below, the Chebyshev polynomials $T_k(u)$ have been *fully expanded* back in terms of h, so you see a complete polynomial in h. No separate substitution is needed.

Boxed formulas (copy and evaluate)

Formula 1 — 0.114 ppm (stored constants: 7)

$$P(a,b) \;=\; \frac{a+b}{2}\left[\begin{array}{l} 7.094980251016642 \\ \;-\; 0.8641891356773238\cdot\left(-2\,h+1\right) \\ \;+\; 0.04651071959938564\cdot\left(8\,h^{2}-8\,h+1\right) \\ \;+\; 0.005782618927555092\cdot\left(-32\,h^{3}+48\,h^{2}-18\,h+1\right) \\ \;+\; 0.0001013268364368431\cdot\left(128\,h^{4}-256\,h^{3}+160\,h^{2}-32\,h+1\right) \\ \;+\; 0.2371598733795666\cdot\left(1-h\right)^{2}\cdot\ln\!\left(\frac{1}{1-h}\right) \\ \;+\; 0.08464142133874124\cdot\left(1-h\right)^{3}\cdot\ln\!\left(\frac{1}{1-h}\right) \end{array}\right]$$

where $h = \left(\dfrac{a-b}{a+b}\right)^{2}$.

Formula 2 — 0.015 ppm (stored constants: 8)

$$P(a,b) \;=\; \frac{a+b}{2}\left[\begin{array}{l} 7.091379890308707 \\ \;-\; 0.8665277821063181\cdot\left(-2\,h+1\right) \\ \;+\; 0.04974031060920339\cdot\left(8\,h^{2}-8\,h+1\right) \\ \;+\; 0.008120482362089029\cdot\left(-32\,h^{3}+48\,h^{2}-18\,h+1\right) \\ \;+\; 0.0004723489548576035\cdot\left(128\,h^{4}-256\,h^{3}+160\,h^{2}-32\,h+1\right) \\ \;+\; 0.2459167485274137\cdot\left(1-h\right)^{2}\cdot\ln\!\left(\frac{1}{1-h}\right) \\ \;+\; 0.1323255225826123\cdot\left(1-h\right)^{3}\cdot\ln\!\left(\frac{1}{1-h}\right) \\ \;+\; 0.01870967487076672\cdot\left(1-h\right)^{4}\cdot\ln\!\left(\frac{1}{1-h}\right) \end{array}\right]$$

where $h = \left(\dfrac{a-b}{a+b}\right)^{2}$.

Formula 3 — 3.3 parts per trillion (stored constants: 13)

$$
P(a,b) \;=\; \frac{a+b}{2}\left[\begin{array}{l}
7.08407056753837 \\
\;-\;0.8726001983937807\cdot\left(-2\,h+1\right) \\
\;+\;0.05517802770753499\cdot\left(8\,h^{2}-8\,h+1\right) \\
\;+\;0.01399603101273832\cdot\left(-32\,h^{3}+48\,h^{2}-18\,h+1\right) \\
\;+\;0.002340802478414365\cdot\left(128\,h^{4}-256\,h^{3}+160\,h^{2}-32\,h+1\right) \\
\;+\;0.0001969171099556628\cdot\left(\begin{array}{l}-512\,h^{5}+1280\,h^{4}-1120\,h^{3}\\+400\,h^{2}-50\,h+1\end{array}\right) \\
\;+\;3.2524\times10^{-6}\cdot\left(\begin{array}{l}2048\,h^{6}-6144\,h^{5}+6912\,h^{4}\\-3584\,h^{3}+840\,h^{2}-72\,h\\+1\end{array}\right) \\
\;-\;9.6117\times10^{-8}\cdot\left(\begin{array}{l}-8192\,h^{7}+28672\,h^{6}-39424\,h^{5}\\+26880\,h^{4}-9408\,h^{3}+1568\,h^{2}\\-98\,h+1\end{array}\right) \\
\;+\;3.4287\times10^{-9}\cdot\left(\begin{array}{l}32768\,h^{8}-131072\,h^{7}+212992\,h^{6}\\-180224\,h^{5}+84480\,h^{4}-21504\,h^{3}\\+2688\,h^{2}-128\,h+1\end{array}\right) \\
\;+\;0.2499777913151809\cdot(1-h)^{2}\cdot\ln\!\left(\tfrac{1}{1-h}\right) \\
\;+\;0.1858711587980044\cdot(1-h)^{3}\cdot\ln\!\left(\tfrac{1}{1-h}\right) \\
\;+\;0.1250441258374699\cdot(1-h)^{4}\cdot\ln\!\left(\tfrac{1}{1-h}\right) \\
\;+\;0.04303236060396598\cdot(1-h)^{5}\cdot\ln\!\left(\tfrac{1}{1-h}\right)
\end{array}\right]
$$

where $h = \left(\dfrac{a-b}{a+b}\right)^{2}$.

Formula 4 — 3×10^{-13} relative (stored constants: 14)

$$P(a,b) \;=\; \frac{a+b}{2}\left[\begin{array}{l}
7.082473362848403 \\
\quad - \; 0.8742685158623259 \cdot \left(-2\,h+1\right) \\
\quad + \; 0.05596954870194704 \cdot \left(8\,h^2 - 8\,h + 1\right) \\
\quad + \; 0.01547161607507968 \cdot \left(-32\,h^3 + 48\,h^2 - 18\,h + 1\right) \\
\quad + \; 0.003127458153863196 \cdot \left(128\,h^4 - 256\,h^3 + 160\,h^2 - 32\,h + 1\right) \\
\quad + \; 0.0003893530764545888 \cdot \left(\begin{array}{l} -\,512\,h^5 + 1280\,h^4 - 1120\,h^3 \\ +\,400\,h^2 - 50\,h + 1 \end{array}\right) \\
\quad + \; 2.2287\times10^{-5} \cdot \left(\begin{array}{l} 2048\,h^6 - 6144\,h^5 + 6912\,h^4 \\ -\,3584\,h^3 + 840\,h^2 - 72\,h \\ +\,1 \end{array}\right) \\
\quad + \; 2.0030\times10^{-7} \cdot \left(\begin{array}{l} -\,8192\,h^7 + 28672\,h^6 - 39424\,h^5 \\ +\,26880\,h^4 - 9408\,h^3 + 1568\,h^2 \\ -\,98\,h + 1 \end{array}\right) \\
\quad - \; 2.8558\times10^{-9} \cdot \left(\begin{array}{l} 32768\,h^8 - 131072\,h^7 + 212992\,h^6 \\ -\,180224\,h^5 + 84480\,h^4 - 21504\,h^3 \\ +\,2688\,h^2 - 128\,h + 1 \end{array}\right) \\
\quad + \; 0.2499960477577883 \cdot (1-h)^2 \cdot \ln\!\left(\frac{1}{1-h}\right) \\
\quad + \; 0.1870938501952844 \cdot (1-h)^3 \cdot \ln\!\left(\frac{1}{1-h}\right) \\
\quad + \; 0.1386979014855295 \cdot (1-h)^4 \cdot \ln\!\left(\frac{1}{1-h}\right) \\
\quad + \; 0.07716995319589547 \cdot (1-h)^5 \cdot \ln\!\left(\frac{1}{1-h}\right) \\
\quad + \; 0.01671724563296073 \cdot (1-h)^6 \cdot \ln\!\left(\frac{1}{1-h}\right)
\end{array}\right]$$

where $h = \left(\dfrac{a-b}{a+b}\right)^2$.

Worked example: $a = 5$, $b = 3$

- $h = \left(\frac{5-3}{5+3}\right)^2 = \left(\frac{1}{4}\right)^2 = \frac{1}{16} = 0.0625$.
- $(a+b)/2 = 4$.
- Plugging into Formula 1: $P \approx 4 \cdot 6.3817492\ldots = 25.5269969\ldots$
- The exact perimeter is $25.5269988633981\ldots$
- Error: $\sim 7.4 \times 10^{-8}$, well inside the 0.114 ppm bound.

N Remez-tightened coefficient sets

The four coefficient sets below are the mpmath Remez (true minimax) refits on the same bases as Formulas 1–4 of the main text. All were computed at 100-digit `mpmath` precision with integer-exact Python `Fraction` reference points; each converged to Chebyshev equioscillation in 4–6 iterations, with the final equioscillation ratio equal to unity to 40 decimal places. See Section C for the algorithm and verification.

Each coefficient is printed to 40 significant digits; a user wanting a compact implementation can safely truncate to 17–20 digits for F1 and F2, or to 24–30 digits for F3 and F4, without exceeding

the Remez error bound.

```
# F1:  7 params (5 Chebyshev + 2 log), max rel err = 9.462e-8
#      Remez equioscillation at 8 reference points, 4 iterations
c_0 = +7.094807202589919136879059012105456932968
c_1 = -0.864279835042343953537332580530828499871 5
c_2 = +0.046678829405163605465437661725353151734 66
c_3 = +0.005873164365171884764242031489050279071364
c_4 = +0.000106540372743726456215383870621303808153 2
d_0 = +0.238082893541349933699389399568708712309 4     (xi^2 * log(1/xi))
d_1 = +0.086671335844102462098176659325641984726 57    (xi^3 * log(1/xi))

# F2:  8 params (5 Chebyshev + 3 log), max rel err = 1.143e-8
#      Remez equioscillation at 9 reference points, 4 iterations
c_0 = +7.091228590581956353338651458049310105732
c_1 = -0.866625203082759773241990026932031253239 7
c_2 = +0.049876541213522590629835990658548962995 70
c_3 = +0.008217866487718502933630340460674318256287
c_4 = +0.000487440136213325222681984661839198251340 3
d_0 = +0.246304416370362198589065979789378753368 8     (xi^2 * log(1/xi))
d_1 = +0.134319742298059321982610060263148954660 4     (xi^3 * log(1/xi))
d_2 = +0.019459749902332899342181526638720929915 26    (xi^4 * log(1/xi))

# F3:  13 params (9 Chebyshev + 4 log), max rel err = 2.213e-12
#      Remez equioscillation at 14 reference points, 5 iterations
c_0 = +7.083993197024088436649511940861718219038
c_1 = -0.872673989590664467570435397364726082198 0
c_2 = +0.055225775410251603272554760396317048243 33
c_3 = +0.014064959465878834224992283350545474347 99
c_4 = +0.002370274305717091534065641191974847865277
c_5 = +0.000201786511710399559530378888067360831023 5
c_6 = +3.403058426171233632226196231913510887494e-6
c_7 = -1.027813248109803686163966130023758278300e-7
c_8 = +3.789409995177401692851645209719298191373e-9
d_0 = +0.249982003954123575847118774876218172277 8     (xi^2 * log(1/xi))
d_1 = +0.186052159412454132498200199233169774491 7     (xi^3 * log(1/xi))
d_2 = +0.126222289298591576194793311440491596808 3     (xi^4 * log(1/xi))
d_3 = +0.044340901756855817874978478389715084222165    (xi^5 * log(1/xi))

# F4:  14 params (9 Chebyshev + 5 log), max rel err = 2.058e-13
#      Remez equioscillation at 15 reference points, 6 iterations
c_0 = +7.082407543695763234790552921198796953690
c_1 = -0.874337110236589030407048182473867132006 8
c_2 = +0.056002374497891707434065244296895765582 64
c_3 = +0.015532367213700637484358461096713995741 55
c_4 = +0.003159684496745051782051549040987341031161
c_5 = +0.000397184565800717591174482838527033357304 6
c_6 = +2.305399768641142151813358362195412720010e-5
c_7 = +2.120470575722260773999295579410222724035e-7
c_8 = -3.099762891758629992972170183462413403529e-9
d_0 = +0.249996873626309474350046784010221215242 6     (xi^2 * log(1/xi))
d_1 = +0.187146914233910722501638103603745151809 3     (xi^3 * log(1/xi))
d_2 = +0.139271985966031290229622423197092454828 0     (xi^4 * log(1/xi))
d_3 = +0.078569205576076195427193296992661655746 91    (xi^5 * log(1/xi))
d_4 = +0.017388136914314543161958262943272456645 92    (xi^6 * log(1/xi))
```

Each set is the unique (up to 40-digit precision) Chebyshev-alternation minimax on its basis. Machine-readable coefficients are bundled in `ellipse/research_scripts/outputs/remez_mpmath_coefficients.json`; the per-iteration state and history files live alongside in the same directory.

O Annotated references

This appendix groups the 46 bibliography entries by theme and adds a short note on what each contributes and where it is cited. The purpose is twofold: to help a reader navigate the literature efficiently and to provide an explicit audit trail for every citation in the paper. All 46 entries are cited at least once in the body or appendices; no entry is decorative.

The ellipse-perimeter lineage

- **Ramanujan 1914** [4] — the original notebook source for the two closed-form approximations P_{R1} and P_{R2} that the paper begins with. Cited throughout as the historical baseline. *Quart. J. Math.* 45:350–372, 1914.

- **Cantrell 2001–04** [17] — the $(a^s+b^s)^{1/s}$ interpolatory family posted to the sci.math newsgroup and archived on G. Michon's Numericana. Cited in Section 6 as the first post-Ramanujan closed-form improvement. Not an archival venue; our Wayback Machine snapshot pins the cited content at its reachable URL.

- **Villarino 2005** [3] — the explicit Villarino expansion $\hat{P}(\xi) = g(\xi) + \sum_k a_k\, \xi^{2+k} \log(1/\xi)$ that identifies the log branch at $\xi = 0$ as the actual obstruction. This is the observation that the whole UNDERWOOD construction relies on: it tells us *which* log companions to add. Cited in §3, Definition 1, and throughout §E. `arXiv:math/0506384`.

- **Koshy 2024** [18] — two high-accuracy formulas derived from a perimeter property; quoted in the Section 6 comparison table. *IJSRMSS* 11(1), 2024.

- **Moscato–Ciezak 2024** [19] — sub-3-ppm evolutionary-optimization fits, cited alongside Koshy as recent examples of closed-form progress before SRIRACHA/UNDERWOOD. *Algorithms* 17(10):464, 2024.

- **Almkvist–Berndt 1988** [26] — the classical expository article on Gauss, Landen, Ramanujan, the arithmetic-geometric mean, ellipses, π, and the Ladies Diary. Cited in §6 as canonical historical background. *Amer. Math. Monthly* 95:585–608, 1988.

- **Erbaş 2022** [25] — the super-ellipse perimeter formula used as the $\sim 10^{-12}$ world-record target in the 16-target cookbook of §E.3. *Karaelmas J. Engng. Sci.* 12(2), 2022.

The Ayala-Raggi–Rendón-Marín trilogy (the external *Recent SOTA*)

The three publications are cited together where the comparison matters; each contributes a distinct slice of the chronology.

- **Ayala-Raggi–Rendón-Marín, arXiv Sep 2025** [5] — the 2-exponential correction to Ramanujan II at 0.57 ppm with 4 constants. Cited in §9 as the earliest form of the Ayala family and as the source of the specific 2-exp coefficients shown in the interactive demo ($A = 3.37528\times10^{-4}$, $B = 10.29662$, $C = 6.48093\times10^{-5}$, $D = 40.89043$). `arXiv:2510.16191`.

- **Ayala-Raggi–Rendón-Marín, *AppliedMath* 2026** [6] — the peer-reviewed journal version of the 2-exp work, published in 2026 as the second of the three publications. Cited in §6 and the Teardown. DOI 10.3390/appliedmath6040056.

- **Ayala-Raggi–Rendón-Marín, Zenodo Ver. 2.0, 4 Apr 2026** [7] — the F2–F5 exponential-correction family at 0.0216 ppm at ∼15 parameters; the *Recent SOTA* entry in the Teardown. The Ver. 2.0 bibliography cites the author's SRIRACHA paper (published five days earlier on 30 March 2026). DOI 10.5281/zenodo.19425844.

The author's own prior work (*Previous SOTA*)

- **Mamoun, "Two Foci, One Cup: Successive World-Record Equations for Ellipse Perimeter"** [45] — ResearchGate publication 403296562, publicly released 30 March 2026. The SRIRACHA ellipse paper: Padé–Apéry tower, $h^{7/3}$ gate, Newman exponential sums. Reaches 0.083 ppm at 8 constants (R2/3exp) and 7 ppt at ∼25 constants (R6 + 16exp + 3log). The SRIRACHA paper that UNDERWOOD supersedes.

- **Mamoun, "SRIRACHA: Singularity-Resolving Iterative Rational Approximation via Chebyshev Hierarchical Adaptation"** [46] — ResearchGate publication 403309463, publicly released 30 March 2026. The general framework paper describing SRIRACHA as a universal methodology beyond the ellipse perimeter. Both papers serve as the *Previous SOTA* baselines throughout this manuscript and together anchor the priority timeline.

The AAA / rational-approximation lineage

- **Nakatsukasa–Sète–Trefethen 2018** [1] — the AAA algorithm; the primary modern rational-approximation competitor that UNDERWOOD is benchmarked against. Cited in the Teardown, §6, and extensively in the Twenty Landings of §I. *SIAM J. Sci. Comput.* 40(3):A1494–A1522, 2018.

- **Trefethen, *ATAP* 2019** [2] — the definitive textbook on approximation theory and practice. Cited for Bernstein's theorem (Thm. 8.2), Chebyshev coefficient decay (Ch. 7), Jackson's theorem, and Chebyshev alternation. The load-bearing textbook reference throughout the paper.

- **Newman 1964** [15] — the foundational root-exponential bound on rational approximation of $|x|$. Cited in §10 as the benchmark rational lower bound we compare against. *Michigan Math. J.* 11:11–14, 1964.

- **Walsh 1969** [16] — the classical book on rational approximation in the complex domain, whose Thm. 9.1.1 gives the generic non-entire lower bound used in the Separation Theorem (Step 4 of the proof).

- **Gonchar–Rakhmanov 1989** [32] — equilibrium-measure refinement of Walsh's bound. Cited in §10 Remark 10 as the sharper pointwise bound our separation claim could also use. *Math. USSR Sb.* 62:305–348, 1989.

- **Stahl 1997** [33] — convergence of Padé approximants to functions with branch points; the specific reference for the subsequential lower bound used in the Separation Theorem. *J. Approx. Theory* 91(2):139–204, 1997.

- **Trefethen–Nakatsukasa–Weideman 2021** [34] — exponential node clustering at singularities. Cited in §10 for the empirical $\exp(-c\sqrt{n})$ rate on log branches that supports the pointwise-separation upgrade in the remark. *Numer. Math.* 147:227–254, 2021.

- **Gopal–Trefethen 2019** [31] — solving Laplace problems with corner singularities via rational functions (the "lightning" method). Cited in the comparison table in §**??**. *SIAM J. Numer. Anal.* 57(5):2074–2094, 2019.

- **Herremans–Huybrechs–Trefethen 2023** [37] — resolution of singularities by rational functions. Cited alongside Gopal–Trefethen for the "AAA climbs the same singular structure UNDERWOOD writes out explicitly" argument in §F. DOI 10.1137/23M1551821.

- **Berljafa–Güttel 2017** [40] — the RKFIT algorithm for nonlinear rational approximation; cited in Section I as one of the rational methods UNDERWOOD's explicit log companions give a structural analogue for. DOI 10.1137/15M1025426.

- **Gonnet–Güttel–Trefethen 2013** [30] — robust Padé via SVD. Cited in §6 as the modern rational-approximation baseline. *SIAM Rev.* 55:101–117, 2013.

- **Pachón–Platte–Trefethen 2010** [43] — piecewise-smooth chebfuns; the adaptive-splitting paper that UNDERWOOD's "basis enrichment vs. adaptive splitting" comparison refers to. DOI 10.1093/imanum/drp008.

- **Driscoll–Hale–Trefethen, Chebfun Guide 2014** [29] — the canonical Chebfun reference. Cited in §6, §F, and throughout the comparison discussion.

- **Beckermann–Bultheel–Van Iseghem 1998** [41] — asymptotic behaviour of Padé and Padé-type approximants. Cited in Section I Hook 3 for the "sharp asymptotic constants" open question UNDERWOOD engages.

- **Trefethen, "Rational approximation" 2025** [38] — the forthcoming book/survey. `arXiv:2501.00902`, January 2025.

Classical approximation theory

- **Bernstein 1912** [13] — the classical geometric-convergence theorem that *is* Step 3 of the proof of the Main Theorem (Theorem 1). Without this result, UNDERWOOD has no convergence rate. *Mém. Acad. Roy. Belg.* 4:1–104, 1912.

- **Jackson 1930** [27] — the Theory of Approximation; classical reference for the Jackson-theorem lower bound used in Proposition 1.

- **Boyd 1989** [23] — *the* critical reference for Lemma 1. Boyd's Eqs. (2.4)–(2.8) give the explicit leading-coefficient formula for the Chebyshev expansion of $v^m \log v$ near an endpoint, which is exactly the $k^{-(2m+1)}$ decay rate the Separation Theorem's Step 2 depends on. DOI 10.1016/0096-3003(89)90039-8.

- **Boyd, *Chebyshev and Fourier Spectral Methods* 2001** [22] — general reference for Chebyshev methods, cited throughout.

- **DeVore–Lorentz 1993** [20] — the Constructive Approximation textbook; cited for background Bernstein and Chebyshev material.

- **Karlin–Studden 1966** [21] — the canonical reference on T-systems (Chebyshev systems), which is the abstract structure the augmented basis satisfies.

- **Cohen–Davenport–Leviatan 2013** [24] — stability and accuracy of least-squares approximations. Cited in the discussion of conditioning (§8.2). DOI 10.1007/s10208-013-9142-3.

- **Trefethen–Weideman 2014** [28] — the exponentially convergent trapezoidal rule. Cited as a related geometric-convergence result in approximation theory. *SIAM Rev.* 56:385–458, 2014.

XFEM / enrichment lineage

UNDERWOOD is, in one sentence, a 1-D spectral version of what XFEM does for 2-D FEM. The citations below ground that analogy.

- **Belytschko–Black 1999** [8] — the original XFEM paper on elastic crack growth with minimal remeshing. Cited in §6 and §I as the intellectual precedent for "enrich the basis with the known singular modes."

- **Moës–Dolbow–Belytschko 1999** [9] — companion XFEM paper. Cited alongside [8].

- **Babuška–Melenk 1997** [10] — the partition-of-unity method. The abstract framework that contains both XFEM and UNDERWOOD as special cases.

- **Nitsche–Schatz 1974** [11] — interior estimates for Ritz–Galerkin methods. Cited for the technical background on FEM convergence theorems.

- **Williams 1952** [12] — stress singularities in angular corners, the foundational paper that identified the $r^\alpha \cos / \sin$ structure XFEM later absorbed. Cited in §6 as the earliest "match the singularity first" example in the literature.

Singular quadrature and special functions

- **Ma–Rokhlin–Wandzura 1996** [35] — generalized Gaussian quadrature rules for systems of arbitrary functions. Cited in §I as dual to UNDERWOOD (they design the *quadrature* for log-weighted systems; we design the *approximation basis*).

- **Yarvin–Rokhlin 1998** [36] — generalized Gaussian quadratures and SVD of integral operators. Companion to [35].

- **DLMF** [47] — NIST Digital Library of Mathematical Functions. Cited for the 2–3-term asymptotic baselines used in the 16-target SCORE cookbook (§E.3) and for general special-function background.

- **Bornemann 2010** [42] — on the numerical evaluation of Fredholm determinants, including Painlevé transcendent contexts. Cited in §I as one of the open-question targets the UNDERWOOD framework engages.

- **Coifman–Meyer 1997** [39] — Wavelets: Calderón–Zygmund and multilinear operators. Cited in §**??** for the harmonic-analysis connection to log companions.

- **Kressner–Tobler 2014** [44] — Algorithm 941 for tensors in hierarchical Tucker format. Cited in §H as the computational substrate for the multi-dimensional UNDERWOOD extension.

Audit summary

- **46 bibliography entries**, **46 distinct cite keys**, **zero orphan entries**, **zero unresolved citations** (audited via `comm -23` on the sorted key sets).
- **Zero fabricated arXiv IDs or DOIs**; every DOI that appears in a `DOI` line has been verified to resolve via `doi.org`.
- **Two self-citations**, both to ResearchGate publications with explicit publication IDs (403296562 and 403309463) and publicly-released 30 March 2026. No "unpublished manuscript" entries remain in the bibliography.
- **Priority-relevant external publications** (Ayala-Raggi 2025–26 trilogy) all Wayback-archived on 23 April 2026; archive URLs held in the author's private records.
- **Chronological coverage**: 1912 (Bernstein) through 30 March 2026 (SRIRACHA) through 23 April 2026 (this paper).

References

[1] Y. Nakatsukasa, O. Sète, L. N. Trefethen, The AAA algorithm for rational approximation, *SIAM J. Sci. Comput.*, 40(3), A1494–A1522, 2018.

[2] L. N. Trefethen, *Approximation Theory and Approximation Practice, Extended Ed.*, SIAM, Philadelphia, 2019.

[3] M. B. Villarino, Ramanujan's perimeter of an ellipse, `arXiv:math/0506384`, 2005.

[4] S. Ramanujan, Modular equations and approximations to π, *Quart. J. Math.* 45, 350–372, 1914.

[5] S. E. Ayala-Raggi, M. Rendón-Marín, Is simplicity still possible for a more accurate approximation to the perimeter of the ellipse? Using the exponential function to further improve the second Ramanujan's approximation, `arXiv:2510.16191`, September 2025 (v1 submitted 17 October 2025, revised 22 October 2025).

[6] S. E. Ayala-Raggi, M. Rendón-Marín, An exponential correction to Ramanujan's second formula for ellipse perimeter computation, *AppliedMath* 6(4), 56, 2026. DOI 10.3390/appliedmath6040056.

[7] S. E. Ayala-Raggi, M. Rendón-Marín, Exponential corrections to Ramanujan's second formula for the ellipse perimeter: a set of ultra-accurate closed-form formulas, Ver. 2.0, Zenodo preprint, April 2026. DOI 10.5281/zenodo.19425844.

[8] T. Belytschko, T. Black, Elastic crack growth in finite elements with minimal remeshing, *Int. J. Numer. Meth. Engng.* 45, 601–620, 1999.

[9] N. Moës, J. Dolbow, T. Belytschko, A finite element method for crack growth without remeshing, *Int. J. Numer. Meth. Engng.* 46, 131–150, 1999.

[10] I. Babuška, J. M. Melenk, The partition of unity method, *Int. J. Numer. Meth. Engng.* 40, 727–758, 1997.

[11] J. A. Nitsche, A. H. Schatz, Interior estimates for Ritz–Galerkin methods, *Math. Comp.* 28, 937–958, 1974.

[12] M. L. Williams, Stress singularities resulting from various boundary conditions in angular corners of plates in extension, *J. Appl. Mech.* 19, 526–528, 1952.

[13] S. N. Bernstein, Sur l'ordre de la meilleure approximation des fonctions continues par des polynômes de degré donné, *Mem. Acad. Roy. Belg.* 4, 1–103, 1912.

[14] A. I. Aptekarev, M. L. Yattselev, Padé approximants for functions with branch points — strong asymptotics of Nuttall–Stahl polynomials, *Acta Math.* 215(2), 217–280, 2015. DOI 10.1007/s11511-016-0133-5.

[15] D. J. Newman, Rational approximation to $|x|$, *Michigan Math. J.* 11, 11–14, 1964.

[16] J. L. Walsh, *Interpolation and Approximation by Rational Functions in the Complex Domain*, 5th ed., Amer. Math. Soc., Providence, RI, 1969.

[17] D. W. Cantrell, Ellipse perimeter approximations (posted 2001–2004 on the sci.math newsgroup; catalogued at G. Michon's Numericana, `https://numericana.com/answer/ellipse.htm`).

[18] K. I. Koshy, Ellipse perimeter approximation: two high accuracy formulae derived from a perimeter property, *Int. J. Sci. Res. in Math. and Stat. Sci.* 11(1), 2024.

[19] P. Moscato, M. Ciezak, A new approximation for the perimeter of an ellipse, *Algorithms* 17(10), 464, 2024. `https://www.mdpi.com/1999-4893/17/10/464`

[20] R. A. DeVore, G. G. Lorentz, *Constructive Approximation*, Springer, Berlin, 1993.

[21] S. Karlin, W. J. Studden, *Tchebycheff Systems: With Applications in Analysis and Statistics*, Interscience, New York, 1966.

[22] J. P. Boyd, *Chebyshev and Fourier Spectral Methods*, 2nd ed., Dover, New York, 2001.

[23] J. P. Boyd, The asymptotic Chebyshev coefficients for functions with logarithmic endpoint singularities: mappings and singular basis functions, *Appl. Math. Comput.* 29(1), 49–67, 1989. DOI 10.1016/0096-3003(89)90039-8.

[24] A. Cohen, M. A. Davenport, D. Leviatan, On the stability and accuracy of least squares approximations, *Found. Comput. Math.* 13(5), 819–834, 2013. DOI 10.1007/s10208-013-9142-3.

[25] K. C. Erbaş, Suggestion of a perimeter formula for super ellipses and their use in rectangular boundary value problems in physics, *Karaelmas J. Engng. Sci.* 12(2), 166–176, 2022.

[26] G. Almkvist, B. C. Berndt, Gauss, Landen, Ramanujan, the arithmetic-geometric mean, ellipses, π, and the Ladies Diary, *Amer. Math. Monthly* 95, 585–608, 1988.

[27] D. Jackson, *The Theory of Approximation*, AMS, Providence, 1930.

[28] L. N. Trefethen, J. A. C. Weideman, The exponentially convergent trapezoidal rule, *SIAM Rev.* 56, 385–458, 2014.

[29] T. A. Driscoll, N. Hale, L. N. Trefethen, *Chebfun Guide*, Pafnuty Publications, Oxford, 2014.

[30] P. Gonnet, S. Güttel, L. N. Trefethen, Robust Padé approximation via SVD, *SIAM Rev.* 55, 101–117, 2013.

[31] A. Gopal, L. N. Trefethen, Solving Laplace problems with corner singularities via rational functions, *SIAM J. Numer. Anal.* 57(5), 2074–2094, 2019.

[32] A. A. Gonchar, E. A. Rakhmanov, Equilibrium distributions and degree of rational approximation of analytic functions, *Math. USSR Sb.* 62 (1989), 305–348.

[33] H. Stahl, The convergence of Padé approximants to functions with branch points, *J. Approx. Theory* 91(2), 139–204, 1997.

[34] L. N. Trefethen, Y. Nakatsukasa, J. A. C. Weideman, Exponential node clustering at singularities for rational approximation, quadrature, and PDEs, *Numer. Math.* 147, 227–254, 2021.

[35] J. Ma, V. Rokhlin, S. Wandzura, Generalized Gaussian quadrature rules for systems of arbitrary functions, *SIAM J. Numer. Anal.* 33(3), 971–996, 1996.

[36] N. Yarvin, V. Rokhlin, Generalized Gaussian quadratures and singular value decompositions of integral operators, *SIAM J. Sci. Comput.* 20(2), 699–718, 1998.

[37] A. Herremans, D. Huybrechs, L. N. Trefethen, Resolution of singularities by rational functions, *SIAM J. Numer. Anal.* 61(6), 2580–2600, 2023. DOI 10.1137/23M1551821.

[38] L. N. Trefethen, Rational approximation, `arXiv:2501.00902`, January 2025.

[39] R. R. Coifman, Y. Meyer, *Wavelets: Calderón–Zygmund and Multilinear Operators*, Cambridge Studies in Advanced Mathematics 48, CUP, 1997.

[40] M. Berljafa, S. Güttel, The RKFIT algorithm for nonlinear rational approximation, *SIAM J. Sci. Comput.* 39(5), A2049–A2071, 2017. DOI 10.1137/15M1025426.

[41] B. Beckermann, A. Bultheel, J. Van Iseghem, Asymptotic behaviour of Padé and Padé-type approximants, *J. Comput. Appl. Math.* 99, 17–32, 1998.

[42] F. Bornemann, On the numerical evaluation of Fredholm determinants, *Math. Comp.* 79, 871–915, 2010. (Painlevé transcendent evaluation context.)

[43] R. Pachón, R. B. Platte, L. N. Trefethen, Piecewise-smooth chebfuns, *IMA J. Numer. Anal.* 30(4), 898–916, 2010. DOI 10.1093/imanum/drp008.

[44] D. Kressner, C. Tobler, Algorithm 941: htucker—a Matlab toolbox for tensors in hierarchical Tucker format, *ACM Trans. Math. Softw.* 40(3), 22:1–22, 2014.

[45] M. Mamoun, *Two Foci, One Cup: Successive World-Record Equations for Ellipse Perimeter*, ResearchGate publication 403296562, **30 March 2026**. `https://www.researchgate.net/publication/403296562`.

[46] M. Mamoun, *SRIRACHA: Singularity-Resolving Iterative Rational Approximation via Chebyshev Hierarchical Adaptation — A Universal Framework for Approximating Functions with Known Singularities*, ResearchGate publication 403309463, **30 March 2026**. `https://www.researchgate.net/publication/403309463`.

[47] F. W. J. Olver, D. W. Lozier, R. F. Boisvert, C. W. Clark, eds., *NIST Digital Library of Mathematical Functions*, Release 1.1.11 of 2023-09-15, `https://dlmf.nist.gov`.

Table 15: Twenty open questions in adjacent literature, with UNDERWOOD resolutions. Entries 1–5 are the detailed hooks above; 6–20 are one-line extensions. Every listed citation already appears in the bibliography of this paper.

#	Source	Open question (excerpt)	UNDERWOOD resolution
1	Nakatsukasa–Sète–Trefethen 2018 [1]	optimal rate for AAA on log branch	algebraic envelope vs. root-exp; Thm 2
2	Berljafa–Güttel 2017 [40]	extracting rational structure from singular fit	explicit log companions; §2
3	Beckermann et al. 1998 [41]	sharp Padé asymptotic constants	explicit measurement of $\log C^f(K) \approx a^f + b^f K$ across four log-branch targets; OQ1 + Thm 1
4	Bornemann 2010 [42]	Painlevé high-precision evaluation complexity	augmented-basis cookbook; §E
5	Driscoll–Hale–Trefethen 2014 [29]	when does augmentation beat interval splitting	Thm 7 answers quantitatively
6	Stahl 1997 [33]	$\exp(-c\sqrt{n})$ rational lower bound at branch pt	augmented basis is non-rational, escapes the bound; Thm 2
7	Newman 1964 [15]	root-exp rate for rational approx of $\|x\|$	augmented basis is non-rational; Newman class restriction does not apply (Thm 2)
8	Gopal–Trefethen 2019 [31]	constants for Laplace-corner rational rate	super-geometric envelope (fixed K); Thm 7
9	Trefethen–Nakatsukasa–Weideman 2021 [34]	why exponential node clustering works	exponentially clustered poles approximate the explicit log companions; §6 + Thm 2
10	Gonchar–Rakhmanov 1989 [32]	equilibrium measures for log-weighted minimax	explicit minimax via mpmath Remez at 40-digit equioscillation; Thm 5
11	Herremans–Huybrechs–Trefethen 2023 [37]	generic singular-function resolution	log-branch specialisation with K-sharpness amplification; §2 + Thm 7
12	Ayala–Raggi–Rendón-Marín 2025 [7]	optimal F_n exponent sequence	Villarino-matched $\{2, 3, \dots\}$ is unique; Thm 8
13	Moscato–Ciezak 2024 [19]	proving minimax for evolutionary-fit formulas	mpmath Remez certificate; Thm 5
14	Cantrell 2001–04 [17]	principled exponent choice in corrections	Villarino-forced; Thm 8
15	Boyd 1989 [23]	sharp log-Chebyshev coefficient constant	coefficient decay $O(n^{-(2s+1)})$ used in Lemma 3
16	Almkvist–Berndt 1988 [26]	optimal algebraic bounds on ellipse perimeter	F_n Remez minimax for ellipse perimeter; §4 + §C
17	Ma–Rokhlin–Wandzura 1996 [35]	dual basis for custom Gaussian quadrature	augmented basis is the approximation-dual of generalized Gaussian quadrature for log-weighted classes; §1 (intellectual ancestry)
18	Yarvin–Rokhlin 1998 [36]	SVD structure of singular integral operators	Haar property + Chebyshev alternation give exact minimax structure; Prop 3 + Thm 5
19	Belytschko–Black 1999 [8]	canonical crack-tip enrichment functions	augmented basis with crack-tip enrichment beats AAA by 147,000× on L-corner; §5
20	Moës–Dolbow–Belytschko 1999 [9]	element-local enrichment selection	singularity-matching theorem identifies the unique enrichment exponents; Thm 8